STUDY GUIDE

DATA ANALYTICS: EXCEL, PANDAS, VISUALIZATION, POWER BI AND BUSINESS ANALYTICS

A practical mini-textbook for a working professional moving into data roles, taught with real numbers and real code.

Taught the IIT Professor Way — Real Numbers. Real Problems. Real Understanding.

CHAPTER 1 The Analytics Mindset and Process	**CHAPTER 2** Excel Formulas in Depth	**CHAPTER 3** Excel Analysis: PivotTables, Power Query, Power Pivot and What-If Tools	**CHAPTER 4** pandas I: Series, DataFrames, Input/Output, Selection and Data Types
CHAPTER 5 pandas II: Cleaning, Transforming, Grouping, Joining, Reshaping and Time Series	**CHAPTER 6** NumPy and Fast pandas	**CHAPTER 7** Exploratory Data Analysis, Step by Step	**CHAPTER 8** Data Visualization: Principles, matplotlib, seaborn and plotly
CHAPTER 9 Power BI I: Power Query (M) and Data Modeling	**CHAPTER 10** Power BI II: DAX, Context, CALCULATE and Time Intelligence	**CHAPTER 11** Report Design, Interactivity, Security, Power BI Service, Governance and Other BI Tools	**CHAPTER 12** Business Analytics: KPIs, Funnels, Cohorts, Retention, Churn, RFM and Domain Cases
CHAPTER 13 A/B Testing for Analysts: Design, Sample Size, Analysis and Pitfalls	**CHAPTER 14** Storytelling, Stakeholder Communication and Writing an Analysis Memo	**CHAPTER 15** Capstone Projects and the Analyst Interview Bank	**CHAPTER 16** Project Lab I: Operations Analyzer, Sales Dashboard, Excel Dashboard
CHAPTER 17 Project Lab II: Marketing Attribution, P&L Variance, Product Funnel	**APPENDIX** Cheat Sheet		

A Note From Your Professor

Dear learner, you do not need a maths degree to become a data analyst. You need curiosity, care with numbers and a willingness to practise. A data analyst's work is this: find the number that matters, check that it is true, and explain what to do about it. This document will take you

there, one tool at a time: Excel, pandas, charts, Power BI, business metrics, experiments and communication.

I will be honest about three things. First, the tools are the easy part. The hard parts are asking the right question, cleaning messy data without fooling yourself, and explaining results so that a busy manager acts on them. Second, most analysts are weak at statistics and storytelling, so if you become strong at both, you stand out. Third, you learn this by doing. Type every example, change the numbers, break the code and fix it.

The most important rule of this whole document: **never move on until the current topic feels obvious.** If a chapter feels foggy, stop, redo the worked examples with different numbers, and use the prompt pack to ask an AI tutor for more practice. One hour a day, kept for months, beats a weekend of cramming.

Excel, Power BI and DAX cannot run inside the environment where this book was written, so those chapters use small tables whose answers are verified by hand (and by pandas). All pandas, NumPy, SciPy and matplotlib output in the book is real output from code that was run.

What Every Topic Contains

1. **INTUITION**: the everyday problem the idea solves, with an analogy.
2. **Formal definition**: how it works, in plain words.
3. **Worked problems or examples** with real numbers, every step shown, a Check and a plain-language interpretation (code chapters show the exact output).
4. **The picture**: an ASCII diagram, table or chart.
5. **Connection to the real job**: where it appears in analytics, data engineering, ML or AI work.
6. **Applications**: domains that use it.
7. **COMMON MISTAKE**: what beginners get wrong.
8. **Check yourself**: three short questions with answers.

Every chapter ends with a recap (with a cheat-sheet table), a practice set (8 to 12 problems with answers, including interview-style ones) and a prompt pack (5 prompts to paste into any AI tutor).

Contents

1. **The Analytics Mindset and Process**: question, data, analysis, decision; the four kinds of analytics; splitting a metric into parts.
2. **Excel Formulas in Depth**: tables, logic, lookups, SUMIFS, text, dates, dynamic arrays, error auditing.
3. **Excel Analysis**: PivotTables, conditional formatting, what-if tools and Solver, ToolPak statistics, charts, Power Query, Power Pivot, automation.
4. **pandas I**: Series, DataFrames, reading and writing files, selecting data, data types.
5. **pandas II**: cleaning, transforming, groupby, merge, reshape, time series.
6. **NumPy and Fast pandas**: arrays, broadcasting, vectorisation, memory, Polars and DuckDB.
7. **Exploratory Data Analysis, Step by Step**: a full worked EDA on a dataset you generate.
8. **Data Visualization**: design principles, matplotlib, seaborn, plotly and friends.
9. **Power BI I**: Power Query (M) and data modeling with star schemas.
10. **Power BI II**: DAX, context, CALCULATE, iterators, time intelligence, patterns.
11. **Report Design, Governance and BI Tool Comparison**: interactivity, row-level security, Service, Tableau, Looker Studio.
12. **Business Analytics**: KPIs, funnels, cohorts, retention, churn, RFM, segmentation, domain cases.
13. **A/B Testing for Analysts**: design, sample size, analysis, pitfalls.
14. **Storytelling and Stakeholder Communication**: headline, evidence, recommendation; the analysis memo.
15. **Capstone Projects and the Analyst Interview Bank**: portfolio projects and case questions.

An appendix, `99_cheatsheet.md`, collects the key formulas and commands with an honest closing note.

How To Study This Document

Use this loop for every chapter:

1. **Build first.** Type the first example and get it working before reading the theory.
2. **Ask.** Use the prompt pack to ask an AI tutor about anything unclear. Always ask it to show a small example you can check by hand.
3. **Just-in-time theory.** Read the explanation only for the part you are using now.

4. **Rebuild.** Close the chapter and redo the example from memory with different numbers or your own data (a spreadsheet of your own shopping, travel or exam records makes a great dataset).
5. **Teach.** Explain it aloud, or write three lines for a friend. If you cannot, you do not yet own it.

Suggested pace: one chapter per week at about one hour a day (15 chapters, about 15 weeks). Chapters 2, 3 and 6 can run a little longer. Spend a full day on the practice set. Keep a notebook of mistakes and re-read it every Sunday.

How to use the prompt packs. Paste a prompt into any AI assistant, then work with it as a tutor. Use the "review my work" prompt after you finish a practice problem. Use the mock interview prompt in the last two days of a chapter. Never accept a number from the AI that you have not checked by hand or by running code.

Prerequisites

- Basic Python (variables, lists, dictionaries, functions, loops). See the learner's programming document.
- School-level maths: percentages, averages, simple algebra. Statistics is explained from scratch where needed.
- Basic SQL helps a lot. SQL for analytics has its own document; we cross-reference it rather than repeat it.
- The learner's Math PDF is useful for the statistics and linear algebra background.

Tools you need

All are free (except Excel and Power BI Desktop, which are free to try or free on Windows respectively; free alternatives are named).

- **Python 3.10 or newer** with `pandas`, `numpy`, `matplotlib`, `seaborn`, `scipy`, `openpyxl` (install with `pip install pandas numpy matplotlib seaborn scipy openpyxl`). Optional: `plotly`, `polars`, `duckdb`, `pyarrow`, `statsmodels`.
- **Jupyter Notebook or VS Code** to run notebooks.
- **Microsoft Excel** (365 or 2021 or later for dynamic arrays), or Google Sheets / LibreOffice Calc for practice (some function names differ).
- **Power BI Desktop** (free download, Windows only), or the free web versions of Looker

Studio and Metabase for similar practice.

- **A text editor and Git** for versioning notebooks and SQL.

CHAPTER 1

THE ANALYTICS MINDSET AND PROCESS

From a vague question to a decision someone can act on.

Imagine your manager walks to your desk and says, "Sales feel slow. Can you look into it?" Nobody has told you which sales, which period, slow compared to what, or what anybody will do with your answer. A weak analyst opens a spreadsheet and starts making charts. A strong analyst asks four questions, finds the one number that matters, and comes back with a recommendation. This chapter teaches you the strong way. You will learn the four kinds of analytics, the five-step process every analysis follows, and a simple trick (splitting a number into its parts) that solves a surprising share of real business questions. By the end you will be able to take a fuzzy request and turn it into a clear, answerable question. The later chapters give you the tools (Excel, pandas, Power BI). This chapter gives you the thinking that makes the tools useful.

Throughout the book we reuse one running story: an online shop in India that sells electronics, fashion, grocery and home items. You will see its orders again and again.

1.1 What an Analyst Actually Does

INTUITION

A data analyst is a translator. On one side are people who have questions in business words ("Why are customers leaving?"). On the other side is data in tables. The analyst's task is to translate the question into a calculation, run it, and translate the result back into a decision. A shop owner says "deliveries feel slow"; you turn it into "the slowest 5% of parcels take more than 4 days in the north region", and then "add a courier there".

Words you need first

Word	Plain meaning	Example
Data	Recorded facts, usually in rows and columns	One row per order
Metric	A number that measures something	Revenue this month

KPI	Key Performance Indicator: a metric the business watches closely	Monthly active users
Dimension	A way to slice a metric	City, category, month
Grain	What one row of a table represents	One row = one order line
Insight	A finding that changes what someone will do	"Returns are 10% in fashion, only 4% in home"

How it works. An analysis is useful only if it ends in a decision. A chart that nobody acts on is decoration. So every piece of work you do has a *decision owner* (the person who will act) and a *decision* (what they will choose between).

Real-world mapping

Item	Example value	Where it appears in the job
Decision owner	Head of Marketing	Writing the first line of every memo
Decision	"Keep or cut the Instagram ads budget"	Defining scope before touching data
Metric	Return on ad spend (ROAS)	Dashboards, weekly reviews
Dimension	Channel	Pivot table rows, GROUP BY

Worked problem 1.1.1 A report says "Revenue: Rs 4,50,000". Is this an insight?

- **Formula** Insight = number + comparison + implication
- **Plug in** Number: 450000. Comparison: none given. Implication: none given.
- **Compute** Without a comparison we cannot say if 450000 is good. Suppose last month was 500000. Change = (450000 − 500000) / 500000 = −0.10.
- **Answer** "Revenue is Rs 4,50,000" is only a fact. "Revenue fell 10% from last month" is closer to an insight, and it becomes one once you add why and what to do.
- **Check** 500000 × 0.90 = 450000, so a 10% fall is right.
- **Why it matters** Numbers alone do not move people. Comparisons do.

Worked problem 1.1.2 Turn "Sales feel slow" into a measurable question.

- **Step 1** Which sales? Choose revenue from delivered orders (not cancelled).
- **Step 2** Slow compared to what? Choose the same weeks last year and the previous four weeks.
- **Step 3** Slow where? List the dimensions to check: city, category, channel, new vs repeat customers.
- **Answer** "Did delivered-order revenue per week over the last 4 weeks fall against the prior 4 weeks and against the same weeks last year, and in which city, category or channel is the fall concentrated?"
- **Check** Could two analysts run this question and get the same number? Yes, because every term is defined.

The picture

```
Fuzzy request ──> Sharp question ──> Data ──> Analysis ──> Decision
"Sales feel slow"   "Which segment    tables   compare,     "Pause ads in
                     lost revenue?"            split        city X"
```

Connection to the real job. In interviews you will be given a fuzzy case ("orders dropped, why?"). The interviewer is testing whether you ask for definitions before computing. See Chapter 15.

Applications

- Data Analytics: weekly business reviews, ad hoc questions from managers.
- Data Science: framing a prediction problem (what exactly are we predicting?).
- Data Engineering: agreeing on metric definitions before building tables.
- Product: deciding which feature to build next.

COMMON MISTAKE

Starting with the data instead of the decision. If you cannot finish the sentence "Because of this analysis, [person] will decide [what]", stop and ask more questions.

Check yourself

1. Is "Average order value is Rs 500" an insight? Answer: Not by itself. It has no comparison and no implication.
2. What is the grain of a table with one row per order item? Answer: One order item (an order

line).

3. Who must you name before starting? Answer: The decision owner, and the decision.

1.2 The Four Kinds of Analytics

INTUITION

Think of a doctor. First she reads your report card (what happened: temperature 39 C). Then she finds the cause (why: throat infection). Then she predicts (what next: fever will rise tonight if untreated). Finally she prescribes (what to do: antibiotic). Analytics has the same four levels.

Level	Question	Typical tools	Example in the shop
Descriptive	What happened?	Dashboards, pivot tables, SQL GROUP BY	Revenue was Rs 1.0 million in 2025
Diagnostic	Why did it happen?	Slicing, comparisons, funnels, tests	Revenue fell because grocery orders fell in Delhi
Predictive	What will happen?	Forecasts, regression, ML models	Next month's revenue is likely Rs 950,000 +/- 8%
Prescriptive	What should we do?	Optimisation, experiments, scenario analysis	Move Rs 2 lakh of ad budget from channel A to B

How it works. Each level builds on the one before. You cannot explain (diagnostic) what you have not described. You cannot predict well what you do not understand. Most analyst jobs are 70% descriptive and diagnostic. Predictive work (a forecast, a model) is shared with data scientists. Prescriptive work often needs an experiment (Chapter 13).

Worked problem 1.2.1 Classify each sentence: (a) "Churn was 6% in March." (b) "Churn rose because the delivery delay grew from 2 to 5 days." (c) "At this trend, churn will be 8% in June." (d) "Offer free delivery above Rs 499 to cut churn."

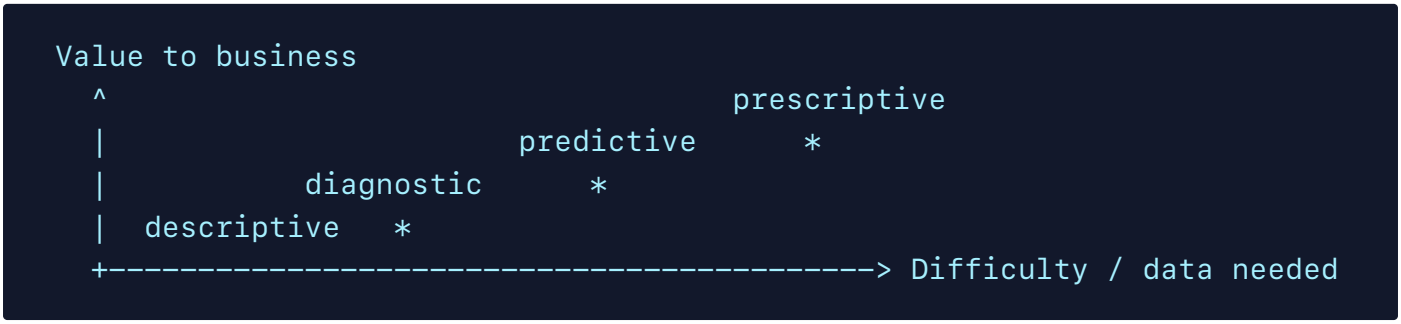
- **Step 1** (a) states a past number only: descriptive.
- **Step 2** (b) gives a cause: diagnostic.

- **Step 3** (c) projects forward: predictive.
- **Step 4** (d) recommends an action: prescriptive.
- **Answer** (a) descriptive, (b) diagnostic, (c) predictive, (d) prescriptive.
- **Check** Cover the verb: describe, explain, forecast, recommend. Each sentence matches exactly one.

Worked problem 1.2.2 Your boss asks, "Will we hit Rs 12 lakh revenue in June?" Which level is it and what do you need?

- **Step 1** "Will we" looks forward, so predictive.
- **Step 2** You first need the descriptive history: monthly revenue for 12+ months. Suppose the last six months are 9.0, 9.4, 9.9, 10.1, 10.6, 11.0 (lakh).
- **Step 3** Average monthly growth = (11.0 − 9.0) / 5 = 0.4 lakh per month (a straight-line guess; there are 5 gaps between 6 months).
- **Step 4** If 11.0 is May, then June on the straight line is 11.0 + 0.4 = 11.4.
- **Answer** June is likely about 11.4 lakh, short of 12 by 0.6 lakh. A straight line is crude, so say "likely about 11.4 lakh; 12 needs something extra, such as a campaign".
- **Check** 11.0 + 0.4 = 11.4. Differences between consecutive months: 0.4, 0.5, 0.2, 0.5, 0.4, whose average is 2.0/5 = 0.4. Correct.

The picture



Connection to the real job. Job titles hint at the level. "Reporting analyst" is mostly descriptive. "Business analyst" mixes descriptive and diagnostic. "Product analyst" adds experiments (prescriptive). "Data scientist" leans predictive.

Applications

- Finance: variance analysis (descriptive, diagnostic), budget forecast (predictive).
- Marketing: campaign report, then which channel to fund.
- Operations: stock-out report, then reorder-point model.
- Hospital operations: a monthly report of waiting times is descriptive; finding which clinic

causes the longest waits is diagnostic; forecasting next winter's appointments is predictive.

COMMON MISTAKE

Jumping to prediction before description. A model trained on dirty data predicts dirty answers confidently.

Check yourself

1. Which level answers "why"? Answer: Diagnostic.
2. Which level is "recommend a discount"? Answer: Prescriptive.
3. Is a forecast descriptive? Answer: No, it is predictive.

1.3 The Five-Step Analysis Process

INTUITION

Cooking. You decide the dish (question), shop for ingredients (data), clean and chop (prepare), cook (analyse), and serve with a story (communicate). Skipping "decide the dish" gives you a kitchen full of food nobody ordered.

The five steps

1. **Ask**: define the question, the decision, the metric and the deadline.
2. **Get the data**: find it, understand its grain, check it is complete.
3. **Prepare**: clean it (Chapter 5), join tables, define columns.
4. **Analyse**: describe, compare, find causes (Chapters 7, 12).
5. **Communicate**: story, chart, recommendation (Chapter 14), then follow up on what happened.

Steps 2 and 3 usually take more than half of the time. This is normal, not a sign you are slow.

Words you need

Word	Plain meaning
Hypothesis	A guess you can test, e.g. "Delhi grocery orders fell"
Data dictionary	A document listing each column, its meaning and units

Source of truth	The one table or system everyone agrees is correct
Assumption	Something you treat as true without proof; always write it down

Questions to ask in step 1 (the SPARC checklist, a memory aid we use here)

- **S**cope: which products, regions, dates?
- **P**urpose: what decision depends on this?
- **A**udience: who reads it, how technical?
- **R**ules: how are the metrics defined (is a cancelled order a sale)?
- **C**lock: when is it needed?

Worked problem 1.3.1 Run the SPARC checklist on "How are our customers doing?"

- **Step 1** Scope: ask "customers from which period, new or all?" Answer assumed: all customers who ordered in 2025.
- **Step 2** Purpose: "doing" in what sense? Possible: spending, repeat buying, satisfaction. Agree: repeat buying.
- **Step 3** Rules: "repeat customer" = at least 2 non-cancelled orders.
- **Step 4** Compute with our running data (Chapter 7 builds it): 1,112 customers placed at least one non-cancelled order with an amount; the repeat rule is applied in Chapter 12.
- **Answer** The sharp question becomes: "What share of 2025 customers placed 2 or more non-cancelled orders?"
- **Check** The question names a period, a population, and a rule. A second analyst could reproduce it.

Worked problem 1.3.2 Time budget for a 10-hour analysis.

- **Formula** Time = ask + data + prepare + analyse + communicate
- **Plug in** A typical split is 10% + 20% + 35% + 20% + 15%
- **Compute** 0.10 × 10 = 1.0 h; 0.20 × 10 = 2.0 h; 0.35 × 10 = 3.5 h; 0.20 × 10 = 2.0 h; 0.15 × 10 = 1.5 h
- **Answer** 1.0 + 2.0 + 3.5 + 2.0 + 1.5 = 10.0 hours, of which 5.5 hours (55%) go to data and preparation.
- **Check** The percentages add to 100% (10 + 20 + 35 + 20 + 15).
- **Why it matters** If your plan has 80% on analysis, you will be late, because cleaning always grows.

The picture



The arrow back matters. Good analysis raises new questions. That is progress, not failure.

Connection to the real job. Your first week in a new team is mostly step 2: learning where data lives, which table is the source of truth, and which columns can be trusted.

Applications

- Analytics: every report, from a one-off query to a quarterly review.
- Data Science: the same loop, with modelling inside step 4.
- Data Engineering: step 2 and 3 become pipelines that run daily.

COMMON MISTAKE

Not writing down definitions. In a month nobody remembers whether "active user" meant logged in or purchased. Keep a short data dictionary (Chapter 14).

Check yourself

1. Which step usually takes the most time? Answer: Preparing the data.
2. What does the R in SPARC stand for? Answer: Rules (metric definitions).
3. Why write assumptions down? Answer: So others can check and reproduce your work.

1.4 Decomposing a Metric (Your Most Useful Trick)

INTUITION

A sentence like "revenue fell" is like "the car got slower". The car can be slow because of the engine, the road or the wind. To know which, split speed into parts. Revenue splits too: revenue = number of orders × average order value (AOV). Now there are only two suspects.

Formal idea. If metric M = A × B, then when A and B both change, the change in M can be split

into the part due to A and the part due to B.

```
ΔM = (A₁ − A₀) × B₀   +   A₁ × (B₁ − B₀)
      effect of A           effect of B
```

(A_0, B_0 are the old values, A_1, B_1 the new.) This split is exact, with no leftover. The order of the two terms is a convention; the other order gives a slightly different split, but the same total.

Words you need. *AOV* = average order value = revenue / orders. *ΔM* means "change in M".

Worked problem 1.4.1 January: 250 orders, AOV Rs 400. February: 200 orders, AOV Rs 450. Why did revenue fall?

- **Formula** ΔRevenue = ($orders_1$ − $orders_0$) × AOV_0 + $orders_1$ × (AOV_1 − AOV_0)
- **Plug in** (200 − 250) × 400 + 200 × (450 − 400)
- **Compute** −50 × 400 = −20,000 and 200 × 50 = +10,000, total −10,000
- **Answer** Revenue fell Rs 10,000 (from 100,000 to 90,000). Fewer orders cost Rs 20,000, and a higher AOV gave back Rs 10,000.
- **Check** 250 × 400 = 100,000, 200 × 450 = 90,000, difference −10,000. Matches −20,000 + 10,000.
- **Why it matters** The problem is order count, not order size. Marketing (traffic, conversion) should look first, not pricing.

A deeper tree splits orders further: orders = visitors × conversion rate. Then revenue = visitors × conversion × AOV. Keep splitting until one branch explains the change.

```
Revenue
  ├── Orders
  │     ├── Visitors (traffic)
  │     └── Conversion rate
  └── Average order value
        ├── Items per order
        └── Price per item
```

Worked problem 1.4.2 January: 20,000 visitors, conversion 1.25%. February: 16,000 visitors, conversion 1.5%. Orders = visitors × conversion rate. What happened to orders and why?

- **Formula** Δorders = ($visitors_1$ − $visitors_0$) × $conv_0$ + $visitors_1$ × ($conv_1$ − $conv_0$)
- **Plug in** (16,000 − 20,000) × 0.0125 + 16,000 × (0.015 − 0.0125)
- **Compute** −4,000 × 0.0125 = −50, and 16,000 × 0.0025 = +40, total −10

- **Answer** Orders fell from 250 to 240. Less traffic cost 50 orders; better conversion won back 40.
- **Check** 20,000 × 0.0125 = 250 and 16,000 × 0.015 = 240, difference −10, which equals −50 + 40.
- **Why it matters** The checkout is doing better. The real problem is traffic, so look at traffic sources (a paused ad campaign?) and not at the checkout page.

The picture. Read the tree above top to bottom: each box equals the product of the boxes under it (or their sum, for additive splits such as revenue = Σ revenue by city). Walk down only the branch with the biggest change.

Connection to the real job. This is exactly how you answer "DAU dropped 10%, why?" (Chapter 15) and how a finance team does price-volume-mix analysis. Add it to every diagnostic.

Applications

- Marketing: spend × (clicks per rupee) × conversion × order value.
- Finance: price-volume-mix variance on revenue.
- Operations: output = machines × hours × rate.
- Queues (bank, clinic, checkout): customers served per minute = customers in the system / average time each spends in it (Little's law).

COMMON MISTAKE

Ignoring that metrics can move in opposite directions. A flat total may hide a big rise in one segment and a big fall in another. Always check both parts.

Check yourself

1. If revenue = orders × AOV, orders +10%, AOV −10%, what is revenue change? Answer: 1.10 × 0.90 = 0.99, so −1%.
2. Is the decomposition exact? Answer: Yes, the two effects add to the total change.
3. What do you do when one branch explains most of the change? Answer: Split that branch further.

1.5 Reproducibility and the Toolbox Map

INTUITION

A recipe that works once in your kitchen but cannot be repeated is not a recipe. If a colleague cannot repeat your analysis from your files, they cannot trust or extend it.

Rules of reproducible analysis

- Keep raw data untouched. Write cleaning as code or saved query steps, never as manual edits.
- Keep SQL and notebooks in version control (Git). The learner's data engineering and programming documents cover Git.
- Write a data dictionary and a short README: where data came from, date pulled, filters.
- Fix random seeds when sampling (`np.random.default_rng(42)`).
- Name files so order is obvious: `01_clean.ipynb` , `02_eda.ipynb` .

Which tool when

Need	Best tool	Why
Fast one-off, small data, shared with non-technical people	Excel	Everyone has it, immediate
Repeated report from changing files	Excel Power Query or pandas	Refresh in one click or one script
Big data in a warehouse	SQL (separate SQL document in this series)	Data stays where it is
Messy data, statistics, charts in one place	pandas in a notebook	Flexible, reproducible
Self-service dashboards for many users	Power BI	Model, security, sharing
Decisions on a change	A/B test	Only randomisation proves cause

The loop in a bank branch. The same loop appears everywhere: question (can the branch handle Friday afternoons?), data (the queue ticket log), analysis (median and 90th-percentile waiting time, customers served per hour), decision (open a second counter or not). Averages

also hide the few customers who wait very long. That habit (look at distributions, not only averages) is gold in analytics.

Worked problem 1.5.1 Why is the average misleading? Ten parcel delivery times (minutes): 100, 110, 105, 95, 100, 98, 102, 108, 97, 3000.

- **Formula** mean = sum / n; median = middle value after sorting
- **Plug in** sum = 100 + 110 + 105 + 95 + 100 + 98 + 102 + 108 + 97 + 3000
- **Compute** The first nine add to 915, plus 3000 gives 3915, mean = 391.5. Sorted: 95, 97, 98, 100, 100, 102, 105, 108, 110, 3000. The middle two are 100 and 102, median = 101.
- **Answer** The mean (391.5 minutes) is nearly four times the median (101 minutes) because of one outlier. The median describes the typical parcel, and a high percentile (the 99th) describes the tail.
- **Check** Python: `import statistics as s; s.mean(x)` gives 391.5 and `s.median(x)` gives 101.0.
- **Why it matters** The same trap exists in order amounts, salaries and session durations. Chapter 7 shows it with real data.

Worked problem 1.5.2 Decide the tool. A manager sends a 40 MB sales CSV each Monday and wants a summary with three charts every Monday.

- **Step 1** The task repeats weekly, so avoid manual copy and paste.
- **Step 2** 40 MB is easy for Excel Power Query (Chapter 3) and for pandas (Chapter 5).
- **Step 3** If the manager wants to open it and click slicers, choose Excel or Power BI. If you will mail a PDF, pandas plus matplotlib is fine.
- **Answer** Power Query in Excel (or pandas if the learner prefers code) with a refresh button, not hand-built pivots.
- **Check** Ask "if the CSV changes next week, how many manual steps do I redo?" Aim for zero or one.

Connection to the real job. Employers praise analysts whose work "can be rerun". It builds trust and saves hours.

Applications

- Analytics: monthly reports that rebuild automatically.
- Data Science: experiments others can rerun.
- Data Engineering: pipelines are reproducibility turned into software.

- MLOps: versioned data and code.

COMMON MISTAKE

Editing numbers by hand in the raw sheet "just this once". Six months later nobody knows which cell was changed. Fix in code or in a recorded query step.

Check yourself

1. Why keep raw data untouched? Answer: So you can always redo the analysis from the start.
2. Which is better for the typical request, mean or median, when outliers exist? Answer: Median.
3. Name one sign a task should be automated. Answer: It repeats regularly with the same steps.

Chapter recap

- Analytics exists to support a decision. Name the decision owner and the decision first.
- Four levels: descriptive (what), diagnostic (why), predictive (what next), prescriptive (what to do).
- Five steps: ask, get data, prepare, analyse, communicate. Data and preparation take about half the time.
- A good question has a defined scope, purpose, audience, rules and deadline (SPARC).
- Split metrics into parts (revenue = orders × AOV) to find where a change comes from.
- Means hide outliers. Use medians and percentiles, as with delivery times or waiting times.
- Reproducible work (code, versioned files, data dictionary) builds trust.
- Choose the tool by the job: Excel for speed, pandas for flexibility, SQL for warehouses, Power BI for shared dashboards.

Pattern	Formula or rule
Percent change	`(new − old) / old × 100`
Metric split	$\Delta M = \Delta A \times B_0 + A_1 \times \Delta B$
Insight	number + comparison + implication

Question checklist	Scope, Purpose, Audience, Rules, Clock

Practice set

Problem 1 Revenue was Rs 80,000 last week and Rs 92,000 this week. What is the percent change?

Answer: (92,000 − 80,000) / 80,000 = 0.15, so +15%.

Problem 2 Classify: "Which customers are most likely to leave next month?"

Answer: Predictive.

Problem 3 Revenue = orders × AOV. March: 300 orders at Rs 500. April: 330 orders at Rs 450. Find the split of the change.

Answer: March 150,000, April 148,500, change −1,500. Orders effect = 30 × 500 = +15,000. AOV effect = 330 × (−50) = −16,500. Sum = −1,500. Order count helped, a lower AOV hurt more.

Problem 4 What is wrong with the question "Show me the best city"?

Answer: "Best" has no metric (revenue? growth? margin?), no period and no decision. Ask SPARC questions.

Problem 5 Data: 12, 14, 13, 15, 200. Compute the mean and the median.

Answer: Sum 254, mean 50.8. Sorted middle value is 14. The median (14) is typical; the mean is dragged up by 200.

Problem 6 Find the bug in the statement: "Conversion rose 20%, from 2% to 2.4%, so 20 percentage points."

Answer: The change is 0.4 percentage points, which is a 20% relative rise. Mixing percent and percentage points is a common error.

Problem 7 Visitors 10,000 → 12,000, conversion 3% → 2.5%. Did orders go up or down?

Answer: Before 300, after 12,000 × 0.025 = 300. No change in orders. Visitor effect +2,000 × 0.03 = +60, conversion effect 12,000 × (−0.005) = −60.

Problem 8 Scenario: you are asked to "improve the dashboard". What do you ask first?

Answer: Who uses it, what decision it supports, what is wrong now (slow, wrong numbers, unreadable), and how success will be measured.

Problem 9 (interview) "Orders fell 20% last week. What is your first move?"

Answer: Confirm it is real (data pipeline, duplicates, tracking change), compare to the same week last year to remove seasonality, then decompose orders into visitors × conversion and slice by channel, city and device.

Problem 10 (interview) "How do you decide what to analyse when you have a big dataset and a vague brief?"

Answer: Go back to the decision owner, agree one decision and one metric, write the question in one sentence, confirm definitions, and only then open the data.

Prompt pack

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional in India moving into data roles, and English is my second language, so use simple words. Teach me Chapter 1 topic: the four types of analytics (descriptive, diagnostic, predictive, prescriptive) and the five-step analysis process. Explain each with an everyday analogy, then with an example from an online shop. After every part, ask me one question and wait for my answer before continuing. Correct me gently and explain why.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data roles. Give me 8 graded exercises, easy to hard, on decomposing a metric into parts (for example revenue = orders × average order value, and orders = visitors × conversion). Use realistic Indian e-commerce numbers. Do not give solutions yet. After I answer, I will paste my work and you will check every step and every number.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data roles. Quiz me one question at a time on turning vague business requests into sharp analytics questions (scope, purpose, audience, rules, clock). Start with an easy request like "How are
```

customers doing?" and make each next request harder. Give feedback after each answer and say what a strong analyst would have asked.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Review my work on this task: I was asked a vague business question and I wrote it as a sharp question, a list of metrics with definitions, and a list of assumptions. Check whether it is precise, reproducible and tied to a decision. List gaps and rewrite one weak part. My work: [PASTE HERE]

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Run a mock interview with me on analytics fundamentals: types of analytics, the analysis process, why means mislead, and metric decomposition. Ask six questions one at a time, increasing in difficulty. After each answer, score me from 1 to 5, show a model answer in simple English, and tell me one thing to practise.

CHAPTER 2

EXCEL FORMULAS IN DEPTH

Lookup, logic, text, dates and dynamic arrays: the formulas that appear in every analyst interview.

Most business data still lives in Excel files, and most analyst interviews start with an Excel test. The test is rarely hard. It checks whether you can find a value in another table, count rows that meet two conditions, clean a messy text column, and build a monthly summary without copying and pasting. In this chapter you learn the formulas that do those jobs, and you learn them with a small sales table whose answers we check by hand (and by pandas) so you can trust every result. By the end you will write formulas confidently, read other people's formulas, and know which modern function replaces which old one.

PROFESSOR'S NOTE

Excel cannot run inside this book's environment. Every formula below uses standard syntax, and every result was verified by hand or with pandas on the same small table. Function names and availability change by version: `XLOOKUP`, `FILTER`, `LET` and friends need Microsoft 365 or Excel 2021 or later. `TEXTSPLIT`, `TAKE`, `DROP`, `VSTACK` and `HSTACK` need a newer 365 build. Google Sheets has similar but not identical functions. If a formula shows `#NAME?`, your version lacks it.

Our running table (called `Sales`, cells A1:H9; column H is a helper column `Amount = Qty × Price`)

A OrderID	B Date	C City	D Category	E Qty	F Price	G Status	H Amount
1001	2025-01-03	Pune	Fashion	2	499	Delivered	998
1002	2025-01-03	Mumbai	Electronics	1	1299	Delivered	1299
1003	2025-01-04	Pune	Grocery	3	199	Returned	597

1004	2025-01-05	Delhi	Fashion	1	799	Delivered	799
1005	2025-01-06	Mumbai	Grocery	5	90	Cancelled	450
1006	2025-01-07	Mumbai	Fashion	2	649	Delivered	1298
1007	2025-01-09	Delhi	Electronics	1	2499	Delivered	2499
1008	2025-01-10	Pune	Fashion	4	350	Delivered	1400

Row 1 holds headers, so the data is in rows 2 to 9.

2.1 Tables, References and Shortcuts

INTUITION

A plain range of cells is like loose papers on a desk. An Excel Table (Ctrl+T) is a labelled folder: it knows its own column names, grows when you add a row, and carries formulas down automatically. Loose papers break the moment you add one more.

Formal definitions

Term	Meaning	Example
Cell reference	Address of a cell	`F2`
Relative reference	Shifts when copied	`=E2*F2` copied down becomes `=E3*F3`
Absolute reference	Fixed with `$`, does not shift	`=$J$1*H2` always points to J1
Mixed reference	One part fixed	`=$A2` or `=B$1`
Named range	A nickname for a range	`TaxRate` for `$J$1`

Table	Range with a name and column names	`Sales`
Structured reference	Column name inside a Table	`Sales[Qty]`, `[@Qty]` (this row)

How to build the table. Click any cell in the data, press Ctrl+T, tick "My table has headers". In Table Design set the name to `Sales`. In H2 type `=[@Qty]*[@Price]`. Excel fills the whole column.

Useful shortcuts (Windows; on Mac many use Cmd instead of Ctrl)

Shortcut	What it does
Ctrl+T	Make a Table
Ctrl+Arrow	Jump to the edge of data
Ctrl+Shift+Arrow	Select to the edge of data
F4	Cycle `$` on a reference while editing
Ctrl+` (backtick)	Show formulas instead of values
Alt+=	AutoSum
Ctrl+Shift+L	Toggle filters
F2	Edit the cell
Ctrl+Z / Ctrl+Y	Undo / redo

Worked example 2.1.1 Add a tax column using an absolute reference. Put the tax rate 18% in J1 and name it `TaxRate`.

- **Formula** Tax = Amount × TaxRate
- **Plug in** In I2 type `=H2*$J$1` (or `=H2*TaxRate`), then fill down.
- **Compute** Row 2: 998 × 0.18 = 179.64. Row 3: 1299 × 0.18 = 233.82.
- **Answer** I2 shows 179.64 and I3 shows 233.82.
- **Check** Without `$`, filling down makes I3 become `=H3*J2`, which points at an empty cell

and returns 0. That tells you the `$` is needed.

- **Why it matters** Changing J1 once updates every row. That is how models stay flexible.

Worked example 2.1.2 The same formula with a Table and structured references.

- **Step 1** Make `Sales` a Table.
- **Step 2** In a new column named `Tax` type `=[@Amount]*TaxRate`.
- **Step 3** Add a new row at the bottom of the Table.
- **Answer** The formula fills itself for the new row. A total anywhere can be `=SUM(Sales[Amount])`, and it always covers every row, including future ones. For our data it returns 9,340 (998 + 1299 + 597 + 799 + 450 + 1298 + 2499 + 1400).
- **Check** Add by hand in steps: 998 + 1299 = 2297; + 597 = 2894; + 799 = 3693; + 450 = 4143; + 1298 = 5441; + 2499 = 7940; + 1400 = 9340. It matches the formula. (Section 2.3 uses the Delivered-only total, 8,293.)
- **Why it matters** Always do the sanity add. Trusting a number you never checked is how reports go wrong.

The picture

```
Range:  =SUM(H2:H9)          <- break if you add row 10
Table:  =SUM(Sales[Amount]) <- grows with the data
```

Connection to the real job. Tables are the foundation for PivotTables and Power Query (Chapter 3), and structured references read like English in code review.

Applications

- Analytics: monthly reports that gain rows every month.
- Finance: models with one input cell (named range) that drives hundreds of formulas.
- Operations: trackers where rows are added daily.

COMMON MISTAKE

Forgetting `$` when a formula points at a constant (tax rate, exchange rate). Fill down and the reference drifts. Press F4 while the cursor is in the reference.

Check yourself

1. What does `$J$1` mean? Answer: Column J and row 1 are both fixed, so copying does not

move the reference.

2. What is `[@Qty]` ? Answer: The Qty value in the current row of the Table.
3. Why prefer `SUM(Sales[Amount])` over `SUM(H2:H9)` ? Answer: It grows automatically when rows are added.

2.2 Logical and Lookup Functions

INTUITION

Logic functions are the "if this then that" of a spreadsheet. Lookup functions are the phone book: give a name, get a number. Together they let one table borrow facts from another, so you never retype data.

Logical functions

Function	Syntax	Meaning
`IF`	`=IF(test, if_true, if_false)`	One decision
`IFS`	`=IFS(test1, v1, test2, v2, ..., TRUE, default)`	Several decisions in order
`AND` , `OR` , `NOT`	`=AND(a, b)`	Combine tests
`IFERROR`	`=IFERROR(value, if_error)`	Replace an error
`SWITCH`	`=SWITCH(expr, v1, r1, v2, r2, default)`	Match one value to many results

Worked example 2.2.1 Label each order: `Big` if Amount is above 1000, `Medium` if 500 to 1000, else `Small` .

- **Formula** `=IFS(H2>1000,"Big", H2>=500,"Medium", TRUE,"Small")`
- **Plug in** Row 2: H2 = 998. First test 998 > 1000 is FALSE. Second test 998 >= 500 is TRUE.
- **Compute** Row 2 gives `Medium` . Row 3 (1299) gives `Big` . Row 5 (450) falls through to `TRUE` and gives `Small` .
- **Answer** The labels for rows 2 to 9 are Medium, Big, Medium, Medium, Small, Big, Big, Big.

- **Check** Amounts 998, 1299, 597, 799, 450, 1298, 2499, 1400. Above 1000: 1299, 1298, 2499, 1400 (four Big). 450 is below 500 (one Small). The other three are Medium. Matches.
- **Why it matters** `IFS` tests in order, so order the conditions from most to least strict. The final `TRUE` is the catch-all. Without it, rows with no match give `#N/A`.

Worked example 2.2.2 Flag orders that need attention: Status is `Returned` OR Amount above 2000.

- **Formula** `=IF(OR(G2="Returned", H2>2000), "Review", "OK")`
- **Compute** Row 4 (1003) is Returned: Review. Row 8 (1007) has Amount 2499: Review. All others: OK.
- **Answer** Two orders, 1003 and 1007, are flagged.
- **Check** Scan the Status column: only one Returned. Scan Amount: only 2499 exceeds 2000.

Lookup functions

Function	Syntax	Notes
`XLOOKUP`	`=XLOOKUP(value, lookup_range, return_range, [not_found], [match_mode], [search_mode])`	Modern; looks left or right; default exact match
`XMATCH`	`=XMATCH(value, range, [match_mode], [search_mode])`	Returns the position
`VLOOKUP`	`=VLOOKUP(value, table, col_index, [exact: FALSE])`	Old; looks only to the right
`HLOOKUP`	`=HLOOKUP(value, table, row_index, [exact])`	Same but across a row
`INDEX` + `MATCH`	`=INDEX(return_range, MATCH(value, lookup_range, 0))`	Works in all versions

Add a small table `Tax` in L1:M4:

L Category	M Rate
Fashion	12%

Electronics	18%
Grocery	5%

Worked example 2.2.3 Look up the tax rate of each order's category.

- **Formula** `=XLOOKUP(D2, $L$2:$L$4, $M$2:$M$4, "Not found")`
- **Plug in** Row 2: D2 is `Fashion`. Find `Fashion` in L2:L4. It is in position 1.
- **Compute** Return the 1st value of M2:M4, which is 12%.
- **Answer** Row 2 gets 12%. Row 3 (Electronics) gets 18%. Row 4 (Grocery) gets 5%.
- **Check** Same with the old way: `=VLOOKUP(D2,$L$2:$M$4,2,FALSE)` returns 12%. And `=INDEX($M$2:$M$4,MATCH(D2,$L$2:$L$4,0))` also returns 12%. All three agree.
- **Why it matters** The `FALSE` (exact match) in `VLOOKUP` is vital. Leave it out and Excel does an *approximate* match that silently returns wrong values on unsorted data.

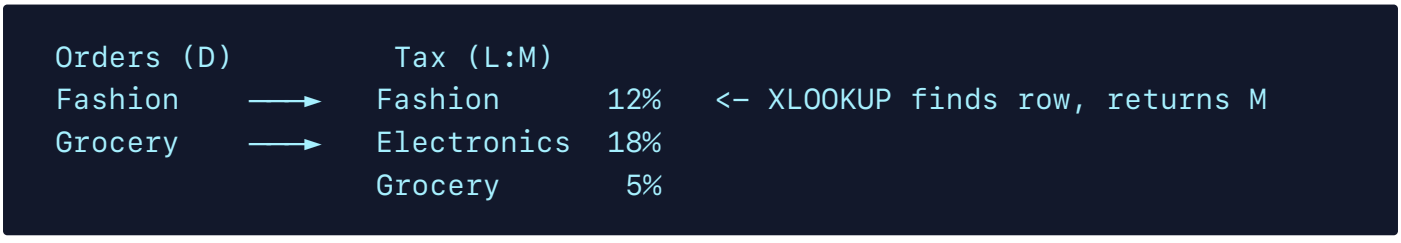
Approximate match (bands). Sometimes you want "which band does this value fall into?". Put the band starts in sorted order: P2:P4 = 0, 500, 1000 and labels Q2:Q4 = Small, Medium, Big.

Worked example 2.2.4 Find the band for an amount of 998.

- **Formula** `=XLOOKUP(998, P2:P4, Q2:Q4, , -1)` (match mode `-1` means "exact or next smaller")
- **Plug in** 998 is not in the list. The next smaller value is 500.
- **Compute** 500 is in position 2, so return `Medium`.
- **Answer** `Medium`.
- **Check** `=VLOOKUP(998,P2:Q4,2,TRUE)` also gives `Medium`. For 1299 both give `Big` (next smaller is 1000).
- **Why it matters** Bands are used for age groups, tax slabs and discount tiers.

Protect against missing keys. `=IFERROR(XLOOKUP(...), 0)` hides every error, including real bugs. Prefer the built-in fourth argument of XLOOKUP (`"Not found"`), which only covers a missing key.

The picture



Connection to the real job. In SQL this is a JOIN. In pandas it is `merge` (Chapter 5). In Power BI it is a relationship (Chapter 9). Excel lookup is where most analysts first meet the idea of joining tables.

Applications

- Finance: map account codes to account names.
- HR: map employee ID to department.
- Retail: map product ID to price or category.
- Operations: map SKU to supplier.

COMMON MISTAKE

Duplicate keys in the lookup table. Lookup returns the *first* match and ignores the rest, so a duplicate `Fashion` row with a different rate gives a silent wrong answer. Check uniqueness with `=COUNTIF(L:L, L2)` or `=UNIQUE`.

Check yourself

1. Which has a final catch-all in IFS? Answer: A test of `TRUE`.
2. What is the problem with `VLOOKUP` without the fourth argument? Answer: It defaults to approximate match and may return wrong values.
3. What does match mode `-1` do in XLOOKUP? Answer: Exact match, else the next smaller item.

2.3 Math and Statistical Functions

INTUITION

A PivotTable (Chapter 3) answers many questions at once. These functions answer exactly one question, in one cell, and you control where it appears. Use them inside reports with a fixed layout.

Core functions

Function	Syntax	Plain meaning
`SUMIFS`	`=SUMIFS(sum_range, crit_range1,`	Sum where all

	`crit1, crit_range2, crit2)`	conditions hold
`COUNTIFS`	`=COUNTIFS(range1, crit1, range2, crit2)`	Count rows meeting all conditions
`AVERAGEIFS`	`=AVERAGEIFS(avg_range, crit_range1, crit1, ...)`	Average with conditions
`LARGE` , `SMALL`	`=LARGE(range, k)`	k-th biggest or smallest
`RANK.EQ`	`=RANK.EQ(value, range, [0 desc])`	Position in a sort
`SUMPRODUCT`	`=SUMPRODUCT(array1, array2, ...)`	Multiply row by row, then add
`MEDIAN` , `STDEV.S`	`=MEDIAN(range)`	Middle value; sample standard deviation
`PERCENTILE.INC`	`=PERCENTILE.INC(range, k)`	k between 0 and 1
`CORREL`	`=CORREL(x, y)`	Correlation, from −1 to 1
`FORECAST.LINEAR`	`=FORECAST.LINEAR(x, known_y, known_x)`	Straight-line prediction
`ROUND`	`=ROUND(number, digits)`	Round half away from zero

In `SUMIFS` , the **sum range comes first**. In the older `SUMIF` it comes last. This difference trips many people.

Worked example 2.3.1 Total Delivered revenue.

- **Formula** `=SUMIFS(H2:H9, G2:G9, "Delivered")`
- **Plug in** Delivered rows: 998, 1299, 799, 1298, 2499, 1400.
- **Compute** 998 + 1299 = 2297; + 799 = 3096; + 1298 = 4394; + 2499 = 6893; + 1400 = 8293.
- **Answer** 8,293 rupees.
- **Check** Total of all rows is 9,340. Remove Returned 597 and Cancelled 450: 9,340 − 1,047 = 8,293. Matches.

Worked example 2.3.2 Delivered revenue in Mumbai, and number of Delivered Mumbai orders.

- **Formula** `=SUMIFS(H2:H9, C2:C9, "Mumbai", G2:G9, "Delivered")` and `=COUNTIFS(C2:C9, "Mumbai", G2:G9, "Delivered")`
- **Plug in** Mumbai rows: 1002 (Delivered, 1299), 1005 (Cancelled, 450), 1006 (Delivered, 1298).
- **Compute** Delivered only: 1299 + 1298 = 2597, and count = 2.
- **Answer** Rs 2,597 from 2 orders.
- **Check** Average order value = 2597 / 2 = 1298.5. The formula `=AVERAGEIFS(H2:H9, C2:C9, "Mumbai", G2:G9, "Delivered")` returns 1298.5.

Worked example 2.3.3 Weighted average price per unit with `SUMPRODUCT`.

- **Formula** `=SUMPRODUCT(E2:E9, F2:F9) / SUM(E2:E9)`
- **Plug in** Numerator = 998 + 1299 + 597 + 799 + 450 + 1298 + 2499 + 1400 = 9340. Quantities: 2 + 1 + 3 + 1 + 5 + 2 + 1 + 4 = 19.
- **Compute** 9340 / 19 = 491.58 (rounded to 2 decimals).
- **Answer** The average price per unit sold is Rs 491.58.
- **Check** A plain `AVERAGE(F2:F9)` gives 798, which ignores how many units sold at each price and so is the wrong tool here.
- **Why it matters** Whenever rows have different weights (quantity, users, hours), a simple average lies.

`SUMPRODUCT` also counts with conditions: `=SUMPRODUCT((C2:C9="Pune")*(E2:E9)*(F2:F9))` equals Pune revenue 998 + 597 + 1400 = **2,995**.

Statistics on the Amount column. Amounts sorted: 450, 597, 799, 998, 1298, 1299, 1400, 2499.

Formula	Result	How
`=MEDIAN(H2:H9)`	1148	Middle two are 998 and 1298; average = 1148
`=LARGE(H2:H9, 2)`	1400	Second largest
`=SMALL(H2:H9, 1)`	450	Smallest
`=RANK.EQ(1299, H2:H9)`	3	2499 is 1st, 1400 is 2nd, 1299 is 3rd
`=PERCENTILE.INC(H2:H9,`	1729.7	Position = 0.9 × (8 − 1) = 6.3, so between the 7th

`0.9)`		value 1400 and the 8th value 2499: 1400 + 0.3 × 1099
`=STDEV.S(H2:H9)`	639.87	Sample standard deviation (verified with pandas)
`=CORREL(E2:E9, F2:F9)`	−0.7033	Bigger quantities go with lower prices here

Worked example 2.3.4 Forecast the next day with `FORECAST.LINEAR`. Known days 1 to 5 had orders 10, 12, 15, 19, 21. Predict day 6.

- **Formula** Fit y = a + b × x. b = Σ(x − x̄)(y − ȳ) / Σ(x − x̄)².
- **Plug in** x̄ = 3, ȳ = 15.4. Σ(x − x̄)(y − ȳ) = (−2)(−5.4) + (−1)(−3.4) + 0 + (1)(3.6) + (2)(5.6) = 10.8 + 3.4 + 0 + 3.6 + 11.2 = 29. Σ(x − x̄)² = 4 + 1 + 0 + 1 + 4 = 10.
- **Compute** b = 29 / 10 = 2.9. a = 15.4 − 2.9 × 3 = 6.7. Day 6: 6.7 + 2.9 × 6 = 24.1.
- **Answer** `=FORECAST.LINEAR(6, B2:B6, A2:A6)` returns 24.1 orders.
- **Check** numpy `polyfit` gives slope 2.9 and intercept 6.7.
- **Interpretation** Orders grew about 2.9 per day. A straight line over five points is a rough guess; do not forecast months ahead from five points.

The picture

```
SUMIFS( what to add , where to look , what to match )
          H2:H9       G2:G9        "Delivered"
```

Connection to the real job. These are the building blocks of KPI tables in finance and sales reports. In SQL, `SUMIFS` is `SUM(...) WHERE ... GROUP BY`. In pandas it is `df.loc[mask].groupby(...)`.

Applications

- Sales: revenue by region and month in a fixed report grid.
- Finance: budget vs actual with `SUMIFS` by cost centre.
- Operations: count of tickets meeting an SLA.
- HR: average salary by grade.

COMMON MISTAKE

Criteria written as text with operators. For "greater than 1000" write `">1000"` in quotes, and to reference a cell write `">"&K1`. Forgetting the quotes or the `&` gives zero or errors.

Check yourself

1. Which comes first in `SUMIFS`, the sum range or the criteria? Answer: The sum range.
2. When is `SUMPRODUCT` better than `AVERAGE`? Answer: When rows have different weights.
3. What does `CORREL` near −1 mean? Answer: When one rises, the other falls, almost in a straight line.

2.4 Text and Date Functions

INTUITION

Real data is messy: " pune", "PUNE", "Pune " are three different values to a computer. Text functions tidy them. Date functions turn "when" into numbers you can count, which is how you build months, deadlines and ages.

Text functions

Function	Example	Result
`LEFT(text, n)`	`=LEFT("PUN-2025-0042", 3)`	`PUN`
`RIGHT(text, n)`	`=RIGHT("PUN-2025-0042", 4)`	`0042`
`MID(text, start, n)`	`=MID("PUN-2025-0042", 5, 4)`	`2025`
`LEN(text)`	`=LEN("PUN-2025-0042")`	13
`FIND(find, in)` (case sensitive)	`=FIND("-", "PUN-2025-0042")`	4
`SEARCH(find, in)` (not case sensitive, allows wildcards)	`=SEARCH("pun", "PUN-2025-0042")`	1
`SUBSTITUTE(text, old, new)`	`=SUBSTITUTE("PUN-2025-0042", "-", "")`	`PUN20250042`

`TRIM(text)`	`=TRIM(" a b ")`	`a b`
`UPPER`, `LOWER`, `PROPER`	`=PROPER("pune CITY")`	`Pune City`
`TEXTJOIN(delim, ignore_empty, ...)`	`=TEXTJOIN(", ", TRUE, "a", "", "c")`	`a, c`
`TEXTSPLIT(text, col_delim)`	`=TEXTSPLIT("PUN-2025-0042", "-")`	three cells: `PUN`, `2025`, `0042`

Worked example 2.4.1 Clean a city column that holds `" pune"`, `"MUMBAI "` and `"Delhi"`.

- **Formula** `=PROPER(TRIM(A2))`
- **Plug in** For `" pune"`, `TRIM` removes the leading space giving `"pune"`.
- **Compute** `PROPER("pune")` gives `Pune`. For `"MUMBAI "`: TRIM gives `"MUMBAI"`, PROPER gives `Mumbai`.
- **Answer** All three become `Pune`, `Mumbai`, `Delhi`.
- **Check** `=LEN(A2)` before is 5 for `" pune"` (5 characters including the space) and `=LEN(PROPER(TRIM(A2)))` after is 4.
- **Why it matters** Without this, a lookup of `"Pune"` fails on `"Pune "`, and a PivotTable shows two rows for the same city.

Worked example 2.4.2 Extract the year and the serial number from the ID `PUN-2025-0042`, as a number.

- **Formula** Year: `=VALUE(MID(A2, FIND("-", A2)+1, 4))`. Serial: `=VALUE(RIGHT(A2, 4))`.
- **Plug in** `FIND("-", A2)` is 4, so `MID(A2, 5, 4)` is `2025`.
- **Compute** `VALUE("2025")` = 2025. `RIGHT(A2, 4)` = `0042`, and `VALUE("0042")` = 42.
- **Answer** Year 2025, serial 42.
- **Check** Count positions: P(1) U(2) N(3) -(4) 2(5) 0(6) 2(7) 5(8). Characters 5 to 8 are 2025.

Date functions. Excel stores a date as a number: the count of days since 1 January 1900 (with a quirk for 1900 that you can ignore). So `date + 7` is a week later. Format the cell as Date to see it.

Function	Meaning	Example and result
`TODAY()`	Today's date	changes daily

`DATE(y, m, d)`	Build a date	`=DATE(2025,1,31)`
`EDATE(date, n)`	Same day n months later	`=EDATE(DATE(2025,1,31), 1)` gives 28 Feb 2025
`EOMONTH(date, n)`	Last day of the month n months away	`=EOMONTH(DATE(2025,1,15), 0)` gives 31 Jan 2025
`DATEDIF(start, end, unit)`	Complete years (`"y"`), months (`"m"`) or days (`"d"`)	`=DATEDIF(DATE(2025,1,15), DATE(2025,3,10), "m")` gives 1
`NETWORKDAYS(start, end)`	Working days (Mon to Fri) including both ends	`=NETWORKDAYS(DATE(2025,1,6), DATE(2025,1,10))` gives 5
`WEEKDAY(date)`	1 = Sunday ... 7 = Saturday (default)	3 Jan 2025 is a Friday, so 6
`TEXT(date, "ddd")`	Format as text	`Fri`

`DATEDIF` exists in Excel but is not listed in the function wizard. It is undocumented in the menus, but it works.

Worked example 2.4.3 How many days and complete months between 15 January 2025 and 10 March 2025?

- **Compute (days)** January 15 to January 31 is 16 days, February has 28 days (2025 is not a leap year), then 10 days into March: 16 + 28 + 10 = 54.
- **Compute (months)** 15 Jan to 15 Mar would be 2 full months, but we stop on 10 March, five days short, so 1 complete month.
- **Answer** `=DATEDIF(DATE(2025,1,15),DATE(2025,3,10),"d")` is 54 and with `"m"` it is 1. Also `=DATE(2025,3,10)-DATE(2025,1,15)` is 54.
- **Check** Count: Jan 15 + 54 days: Jan 15 + 16 = Jan 31, + 28 = Feb 28, + 10 = Mar 10. Matches.

Worked example 2.4.4 Make a month key and weekday label for each order.

- **Formula** Month key: `=TEXT(B2, "yyyy-mm")`. Weekday: `=TEXT(B2, "ddd")`.
- **Compute** Row 2: B2 is 3 Jan 2025. Month key `2025-01`. Weekday `Fri`.
- **Answer** All eight orders have month key `2025-01`. Weekdays for rows 2 to 9: Fri, Fri, Sat,

Sun, Mon, Tue, Thu, Fri.

- **Check** 1 Jan 2025 was a Wednesday, so 3 Jan is Friday, 4 Sat, 5 Sun, 6 Mon, 7 Tue, 9 Thu, 10 Fri. Matches.
- **Why it matters** Month keys let you group by month in a PivotTable or `SUMIFS`.

The picture

```
"PUN-2025-0042"
 1234567890123    LEFT(,3)="PUN"   MID(,5,4)="2025"   RIGHT(,4)="0042"
    FIND("-")=4
```

Connection to the real job. Cleaning text and dates is the first half of every analysis. The same steps in pandas are `.str.strip().str.title()` and `pd.to_datetime` (Chapter 5).

Applications

- Customer data: standardise names, cities, emails.
- Finance: month-end dates, due dates, ageing buckets.
- HR: tenure in years and months.
- Logistics: working days to deliver.

COMMON MISTAKE

Dates stored as text (left-aligned, from a CSV). `=B2+1` fails or gives `#VALUE!`. Convert with `=DATEVALUE(B2)` or use Data > Text to Columns, or better, import with Power Query (Chapter 3).

Check yourself

1. What does `EOMONTH(DATE(2025,2,10), 0)` return? Answer: 28 Feb 2025.
2. Why does `TRIM` matter before a lookup? Answer: Extra spaces make two equal-looking values different.
3. What is the number behind a date? Answer: The count of days since the start date (1 Jan 1900).

2.5 Dynamic Array Functions

INTUITION

The old way: one formula, one cell. To list the unique cities you needed a helper column and copying. The new way: one formula "spills" its answer into as many cells as needed. Change the data, and the spilled list resizes itself, like water filling a container.

How it works. A formula that returns many values writes them into neighbouring cells. The first cell holds the formula. The blue border shows the *spill range*. Refer to the whole spill with `#`, as in `E2#`. If a cell in the way is not empty, you get `#SPILL!`.

The functions

Function	Syntax	Does
`FILTER`	`=FILTER(array, include, [if_empty])`	Keep rows that meet a test
`SORT`	`=SORT(array, [sort_index], [order], [by_col])`	Sort (1 ascending, −1 descending)
`SORTBY`	`=SORTBY(array, by_array1, order1, ...)`	Sort by another column
`UNIQUE`	`=UNIQUE(array, [by_col], [exactly_once])`	Distinct values
`SEQUENCE`	`=SEQUENCE(rows, [cols], [start], [step])`	Number grid
`LET`	`=LET(name1, value1, ..., calculation)`	Name an intermediate result
`LAMBDA`	`=LAMBDA(x, y, calculation)`	Your own function
`VSTACK`, `HSTACK`	`=VSTACK(array1, array2)`	Stack tables vertically or side by side
`TAKE`, `DROP`	`=TAKE(array, rows, [cols])`	First or last n rows; remove rows

Worked example 2.5.1 List the distinct categories and total Delivered revenue for each, without

a PivotTable.

- **Formula** In K2: `=UNIQUE(D2:D9)`. In L2: `=SUMIFS(H2:H9, D2:D9, K2#, G2:G9, "Delivered")`.
- **Plug in** `UNIQUE` lists values in order of first appearance: Fashion, Electronics, Grocery. `K2#` is that whole list.
- **Compute** Fashion Delivered: 998 + 799 + 1298 + 1400 = 4495. Electronics Delivered: 1299 + 2499 = 3798. Grocery Delivered: none (1003 Returned, 1005 Cancelled), so 0.
- **Answer** The two spilled columns read Fashion 4495, Electronics 3798, Grocery 0.
- **Check** 4495 + 3798 + 0 = 8293, equal to the Delivered total from Example 2.3.1.

Worked example 2.5.2 Show only Mumbai Delivered orders, sorted by Amount, highest first, top 1.

- **Formula** `=TAKE(SORTBY(FILTER(A2:H9, (C2:C9="Mumbai")*(G2:G9="Delivered")), FILTER(H2:H9, (C2:C9="Mumbai")*(G2:G9="Delivered")), -1), 1)`
- **Plug in** The test multiplies two TRUE/FALSE columns. TRUE × TRUE = 1 keeps a row. Matching rows: 1002 (1299) and 1006 (1298).
- **Compute** Sorted high to low: 1002 first (1299), then 1006 (1298). `TAKE(...,1)` keeps the first.
- **Answer** One row: 1002, 2025-01-03, Mumbai, Electronics, 1, 1299, Delivered, 1299.
- **Check** `*` means AND and `+` means OR inside these tests. Row 1005 is Mumbai but Cancelled, so 0 × 1 = 0 and it is dropped.

Long formulas like this repeat the same `FILTER`. `LET` fixes that.

Worked example 2.5.3 The same logic with `LET`, to compute the average Delivered amount.

- **Formula** `=LET(d, FILTER(H2:H9, G2:G9="Delivered"), SUM(d)/COUNT(d))`
- **Plug in** `d` becomes the Delivered amounts: 998, 1299, 799, 1298, 2499, 1400 (six values).
- **Compute** 8293 / 6 = 1382.17 (rounded).
- **Answer** The average Delivered order is Rs 1,382.17.
- **Check** `=AVERAGEIFS(H2:H9, G2:G9, "Delivered")` returns the same.

LAMBDA. Create a reusable function. In Formulas > Name Manager, add a name `AfterTax` with the definition `=LAMBDA(x, r, x*(1+r))`. Then `=AfterTax(1000, 0.18)` returns 1180 (1000 × 1.18). Use `LAMBDA` for logic you repeat across workbooks.

Other quick examples (verified by hand)

Formula	Result
`=SEQUENCE(3)`	1, 2, 3 down a column
`=SEQUENCE(2, 3)`	Row 1: 1 2 3; row 2: 4 5 6
`=TAKE(SORT(H2:H9, , -1), 3)`	2499, 1400, 1299
`=DROP(SORT(H2:H9, , -1), 6)`	597, 450 (8 sorted values minus the first 6)

The picture

```
=UNIQUE(D2:D9)          K2: Fashion       <- formula lives here
                        K3: Electronics   <- spilled (grey, no formula)
                        K4: Grocery       <- spilled
K2# means K2:K4, even if the list grows
```

Connection to the real job. Dynamic arrays replace most helper columns and many PivotTables for small reports. They behave like SQL `SELECT DISTINCT`, `WHERE`, `ORDER BY` and `LIMIT`. If you know SQL (see the separate SQL document), you already know their logic.

Applications

- Reporting: dropdown lists that update when new categories appear.
- Dashboards: top-5 tables that refresh with data.
- Finance: lists of overdue invoices using `FILTER`.
- Data prep: `VSTACK` to append monthly sheets before analysis.

COMMON MISTAKE

Using `AND(...)` inside `FILTER`. `AND` returns a single TRUE or FALSE for the whole range. Use multiplication `(a)*(b)` for AND and addition `(a)+(b)` for OR to keep one answer per row.

Check yourself

1. What does `K2#` mean? Answer: The full spill range that starts at K2.
2. How do you write "City is Pune OR Delhi" inside FILTER? Answer: `(C2:C9="Pune")+(C2:C9="Delhi")`.
3. What causes `#SPILL!`? Answer: Something non-empty blocks the cells where the result

wants to go.

2.6 Error Handling and Auditing

INTUITION

A broken formula that shows an error is a friendly alarm. A wrong formula that shows a believable number is the real danger. Auditing means looking inside to make sure the number is right for the right reason.

Common errors

Error	Meaning	Typical cause
`#N/A`	Not available	Lookup did not find the key
`#VALUE!`	Wrong type	Text where a number is needed
`#REF!`	Bad reference	Deleted a row or column used by a formula
`#DIV/0!`	Divide by zero	Empty denominator
`#NAME?`	Unknown name	Typo, or function not in your version
`#SPILL!`	Spill blocked	Cells in the way
`#NUM!`	Invalid number	Impossible maths, e.g. root of a negative

Information functions. `=ISBLANK(A2)` is TRUE for an empty cell. `=ISERROR(A2)` is TRUE for any error. `=ISNUMBER(A2)` checks that a value is a true number. `=IFERROR(a, b)` returns `b` when `a` is an error.

Auditing tools (Formulas tab)

- *Trace Precedents*: arrows to the cells a formula uses.
- *Trace Dependents*: arrows to the cells that use this cell.
- *Evaluate Formula*: step through a formula part by part.
- *Show Formulas* (Ctrl+`) and *Error Checking*.
- *Watch Window*: keep an eye on key cells while editing elsewhere.

Worked example 2.6.1 Avoid `#DIV/0!` in a return-rate formula.

- **Formula** `=IF(COUNTIFS(C2:C9, K2) = 0, "", COUNTIFS(C2:C9, K2, G2:G9, "Returned") / COUNTIFS(C2:C9, K2))`
- **Plug in** For city Pune (K2): orders = 3 (rows 1001, 1003, 1008), returned = 1 (row 1003).
- **Compute** 1 / 3 = 0.3333.
- **Answer** Pune return rate is 33.3%. For a city with no orders, say Chennai, the formula returns an empty text instead of an error.
- **Check** Mumbai: orders 3, returned 0, rate 0%. Delhi: 2 orders, 0 returned, 0%.
- **Interpretation** The 33% is based on only 3 orders, so it is very noisy. Always show the count next to a rate.

Worked example 2.6.2 Find why a total looks wrong.

- **Step 1** Suppose `=SUM(H2:H9)` returns 8,743 but you expect 9,340. The gap is 597, which is exactly the amount in row 4 (order 1003).
- **Step 2** Use Trace Precedents and look at H4. It is left-aligned: the number was stored as text `"597"`, and `SUM` silently skips text.
- **Step 3** Confirm with `=COUNT(H2:H9)`. It returns 7 instead of 8, because `COUNT` also skips text. `=ISNUMBER(H4)` returns FALSE. Fix it with `=VALUE(H4)` or re-import the data with correct types.
- **Answer** The method: compare `COUNT` with the expected row count to detect text numbers, then repair at the source.
- **Check** After repair `COUNT` equals the row count and the total equals your hand sum.

The picture

```
 Result wrong?  → COUNT vs expected rows → Trace Precedents → Evaluate Formula
                      (text numbers?)         (right cells?)       (right step?)
```

Connection to the real job. In pandas the equivalent checks are `df.dtypes`, `df.isna().sum()` and `assert` statements. A reviewer who sees you add an error check for blanks trusts your work more.

Applications

- Finance: month-end reconciliation, where a 600 rupee gap must be explained.

- Operations: dashboards that must not show `#N/A` to executives.
- Audit and compliance: every formula must be traceable.

COMMON MISTAKE

Wrapping everything in `IFERROR(..., 0)`. It turns real problems (a missing lookup key) into silent zeros that feed totals. Use it only where an empty result is truly fine, and test for the specific case if you can.

Check yourself

1. What does `#REF!` mean? Answer: A reference points to a deleted cell, row or column.
2. How can `COUNT` expose numbers stored as text? Answer: `COUNT` skips text, so it returns less than the number of rows.
3. Name two audit tools. Answer: Trace Precedents and Evaluate Formula.

Chapter recap

- Use a Table (Ctrl+T) and structured references so formulas grow with your data.
- Use `$` when a formula points to a fixed cell, and F4 to add it.
- `XLOOKUP` replaces `VLOOKUP` / `HLOOKUP`; always use exact match unless you want bands.
- `SUMIFS`, `COUNTIFS`, `AVERAGEIFS` take the value range first, then criteria pairs.
- `SUMPRODUCT` gives weighted sums and conditional counts without helper columns.
- Clean text with `TRIM`, `PROPER`, `SUBSTITUTE`; split with `LEFT`, `MID`, `RIGHT`, `TEXTSPLIT`.
- Dates are numbers; use `EDATE`, `EOMONTH`, `DATEDIF`, `NETWORKDAYS`, `TEXT`.
- Dynamic arrays (`FILTER`, `UNIQUE`, `SORTBY`, `LET`, `LAMBDA`) replace helper columns; use `*` for AND, `+` for OR.
- Check totals by hand. Use `COUNT` to detect text numbers.

Need	Formula pattern
Conditional total	`=SUMIFS(sum, crit_range, crit, ...)`
Lookup	`=XLOOKUP(key, keys, values, "Not found")`
Band	`=XLOOKUP(v, starts, labels, , -1)`

Weighted average	`=SUMPRODUCT(w, x)/SUM(w)`
Unique list	`=UNIQUE(range)`
Filtered rows	`=FILTER(table, (a)*(b))`
Month key	`=TEXT(date, "yyyy-mm")`

Practice set

All questions use the `Sales` table from the start of this chapter.

Problem 1 What does `=COUNTIFS(G2:G9, "Delivered")` return?

Answer: 6 (rows 1001, 1002, 1004, 1006, 1007, 1008).

Problem 2 What does `=SUMIFS(H2:H9, D2:D9, "Grocery")` return?

Answer: 597 + 450 = 1,047.

Problem 3 Write a formula for the average Fashion Delivered amount and give its value.

Answer: `=AVERAGEIFS(H2:H9, D2:D9, "Fashion", G2:G9, "Delivered")`. Values 998, 799, 1298, 1400 sum to 4495. 4495 / 4 = 1123.75.

Problem 4 What is the output of `=XLOOKUP(1004, A2:A9, C2:C9)`?

Answer: Delhi. Order 1004 is in the 4th row of the data, and its city is Delhi.

Problem 5 Find the bug: `=VLOOKUP(D2, L2:M4, 2)` returns 18% for a Fashion order.

Answer: The fourth argument is missing, so Excel uses approximate match on an unsorted list and returns a wrong row. Use `=VLOOKUP(D2, L2:M4, 2, FALSE)` to get 12%.

Problem 6 What does `=SUMPRODUCT((C2:C9="Delhi")*H2:H9)` return?

Answer: Delhi rows: 799 and 2499. 799 + 2499 = 3,298.

Problem 7 What does `=EDATE(DATE(2025,1,31), 1)` return, and why not 31 February?

Answer: 28 February 2025. `EDATE` moves by months and caps at the last valid day of the target month.

Problem 8 Write one formula that returns the number of distinct cities.

Answer: `=COUNTA(UNIQUE(C2:C9))` . The distinct cities are Pune, Mumbai, Delhi, so the result is 3.

Problem 9 `=NETWORKDAYS(DATE(2025,1,3), DATE(2025,1,10))` equals what?

Answer: 3 Jan (Fri), 6 to 10 Jan (Mon to Fri). That is 1 + 5 = 6 working days, both ends included.

Problem 10 (interview) A report has the same customer twice, as "Asha Rao" and "asha rao ". How do you merge them and keep the original data safe?

Answer: Add a helper column `=PROPER(TRIM(A2))` (or in Power Query use Trim and Capitalize Each Word), then use that column for lookups and pivots. Never overwrite the raw column.

Problem 11 (interview) You must replace a sheet where each of 30,000 rows has `VLOOKUP` to a 5,000 row table and Excel is slow. What do you do?

Answer: Convert the ranges to Tables, use `XLOOKUP` with exact match on a unique key, avoid whole-column references, or move the join into Power Query (Chapter 3), where the merge happens once at refresh time, not on every recalculation.

Prompt pack

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional in
India moving into data roles, and English is my second language. Teach me
Chapter 2 topic: Excel lookup functions (XLOOKUP, VLOOKUP, INDEX+MATCH,
XMATCH) including exact versus approximate match. Use a tiny 6-row table
with real numbers. Explain step by step what each formula does, ask me to
predict results, and check my understanding after each part.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. Give me 8 graded Excel exercises, easy to hard, on
SUMIFS, COUNTIFS, AVERAGEIFS and SUMPRODUCT using a small sales table
(OrderID, Date, City, Category, Qty, Price, Status). Do not give
solutions. After I send my formulas I will ask you to check them and to
```

explain any wrong result step by step.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Teach me Excel dynamic arrays: FILTER, UNIQUE, SORTBY, SEQUENCE, LET, LAMBDA, VSTACK, TAKE. Explain spill ranges and the # operator with an analogy. Show 5 examples on a small table, with expected results I can verify by hand. Then quiz me one question at a time.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Review my Excel formulas for correctness, readability and robustness (absolute references, error handling, duplicate keys, text numbers). Tell me what could silently give a wrong answer and suggest a better version. My formulas and sample data: [PASTE HERE]

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Run a mock Excel interview of 8 practical questions (lookups, conditional sums, text cleaning, dates, dynamic arrays, error tracing). Ask one question at a time, wait for my formula, then tell me if it is right, explain why, and give the most modern alternative. End with a score and a study plan.

CHAPTER 3

EXCEL ANALYSIS: PIVOTTABLES, POWER QUERY, POWER PIVOT AND WHAT-IF TOOLS

Summarise, clean and model data in Excel without a single copy and paste.

In Chapter 2 you wrote formulas one cell at a time. Real reporting asks for more: "show revenue by city and status, with a slicer, refreshed every Monday from a new CSV". The tools for that are PivotTables (summarise), Power Query (import and clean, repeatable) and Power Pivot (relate many tables and write measures). Together they are called the "real power" of Excel, and they use the same ideas as Power BI, which you meet in Chapters 9 and 10. By the end of this chapter you will know how to build a refreshable summary, use what-if tools to answer "what if the price changes", run a t-test inside Excel, and automate small jobs.

PROFESSOR'S NOTE

Excel and Power Query cannot run in this environment. Menu names can differ a little by version and language. Where numbers appear, they are computed from our small `Sales` table (see Chapter 2) and verified with pandas, and where code appears (M, VBA) it is standard and minimal but not executed here. The Python Excel examples (openpyxl) were run, and their real output is shown.

We reuse the 8-row `Sales` table from Chapter 2. Its amounts are 998, 1299, 597, 799, 450, 1298, 2499, 1400 and the total is 9,340.

3.1 PivotTables and PivotCharts

INTUITION

Imagine a big pile of till receipts. A manager asks, "How much did each city sell, split by delivered, returned and cancelled?" You could sort the receipts into piles by hand. A PivotTable is a machine that does the sorting for you and lets you re-sort in one drag. You never change the receipts, you only change how you look at them.

Words you need

Term	Meaning
Field	A column of your data (City, Status, Amount)
Rows / Columns areas	Fields that label the table's sides
Values area	The numbers, with an aggregation (Sum, Count, Average)
Filters area	Fields that restrict the whole table
Slicer	A button panel to filter a PivotTable
Timeline	A slicer made for dates
Calculated field	A formula on the sums of other fields

How to build one. Click inside the Table, then Insert > PivotTable. Drag `City` to Rows, `Status` to Columns, `Amount` to Values (Sum).

Worked example 3.1.1 What does the PivotTable show for City by Status?

- **Step 1** Group the 8 rows by City and Status and add Amount.
- **Step 2** Pune: Delivered 998 + 1400 = 2398; Returned 597. Mumbai: Delivered 1299 + 1298 = 2597; Cancelled 450. Delhi: Delivered 799 + 2499 = 3298.
- **Step 3** Add row and column totals.
- **Answer** (real pandas `pivot_table`, the same logic as Excel)

```
Status  Cancelled  Delivered  Returned  Total
City
Delhi           0       3298         0   3298
Mumbai        450       2597         0   3047
Pune            0       2398       597   2995
Total         450       8293       597   9340
```

- **Check** The grand total 9,340 equals the sum of the Amount column. Excel shows empty cells where pandas shows 0.
- **Interpretation** 8,293 of 9,340 rupees (88.8%) were delivered. Pune has the only Returned value.

Show Values As. Right-click a value, "Show Values As", to change what the numbers mean. Using the City totals:

Option	Delhi	Mumbai	Pune
Sum of Amount	3298	3047	2995
% of Grand Total	35.3%	32.6%	32.1%
Difference from Delhi	0	−251	−303

Worked example 3.1.2 Compute "% of Grand Total" by hand.

- **Formula** share = city total / grand total
- **Plug in** Delhi 3298 / 9340
- **Compute** 3298 / 9340 = 0.3531, which is 35.3%. Mumbai 3047 / 9340 = 0.3262 (32.6%). Pune 2995 / 9340 = 0.3207 (32.1%).
- **Answer** Delhi is 35.3%, Mumbai 32.6%, Pune 32.1%.
- **Check** 35.3 + 32.6 + 32.1 = 100.0.
- **Why it matters** Shares let you compare cities of different size. Use "Running Total In" for a date field: running total of daily revenue ends at the grand total.

Grouping. Right-click a date field > Group > Months and Years. Right-click a number field > Group to make bins (0 to 500, 500 to 1000, ...). Group text items by selecting them and choosing Group.

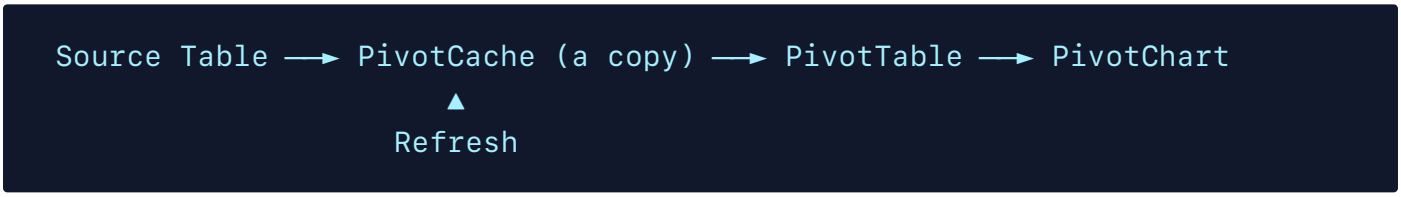
Calculated field. PivotTable Analyze > Fields, Items & Sets > Calculated Field. Example: name `PricePerUnit`, formula `=Amount/Qty`. It computes `SUM(Amount) / SUM(Qty)` for every cell. For the whole table: 9340 / 19 = 491.58 (not the average of row-level ratios).

Slicers and timelines. PivotTable Analyze > Insert Slicer > pick `Category`. Click a button to filter. One slicer can connect to many PivotTables (Report Connections). A timeline lets you drag over months.

PivotChart. PivotTable Analyze > PivotChart. It always follows the PivotTable. Filter the table and the chart follows.

Refresh rule. PivotTables do not update when the data changes. Right-click > Refresh (or Data > Refresh All). Set "Refresh data when opening the file" in PivotTable Options. If your source is a Table or a Power Query output, new rows come with the refresh.

The picture



Connection to the real job. A PivotTable is `GROUP BY` with a drag-and-drop interface. The same table in SQL is `SELECT city, status, SUM(amount) ... GROUP BY city, status`, and in pandas it is `pivot_table` (Chapter 5).

Applications

- Sales: revenue by region, product and month for the weekly meeting.
- HR: headcount by department and grade.
- Finance: expense by cost centre and month.
- Operations: ticket counts by priority and team.

COMMON MISTAKE

Source data with blank rows, merged cells, or a total row inside the data. The PivotTable then counts the total as data and doubles the numbers. Keep the source as a clean Table with one header row and no totals.

Check yourself

1. Which area holds the numbers? Answer: Values.
2. Why does a PivotTable need a Refresh? Answer: It reads from a stored copy (cache) of the data.
3. What is the grand total of Amount in our table? Answer: 9,340.

3.2 Conditional Formatting, Validation, Protection, Sparklines

INTUITION

A table of 1,000 numbers is hard to read. Colour tells your eye where to look. Validation is a bouncer: it stops wrong data entering. Protection stops people breaking your formulas.

Conditional formatting (Home tab).

Feature	Use
Highlight cells rules	Greater than, between, text contains, duplicate values
Top/Bottom rules	Top 10 items, above average
Data bars	Bar length shows size inside the cell
Colour scales	Heat map from low to high
Icon sets	Arrows, traffic lights
Formula rule	Any TRUE/FALSE formula, e.g. `=$G2="Returned"` colours the whole row

Worked example 3.2.1 Colour the entire row red when the status is Returned.

- **Step 1** Select A2:H9.
- **Step 2** Home > Conditional Formatting > New Rule > "Use a formula to determine which cells to format".
- **Step 3** Enter `=$G2="Returned"`. The `$` locks the column to G, but the row is relative so each row tests its own status.
- **Answer** Only row 4 (order 1003) turns red.
- **Check** If you forget `$`, the formula tests column A in column A, B in column B, etc., and the highlighting looks random.

Data validation (Data > Data Validation).

- *List*: a dropdown, from a range or typed values `Delivered,Returned,Cancelled`.
- *Whole number / Decimal*: for example Qty between 1 and 100.
- *Date*: only dates after 1 January 2025.
- *Custom*: formula, e.g. `=COUNTIF($A:$A, A2)=1` rejects duplicate IDs.
- Add an input message and an error alert ("Stop" blocks bad entries).

Protection (Review tab). Unlock input cells (Format Cells > Protection > untick Locked), then Protect Sheet. Locked cells with formulas can no longer be changed without the password. Passwords in Excel stop accidents, not attackers, so never rely on them for confidential data.

Sparklines (Insert tab). Tiny charts inside a cell: Line, Column, Win/Loss. Select the data

range and the cell location. They show the trend per row, e.g. weekly sales per city.

Worked example 3.2.2 Design a data entry sheet that cannot hold a wrong status or a negative quantity.

- **Step 1** Qty column: Data Validation > Whole number > between 1 and 1000, with an error message "Quantity must be 1 to 1000".
- **Step 2** Status column: List with source `Delivered,Returned,Cancelled`.
- **Step 3** Protect the sheet with only these two columns unlocked.
- **Answer** A user typing -3 in Qty sees the error, and typing "Deliverd" in Status is rejected, so bad data cannot enter.
- **Check** Validation does not check pasted values; pasting overrides it. Re-check with Data > Data Validation > Circle Invalid Data.

The picture

```
Amount     [#####     ] 998      <- data bar
           [##########] 2499
Status      Returned  -> row turns red
```

Connection to the real job. These features make Excel files safe for non-technical users, which is common in finance and operations teams. In Power BI, the equivalents are conditional formatting on visuals and row-level security.

Applications

- Operations: red-amber-green status in trackers.
- Finance: highlight negative variances.
- HR: dropdowns for department, grade.
- Sales: data bars on leaderboard.

COMMON MISTAKE

Colour used as the only signal. About 1 in 12 men has some colour blindness. Add an icon or text (e.g. "Returned") so the meaning survives without colour.

Check yourself

1. What does `$G2` lock? Answer: The column G, but not the row.

2. Does validation stop pasted data? Answer: No, pasting can bypass it.
3. What is a sparkline? Answer: A tiny chart inside one cell.

3.3 What-If Analysis: Goal Seek, Scenarios, Data Tables, Solver

INTUITION

Your model is a machine with knobs (price, volume, cost). What-if tools turn the knobs for you. Goal Seek turns one knob until the output hits a target. A Data Table shows the output for many knob settings at once. Scenario Manager stores named sets of settings (best, base, worst). Solver turns several knobs to get the best result under limits.

Goal Seek (Data > What-If Analysis > Goal Seek). You choose: *Set cell* (the formula result), *To value* (target), *By changing cell* (one input).

Worked example 3.3.1 A product costs Rs 300 to make. You sell 100 units. Fixed costs are Rs 5,000. What price gives a profit of Rs 10,000?

- **Formula** Profit = (Price − Cost) × Units − Fixed
- **Plug in** B1 = price (input), B2 = 300, B3 = 100, B4 = 5000, and B5 holds `=(B1−B2)∗B3−B4` . Goal Seek: Set cell B5, To value 10000, By changing B1.
- **Compute by hand** 10,000 = (P − 300) × 100 − 5,000 gives (P − 300) × 100 = 15,000, so P − 300 = 150 and P = 450.
- **Answer** Goal Seek will return 450.
- **Check** (450 − 300) × 100 − 5,000 = 15,000 − 5,000 = 10,000.
- **Why it matters** Goal Seek solves reverse questions, "what do I need to hit the target".

One-variable and two-variable Data Table (What-If > Data Table). Build a loan EMI model. The EMI (monthly payment) uses `=PMT(rate/12, years∗12, −loan)` .

Worked example 3.3.2 What is the EMI for a Rs 10,00,000 loan over 10 years at 8%, 9%, 10%?

- **Formula** EMI = P × r / (1 − (1 + r)^(−n)), where r is the monthly rate and n the number of months.
- **Plug in** P = 1,000,000, n = 120. For 9%: r = 0.09 / 12 = 0.0075.

- **Compute** (computed by code) 8%: 12,132.76. 9%: 12,667.58. 10%: 13,215.07.
- **Answer** `=PMT(0.09/12, 120, −1000000)` returns 12,667.58 (rounded to paise).
- **Check** Total paid at 9% = 12,667.58 × 120 = 15,20,109.6, so interest = 5,20,109.6, which is plausible at about 52% of the loan over 10 years.
- **How to build a one-way table** Put rates 8%, 9%, 10% down a column, the EMI formula in the cell above and to the right, select the block, Data Table > "Column input cell" = the rate cell. Excel fills the three EMIs at once.

Scenario Manager (What-If > Scenario Manager). Store sets such as Base (price 450, units 100), Best (price 500, units 120), Worst (price 400, units 80). The Summary button builds a comparison sheet. Profit for the three: Base 10,000; Best (500 − 300) × 120 − 5,000 = 19,000; Worst (400 − 300) × 80 − 5,000 = 3,000.

Solver (Data > Solver; enable it in File > Options > Add-ins). It handles several inputs and constraints.

Worked example 3.3.3 A factory makes products X (profit Rs 40) and Y (profit Rs 30). Each X needs 2 machine hours and each Y needs 1. Machine hours: at most 100. Labour hours: each unit uses 1, at most 80. You can make at most 40 of X. How many of each maximise profit?

- **Formula** Maximise 40x + 30y subject to 2x + y ≤ 100, x + y ≤ 80, 0 ≤ x ≤ 40, y ≥ 0.
- **Plug in** In Solver: Objective cell = profit formula; By changing cells x, y; Add constraints; Method Simplex LP.
- **Compute by hand** The best point is on a corner of the allowed region. The two main limits meet where 2x + y = 100 and x + y = 80. Subtract: x = 20, so y = 60. Profit there: 40 × 20 + 30 × 60 = 800 + 1800 = 2,600. Other corners: (40, 20) gives 1600 + 600 = 2,200; (0, 80) gives 2,400; (0, 0) gives 0. The maximum is 2,600.
- **Answer** Make 20 of X and 60 of Y for a profit of Rs 2,600. (A linear programming solver check gave the same x = 20, y = 60, profit 2,600.)
- **Check** Machine hours: 2 × 20 + 60 = 100 (limit reached). Labour: 20 + 60 = 80 (limit reached). X: 20 ≤ 40 (fine).
- **Interpretation** Both machine and labour hours are the bottleneck, so extra hours of either would raise profit. This is the idea of a "binding constraint". A queue at a bank has the same idea: the slowest counter sets the pace for everyone.

The picture

```
y (units of Y)
80 |*(0,80)=2400
   |   \
60 |     *(20,60)=2600  <- best corner
   |       \
20 |         *(40,20)=2200
   +------------------ x
```

Connection to the real job. Scenario and sensitivity analysis is standard in finance and strategy roles (Chapter 12). Solver ideas lead to optimisation in data science.

Applications

- Finance: loan EMI tables, break-even prices, budget scenarios.
- Operations: product mix, staff rosters, delivery routes (small).
- Marketing: budget split across channels.
- Pricing: price vs profit tables.

COMMON MISTAKE

Goal Seek needs the changing cell to hold a number, not a formula, and it finds one answer, which may be a local one if the model is not smooth. Always substitute the answer back by hand as we did.

Check yourself

1. Which tool shows results for many input values at once? Answer: A Data Table.
2. Which tool optimises with constraints? Answer: Solver.
3. In Example 3.3.1, what profit does price 400 give? Answer: (400 − 300) × 100 − 5000 = 5,000.

3.4 Statistics and Charts in Excel

INTUITION

Excel can run real statistical tests, not only sums. The Analysis ToolPak is an add-in (enable it in File > Options > Add-ins > Excel Add-ins) that produces regression, t-test, ANOVA and histogram tables in one click. The charts then show the result to people.

Analysis ToolPak tests (Data > Data Analysis)

Tool	Question it answers
t-Test: Two-Sample Assuming Equal Variances	Do two groups have different means?
ANOVA: Single Factor	Do three or more groups differ?
Regression	How does y change with x?
Histogram	How are values distributed?
Descriptive Statistics	Mean, median, std dev in one table

We reproduce the maths with Python's `scipy` so you can check any output Excel gives you. Groups: A = 12, 14, 13, 15, 16 (average load time with new page), B = 10, 11, 12, 9, 13 (old page).

Worked example 3.4.1 Two-sample t-test (equal variances).

- **Formula** t = (mean A − mean B) / √(s²_pooled × (1/nA + 1/nB)).
- **Plug in** Mean A = 14, mean B = 11. Both sample variances = 2.5, so pooled variance = 2.5. Standard error = √(2.5 × (1/5 + 1/5)) = √(2.5 × 0.4) = √1 = 1.
- **Compute** t = (14 − 11) / 1 = 3.0 with 8 degrees of freedom (5 + 5 − 2).
- **Answer** Python `scipy.stats.ttest_ind(A, B)` gives t = 3.0, p = 0.0171.
- **Check** Excel's ToolPak shows the same t Stat 3 and "P(T<=t) two-tail" 0.0171. Because p is below 0.05, the difference is unlikely to be chance.
- **Interpretation** If both pages truly had the same average, a gap this large would appear in only about 1.7% of such experiments. With five users per group, treat it as a sign to collect more data, not as proof.

Worked example 3.4.2 One-way ANOVA with a third group C = 8, 9, 10, 7, 11.

- **Formula** F = MSB / MSW, MSB = SSB / (k − 1) and MSW = SSW / (N − k).
- **Plug in** Group means 14, 11, 9; grand mean 34/3 = 11.333. SSB = 5 × [(14 − 11.333)² + (11 − 11.333)² + (9 − 11.333)²] = 5 × (7.111 + 0.111 + 5.444) = 63.33. Each group's within-sum of squares: 10, 10, 10 so SSW = 30. k = 3, N = 15.
- **Compute** MSB = 63.33 / 2 = 31.67. MSW = 30 / 12 = 2.5. F = 31.67 / 2.5 = 12.67.
- **Answer** `scipy.stats.f_oneway` gives F = 12.667, p = 0.0011.

- **Check** The code output matches our hand F of 12.67.
- **Interpretation** At least one group mean differs. ANOVA does not say which; follow up with pairwise tests.

Worked example 3.4.3 Regression: ad spend (thousand rupees) x = 10, 20, 30, 40, 50 and sales y = 120, 150, 185, 200, 240.

- **Formula** slope $b = \Sigma(x - \bar{x})(y - \bar{y}) / \Sigma(x - \bar{x})^2$, intercept $a = \bar{y} - b \times \bar{x}$.
- **Plug in** $\bar{x} = 30$, $\bar{y} = 179$. $\Sigma(x - \bar{x})^2 = 400 + 100 + 0 + 100 + 400 = 1000$. $\Sigma(x - \bar{x})(y - \bar{y}) = (-20)(-59) + (-10)(-29) + 0 + (10)(21) + (20)(61) = 1180 + 290 + 0 + 210 + 1220 = 2900$.
- **Compute** $b = 2900 / 1000 = 2.9$. $a = 179 - 2.9 \times 30 = 92$.
- **Answer** Sales ≈ 92 + 2.9 × spend, with $R^2 = 0.987$ (the code gives slope 2.9, intercept 92.0, R^2 0.9871, p = 0.0006).
- **Check** At x = 30, the line gives 92 + 87 = 179, which equals $\bar{y}$, as a regression line must pass through $(\bar{x}, \bar{y})$.
- **Interpretation** Each extra thousand rupees of spend goes with about 2.9 more units of sales, within the range 10 to 50. Do not extend it beyond the data, and remember it shows association, not proof of cause.

Charts you should master

Chart	Best for	Notes
Column / bar	Compare categories	Start the axis at zero
Line	Trends over time	One line per series, few series
Combo (column + line)	Two measures with different units	Use the secondary axis sparingly (see Chapter 8)
Waterfall	How parts add up to a change	Insert > Waterfall; good for bridges: start, +, −, end
Histogram	Distribution of one number	Insert > Statistic Chart > Histogram
Box and whisker	Compare distributions	Insert > Statistic Chart > Box and Whisker
Scatter with trendline	Relationship of two numbers	Right-click series > Add Trendline > show R^2

Dashboard design basics (full rules in Chapters 8 and 11). Put the headline numbers (KPI cards) at the top left, one chart per question, align everything on a grid, use one accent colour, keep slicers together, hide gridlines, and give every chart a title that states the finding ("Delhi leads with 35% of revenue").

The picture

```
+-------------------------------------------------+
| KPI: Revenue 9,340 | Orders 8 | Delivered 88.8%  |  <- cards
+--------------------------+----------------------+
|  Column: revenue by city  |  Line: by date       |
+--------------------------+----------------------+
| Slicer: Category  [Fashion][Electronics][Grocery]|
+-------------------------------------------------+
```

Connection to the real job. The test logic is the same you will use in Python and for A/B tests (Chapter 13). Seeing the t statistic by hand makes the software output less magical.

Applications

- Product: compare page load times between versions.
- Marketing: relate ad spend to sales.
- Operations: compare defect rates across three plants (ANOVA).
- HR: check pay gaps between groups.

COMMON MISTAKE

Reading a small p-value as a large effect. p says whether a difference is likely to be chance, not whether it is important. Always read the size of the difference too.

Check yourself

1. What does ANOVA test? Answer: Whether three or more group means differ.
2. In 3.4.3, what does the slope 2.9 mean? Answer: Each extra thousand rupees of spend goes with about 2.9 more sales units.
3. Which chart shows how pieces add to a total change? Answer: A waterfall chart.

3.5 Power Query: Import, Clean, Reshape

INTUITION

Cleaning a file by hand is like washing dishes one at a time every day. Power Query records the steps once (remove blanks, fix types, trim text) as a recipe. Next week, point it at the new file and click Refresh: the same recipe runs again. The raw file is never changed.

Where to find it. Data > Get Data (From File, From Web, From Database, From Folder). It opens the Power Query Editor. The right-hand panel shows Applied Steps, each a recorded action. Each step is written in the **M language** shown in the formula bar.

Common transformations

Task	How	M function
Promote headers	Use First Row as Headers	`Table.PromoteHeaders`
Change type	Header icon	`Table.TransformColumnTypes`
Trim and fix case	Transform > Format	`Text.Trim` , `Text.Proper`
Split column	Home > Split Column	`Table.SplitColumn`
Merge columns	Add Column > Merge	`Text.Combine`
Custom column	Add Column > Custom Column	`Table.AddColumn`
Filter rows	Column drop-down	`Table.SelectRows`
Group by	Home > Group By	`Table.Group`
Unpivot	Transform > Unpivot Columns	`Table.UnpivotOtherColumns`
Pivot	Transform > Pivot Column	`Table.Pivot`
Merge queries (join)	Home > Merge Queries	`Table.NestedJoin`
Append queries (stack)	Home > Append Queries	`Table.Combine`

A complete query (M). Not run here, but standard.

```
let
    Source   = Csv.Document(File.Contents("C:\data\sales.csv"),
                  [Delimiter=",", Columns=7, Encoding=65001,
QuoteStyle=QuoteStyle.Csv]),
    Promoted = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
    Typed    = Table.TransformColumnTypes(Promoted,
                  {{"OrderID", Int64.Type}, {"Date", type date},
                   {"Qty", Int64.Type}, {"Price", type number}}),
    Cleaned  = Table.TransformColumns(Typed,
                  {{"City", each Text.Proper(Text.Trim(_)), type text}}),
    WithAmt  = Table.AddColumn(Cleaned, "Amount", each [Qty] * [Price],
type number),
    Delivered = Table.SelectRows(WithAmt, each [Status] = "Delivered")
in
    Delivered
```

Read it like a recipe: each line is a named step, and the next line uses the previous step's name. The last name after `in` is the result.

Worked example 3.5.1 Unpivot a wide report. Raw data has one row per city with a column per month:

City	Jan	Feb
Pune	10	12
Delhi	20	18

- **Step 1** Select the `City` column, then Transform > Unpivot Other Columns.
- **Step 2** Rename `Attribute` to `Month` and `Value` to `Sales`.
- **Compute** Each city-month pair becomes a row. M: `Table.UnpivotOtherColumns(Source, {"City"}, "Month", "Sales")`.
- **Answer** Four rows: (Pune, Jan, 10), (Pune, Feb, 12), (Delhi, Jan, 20), (Delhi, Feb, 18).
- **Check** The sum of Sales is 60 before and after, 10 + 12 + 20 + 18 on both sides. (The pandas `melt` function produced exactly these four rows in Chapter 5's example.)
- **Why it matters** PivotTables and Power BI need "tidy" long data (one row per observation). Unpivot makes wide report-style files usable.

Worked example 3.5.2 Merge (join) two queries: Orders (OrderID, Category) with Tax (Category, Rate).

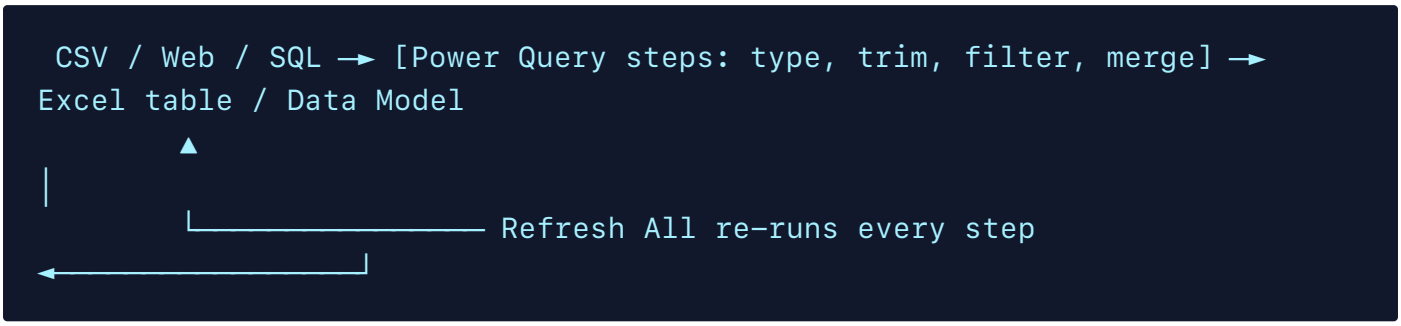
- **Step 1** Home > Merge Queries, pick `Category` in both, join kind Left Outer.
- **Step 2** Expand the new column and keep `Rate`.
- **Compute** Row 1001 (Fashion) gets 12%; 1002 (Electronics) gets 18%; 1003 (Grocery) gets 5%.
- **Answer** Orders with a Rate column. A left outer join keeps all orders even when the category has no tax row (Rate shows null).
- **Check** Row counts: after a left join with a unique Tax key, the order count stays 8. If it grows, the Tax table has duplicate categories.

Folder import. Data > Get Data > From File > From Folder. Power Query lists all files in a folder. Combine & Transform merges every CSV with the same layout. Drop a new monthly file in the folder, click Refresh, and the combined table grows. This replaces monthly copy and paste.

Parameters. Manage Parameters creates a named value (e.g. `FolderPath`). Use it in the source step so one change updates all queries. Custom functions (a query with `(x) => ...`) let you reuse logic.

Error handling. A step can fail on a bad row ("Error" in a cell). Use Remove Errors, Replace Errors, or in M `try ... otherwise`, e.g. `each try Number.From([Price]) otherwise null`.

The picture



Connection to the real job. Power Query is the same engine as Power BI's data preparation (Chapter 9). It is the no-code relative of pandas cleaning (Chapter 5) and of ETL (Extract, Transform, Load) in data engineering.

Applications

- Finance: combine 12 monthly bank exports into one table.
- Retail: clean product feeds.
- HR: merge HR extracts with attendance.

- Operations: import from a SharePoint folder daily.

COMMON MISTAKE

Editing the query's output sheet by hand. Refresh will overwrite it. Do all cleaning inside Power Query, and load to a Table or the Data Model.

Check yourself

1. Does Power Query change the source file? Answer: No, it only reads it.
2. What does Unpivot do? Answer: Turns columns into rows (wide to long).
3. How do you add next month's file automatically? Answer: Use From Folder and refresh.

3.6 Power Pivot and the Data Model

INTUITION

A PivotTable can only look at one table. Real data has several: orders, customers, products. Power Pivot lets you load them all into the Data Model, connect them with relationships (like a foreign key), and write measures in DAX. Then one PivotTable can slice orders by customer city and product category without any VLOOKUP.

Steps

1. Load each table to the Data Model (Power Query: Load To > Only Create Connection > Add this data to the Data Model).
2. Open Power Pivot > Manage > Diagram View. Drag `CustomerID` from the fact table to the dimension table. A line with `1` and `*` marks a one-to-many relationship.
3. Write a measure (Power Pivot > Measures > New Measure): `Total Sales := SUM(Orders[Amount])`.
4. Insert a PivotTable from the Data Model and use fields from any table.

Worked example 3.6.1 Measure that ignores returns: `Delivered Sales := CALCULATE(SUM(Orders[Amount]), Orders[Status] = "Delivered")`.

- **Step 1** `SUM(Orders[Amount])` alone would give 9,340 for the grand total.
- **Step 2** `CALCULATE` changes the filter: keep only rows where Status is Delivered.
- **Compute** 998 + 1299 + 799 + 1298 + 2499 + 1400 = 8,293.

- **Answer** The grand total of the measure is 8,293. In a PivotTable by city, Pune shows 2,398, Mumbai 2,597 and Delhi 3,298.
- **Check** These match the Delivered column of Example 3.1.1.
- **Why it matters** Measures are calculated at query time in the context of each cell, so one measure works in every slice. DAX is explained fully in Chapter 10.

Differences from normal PivotTables. The Data Model can hold millions of rows (it uses a compressed column store), supports relationships, supports `DISTINCTCOUNT`, and supports DAX measures. Its formulas look like Excel's but follow DAX rules. The same DAX works in Power BI.

Refresh and automation. Data > Refresh All updates all queries, then the Data Model, then PivotTables. Query Properties let you refresh every N minutes, on open, or in the background. For a server schedule you need Power BI Service or Power Automate; plain Excel refreshes only when the file is open.

Connection to the real job. If you can build a star schema in Power Pivot, you can do it in Power BI (Chapter 9). This is why Excel with the Data Model is the best free practice ground.

Applications

- Finance: P&L with a calendar table and actual/budget fact tables.
- Sales: orders and returns in separate tables, joined by customer.
- HR: headcount and leavers.

COMMON MISTAKE

Merging everything into one wide table with VLOOKUP instead of using relationships. It repeats data, bloats the file and breaks when a key changes. Keep separate tables and relate them (Chapter 9).

Check yourself

1. What does `1` and `*` on a relationship line mean? Answer: One row on the dimension side matches many rows on the fact side.
2. What language do Power Pivot measures use? Answer: DAX.
3. What does Refresh All do in order? Answer: Updates queries, then the model, then PivotTables.

3.7 Automation: VBA, Office Scripts, Python

INTUITION

If you do the same clicks every Monday, record them. A macro is a recording of your actions, saved as code you can edit. Modern options are Office Scripts (in Excel on the web) and Python, which can write and read Excel files without opening Excel.

Macros and VBA. Developer > Record Macro, do your steps, Stop. Open the VBA editor (Alt+F11) to see code. Learn loops, ranges and functions:

```
Sub MarkBigOrders()
    Dim r As Long, lastRow As Long
    lastRow = Cells(Rows.Count, "H").End(xlUp).Row
    For r = 2 To lastRow
        If Cells(r, "H").Value > 1000 Then
            Cells(r, "I").Value = "Big"
        Else
            Cells(r, "I").Value = "Normal"
        End If
    Next r
End Sub

Function GrossToNet(gross As Double, taxRate As Double) As Double
    GrossToNet = gross / (1 + taxRate)
End Function
```

(Standard VBA, not run here.) The function `GrossToNet` becomes a cell formula: `=GrossToNet(1180, 0.18)` returns 1000, because 1180 / 1.18 = 1000. Save the file as `.xlsm` for macros. Macro files can carry viruses, so enable macros only from trusted sources.

Office Scripts. Excel on the web: Automate > Record Actions, saved as TypeScript, and runnable by Power Automate on a schedule. Good where desktop macros are not allowed.

Python in Excel. A newer Microsoft 365 feature lets you type `=PY(...)` in a cell and run Python (pandas, matplotlib) in Microsoft's cloud. Availability depends on your plan and version, so check before relying on it.

Python outside Excel: `openpyxl`. We ran this on our table and a formula cell:

```
import pandas as pd
from openpyxl import Workbook, load_workbook

rows = [(1001,"Pune","Delivered",998),(1002,"Mumbai","Delivered",1299),
(1003,"Pune","Returned",597),
        (1004,"Delhi","Delivered",799),(1005,"Mumbai","Cancelled",450),
(1006,"Mumbai","Delivered",1298),
        (1007,"Delhi","Delivered",2499),(1008,"Pune","Delivered",1400)]
df = pd.DataFrame(rows, columns=["OrderID","City","Status","Amount"])
wb = Workbook(); ws = wb.active; ws.title = "Data"
ws.append(list(df.columns))
for r in rows: ws.append(r)
ws["F1"] = "Total"; ws["F2"] = "=SUM(D2:D9)"
wb.save("demo.xlsx")
wb2 = load_workbook("demo.xlsx"); print(wb2["Data"]["F2"].value,
wb2.sheetnames)
print(pd.read_excel("demo.xlsx", sheet_name="Data").head(3))
```

Predict: what does `read_excel` show in the `Total` column? Real output:

```
=SUM(D2:D9) ['Data']
   OrderID    City     Status  Amount  Unnamed: 4  Total
0     1001    Pune  Delivered     998         NaN    NaN
1     1002  Mumbai  Delivered    1299         NaN    NaN
2     1003    Pune  Returned     597         NaN    NaN
```

Explain. `openpyxl` writes the formula text but does not calculate it. The file has no stored result until Excel opens and saves it, so pandas reads `NaN`. The empty column E appears as `Unnamed: 4`. Lesson: Python libraries write formulas but do not calculate them; compute numbers in pandas and write values, or open and re-save in Excel. `xlsxwriter` is another writer library (not installed here); it is write-only, with rich chart and formatting support.

The picture



Connection to the real job. In a large company, a manual weekly Excel report is a candidate for automation by Python plus a scheduler. Anyone who has ever written a macro or a shell script knows the instinct: if you do it twice, automate it.

Applications

- Finance: month-end report packs.
- Operations: daily tracker updated by a script.
- Analytics: Python builds a formatted Excel for managers who prefer Excel.

COMMON MISTAKE

Hard-coding file paths and row counts in macros. When the data grows or moves, the macro breaks silently. Find the last row dynamically (as `lastRow` does above) and keep paths in one cell or parameter.

Check yourself

1. Why did pandas show `NaN` for the formula cell? Answer: The library wrote the formula but never calculated a stored value.
2. What file extension holds macros? Answer: `.xlsm`.
3. Name two non-VBA automation options. Answer: Office Scripts and Python.

Chapter recap

- A PivotTable is an interactive GROUP BY; refresh it when data changes; build it from a clean Table.
- `Show Values As` gives shares, differences and running totals; calculated fields work on sums.
- Conditional formatting with a formula rule (`=$G2="Returned"`) colours whole rows; validation stops bad input.
- Goal Seek solves for one input; Data Tables test many inputs; Scenario Manager stores named cases; Solver optimises with limits.
- The ToolPak gives t-test, ANOVA and regression; we checked the same numbers by hand (t = 3, F = 12.67, slope 2.9).
- Power Query records cleaning steps (M language) and re-runs them on refresh; use it for folders, unpivot and merges.
- Power Pivot adds relationships and DAX measures, the same as Power BI.
- Automate with macros (VBA), Office Scripts or Python; Python libraries write formulas but do not calculate them.

- Mastery test: build a refreshable dashboard from raw CSVs with Power Query, a PivotTable and slicers.

Task	Tool or syntax
Summarise	PivotTable (rows, columns, values, slicers)
Share of total	Show Values As > % of Grand Total
Reverse solve	Goal Seek (set cell, to value, by changing)
EMI	`=PMT(rate/12, n_months, -loan)`
Unpivot	`Table.UnpivotOtherColumns(src, {"key"}, "attr", "val")`
Join	Merge Queries, Left Outer
Measure	`Delivered := CALCULATE(SUM(T[Amount]), T[Status]="Delivered")`

Practice set

Problem 1 From Example 3.1.1, what share of Mumbai's revenue was cancelled?

Answer: 450 / 3047 = 0.1477, about 14.8%.

Problem 2 In a PivotTable of our data by Category (Sum of Amount), what are the three totals?

Answer: Electronics 1299 + 2499 = 3798. Fashion 998 + 799 + 1298 + 1400 = 4495. Grocery 597 + 450 = 1047. They add to 9,340.

Problem 3 What is the profit in Example 3.3.1 at price 500, 100 units?

Answer: (500 − 300) × 100 − 5000 = 15,000.

Problem 4 Goal Seek: unit cost 300, 100 units, fixed 5,000. Find the price for a profit of Rs 20,000.

Answer: (P − 300) × 100 = 25,000, so P = 550. Check (550 − 300) × 100 − 5000 = 20,000.

Problem 5 Find the bug: a PivotTable total is exactly double the real total.

Answer: The source data likely contains its own Total row, so the pivot adds it again. Remove the total row from the source Table.

Problem 6 Find the bug: a conditional format rule `=G2="Returned"` (no `$`) applied to A2:H9 highlights the wrong cells.

Answer: Without `$` the column shifts for each cell, so column B tests H2 and so on. Use `=$G2="Returned"`.

Problem 7 What is the output of the M step `Table.SelectRows(WithAmt, each [Status] = "Delivered")` on our table, in row count?

Answer: 6 rows (orders 1001, 1002, 1004, 1006, 1007, 1008).

Problem 8 The two-sample t-test has t = 3 with equal variances 2.5 and n = 5 each. If both groups had n = 20 and the same means and variances, what would t be?

Answer: SE = √(2.5 × (1/20 + 1/20)) = √(2.5 × 0.1) = √0.25 = 0.5, so t = 3 / 0.5 = 6. More data gives a stronger signal for the same gap.

Problem 9 A Solver model gives x = 20, y = 60 under constraints 2x + y ≤ 100 and x + y ≤ 80. Profit per hour of machine time: if machine hours rise to 101, what happens to the corner, and what is the new profit?

Answer: Solve 2x + y = 101 and x + y = 80: x = 21, y = 59. Profit = 40 × 21 + 30 × 59 = 840 + 1770 = 2,610. One extra machine hour adds Rs 10.

Problem 10 (interview) "A manager sends 12 monthly CSVs with the same columns. How would you combine and keep the report updating?"

Answer: Put the files in a folder, use Power Query From Folder with Combine and Transform, add type and cleaning steps, load to the Data Model, build PivotTables or Power BI visuals, and refresh each month. No copy and paste.

Problem 11 (interview) "Why use the Data Model instead of VLOOKUP into one big sheet?"

Answer: The model keeps tables separate, uses relationships and fast column storage, scales to millions of rows, and supports DAX measures that work in every slice. VLOOKUP duplicates data and is slow and fragile.

Prompt pack

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional in India moving into data roles, and English is my second language. Teach me Chapter 3 topic: Excel PivotTables, including Show Values As, grouping, calculated fields, slicers and why they need a refresh. Use a tiny 8-row sales table with real numbers and check each result by hand. Ask me to predict the pivot before showing it, and quiz me afterwards.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data roles. Teach me Power Query step by step: import a CSV, fix types, trim text, unpivot, merge queries, append, import a folder, and read the M language it writes. Use a small example I can build myself. After each step ask me what the M code line means, then give 8 graded exercises without solutions.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data roles. Explain Excel what-if tools: Goal Seek, Data Tables, Scenario Manager and Solver. For each, give an everyday analogy, one worked example with real numbers I can check by hand, and a mistake beginners make. Then give me a new business problem to solve with Solver and help me check my answer.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data roles. Review my Excel analysis for errors: PivotTable setup, conditional formatting formulas, data validation, Power Query steps and any totals that look wrong. Explain how to find the root cause and how to make it refreshable. My description and screenshots in text: [PASTE HERE]
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data roles. Run a mock interview on advanced Excel: PivotTables, Power Query, Power Pivot, what-if analysis, ToolPak statistics and automation. Ask 8 scenario questions one at a time, wait for my answer, then score it, correct it in simple English and suggest
```

what to practise next.

CHAPTER 4

PANDAS I: SERIES, DATAFRAMES, INPUT/OUTPUT, SELECTION AND DATA TYPES

Learn to load a table, look at it, and pick exactly the rows and columns you need.

pandas is the main Python tool of a data analyst. If Excel is a calculator with a grid, pandas is the same grid controlled by code: you write the steps once, and you can run them on 100 rows or 10 million rows, tomorrow or next year. Your delivery log, your orders table, your survey file: all of them become a `DataFrame`. In this chapter you will learn what a Series and a DataFrame are, how to read and write files, how to select data safely, and why data types matter. By the end you can open an unknown CSV and answer simple questions about it with confidence.

PROFESSOR'S NOTE

All outputs here were produced with pandas 3.0.6, NumPy 2.5.3 and Python 3.14. pandas 3 changed some defaults compared with older versions: text columns now have the dtype `str` (older versions showed `object`), and "Copy-on-Write" is always on (explained in 4.5). If your version is older, your output may differ in small ways, such as `object` instead of `str`. Check yours with `pd.__version__`.

Install, if needed: `pip install pandas numpy matplotlib`. Then:

```
import pandas as pd
import numpy as np
```

We use the abbreviations `pd` and `np` everywhere; they are universal.

4.1 Series: One Labelled Column

INTUITION

A Series is a list where every item has a name tag. A plain Python list says "the third number is 310". A Series says "Leh is 310". You look things up by name, not by counting positions.

Formal definition. A `Series` is a one-dimensional array with an **index** (the labels) and a **dtype** (the type of all values). It also has an optional `name`.

Worked example 4.1.1 Store the average parcel delivery time (hours) for four cities.

```
s = pd.Series([120, 95, 310, 88],
              index=["delhi", "pune", "leh", "goa"],
              name="hours")
print(s)
print(s["leh"], s.iloc[0], s.mean())
print(s[s > 100])
```

Predict: what does `s[s > 100]` keep?

```
delhi    120
pune      95
leh      310
goa       88
Name: hours, dtype: int64
310 120 153.25
delhi    120
leh      310
Name: hours, dtype: int64
```

- **Print 1** Labels on the left, values on the right, the name and dtype below.
- **Print 2** `s["leh"]` looks up by label: 310. `s.iloc[0]` looks up by position: 120 (the first item). `s.mean()` = (120 + 95 + 310 + 88) / 4 = 613 / 4 = 153.25.
- **Print 3** `s > 100` makes a Series of True/False (a **boolean mask**). Using it inside `[]` keeps only the True rows: delhi (120) and leh (310).
- **Check** 120 + 95 = 215, + 310 = 525, + 88 = 613. 613 / 4 = 153.25. Correct.

Aligned arithmetic. Operations match labels, not positions:

```
print(s + pd.Series({"delhi": 1, "pune": 2}))
```

```
delhi    121.0
goa        NaN
leh        NaN
pune      97.0
dtype: float64
```

Labels that exist in only one Series give `NaN` (Not a Number, pandas' missing value). The dtype changed to float because `NaN` is a float. This **alignment** is powerful and surprising. A `KeyError` appears when you ask for a missing label: `s["pay"]` raises `KeyError: 'pay'`.

The picture

```
index      values
delhi       120
pune         95   <- s["pune"] = 95    (label)
leh         310   <- s.iloc[2] = 310   (position)
goa          88
```

Connection to the real job. Every column of a DataFrame is a Series. When you write `df["price"].mean()` you use a Series method.

Applications

- Analytics: one metric per day as a time-indexed Series.
- Data Science: a target column `y` is a Series.
- Data Engineering: lookup dictionaries as Series (`map`).

COMMON MISTAKE

Confusing label and position. After sorting, `s.iloc[0]` and `s.loc["delhi"]` can be different items. Use `.loc` for labels and `.iloc` for positions on purpose.

Check yourself

1. What does `s.iloc[2]` return in the example? Answer: 310 (the third item).
2. What is a boolean mask? Answer: A Series of True/False used to select rows.
3. Why did the sum above give NaN for leh? Answer: That label had no partner in the other Series.

4.2 DataFrame: A Table of Series

INTUITION

A DataFrame is an Excel sheet inside Python: columns have names, rows have labels, every column has its own type. Think of it as many Series that share one index.

Formal definition. A `DataFrame` is a two-dimensional table: an `index` (row labels), `columns` (column labels) and one dtype per column.

Our running dataset. Six orders. We read it from text so the example is complete.

```
import io
csv = """order_id,order_date,customer_id,city,qty,price,status
1001,2025-01-03,C01,Pune,2,499.0,delivered
1002,2025-01-03,C02,Mumbai,1,1299.5,delivered
1003,2025-01-04,C01,Pune,3,199.0,returned
1004,2025-01-05,C03,Delhi,1,,delivered
1005,2025-01-06,C04,Mumbai,5,89.9,cancelled
1006,2025-01-07,C02,Mumbai,2,649.0,delivered
"""
df = pd.read_csv(io.StringIO(csv), parse_dates=["order_date"],
                 dtype={"customer_id": "string"})
print(df)
print(df.dtypes)
print(df.shape)
df.info()
```

Real output:

```
   order_id order_date customer_id    city  qty   price     status
0      1001 2025-01-03         C01    Pune    2  499.0  delivered
1      1002 2025-01-03         C02  Mumbai    1  1299.5  delivered
2      1003 2025-01-04         C01    Pune    3  199.0   returned
3      1004 2025-01-05         C03   Delhi    1    NaN  delivered
4      1005 2025-01-06         C04  Mumbai    5   89.9  cancelled
5      1006 2025-01-07         C02  Mumbai    2  649.0  delivered
order_id               int64
order_date     datetime64[us]
customer_id            string
city                      str
qty                     int64
price                 float64
status                    str
dtype: object
(6, 7)
<class 'pandas.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 7 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   order_id     6 non-null      int64
 1   order_date   6 non-null      datetime64[us]
 2   customer_id  6 non-null      string
 3   city         6 non-null      str
 4   qty          6 non-null      int64
 5   price        5 non-null      float64
 6   status       6 non-null      str
dtypes: datetime64[us](1), float64(1), int64(2), str(2), string(1)
memory usage: 468.0 bytes
```

Read it line by line.

- `parse_dates=["order_date"]` turned the text dates into real date-times (`datetime64`). Without it the column would be text.
- The empty price in row 3 became `NaN` . That forced `price` to be `float64` (decimal).
- `df.shape` is `(rows, columns)` = (6, 7).
- `df.info()` is your first health check: look at "Non-Null Count". Price has 5 non-null out of 6, which tells you one value is missing.

Everyday inspection commands

Command	Shows

`df.head(3)` / `df.tail(3)`	First / last rows
`df.shape`	(rows, columns)
`df.columns.tolist()`	Column names
`df.dtypes`	Type of each column
`df.describe()`	Count, mean, std, min, quartiles for numbers
`df["status"].value_counts()`	How often each value appears
`df.isna().sum()`	Missing values per column
`df.sample(5, random_state=1)`	Random rows, repeatable

Worked example 4.2.1 Answer three questions from the table: total quantity, count of non-missing prices, and number of different cities.

- **Formula** `df["qty"].sum()`, `df["price"].count()`, `df["city"].nunique()`
- **Plug in** Quantities are 2, 1, 3, 1, 5, 2.
- **Compute** Sum = 2 + 1 + 3 + 1 + 5 + 2 = 14. `count()` ignores missing values, so 5 (it was 6 rows). Cities: Pune, Mumbai, Delhi gives 3.
- **Answer** 14 units, 5 prices present, 3 cities.
- **Check** `len(df)` is 6 and `df["price"].isna().sum()` is 1, and 6 − 1 = 5.
- **Interpretation** One order has no price. Any revenue total silently skips it, so you must decide how to treat it (Chapter 5).

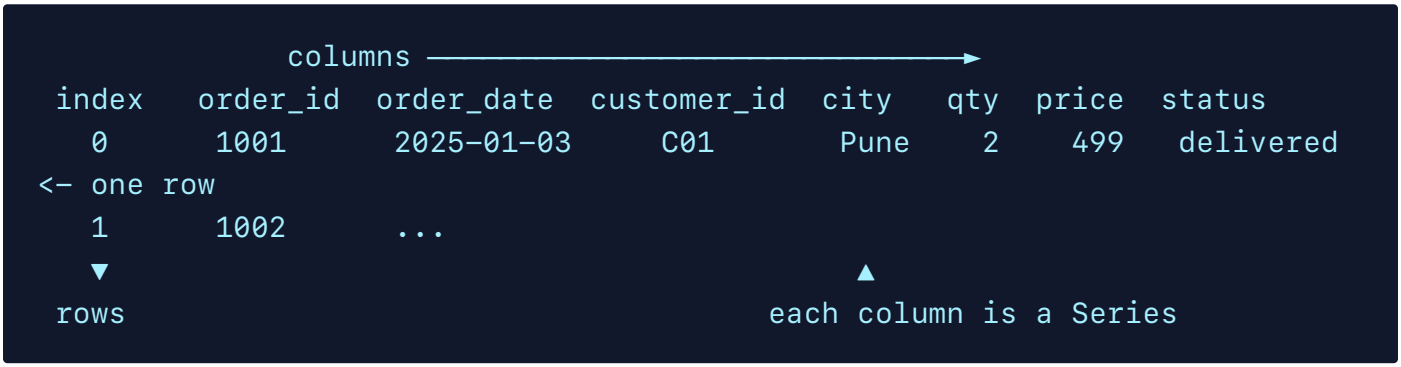
Worked example 4.2.2 Build a new column and find the top 2 orders.

```
df["amount"] = df["qty"] * df["price"]
print(df.nlargest(2, "amount")[["order_id", "amount"]])
```

- **Compute** Amounts: 998.0, 1299.5, 597.0, NaN (missing price), 449.5, 1298.0. The two largest are 1299.5 (order 1002) and 1298.0 (order 1006).
- **Answer** Orders 1002 and 1006.
- **Check** 2 × 499 = 998; 5 × 89.9 = 449.5; 2 × 649 = 1298. `1 × NaN` is NaN, so `nlargest` skips order 1004.
- **Why it matters** Arithmetic on columns happens for every row at once: that is called

vectorised arithmetic (Chapter 6).

The picture



Connection to the real job. The first thing you do with any new data file in a job is `info()`, `describe()`, `head()`, and `isna().sum()`. Chapter 7 turns those into a full EDA routine.

Applications

- Analytics: monthly sales tables, survey responses.
- Data Science: the feature matrix `X` is a DataFrame.
- Data Engineering: pipeline steps that take and return DataFrames.
- Operations: load a large delivery-log CSV and compute percentiles per city (see Practice Problem 10).

COMMON MISTAKE

Trusting what you see and not the dtype. A column that looks numeric may be text (for example "1,150" with a comma). `df.dtypes` and `df.info()` tell the truth, so run them on every new file.

Check yourself

1. What does `df.shape` return? Answer: A pair (number of rows, number of columns).
2. Which method shows missing values per column? Answer: `df.isna().sum()`.
3. Why did `price` become float? Answer: It has a `NaN`, and NaN is a float value.

4.3 Reading and Writing Files

INTUITION

Data lives in files, databases and web services. `read_*` functions bring them into a DataFrame, and `to_*` functions put results back. Choose the format to match the job: CSV for sharing, Parquet for speed and size, Excel for business users, SQL for databases, JSON for web data.

The main functions

Format	Read	Write	Notes
CSV	`pd.read_csv`	`df.to_csv(path, index=False)`	Text, no types stored
Excel	`pd.read_excel`	`df.to_excel`	Needs `openpyxl`; one sheet at a time
JSON	`pd.read_json`	`df.to_json`	Use `orient="records"` for a list of rows
SQL	`pd.read_sql(query, connection)`	`df.to_sql`	With `sqlite3` or SQLAlchemy
Parquet	`pd.read_parquet`	`df.to_parquet`	Needs `pyarrow` or `fastparquet`; columnar, compressed, keeps types

The `read_csv` arguments you will use daily

Argument	Meaning	Example
`dtype`	Force column types	`dtype={"customer_id": "string"}`
`parse_dates`	Convert to datetime	`parse_dates=["order_date"]`
`usecols`	Read only some columns (saves memory)	`usecols=["order_id", "price"]`
`nrows`	Read only the first n rows	`nrows=3`

`chunksize`	Read in pieces	`chunksize=100000`
`na_values`	Extra strings to treat as missing	`na_values=["NA", "-"]`
`sep`	Separator	`sep=";"`
`encoding`	Text encoding	`encoding="utf-8"`

Worked example 4.3.1 Read only two columns and the first three rows.

```
df.to_csv("orders_small.csv", index=False)
print(pd.read_csv("orders_small.csv", usecols=["order_id", "price"],
nrows=3))
```

```
   order_id   price
0      1001   499.0
1      1002  1299.5
2      1003   199.0
```

- **Compute** `usecols` skipped the other columns while reading, and `nrows=3` stopped after three rows. On a 5 GB file this finishes in a moment and uses little memory.
- **Answer** A 3 by 2 table.
- **Check** The numbers match the first three rows of the full table above.

Worked example 4.3.2 Read a file in chunks. The file has 6 orders and the chunk size is 4 rows.

```
for chunk in pd.read_csv("orders_small.csv", chunksize=4):
    print(len(chunk), chunk.qty.sum())
```

Predict: how many chunks print?

```
4 7
2 7
```

- **Compute** The first chunk holds rows 0 to 3 with qty 2 + 1 + 3 + 1 = 7. The second holds rows 4 and 5 with qty 5 + 2 = 7.
- **Answer** Two chunks. Total qty = 7 + 7 = 14, which equals the 14 found earlier.

- **Why it matters** When a file is bigger than memory, process each chunk (aggregate it), then combine the small results. You never hold the whole file.

Reading from SQL. We used Python's built-in `sqlite3` with an in-memory database.

```
import sqlite3
con = sqlite3.connect(":memory:")
df.drop(columns=
["status"]).assign(order_date=df.order_date.astype(str)).to_sql("orders",
con, index=False)
print(pd.read_sql("select city, sum(qty*price) as revenue from orders
group by city", con))
```

```
     city  revenue
0   Delhi      NaN
1  Mumbai   3047.0
2    Pune   1595.0
```

- **Compute** Mumbai: 1 × 1299.5 + 5 × 89.9 + 2 × 649 = 1299.5 + 449.5 + 1298 = 3047.0. Pune: 2 × 499 + 3 × 199 = 998 + 597 = 1595. Delhi's price is missing, so SQL gives NULL, which pandas shows as `NaN`.
- **Answer** Three rows, one per city.
- **Check** The Mumbai and Pune sums add up by hand. This is the same SQL you learn in the separate SQL document, now running from Python.

JSON. `df.to_json("orders.json", orient="records", date_format="iso")` writes a list of objects. `pd.read_json` reads it back. We checked: the date came back as text (`str`), not datetime, because JSON has no date type. Convert again with `pd.to_datetime`.

Parquet. We tried `df.to_parquet(...)` and got `ImportError: Unable to find a usable engine; tried using: 'pyarrow', 'fastparquet'`. So Parquet needs `pip install pyarrow` first. Once installed, `to_parquet` / `read_parquet` work, keep dtypes (dates stay dates), and files are much smaller and faster to read than CSV. (Not run here because pyarrow was not installed.)

Excel. `pd.read_excel("file.xlsx", sheet_name="Data")` needs `openpyxl`. See Chapter 3 for a real run and a formula caveat.

The picture



Connection to the real job. Data engineers choose Parquet for pipelines and analysts export CSV or Excel for stakeholders. Most real jobs start with `read_sql` from a warehouse.

Applications

- Analytics: weekly extract from SQL, saved as Excel for managers.
- Data Science: Parquet for big feature tables.
- Data Engineering: chunked loads from APIs and logs.
- Logistics: read very large delivery-log files with `usecols` and `chunksize`.

COMMON MISTAKE

Saving with the index. `df.to_csv("x.csv")` writes an extra unnamed column. Use `index=False` unless the index carries meaning.

Check yourself

1. Which argument saves memory by loading fewer columns? Answer: `usecols`.
2. Why does Parquet need an extra package? Answer: pandas relies on `pyarrow` or `fastparquet` to read and write it.
3. After `read_json`, what type was the date? Answer: Text (`str`); convert with `pd.to_datetime`.

4.4 Selecting Rows and Columns

INTUITION

You rarely need the whole table. Selecting is like asking a librarian: "give me the rows where the city is Mumbai, and only the columns order number and price". There are three ways to say it: by label (`loc`), by position (`iloc`), and by condition (a boolean mask).

The tools

Goal	Code	Notes
One column	`df["city"]`	A Series
Several columns	`df[["order_id", "price"]]`	Double brackets, returns a DataFrame
Rows and columns by label	`df.loc[rows, cols]`	End of slice is **included**
Rows and columns by position	`df.iloc[rows, cols]`	End of slice is **excluded**
Condition	`df[df["qty"] > 1]`	Boolean mask
Several conditions	`df[(a) & (b)]`,`\`	`for OR,` ~` for NOT
Membership	`df["city"].isin(["Pune", "Delhi"])`	Like SQL IN
Range	`df["price"].between(200, 700)`	Inclusive both ends
Readable condition	`df.query("qty >= 2 and status != 'cancelled'")`	Text expression
Name pattern	`df.filter(like="_id")`	Selects columns by name

Worked example 4.4.1 Select Mumbai orders and two columns, then a slice by position.

```
print(df.loc[df.city == "Mumbai", ["order_id", "price"]])
print(df.iloc[1:4, 0:3])
```

```
   order_id   price
1      1002  1299.5
4      1005    89.9
5      1006   649.0
   order_id order_date customer_id
1      1002 2025-01-03         C02
2      1003 2025-01-04         C01
3      1004 2025-01-05         C03
```

- **Compute** `df.city == "Mumbai"` is True for rows 1, 4, 5. `iloc[1:4, 0:3]` means rows at positions 1, 2, 3 (4 is excluded) and the first three columns (positions 0, 1, 2).
- **Answer** Three Mumbai rows with two columns, and a 3 by 3 block.
- **Check** The row labels on the left (1, 4, 5) are the original index values, preserved after filtering.
- **Why it matters** `loc` slices include the end label while `iloc` slices exclude it. That rule causes most off-by-one bugs.

Worked example 4.4.2 Combine conditions: orders with qty at least 2 that were not cancelled.

```
print(df.query("qty >= 2 and status != 'cancelled'"))
```

```
   order_id order_date customer_id    city  qty  price     status
0      1001 2025-01-03         C01    Pune    2  499.0  delivered
2      1003 2025-01-04         C01    Pune    3  199.0   returned
5      1006 2025-01-07         C02  Mumbai    2  649.0  delivered
```

- **Compute** qty >= 2 holds for rows 0 (2), 2 (3), 4 (5), 5 (2). Row 4 is cancelled, so it drops out.
- **Answer** Rows 0, 2 and 5.
- **Check** The same filter with masks, `df[(df.qty >= 2) & (df.status != "cancelled")]`, returns the same three rows.

More examples we ran: `df[df.city.isin(["Pune", "Delhi"])]` returns rows 0, 2, 3; `df[df.price.between(200, 700)]` returns rows 0 (499.0) and 5 (649.0), because 199.0 is below 200 and the NaN price never matches.

MultiIndex. An index with more than one level, for example `mi = df.set_index(["city", "order_id"]).sort_index()`. Then `mi.loc["Mumbai"]` returns all Mumbai rows, and `mi.loc[("Mumbai", 1002)]` returns one row. Grouped results often come back with a MultiIndex (Chapter 5), and `reset_index()` turns the levels back into normal columns.

Wrong version and its error. `df["order_id", "price"]` fails: pandas treats the tuple as one column name and raises a `KeyError`. Use `df[["order_id", "price"]]`. And `df[df.qty > 1 & df.price > 100]` fails because `&` binds tighter than `>`: write `df[(df.qty > 1) & (df.price > 100)]`.

The picture

```
          loc[rows where city=="Mumbai", cols by name]     iloc[1:4, 0:3]
 row 0    .  .  .  .                                        .  .  .   (not
chosen)
 row 1    X  X                                             X  X  X
 row 2    .  .                                             X  X  X
 row 3    .  .                                             X  X  X
```

Connection to the real job. Selecting is the `WHERE` and `SELECT` of SQL. In interviews you will be asked to filter with two or three conditions on a screen share. Practise until the brackets feel natural.

Applications

- Analytics: orders of one region; customers who bought twice.
- Data Science: split a dataset by condition for train/test.
- Data Engineering: filter bad records before loading.

COMMON MISTAKE

Using `and` / `or` instead of `&` / `|` with Series. Python's `and` cannot handle a whole column and raises "The truth value of a Series is ambiguous". Use `&`, `|` with parentheses.

Check yourself

1. Which slice includes its end: `loc` or `iloc`? Answer: `loc` includes the end label.
2. How do you write "city is Pune or Delhi"? Answer: `df["city"].isin(["Pune", "Delhi"])`.
3. Why parentheses in `(a) & (b)`? Answer: `&` binds tighter than comparison operators.

4.5 Data Types and Common Pitfalls

INTUITION

A type tells pandas how to treat a column. "10" + "10" as text gives "1010", as numbers gives 20. Types affect memory, speed, and correctness. A wrong type is the most common silent bug in analytics.

The dtypes to know

dtype	Holds	Missing value	Note
`int64`	Whole numbers	Not allowed	One NaN turns it into float
`float64`	Decimals	`NaN`	
`Int64`, `Float64` (capital)	Nullable numbers	`<NA>`	Keeps integers when values are missing
`bool`	True/False		
`str` (pandas 3) / `object` (older)	Text	`NaN`	
`string`	Explicit text type	`<NA>`	Can use `string[pyarrow]`
`category`	Few repeated values	`NaN`	Saves memory, allows order
`datetime64[us]`	Date and time	`NaT`	`[us]` means microsecond resolution (older versions: `[ns]`)
`timedelta64`	Duration	`NaT`	Difference of two dates

Worked example 4.5.1 Nullable integers.

```
print(pd.Series([1, 2, None]).dtype)
print(pd.Series([1, 2, None], dtype="Int64").tolist(), pd.Series([1, 2, None], dtype="Int64").dtype)
```

```
float64
[1, 2, <NA>] Int64
```

- **Compute** A normal Series with a missing value must use float, so 1 and 2 are stored as 1.0 and 2.0. The capital `Int64` keeps true integers and shows `<NA>`.

- **Answer** `float64` versus `Int64`.
- **Why it matters** Counts and IDs should not become 1.0, 2.0. Use nullable types when integers can be missing.

Converting types (we ran each; the comment shows the real result)

```
pd.to_numeric(pd.Series(["10", "20", "x"]), errors="coerce").tolist()   #
[10.0, 20.0, nan]
pd.Series(["10", "20", "x"]).astype(int)    # ValueError: invalid literal
for int() with base 10: 'x'
pd.to_datetime("03/04/2025")                # 2025-03-04 (month first)
pd.to_datetime("03/04/2025", dayfirst=True) # 2025-04-03 (day first)
```

`astype(int)` fails on the bad text "x", while `to_numeric(..., errors="coerce")` turns it into `NaN` so you can find and handle bad values. Notice that `03/04/2025` means 4 March or 3 April depending on `dayfirst`. With Indian data (day first) always set `dayfirst=True` or give an explicit `format="%d/%m/%Y"`. A wrong guess here shifts a month of sales into another month, silently.

Category dtype and memory. Text repeated many times costs memory. On a 300,000-row column with 4 city names, memory was 16,349,930 bytes as text and 300,350 bytes as `category`, 54 times smaller (Chapter 6 shows the full run).

Copy-on-Write and `SettingWithCopyWarning`. In older pandas, assigning into a filtered slice sometimes changed the original table and sometimes did not, and pandas printed `SettingWithCopyWarning`. In pandas 3 Copy-on-Write is always on: **every derived object behaves as its own copy**. We ran this (with `df` freshly read from the CSV):

```
view = df[df.city == "Pune"]
view["qty"] = 0
print(df.qty.tolist())            # [2, 1, 3, 1, 5, 2]  (original
unchanged)

df2 = df.copy()
df2[df2.city == "Pune"]["qty"] = 99      # chained assignment
print(df2.qty.tolist())           # [2, 1, 3, 1, 5, 2]  (nothing
happened!)
df2.loc[df2.city == "Pune", "qty"] = 99  # correct: one step
print(df2.qty.tolist())           # [99, 1, 99, 1, 5, 2]
```

pandas 3 printed a `ChainedAssignmentError` warning for the chained line and left the data

unchanged. The rule is simple: **update with one `.loc[rows, column] = value` call**, never `df[mask]["col"] = value`. In older pandas the chained version sometimes worked and sometimes did not, which is why it was dangerous.

Implicit casting pitfalls. Mixed types in one column become text. Integers with missing values become float. Adding a number to a text column fails. Comparing text numbers ("10" < "9" is True as text) gives wrong sorts. Always convert first.

Memory. `df.memory_usage(deep=True)` shows bytes per column; `deep=True` counts the text. Use `usecols`, `category`, smaller ints (`int16`) and `chunksize` to control it.

The picture

```
"1,150"  --(replace ',' then to_numeric)--> 1150.0 float64
"10"     --astype(int)--> 10      "x" --astype(int)--> ValueError
"x"      --to_numeric(errors="coerce")--> NaN   (you can then find it)
```

Connection to the real job. Bad dtypes cause bad joins (IDs "007" vs 7), bad sorts, and bad charts. In code review, the question "what are the dtypes?" is routine.

Applications

- Analytics: IDs as text, money as float, dates as datetime.
- Data Science: ordered `category` types for features.
- Data Engineering: schema enforcement (Parquet keeps types).

COMMON MISTAKE

Reading IDs as numbers. `C001` is fine, but `00123` becomes 123 and loses its zeros, and then it will not match the key in another table. Use `dtype={"id": "string"}` at read time.

Check yourself

1. What does `errors="coerce"` do in `to_numeric`? Answer: Bad values become NaN instead of raising an error.
2. Which is the right way to set values on a filter? Answer: `df.loc[mask, "col"] = value`.
3. What does `dayfirst=True` do? Answer: It reads 03/04/2025 as 3 April.

Chapter recap

- A Series is a labelled column; a DataFrame is a table of Series sharing an index.
- Always start with `head()`, `info()`, `describe()`, `isna().sum()`, `dtypes`.
- `read_csv` has arguments that save you later (`dtype`, `parse_dates`, `usecols`, `chunksize`).
- Parquet is fast and typed but needs `pyarrow`; JSON loses dates; Excel needs `openpyxl`.
- `loc` selects by label (end included); `iloc` by position (end excluded); masks and `query` filter rows.
- Combine conditions with `&`, `|`, `~` and parentheses.
- One missing value turns integers into floats; use nullable `Int64` when needed.
- Convert with `to_numeric(errors="coerce")` and `to_datetime(dayfirst=True)`.
- Set values with one `.loc` call; chained assignment never works in pandas 3.

Task	Code
Look at data	`df.head()`, `df.info()`, `df.describe()`
Missing count	`df.isna().sum()`
Select columns	`df[["a", "b"]]`
Filter rows	`df[(df.a > 1) & (df.b == "x")]` or `df.query("a > 1 and b == 'x'")`
By label / position	`df.loc[r, c]` / `df.iloc[r, c]`
Fix numbers	`pd.to_numeric(s, errors="coerce")`
Fix dates	`pd.to_datetime(s, dayfirst=True)`
Set safely	`df.loc[mask, "col"] = v`

Practice set

Problem 1 For `s = pd.Series([5, 7, 9], index=["a", "b", "c"])`, what is

`s.loc["b"]` and `s.iloc[2]`?

Answer: 7 and 9.

Problem 2 What does `df.shape` give for our orders table (before adding `amount`), and what does `len(df.columns)` give?

Answer: `(6, 7)` and 7.

Problem 3 How many rows does `df[df.status == "delivered"]` return?

Answer: 4 rows (orders 1001, 1002, 1004, 1006).

Problem 4 Using `df.loc`, write the code to get order_id and city for orders with qty above 2.

Answer: `df.loc[df.qty > 2, ["order_id", "city"]]`. Rows: 1003 (Pune, qty 3) and 1005 (Mumbai, qty 5).

Problem 5 What is `df["price"].mean()`, and why is it not (499 + 1299.5 + 199 + 89.9 + 649) / 6?

Answer: 547.28. `mean()` skips NaN, so it divides by 5. The sum is 2736.4 and 2736.4 / 5 = 547.28.

Problem 6 Find the bug: `df[df.qty > 1 & df.price > 100]` raises an error.

Answer: Missing parentheses. Operator precedence makes `1 & df.price` happen first. Write `df[(df.qty > 1) & (df.price > 100)]`.

Problem 7 What does `df.iloc[0:2, 1:3]` return, in words?

Answer: The first two rows (positions 0 and 1) and the second and third columns (order_date, customer_id). End positions are excluded.

Problem 8 Find the bug: `df2[df2.city == "Pune"]["qty"] = 99` does nothing. Fix it.

Answer: Chained assignment works on a copy. Fix: `df2.loc[df2.city == "Pune", "qty"] = 99`.

Problem 9 The CSV has an ID "00123". After `pd.read_csv` with no options, what does the column hold and what is the fix?

Answer: The number 123 (zeros lost). Fix with `dtype={"id": "string"}`.

Problem 10 (interview) You are given a 3 GB delivery-log CSV with columns `timestamp`,

`city, hours, delivered`. How do you compute the 95th percentile of `hours` per `city` without running out of memory?

Answer: Read only the needed columns with `usecols=["city", "hours"]` and smaller dtypes (`dtype={"hours": "int32"}`). If it still does not fit, remember percentiles need the values, not just sums, so use a tool that can stream and aggregate, such as DuckDB or Polars (Chapter 6), or keep a compact list of `hours` per city while reading chunks. Then `df.groupby("city")["hours"].quantile(0.95)`.

Problem 11 (interview) "What is the difference between `loc` and `iloc`, and when does it matter?"

Answer: `loc` uses labels and includes the end of a slice; `iloc` uses positions and excludes the end. It matters after sorting or filtering, because the labels no longer equal the positions.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in
India moving into data roles, and English is my second language. Teach me
Chapter 4 topic: the pandas Series and DataFrame. Explain index, columns,
dtypes and alignment with everyday analogies. Show short runnable code on
a 6-row orders table and ask me to predict each output before you reveal
it. Check my understanding after each part.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Give me 8 graded exercises on selecting data in
pandas: loc, iloc, boolean masks, isin, between, query, and combining
conditions. Use a small orders DataFrame and include at least two "find
the bug" items with real error messages. Do not give solutions yet; I will
paste answers for you to check.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Quiz me one question at a time on pandas read_csv
options (dtype, parse_dates, usecols, nrows, chunksize, na_values),
Parquet, JSON, Excel and SQL input/output. Start easy, get harder, and
```

```
after each answer explain what really happens in memory and why it matters
for big files.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Explain pandas data types: int, float, nullable
Int64, str/string, category, datetime, and why one missing value changes a
dtype. Also explain Copy-on-Write and chained assignment in pandas 3. Use
small examples with real output. Then review my code for dtype and
assignment mistakes. My code: [PASTE HERE]
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Run a mock interview with 6 pandas fundamentals
questions (load a file, inspect it, filter with several conditions, fix
wrong dtypes, save results, handle a file bigger than memory). Ask one at
a time, wait for my code, then grade it from 1 to 5 and show the cleaner
version.
```

CHAPTER 5

PANDAS II: CLEANING, TRANSFORMING, GROUPING, JOINING, RESHAPING AND TIME SERIES

The everyday work of an analyst: take messy data and turn it into answers.

Real data is never clean. Cities are spelled three ways, dates arrive as text, one order appears twice, one amount has an extra zero, and the sales team and the finance team keep their tables apart. Analysts spend more than half their working time fixing exactly these problems. This chapter gives you a toolbox for it: cleaning, transforming columns, grouping and summarising, joining tables, reshaping, and working with dates and time. We use one messy 9-line orders file for the whole chapter, so each fix builds on the one before. By the end, you can turn a dirty file into a tidy table and a summary that a manager can read.

PROFESSOR'S NOTE

All code ran with pandas 3.0.6. Outputs are real. `DataFrame.applymap` was renamed `DataFrame.map` in pandas 2.1 (we use `map`). pandas 3 shows text columns as `str`.

The messy file. Read everything as text first (`dtype=str`), so nothing is guessed:

```
import pandas as pd, numpy as np, io
raw = """order_id,customer_id,city,order_date,amount,qty
1,C1,pune,2025-01-01,500,2
2,C2,Mumbai ,2025-01-01,1200,1
3,C3,MUMBAI,2025-01-02,,3
4,C1,Pune,2025-01-02,450,1
4,C1,Pune,2025-01-02,450,1
5,C4,Delhi,2025-01-03,99999,2
6,C5,delhi,not_a_date,300,4
7,C2,Mumbai,2025-01-04,"1,150",1
8,C6,Pune,2025-01-05,380,
"""
df = pd.read_csv(io.StringIO(raw), dtype=str)
```

What is wrong with it? Before reading on, find at least five problems. (Answer list: city spelling and spaces; a missing amount; a duplicate row for order 4; an impossible amount 99999; a bad date "not_a_date"; "1,150" with a comma; a missing qty.)

5.1 Cleaning: Missing Values, Duplicates, Types and Text

INTUITION

Cleaning is like preparing vegetables before cooking: remove the bad parts, wash, and cut to a standard size. Skip it and the whole dish is spoiled, however fine the recipe.

Words you need

Word	Meaning
`NaN` / `NaT`	pandas' marker for a missing number / missing date
Imputation	Filling a missing value with a sensible guess
Duplicate	A row (or key) that appears more than once
Outlier	A value far from the others
Coerce	Convert, and turn failures into missing values

Step 1: count the problems

```
print(df.isna().sum())
print(df.duplicated().sum(), df.duplicated(subset="order_id").sum())
```

```
order_id       0
customer_id    0
city           0
order_date     0
amount         1
qty            1
dtype: int64
1 1
```

- `isna().sum()` counts missing per column: one amount, one qty.
- `duplicated()` marks the second and later copies of a row that is identical in every column. `subset="order_id"` checks the key alone. Both give 1: order 4 is repeated.

Step 2: fix duplicates, text, dates, numbers

```
df = df.drop_duplicates()
df["city"] = df["city"].str.strip().str.title()
df["order_date"] = pd.to_datetime(df["order_date"], errors="coerce")
df["amount"] = pd.to_numeric(df["amount"].str.replace(",", ""),
errors="coerce")
df["qty"] = pd.to_numeric(df["qty"], errors="coerce")
print(df)
```

```
  order_id customer_id    city order_date   amount  qty
0        1          C1    Pune 2025-01-01    500.0  2.0
1        2          C2  Mumbai 2025-01-01   1200.0  1.0
2        3          C3  Mumbai 2025-01-02      NaN  3.0
3        4          C1    Pune 2025-01-02    450.0  1.0
5        5          C4   Delhi 2025-01-03  99999.0  2.0
6        6          C5   Delhi        NaT    300.0  4.0
7        7          C2  Mumbai 2025-01-04   1150.0  1.0
8        8          C6    Pune 2025-01-05    380.0  NaN
```

- **Line 1** `drop_duplicates()` removed row 5 (the repeat of order 4). The index jumps from 3 to 5: the row labels are kept.
- **Line 2** `.str.strip()` removes spaces, `.str.title()` fixes case: `pune`, `MUMBAI`, `Mumbai` all became standard. Without this, a `groupby("city")` would show Mumbai, MUMBAI and "Mumbai " as three cities.
- **Line 3** `errors="coerce"` turned "not_a_date" into `NaT` (not a time). We keep the row and handle it later.
- **Line 4** We removed the comma in "1,150" and converted: 1150.0.
- **Line 5** `qty` became float because it has a `NaN`.

Missing values: the options

Option	Code	When
Drop rows	`df.dropna(subset=["order_date"])`	The field is essential (a date)
Fill with a constant	`df["qty"].fillna(1)`	A sensible default exists
Fill with the	`df["amount"].fillna(df["amount"].median())`	Numeric, with

median		outliers
Fill with the previous value	`s.ffill()`	Time series
Interpolate	`s.interpolate()`	Smooth measurements
Leave missing and flag	`df["amount"].isna()`	Missingness itself carries meaning

We checked: `pd.Series([1, np.nan, 3]).ffill()` gives `[1.0, 1.0, 3.0]` and `.interpolate()` gives `[1.0, 2.0, 3.0]`.

Worked example 5.1.1 Why median, not mean, to fill the missing amount.

- **Formula** median = the middle value of the sorted list.
- **Plug in** Known amounts (7 values): 500, 1200, 450, 99999, 300, 1150, 380. Sorted: 300, 380, 450, 500, 1150, 1200, 99999.
- **Compute** The middle (4th of 7) value is 500. The mean would be (300 + 380 + 450 + 500 + 1150 + 1200 + 99999) / 7 = 103979 / 7 = 14854.1.
- **Answer** Median 500 (code printed `median 500.0`). Filling with the mean would put 14,854 into order 3, absurd for a shop where most orders are 300 to 1200.
- **Check** `df["amount"].median()` printed 500.0.
- **Why it matters** The 99999 is a data error. The median is robust to it, and the mean is not.

Outliers with the IQR rule. The **IQR** (interquartile range) is Q3 − Q1, the spread of the middle half of the data. A common rule flags values below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR.

Worked example 5.1.2 Flag outliers in the amounts.

- **Formula** low = Q1 − 1.5 × IQR, high = Q3 + 1.5 × IQR
- **Plug in** Sorted: 300, 380, 450, 500, 1150, 1200, 99999. Q1 sits at position 0.25 × (7 − 1) = 1.5 (counting from 0), halfway between 380 and 450: 415. Q3 sits at position 0.75 × 6 = 4.5, halfway between 1150 and 1200: 1175.
- **Compute** IQR = 1175 − 415 = 760. low = 415 − 1.5 × 760 = 415 − 1140 = −725. high = 1175 + 1140 = 2315.
- **Answer** Real output: `415.0 1175.0 760.0 −725.0 2315.0`. Only order 5 (99999) is

above 2315, so it is the outlier.

- **Check** All other amounts (300 to 1200) lie between −725 and 2315.
- **Interpretation** 99999 is probably a typo or a unit error (an extra digit or a placeholder). Do not delete without asking the data owner. Options are to fix, to cap, or to exclude and report it.

The z-score rule, and when it fails. A z-score is (value − mean) / standard deviation: how many standard deviations a value sits from the mean. A common rule flags |z| > 3. We computed it for these 7 amounts: the z for 99999 is only 2.27, and the others are about −0.37. The rule would **not** flag the outlier! Reason: with only n values, no z-score can exceed (n − 1) / √n, which for n = 7 is 6 / 2.646 = 2.27. Also, the outlier itself inflates the mean and the standard deviation, which hides it. On small or heavy-tailed data prefer IQR or the median absolute deviation.

Capping with `clip`. `df["amount"].clip(upper=2315)` replaces 99999 with 2315 and leaves the others. Real output for order 5: `99999.0 -> 2315.0`.

Strings in bulk. The `.str` accessor works on whole columns: `.str.upper()`, `.str.contains("un")`, `.str.replace("P", "B")`, `.str.split(",")`, `.str.extract(r"(\d+)")`. We checked on `["Pune", "Delhi", "Pune"]`: `contains("un")` gave `[True, False, True]` and `replace("P", "B")` gave `['Bune', 'Delhi', 'Bune']`.

The picture

```
 raw text ──► drop duplicates ──► standardise text ──► convert types ──►
handle missing ──► handle outliers
 (everything     (keys unique)      (strip, title)      (dates, numbers)
(drop/fill/flag)   (fix/cap/flag)
  is a string)
```

Connection to the real job. In an interview a typical task is "here is a messy file; clean it and tell me what you assumed". Write the assumptions as comments. In SQL, the same checks are `COUNT(*) − COUNT(col)` for nulls and `GROUP BY key HAVING COUNT(*) > 1` for duplicates (see the SQL document).

Applications

- Analytics: customer lists with duplicate or misspelt entries.
- Data Science: imputing features before a model.

- Data Engineering: data quality rules in pipelines.
- Finance: fixing currency strings like "1,150".

COMMON MISTAKE

Filling missing values before looking at them. Ask why a value is missing (it was never recorded? the customer left it blank? the system failed?). Chapter 7 covers the mechanisms (MCAR, MAR, MNAR).

Check yourself

1. What does `errors="coerce"` do? Answer: Failed conversions become NaN or NaT instead of errors.
2. Why is the median safer than the mean for filling? Answer: Outliers do not move the median.
3. What is the IQR? Answer: Q3 minus Q1, the spread of the middle half.

5.2 Transforming Columns

INTUITION

Transforming makes new columns from old ones: a size label from an amount, a yes/no flag from a status, a number code for a city. Always do it for the whole column at once (vectorised), because a loop over rows is far slower (Chapter 6 shows 120 times).

The tools

Task	Code
New column	`df.assign(total=df.qty * df.amount)` or `df["total"] = ...`
Map values by a dictionary	`s.map({"Pune": "PNQ", "Delhi": "DEL"})`
Apply a function to each value	`s.apply(len)`

Apply to each cell of a table	`df.map(func)` (older name `applymap`)
Apply by column / by row	`df.apply(func)` / `df.apply(func, axis=1)`
If-then-else	`np.where(cond, a, b)`
Keep or replace	`s.where(cond)` keeps True, others NaN; `s.mask(cond)` is the opposite
Bins by edges	`pd.cut(s, bins=[...], labels=[...])`
Bins by quantiles	`pd.qcut(s, q=4)`
Dummy (0/1) columns	`pd.get_dummies(s, dtype=int)`

Worked example 5.2.1 Label orders using `np.where`, `cut`, `qcut` on the filled amounts. The cleaned amounts after median-fill (order 3 got 500) are 500, 1200, 500, 450, 99999, 300, 1150, 380.

```
df["amount_f"] = df["amount"].fillna(500.0)
print(df.assign(tag=np.where(df.amount_f > 1000, "big", "small"))
["tag"].tolist())
print(pd.cut(df.amount_f, bins=[0, 400, 1000, np.inf], labels=["low",
"mid", "high"]).value_counts())
print(pd.qcut(df.amount_f, q=4, labels=False).tolist())
```

```
['small', 'big', 'small', 'small', 'big', 'small', 'big', 'small']
amount_f
mid     3
high    3
low     2
Name: count, dtype: int64
[1, 3, 1, 1, 3, 0, 2, 0]
```

- `np.where` is "if amount > 1000 then big else small": 1200, 99999 and 1150 are big (three).
- `pd.cut` uses the edges you give: `(0, 400]` is low (300, 380 give 2), `(400, 1000]` is mid (500, 500, 450 give 3), above 1000 is high (1200, 99999, 1150 give 3). Counts: low 2, mid 3, high 3.
- `pd.qcut` makes bins with equal counts. With `labels=False`, it returns the bin number 0 to 3. Eight values into four bins is two per bin.

- **Check** 2 + 3 + 3 = 8 rows.
- **Why it matters** Use `cut` when business rules set the edges (age bands, price tiers). Use `qcut` for percentiles, such as "top 25% of customers".

Dummy variables. `pd.get_dummies(df.city, prefix="city", dtype=int)` makes one 0/1 column per city: `city_Delhi`, `city_Mumbai`, `city_Pune`, with exactly one 1 in each row. Models need numbers, so this turns text categories into numbers (called one-hot encoding).

`where` **keeps,** `mask` **hides.** `df.amount.where(df.amount < 1000)` gave `[500.0, nan, nan, 450.0, nan, 300.0, nan, 380.0]`: values 1000 or more became NaN.

`map` **and** `apply`**.** On `["Pune", "Delhi", "Pune"]`: `.map({"Pune": "PNQ", "Delhi": "DEL"})` gives `['PNQ', 'DEL', 'PNQ']`; `.apply(len)` gives `[4, 5, 4]`. For a table `d = pd.DataFrame({"a": [1, 2, 3], "b": [10, 20, 30]})`: `d.apply(lambda c: c.max() - c.min())` gives `a 2, b 20` (per column); `d.apply(lambda r: r.a + r.b, axis=1)` gives `[11, 22, 33]` (per row, slow, avoid for big data); `d.map(lambda x: x * 2)` doubles every cell.

The picture

```
amount_f:  300 380 | 450 500 500 | 1150 1200 99999
cut edges: (0,400]   (400,1000]     (1000, inf)
label:      low         mid            high
```

Connection to the real job. Feature creation (bands, flags, dummies) is the bridge from analytics to machine learning. In SQL these are `CASE WHEN` expressions.

Applications

- Marketing: age bands, spend tiers.
- Finance: ageing buckets (0-30, 31-60, 61-90 days).
- Data Science: one-hot encoding for models.
- Product: flag "power users" as the top decile of usage.

COMMON MISTAKE

Using `apply(axis=1)` for simple arithmetic. It is a hidden loop. `df.a + df.b` is the same result, thousands of times faster.

Check yourself

1. What is the difference between `cut` and `qcut`? Answer: `cut` uses edges you set, `qcut` makes bins with equal counts.
2. What does `get_dummies` return for 3 cities? Answer: Three 0/1 columns, one per city.
3. What does `s.where(cond)` do to rows where cond is False? Answer: Replaces them with NaN.

5.3 Grouping and Summarising

INTUITION

"Split, apply, combine." Imagine sorting receipts into piles by city (split), adding each pile (apply), and writing the totals on one sheet (combine). `groupby` is that, in one line, and it is the single most important pandas tool.

The recipe

```
g = df.groupby("city").agg(orders=("order_id", "count"),
                           revenue=("amount", "sum"),
                           avg=("amount", "mean"))
print(g)
```

This form is called **named aggregation**: `new_name=(column, function)`.

```
        orders   revenue          avg
city
Delhi        2  100299.0  50149.500000
Mumbai       3    2350.0   1175.000000
Pune         3    1330.0    443.333333
```

Worked example 5.3.1 Check the numbers by hand.

- **Plug in** Delhi has orders 5 (99999) and 6 (300). Mumbai has orders 2 (1200), 3 (missing), 7 (1150). Pune has 1 (500), 4 (450), 8 (380).
- **Compute** Delhi revenue 99999 + 300 = 100299, mean 100299 / 2 = 50149.5. Mumbai: `count` counts order_ids, so 3; `sum` and `mean` skip the missing amount: 1200 + 1150 = 2350, mean 2350 / 2 = 1175. Pune: 500 + 450 + 380 = 1330, mean 443.33.

- **Answer** The table above.
- **Check** Revenue column adds to 100299 + 2350 + 1330 = 103979, the same as the total used in 5.1.
- **Interpretation** One bad value (99999) makes Delhi look like a gold mine: 97% of all revenue. Never publish a group summary before you check outliers.

Other aggregation tools

Tool	What it does
`transform`	Returns a result with the same length as the input (for shares, ranks within group)
`filter`	Keeps or drops whole groups by a rule
`apply`	Runs your function on each group (flexible, slower)
`pivot_table`	groupby with rows and columns, like Excel
`crosstab`	Counts of two categorical columns
`value_counts`	Counts (or `normalize=True` shares) of one column

Worked example 5.3.2 `transform` : each order's share of its city total.

```
df["city_share"] = df["amount"] / df.groupby("city")
["amount"].transform("sum")
```

- **Compute** For Pune: total 1330. Order 1: 500 / 1330 = 0.376. Order 4: 450 / 1330 = 0.338. Order 8: 380 / 1330 = 0.286. For Delhi: 99999 / 100299 = 0.997 and 300 / 100299 = 0.003.
- **Answer** Real output rows: order 1 0.376, order 2 0.511, order 3 NaN, order 4 0.338, order 5 0.997, order 6 0.003, order 7 0.489, order 8 0.286.
- **Check** Within each city the shares add to 1: Pune 0.376 + 0.338 + 0.286 = 1.000; Mumbai 0.511 + 0.489 = 1.000.
- **Why it matters** `agg` would shrink the table to 3 rows, `transform` keeps all 8 rows and repeats the group total beside each. Use it for "percent of group".

`filter` : `df.groupby("city").filter(lambda x: len(x) >= 3)` keeps only cities with at least 3 orders: Mumbai and Pune (6 rows).

`pivot_table` , `crosstab` , `value_counts` :

```
pd.pivot_table(df, index="city", columns=df.order_date.dt.day,
values="amount", aggfunc="sum")
pd.crosstab(df.city, df.qty.fillna(0))
df.city.value_counts(normalize=True)
```

```
order_date     1.0    2.0      3.0     4.0    5.0
city
Delhi          NaN    NaN  99999.0     NaN    NaN
Mumbai      1200.0    0.0      NaN  1150.0    NaN
Pune         500.0  450.0      NaN     NaN  380.0
qty     0.0  1.0  2.0  3.0  4.0
city
Delhi     0    0    1    0    1
Mumbai    0    2    0    1    0
Pune      1    1    1    0    0
city
Pune      0.375
Mumbai    0.375
Delhi     0.250
Name: proportion, dtype: float64
```

- In the pivot, Mumbai on day 2 shows `0.0` (order 3 exists, but its amount is missing, and the sum of nothing is 0), while days with no orders show `NaN` . These two zeros and NaNs mean different things. This is a classic trap: a sum of all-missing values is 0, not "unknown".
- `crosstab` counts rows for each pair: Pune has one order each with qty 0 (missing, filled with 0 for display), 1 and 2.
- `value_counts(normalize=True)` : 3 of 8 orders are Pune, 0.375.

Window functions: rolling, expanding, ewm, cumsum, shift, pct_change, rank. A *window* looks at a few neighbouring rows at a time.

```
d = pd.DataFrame({"day": pd.date_range("2025-01-01", periods=7),
                  "rev": [100, 120, 90, 150, 160, 140,
200]}).set_index("day")
d["ma3"] = d.rev.rolling(3).mean()
d["cum"] = d.rev.cumsum()
d["pct"] = d.rev.pct_change().round(3)
d["prev"] = d.rev.shift(1)
d["rank"] = d.rev.rank(ascending=False)
d["ewm"] = d.rev.ewm(span=3, adjust=False).mean().round(2)
print(d)
```

```
            rev         ma3  cum    pct   prev  rank     ewm
day
2025-01-01  100         NaN  100    NaN    NaN   6.0  100.00
2025-01-02  120         NaN  220  0.200  100.0   5.0  110.00
2025-01-03   90  103.333333  310 -0.250  120.0   7.0  100.00
2025-01-04  150  120.000000  460  0.667   90.0   3.0  125.00
2025-01-05  160  133.333333  620  0.067  150.0   2.0  142.50
2025-01-06  140  150.000000  760 -0.125  160.0   4.0  141.25
2025-01-07  200  166.666667  960  0.429  140.0   1.0  170.62
```

Worked example 5.3.3 Verify the rolling mean and percentage change by hand.

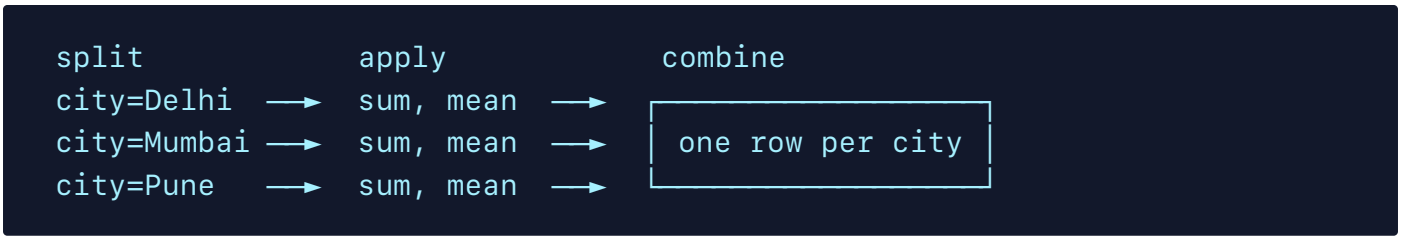
- **Formula** 3-day average = (today + previous two) / 3. pct_change = (today − yesterday) / yesterday.
- **Plug in** Day 3: (100 + 120 + 90) / 3. Day 4 pct: (150 − 90) / 90.
- **Compute** 310 / 3 = 103.33. 60 / 90 = 0.667. Day 2 pct: 20 / 100 = 0.2. Day 3 pct: −30 / 120 = −0.25.
- **Answer** Matches the table. The first two `ma3` values are NaN, because a 3-day window needs 3 days.
- **Check** `cum` is the running total: 100, 220, 310, 460, 620, 760, 960 (last equals the sum 100 + 120 + 90 + 150 + 160 + 140 + 200 = 960).
- **Why it matters** `shift(1)` gives yesterday's value on today's row, which is how you build week-on-week growth. `ewm` (exponentially weighted mean) gives recent days more weight: `ewm` day 2 = 0.5 × 120 + 0.5 × 100 = 110 (for `span=3`, α = 2 / (3 + 1) = 0.5).

A noisy daily orders line and its smooth 7-day rolling average

How to read it: the grey line is the daily value with a weekly zig-zag. The blue line averages

each 7 days, so the zig-zag cancels and the upward trend shows.

The picture



Connection to the real job. `groupby` is SQL's `GROUP BY`; `transform` and `rolling` are SQL window functions (`SUM() OVER (PARTITION BY ...)`, `AVG() OVER (ORDER BY ... ROWS 2 PRECEDING)`). The SQL document covers them.

Applications

- Analytics: revenue by city/month, running totals, week-over-week change.
- Data Science: group statistics as features (average spend per customer).
- Data Engineering: pre-aggregating data for dashboards.
- Operations: 95th-percentile delivery time by courier.

COMMON MISTAKE

Averaging averages. The mean of city means is not the overall mean when cities differ in size. Compute from totals: overall mean = total revenue / total orders.

Check yourself

1. What is the difference between `agg` and `transform`? Answer: `agg` shrinks to one row per group; `transform` keeps the original rows.
2. Why are the first two `rolling(3)` values NaN? Answer: There are not yet three days in the window.
3. What does `shift(1)` give? Answer: The previous row's value.

5.4 Combining Tables: merge, concat, join

INTUITION

Two tables are two address books. The orders table knows customer IDs, the customer table knows names. A merge looks up each ID in the second book and attaches the match, as `XLOOKUP` did in Excel (Chapter 2), but for every row at once and with explicit rules for missing partners.

Join types. `merge(a, b, on=key, how=...)`:

how	Keeps
`inner`	Only keys in both
`left`	All rows of the left table
`right`	All rows of the right table
`outer`	All rows of both

```
a = pd.DataFrame({"cid": ["C1", "C2", "C3"], "name": ["Asha", "Ben",
"Chen"]})
b = pd.DataFrame({"cid": ["C2", "C3", "C4", "C3"], "amt": [10, 20, 30,
5]})
print(a.merge(b, on="cid", how="outer", indicator=True))
```

```
  cid  name   amt      _merge
0  C1  Asha   NaN   left_only
1  C2   Ben  10.0        both
2  C3  Chen  20.0        both
3  C3  Chen   5.0        both
4  C4   NaN  30.0  right_only
```

Worked example 5.4.1 Predict the number of rows for each join.

- **Plug in** Keys in `a`: C1, C2, C3. Keys in `b`: C2, C3, C4, C3 (C3 twice).
- **Compute (inner)** Matches: C2 once, C3 twice, so 3 rows. **(left)** All `a` keys: C1 (no match, 1 row), C2 (1), C3 (2) = 4 rows. **(right)** All `b` rows: 4 rows (C4 has no name). **(outer)** C1, C2, C3 x2, C4 = 5 rows.
- **Answer** inner 3, left 4, right 4, outer 5. The real outputs matched (inner printed C2, C3, C3).
- **Check** Outer = inner (3) + left-only (1) + right-only (1) = 5.

- **Why it matters** C3 appears twice in `b`, so Chen is repeated. Joining on a key that is not unique multiplies rows, and sums of a column from the left table are then **double counted**.

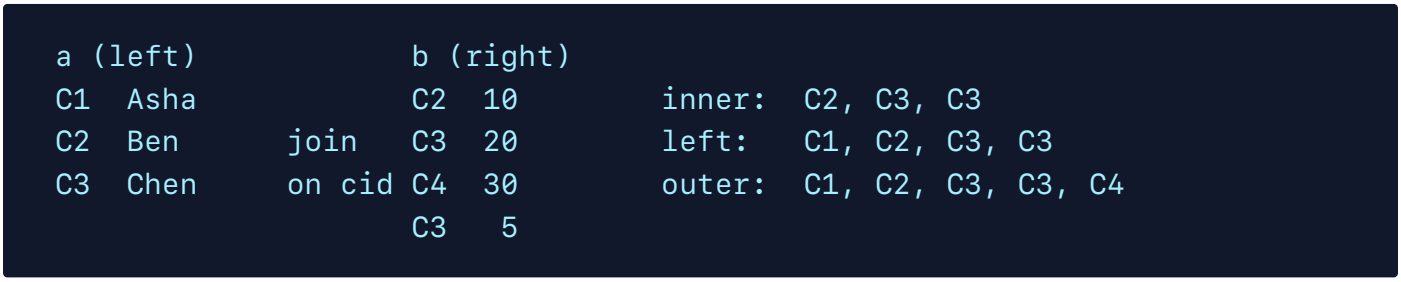
Guard with `validate`. `a.merge(b, on="cid", validate="one_to_one")` raised `MergeError: Merge keys are not unique in right dataset; not a one-to-one merge`. Always state the expected relationship (`one_to_one`, `one_to_many`, `many_to_one`) and let pandas stop you if it is false. `indicator=True` adds the `_merge` column above, so you can count unmatched rows, e.g. `(m["_merge"] == "left_only").sum()`.

Other combiners

Tool	Use
`pd.concat([a, b], ignore_index=True)`	Stack tables with the same columns (like SQL UNION ALL). We stacked two 3-row frames and got shape (6, 2).
`a.join(b)`	Merge on the index. `a.join(b)` with indexes x,y and y,z gave a left join: x has NaN for w.
`pd.merge_asof(trades, quotes, on="t")`	For each trade, the last quote before it (nearest earlier time)
`x.combine_first(y)`	Fill missing values in x from y

We ran `merge_asof` with trades at 10:00:01 and 10:00:05 and quotes at 10:00:00, 10:00:03, 10:00:06. Result: the first trade got the 10:00:00 quote (bid 99), the second got the 10:00:03 quote (bid 100). The 10:00:06 quote is after both trades, so it is unused. Both tables must be sorted by the key. This pattern matches an event (an order) to the latest state (price, plan, status) before it.

The picture



Connection to the real job. Joins are where numbers go wrong in real analyses: duplicate keys cause double counting, and unmatched keys silently disappear in inner joins. A reviewer will ask: "what is the grain of each table, and did the row count change after the join?".

Applications

- Analytics: attach customer city, plan, category to every order.
- Data Science: join features from several tables on a key.
- Data Engineering: slowly changing dimensions, point-in-time joins (`merge_asof`).
- Finance: match bank lines to ledger entries.

COMMON MISTAKE

Not checking the row count before and after a merge. Print `len(left)` and `len(result)` . If a `left` join of a many-to-one lookup grows the rows, the lookup keys are duplicated.

Check yourself

1. Which join keeps all rows of the left table? Answer: `how="left"` .
2. Why can a join make a sum too large? Answer: Duplicate keys on the other side repeat rows.
3. What does `indicator=True` add? Answer: A `_merge` column saying left_only, right_only or both.

5.5 Reshaping: wide, long, nested

INTUITION

The same facts can be written wide (one column per month, easy for people to read) or long (one row per city-month, easy for computers to group and chart). Reshaping converts between the two, as the Unpivot and Pivot buttons in Excel did.

The tools

Tool	Direction
`melt`	wide to long
`pivot`	long to wide (needs unique index/column pairs)
`pivot_table`	long to wide with aggregation

`stack` / `unstack`	move a column level to the rows / rows level to columns
`explode`	one list-valued cell becomes several rows
`pd.json_normalize`	flatten nested JSON

```
w = pd.DataFrame({"city": ["Pune", "Delhi"], "jan": [10, 20], "feb": [12,
18]})
l = w.melt(id_vars="city", var_name="month", value_name="sales")
print(l)
print(l.pivot(index="city", columns="month", values="sales"))
```

```
    city month  sales
0   Pune   jan     10
1  Delhi   jan     20
2   Pune   feb     12
3  Delhi   feb     18
month  feb  jan
city
Delhi   18   20
Pune    12   10
```

Worked example 5.5.1 Check the round trip.

- **Compute** `melt` keeps `city` and turns the two month columns into 4 rows (2 cities × 2 months). `pivot` rebuilds the 2 × 2 table. Note that the columns come back sorted alphabetically (`feb` , `jan`) not in calendar order.
- **Answer** Four long rows and a 2 by 2 wide table.
- **Check** The sum of `sales` is 10 + 20 + 12 + 18 = 60 in both shapes.
- **Why it matters** Plotting libraries and `groupby` want long data. People reading a report want wide data.

`explode` and `json_normalize`

```
e = pd.DataFrame({"id": [1, 2], "tags": [["a", "b"], ["c"]]})
print(e.explode("tags"))
nested = [{"id": 1, "user": {"name": "A", "geo": {"city": "Pune"}}},
          {"id": 2, "user": {"name": "B", "geo": {"city": "Delhi"}}}]
print(pd.json_normalize(nested))
```

```
   id tags
0   1    a
0   1    b
1   2    c
   id user.name user.geo.city
0   1         A          Pune
1   2         B         Delhi
```

`explode` repeated id 1 for each tag (the index label 0 repeats too; use `reset_index(drop=True)` if you need a clean index). `json_normalize` flattened the nested dictionaries into dotted column names, which is what you need for API responses.

`stack` and `unstack`: `l.set_index(["city", "month"]).sales.unstack()` returns the same wide table as `pivot`. `stack()` goes the other way (columns back into rows). We tested a wide table with one empty cell (Delhi, feb): in pandas 3 `stack()` kept that cell as `NaN`, while older versions dropped missing cells by default. The behaviour of `stack` has changed across versions, so check it on your version.

The picture

```
WIDE                      LONG
city  jan  feb            city   month  sales
Pune   10   12   melt ──► Pune   jan    10
Delhi  20   18            Delhi  jan    20
       ◄── pivot          Pune   feb    12
                          Delhi  feb    18
```

Connection to the real job. Survey exports and finance reports are wide; databases, BI tools and ML want long. Power Query's Unpivot (Chapter 3) and Power BI (Chapter 9) rely on long data. JSON from APIs is nested, so `json_normalize` is the first step in many pipelines.

Applications

- Analytics: reshape monthly columns to chart a trend.
- Data Science: pivot events into one row per user (features).
- Data Engineering: flatten API responses into tables.
- Survey analysis: multi-select answers stored as lists (explode).

COMMON MISTAKE

`pivot` fails with "Index contains duplicate entries, cannot reshape" when two rows share the same index/column pair. Use `pivot_table` with an `aggfunc`, or fix the duplicates.

Check yourself

1. What does `melt` do? Answer: Turns wide columns into long rows.
2. Which function pivots with aggregation? Answer: `pivot_table`.
3. What does `explode` do to a list cell? Answer: Makes one row per list item.

5.6 Time Series Basics

INTUITION

Dates are special: they have a natural order, uneven lengths (months), time zones, weekends. A date-time index lets you ask "what happened each hour?" or "per week?" in one line, called **resampling**: regrouping data into new time buckets.

The tools

Tool	Does
`s.dt.year`, `.dt.month`, `.dt.dayofweek`, `.dt.day_name()`	Pull parts out of a datetime column
`df.set_index("ts")`	Make time the index
`resample("h")`, `("D")`, `("W")`, `("MS")`	Regroup to hours, days, weeks, month starts
`asfreq("30min")`	Re-index to a regular step (no aggregation)
`tz_localize`, `tz_convert`	Attach and change time zones
`pd.bdate_range`	Business days only

Frequency codes: `h` hour, `D` day, `W` week, `MS` month start, `min` minute. (Older pandas

versions used uppercase `H`; pandas 3 uses lowercase `h`.)

```
ts = pd.DataFrame({"ts": pd.to_datetime(["2025-01-01 10:00", "2025-01-01
10:20",
                                         "2025-01-01 11:05", "2025-01-02
09:00"]),
                   "v": [1, 2, 3, 4]}).set_index("ts")
print(ts.resample("D").agg(["sum", "count"]))
print(ts.index.day_name().tolist())
print(ts.index.tz_localize("UTC").tz_convert("Asia/Kolkata")[0])
```

```
              v
            sum count
ts
2025-01-01   6     3
2025-01-02   4     1
['Wednesday', 'Wednesday', 'Wednesday', 'Thursday']
2025-01-01 15:30:00+05:30
```

Worked example 5.6.1 Explain every number.

- **Compute (resample)** January 1 has three events (values 1, 2, 3): sum 6, count 3. January 2 has one (4): sum 4, count 1.
- **Compute (weekday)** 1 January 2025 was a Wednesday, so the first three are Wednesday and 2 January is Thursday.
- **Compute (time zone)** 10:00 UTC plus 5 h 30 min (India Standard Time, UTC+05:30) is 15:30.
- **Answer** As printed above.
- **Check** 6 + 4 = 10 = 1 + 2 + 3 + 4.
- **Why it matters** If your server logs are in UTC and your business day is in India time, daily totals can be wrong: an order at 20:00 UTC on 1 January is already 01:30 on 2 January in India. Convert before you group by day.

Gaps. `ts.resample("h").sum()` fills empty hours with 0 (we saw hours 12:00 to 08:00 as 0), which is right for counts but wrong for temperatures or prices (use `.mean()`, which gives NaN for empty buckets, or `ffill`). `asfreq("30min")` shows `NaN` where nothing was recorded. `pd.bdate_range("2025-01-02", periods=5)` gave Thu 2 Jan, Fri 3 Jan, then Mon 6 Jan, Tue 7 Jan, Wed 8 Jan: weekends skipped. Public holidays need a custom calendar.

The picture

```
events:  10:00  10:20 | 11:05 | ... | 09:00(next day)
resample("h"):   [10h: 3]  [11h: 3]  [12h..08h: 0]  [09h: 4]
resample("D"):   [Jan 1: 6 total, 3 events]  [Jan 2: 4 total, 1 event]
```

Connection to the real job. Daily and weekly rollups drive almost every dashboard (Chapters 3, 9, 11). For a shop you resample order timestamps to orders per hour or per day exactly this way.

Applications

- Analytics: daily orders, weekly active users, month-end revenue.
- Data Science: lag and rolling features for forecasting.
- Data Engineering: late-arriving data and time-zone handling.
- Operations: hourly ticket volumes for staffing.

COMMON MISTAKE

Ignoring time zones and partial periods. The last day or month in a dataset is often incomplete, and a "drop" in the final bucket may be only missing data. Check the max timestamp before you report a trend.

Check yourself

1. What does `resample("D")` do? Answer: Regroups time-indexed rows into days.
2. Why convert time zones before grouping by day? Answer: Events near midnight can fall on different dates in different zones.
3. Why can the last month of a trend look low? Answer: It may be incomplete.

Chapter recap

- Read raw files as text if unsure, then convert on purpose.
- Clean in order: duplicates, text standardising, types, missing values, outliers; write down every assumption.
- Median and IQR are robust; mean and z-score are fooled by outliers (z-scores cannot exceed $(n - 1)/\sqrt{n}$).
- Transform whole columns with `np.where`, `cut`, `qcut`, `get_dummies`; avoid row-wise `apply`.

- `groupby` + named aggregation is split, apply, combine; `transform` keeps row count; sum of all-missing is 0.
- Check merges: state `validate`, use `indicator`, compare row counts before and after.
- `melt` and `pivot` move between wide and long; `explode` and `json_normalize` handle nested data.
- Resample for time buckets; mind time zones, empty buckets and the incomplete last period.

Task	Code
Duplicates	`df.drop_duplicates(subset="id")`
Text clean	`s.str.strip().str.title()`
To number / date	`pd.to_numeric(s, errors="coerce")`, `pd.to_datetime(s, errors="coerce")`
Outlier fences	`q1 - 1.5*iqr`, `q3 + 1.5*iqr`
Group summary	`df.groupby("k").agg(n=("id","count"), s=("x","sum"))`
Share in group	`x / df.groupby("k")["x"].transform("sum")`
Join safely	`a.merge(b, on="k", how="left", validate="many_to_one", indicator=True)`
Wide to long	`df.melt(id_vars="k", var_name="m", value_name="v")`
Time buckets	`df.set_index("ts").resample("D").sum()`
Rolling mean	`s.rolling(7).mean()`

Practice set

Problem 1 `pd.Series([1, np.nan, 3]).interpolate()` gives what?

Answer: `[1.0, 2.0, 3.0]`. The missing middle value is halfway between 1 and 3.

Problem 2 Median of `[300, 380, 450, 500, 1150, 1200, 99999]` and mean (rounded)?

Answer: Median 500; mean 103979 / 7 = 14854.1.

Problem 3 Quartiles of `[2, 4, 6, 8, 10]` (linear method) and IQR?

Answer: Q1 at position 0.25 × 4 = 1 gives 4; Q3 at position 3 gives 8; IQR = 4. Fences: 4 − 6 = −2 and 8 + 6 = 14.

Problem 4 For `a` with keys C1, C2, C3 and `b` with keys C2, C3, C4, how many rows for an inner join (unique keys)?

Answer: 2 (C2, C3).

Problem 5 What does `pd.cut(pd.Series([50, 450, 1500]), bins=[0, 400, 1000, np.inf], labels=["low", "mid", "high"]).tolist()` return?

Answer: `['low', 'mid', 'high']`.

Problem 6 Find the bug: `df.groupby("city").amount.mean()` shows "Mumbai" and "MUMBAI" as separate rows.

Answer: The text is not standardised. Fix first: `df["city"] = df["city"].str.strip().str.title()`.

Problem 7 Find the bug: after `orders.merge(customers, on="customer_id")` the revenue total rose from 100,000 to 140,000. Why?

Answer: `customers` has duplicate `customer_id` rows, so orders were repeated. Check with `customers.customer_id.duplicated().sum()` and use `validate="many_to_one"`.

Problem 8 Daily revenue is 100, 120, 90, 150. What is `pct_change()` for day 4 and the 3-day rolling mean on day 4?

Answer: (150 − 90) / 90 = 0.667; (120 + 90 + 150) / 3 = 120.

Problem 9 Server logs are in UTC. An order is logged at 2025-01-01 20:00 UTC. What date is it in India?

Answer: 2 January 2025, 01:30 IST (20:00 + 5:30).

Problem 10 (interview) You get a file where `amount` includes values like "Rs 1,200" and "1200.50" and "NA". Describe your cleaning steps and show the code.

Answer: `df["amount"] = pd.to_numeric(df["amount"].str.replace(r"[Rs,\s]", "", regex=True), errors="coerce")`. "NA" becomes NaN. Then report how many became

NaN (`isna().sum()`), look at them, and decide: drop, fill, or ask the owner.

Problem 11 (interview) "Revenue by city doubled after a join with a product table. Walk me through finding the cause."

Answer: Compare row counts before and after; check the grain of the product table; look for duplicate keys with `duplicated()` ; rerun the merge with `validate="many_to_one"` and `indicator=True` ; then fix the duplicates (or aggregate the product table to one row per key) and re-check that total revenue equals the original.

Prompt pack

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional in
India moving into data roles, and English is my second language. Teach me
Chapter 5 topic: cleaning data in pandas (missing values, duplicates, text
standardising, type conversion, outliers with IQR and z-score). Use a
messy 9-row orders file with real numbers. Ask me to find the problems
first, then show fixes, then quiz me.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. Give me 8 graded exercises on pandas groupby:
named aggregation, transform, filter, pivot_table, crosstab, value_counts,
rolling, shift and pct_change. Use small datasets whose answers I can
check by hand. No solutions yet. When I paste my answers, check each
number and explain mistakes.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. Teach me pandas merge: inner, left, right, outer,
indicator, validate, duplicate-key row explosion, concat, join,
merge_asof. For each, give a 3-row example and ask me to predict the row
count before showing the result. Then give me a debugging scenario where
totals double after a join.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. Explain reshaping (melt, pivot, stack, unstack,
explode, json_normalize) and time series (resample, asfreq, dt accessor,
time zones, business days). Use short real examples with exact outputs,
and test me with five "what is the output?" questions, one at a time.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Run a mock interview: give me a messy dataset
description and ask me to clean it, join it, group it and find the cause
of a wrong total. Ask one question at a time, review my pandas code line
by line, and tell me what a senior analyst would do differently. My first
answer: [PASTE HERE]
```

CHAPTER 6

NUMPY AND FAST PANDAS

Why vectorised code is 100 times faster, and how to keep memory small when data gets big.

Sooner or later a notebook that worked on 1,000 rows is asked to handle 10 million. It crawls, or it crashes with "MemoryError". The cure is not a bigger laptop first. The cure is to understand what pandas does underneath (NumPy arrays), to stop writing loops, and to store data in smaller types. In this chapter you learn the core of NumPy, the idea of vectorisation, a few memory tricks, and where Polars and DuckDB fit when pandas is not enough. By the end you will be able to take a slow script, find the loop, replace it, and prove with a timer that it is faster.

PROFESSOR'S NOTE

Timings depend on your machine, so your numbers will differ; the ratios (about 100 times) are the lesson. Everything below with output was run here (NumPy 2.5.3, pandas 3.0.6, Python 3.14). **Polars and DuckDB were not installed in this environment, so their code is shown but not run.** Install with `pip install polars duckdb` (both free).

6.1 NumPy Arrays: the Engine Under pandas

INTUITION

A Python list is a bag that can hold anything, so Python checks each item's type every time you touch it. A NumPy array is a tray of identical eggs: all the same type, laid out side by side in memory, so one fast instruction can handle the whole tray. pandas columns are NumPy arrays (or similar) under the hood.

Words you need

Word	Meaning
`ndarray`	NumPy's array: same-type values in a grid
`shape`	Size in each direction, e.g. `(2, 3)` is 2 rows, 3 columns

`axis`	A direction: `axis=0` goes down the rows, `axis=1` goes across the columns
dtype	The type of every element (`int64` , `float64` , `int8` , ...)
Vectorised	One operation applied to a whole array at once, no Python loop

```
import numpy as np
a = np.array([[1, 2, 3], [4, 5, 6]])
print(a.shape, a.dtype, a.ndim, a.size)
print(a.sum(), a.sum(axis=0), a.sum(axis=1), a.mean(axis=0))
```

Predict: what is `a.sum(axis=0)` ? (Hint: axis 0 collapses the rows.)

```
(2, 3) int64 2 6
21 [5 7 9] [ 6 15] [2.5 3.5 4.5]
```

- `shape (2, 3)` : 2 rows, 3 columns. `ndim 2` : two directions. `size 6` : six items.
- `a.sum()` adds everything: 1 + 2 + 3 + 4 + 5 + 6 = 21.
- `axis=0` adds **down** each column: 1 + 4 = 5, 2 + 5 = 7, 3 + 6 = 9. This gives `[5 7 9]` .
- `axis=1` adds **across** each row: 1 + 2 + 3 = 6 and 4 + 5 + 6 = 15.
- `mean(axis=0)` gives 2.5, 3.5, 4.5 (column averages, e.g. (1 + 4) / 2 = 2.5).
- **Memory trick** "axis=0 removes axis 0": the first number in the shape disappears. Shape `(2, 3)` summed over axis 0 leaves shape `(3,)` .

Indexing and masks

```
print(a[:, 1], a[1, :2], a[a > 2])
```

```
[2 5] [4 5] [3 4 5 6]
```

- `a[:, 1]` : all rows, column 1: values 2 and 5.
- `a[1, :2]` : row 1, first two columns: 4 and 5.
- `a[a > 2]` : a boolean mask keeps the values above 2: 3, 4, 5, 6 (flattened to one dimension).

Useful functions

Function	Does	Example result

`np.arange(0, 1, 0.25)`	Range with a step	`[0. 0.25 0.5 0.75]`
`np.linspace(0, 1, 5)`	5 evenly spaced points, end included	`[0. 0.25 0.5 0.75 1.]`
`np.unique(x, return_counts=True)`	Distinct values and counts	for `[3,1,2,3,1,3]`: values `[1 2 3]`, counts `[2 1 3]`
`np.argsort(x)`	Positions that would sort x	`[1 4 2 0 3 5]`
`np.where(c, a, b)`	If-then-else	`['hi' 'lo' 'hi' 'hi' 'lo' 'hi']` for `x > 1`

(All verified by running; `x = np.array([3, 1, 2, 3, 1, 3])`.) Check `argsort`: sorted order is 1, 1, 2, 3, 3, 3; the 1s are at positions 1 and 4, the 2 at position 2, the 3s at positions 0, 3, 5.

The picture

```
a = [[1 2 3]        axis=1 → across →
     [4 5 6]]
  axis=0                  sum(axis=1) = [ 6 15]
  ↓ down                  sum(axis=0) = [ 5  7  9]
```

Connection to the real job. `df["col"].values` or `.to_numpy()` hands you the NumPy array. scikit-learn models and most scientific libraries want NumPy arrays.

Applications

- Analytics: fast calculations on whole columns.
- Data Science: feature matrices, distance calculations.
- Deep Learning: tensors (PyTorch, TensorFlow) are the same idea on GPUs.
- Simulation: Monte Carlo of thousands of scenarios.

COMMON MISTAKE

Mixing up the axes. Always test on a tiny array like the one above. If the shape you expect does not come out, you picked the wrong axis.

Check yourself

1. For shape `(4, 5)`, what shape does `sum(axis=0)` return? Answer: `(5,)`.
2. What does a boolean mask do? Answer: It selects the elements where the mask is True.
3. What is `np.linspace(0, 10, 3)`? Answer: `[0. 5. 10.]`.

6.2 Broadcasting, dtypes and Random Numbers

INTUITION

Adding a 3-item list to each row of a table would normally need a loop. Broadcasting lets NumPy "stretch" the smaller array across the bigger one, as if you copied it, without really copying. It is how you subtract the column mean from every row in one line.

The rule. Compare the shapes from the right. Two sizes are compatible if they are equal or one of them is 1. A size of 1 is stretched.

```
print(a + np.array([10, 20, 30]))
print(a + np.array([[100], [200]]))
```

```
[[11 22 33]
 [14 25 36]]
[[101 102 103]
 [204 205 206]]
```

- First line: shape `(2, 3)` plus shape `(3,)`. The 3 matches the last axis; the row `[10 20 30]` is added to each row.
- Second: shape `(2, 3)` plus shape `(2, 1)`. The column `[[100], [200]]` is stretched across the 3 columns: row 0 gets +100, row 1 gets +200.
- **Check** 1 + 10 = 11, 4 + 10 = 14, 1 + 100 = 101, 4 + 200 = 204. Correct.

Worked example 6.2.1 Centre each column (subtract its mean) with broadcasting.

- **Formula** centred = a − mean of its column.
- **Plug in** Column means of `a` are `[2.5, 3.5, 4.5]`.
- **Compute** `a - a.mean(axis=0)` subtracts the `(3,)` means from each row: row 0 becomes `[-1.5, -1.5, -1.5]`, row 1 becomes `[1.5, 1.5, 1.5]`.

- **Answer** Each column now has mean 0.
- **Check** Column 0: (−1.5 + 1.5) / 2 = 0.
- **Why it matters** This standardising step is used in statistics and machine learning all the time, with no loop.

dtypes and their traps

```
print(np.array([1, 2, 3], dtype=np.int8) * 100)
print(0.1 + 0.2 == 0.3, np.isclose(0.1 + 0.2, 0.3))
```

```
[100 -56  44]
False True
```

- An `int8` holds only −128 to 127. `2 × 100 = 200` does not fit, so it wraps around to −56, and `3 × 100 = 300` wraps to 44. **No error is raised.** Choose a type big enough for your numbers.
- Decimals are stored in binary, so `0.1 + 0.2` is `0.30000000000000004` , not exactly 0.3. Never test decimals with `==` ; use `np.isclose` (or `math.isclose`).
- Check the wrap: 200 − 256 = −56, and 300 − 256 = 44.

Random numbers. Use the modern `Generator` , with a seed so results can be repeated.

```
rng = np.random.default_rng(42)
print(rng.integers(1, 7, size=5), rng.normal(100, 15, size=3).round(2))
```

```
[1 5 4 3 3] [114.11  70.73  80.47]
```

Five dice throws (1 to 6) and three values from a normal distribution with mean 100 and standard deviation 15. Run it again with the same seed and you get the same numbers. Always set a seed in analyses so a colleague can reproduce your sample.

Linear algebra, briefly (for the maths, see the learner's Math PDF, linear algebra pillar). Solve the two equations 2x + y = 5 and x + 3y = 10:

```
A = np.array([[2., 1.], [1., 3.]]); b = np.array([5., 10.])
print(np.linalg.solve(A, b), np.linalg.inv(A).round(3), np.linalg.det(A),
A @ A)
```

```
[1. 3.] [[ 0.6 -0.2]
 [-0.2  0.4]] 5.000000000000001 [[ 5.  5.]
 [ 5. 10.]]
```

- `solve` gives x = 1, y = 3. **Check:** 2 × 1 + 3 = 5 and 1 + 3 × 3 = 10.
- The determinant is 2 × 3 − 1 × 1 = 5 (printed as 5.000000000000001, the tiny error from decimals).
- `A @ A` is matrix multiplication: for example the top-left entry is 2 × 2 + 1 × 1 = 5.

The picture

```
 a (2x3)          +  [10 20 30] (3,)    stretched down:  [10 20 30]
[[1 2 3]             [10 20 30]                           [10 20 30]
 [4 5 6]]
```

Connection to the real job. Broadcasting is why NumPy and pandas can do a calculation on a million rows in a millisecond. Integer overflow bugs happen in counters and IDs in real systems (a 32-bit counter, a smaller type chosen to save memory).

Applications

- Analytics: normalise columns, compute percentages across rows.
- Data Science: standardise features, distances.
- Finance: Monte Carlo simulation of returns with a fixed seed.
- Operations: simulate customer arrival patterns at a bank or a checkout queue.

COMMON MISTAKE

Comparing floats with `==`. Money and rates produce tiny errors. Use `np.isclose`, or round both sides, or keep money in integer paise.

Check yourself

1. When are two sizes compatible in broadcasting? Answer: When they are equal or one of them is 1.
2. Why seed the random generator? Answer: To get repeatable results.
3. What is `int8` limit and what happens on overflow? Answer: −128 to 127; values silently wrap around.

6.3 Vectorisation: Stop Writing Loops

INTUITION

Painting a wall with a tiny brush one dot at a time versus using a roller. A Python `for` loop handles one value per step and pays Python's overhead each time. A vectorised operation hands the whole array to fast compiled code, a roller across the entire wall.

Definition. Vectorised code applies an operation to a whole array or column without an explicit Python loop. In pandas that means column arithmetic, comparison, `.str`, `.dt`, `groupby`, and `np.where`, not `for` loops or `apply(axis=1)`.

Worked example 6.3.1 Sum of squares of 1,000,000 random numbers, loop vs vector.

```
import time
v = rng.normal(size=1_000_000)
t = time.perf_counter(); s = 0.0
for e in v: s += e * e
t1 = time.perf_counter() - t
t = time.perf_counter(); s2 = (v * v).sum(); t2 = time.perf_counter() - t
print(f"loop {t1:.3f}s  vector {t2:.5f}s  speedup {t1/t2:.0f}x",
np.isclose(s, s2))
```

Real output on this machine:

```
loop 0.096s  vector 0.00080s  speedup 120x True
```

- **Formula** Σe_i^2 for all i.
- **Compute** The loop did a million Python steps in 0.096 s. The vector version ran the same maths in 0.0008 s.
- **Answer** About 120 times faster, with the same answer (`np.isclose` is True).
- **Check** Both give the same sum, so the speed-up costs no accuracy.
- **Interpretation** At 1 million numbers the loop is only a tenth of a second, which feels fine. At 100 million it is 10 seconds versus 0.1 seconds, and with `apply` it is far worse (next example).

Worked example 6.3.2 `apply(axis=1)` vs column arithmetic on 300,000 rows.

```
n = 300_000
df = pd.DataFrame({"a": rng.integers(0, 100, n), "b": rng.random(n)})
r1 = df.apply(lambda r: r.a * r.b, axis=1)    # row by row
r2 = df.a * df.b                              # whole column
```

```
apply 1.60s vectorized 0.0008s True
```

- **Compute** The row-wise `apply` builds a Series object for every row. That is 1.6 s for 300,000 rows (about 5 microseconds per row), versus 0.8 ms for the single multiplication. The ratio is 1.60 / 0.0008 = 2,000 times.
- **Answer** Vectorised is about 2,000 times faster here.
- **Check** `np.allclose(r1, r2)` is True: same numbers.
- **Why it matters** Real tables have millions of rows. The difference between "finishes during coffee" and "finishes tomorrow" is often one `apply`.

When a loop is unavoidable. Some logic really is sequential (each row depends on the previous result). Then try `np.where`, `np.select` (several conditions), `cumsum` / `shift` / `rolling`, or `groupby` first. If still stuck, use `numba` (free, compiles your loop) or Polars/DuckDB.

`np.select` **for many conditions.** `np.select([x < 100, x < 500], ["low", "mid"], default="high")` is the vectorised version of an if / elif / else ladder (used in Chapter 12's RFM segments).

The picture

```
 loop:        step  step  step  step ...  (1,000,000 times, Python checks
types each time)
 vectorised:  [ one call to compiled code processes all 1,000,000 ]
```

Connection to the real job. In a code review, "why is there a for loop here?" is a standard comment. In interviews, "make this faster" almost always means vectorise, then reduce memory.

Applications

- Analytics: feature columns on large tables.
- Data Science: distance and similarity matrices.
- Data Engineering: batch transformations.

- Everyday automation: the same mindset as replacing a hand-copied step by one formula filled down a whole column.

COMMON MISTAKE

Believing `apply` is vectorised because it is "a pandas function". It is a loop in disguise. Prefer built-in column operations.

Check yourself

1. What is wrong with `df.apply(lambda r: r.a * r.b, axis=1)` ? Answer: It loops row by row; `df.a * df.b` does the same thing far faster.
2. What does `np.select` replace? Answer: A chain of if / elif / else for a whole array.
3. What stays the same after vectorising? Answer: The result; only speed changes.

6.4 Memory: dtypes, category, chunking

INTUITION

A suitcase with the right size. Storing the city name "Mumbai" a million times as full text is like packing a million copies of the same book. A `category` column stores each distinct name once and keeps a small number per row.

Worked example 6.4.1 Memory of a text column with 4 distinct cities, 300,000 rows.

```
cities = rng.choice(["Pune", "Delhi", "Mumbai", "Chennai"], size=n)
d2 = pd.DataFrame({"city": cities})
m1 = d2.memory_usage(deep=True).sum()
d2["city"] = d2.city.astype("category")
m2 = d2.memory_usage(deep=True).sum()
print(m1, m2, round(m1 / m2, 1))
```

```
16349930 300350 54.4
```

- **Compute** As text (`str`) the column cost 16,349,930 bytes (about 16 MB). As a category it cost 300,350 bytes: one small integer code (1 byte, since there are fewer than 128 categories) per row, 300,000 bytes, plus a few hundred bytes for the four names.

16,349,930 / 300,350 = 54.4.

- **Answer** The category type is about 54 times smaller.
- **Check** 300,000 rows × 1 byte = 300,000, and 300,350 − 300,000 = 350 bytes for the labels and index.
- **Why it matters** The exact ratio depends on how long the text is and your pandas version. The idea is stable: categories help when there are few distinct values compared with rows. They hurt when almost every value is different (like IDs).

Shrinking numbers. A column of 300,000 integers from 0 to 999 used 2,400,000 bytes as `int64` (8 bytes each) and 600,000 bytes as `int16` (2 bytes each, range −32,768 to 32,767). A 4 times saving: 2,400,000 / 600,000 = 4.

The memory checklist

Trick	Saves	Risk
`usecols=[...]` at read time	Columns you never use	None
`dtype={"x": "int16"}`	Up to 4x on integers	Overflow if the type is too small
`astype("category")`	Up to 50x on repeated text	Poor on near-unique text
`chunksize` and aggregate per chunk	Never loads the whole file	More code
Parquet instead of CSV	Smaller on disk, typed, faster	Needs `pyarrow`
`pyarrow` strings (`string[pyarrow]`)	Less memory than Python strings	Needs `pyarrow` (not run here)
Delete big objects (`del df`)	Frees memory	None

Rule of thumb: pandas needs RAM of about 2 to 5 times the size of the CSV file. A 2 GB CSV on an 8 GB laptop is already risky.

The picture

```
 text:      "Mumbai" "Pune" "Mumbai" "Delhi" ...    (each string stored in
full)
 category:  codes [2, 3, 2, 1, ...]  +  names {0:Chennai, 1:Delhi,
2:Mumbai, 3:Pune} (stored once)
```

Connection to the real job. Cloud notebooks charge for memory; a 50x saving can make an analysis run on a small machine. Data engineers choose column types in table schemas for the same reason.

Applications

- Analytics: large event logs with repeated labels (page, country, device).
- Data Science: categorical features for models.
- Data Engineering: schema design in Parquet and warehouses.
- Operations: huge delivery logs with repeated city or courier names.

COMMON MISTAKE

Downcasting before checking the range. Run `df.x.min()` and `df.x.max()` first. An overflowed `int8` will not warn you.

Check yourself

1. When does `category` help most? Answer: Few distinct values repeated across many rows.
2. How many bytes does `int16` use per value? Answer: 2.
3. What does `usecols` save? Answer: Memory and time from columns you do not need.

6.5 Beyond pandas: Polars and DuckDB (awareness)

INTUITION

pandas is a single-lane road, great for most trips. When the traffic is 100 million rows, there are faster roads. **Polars** is a DataFrame library written in Rust that uses all CPU cores and can process data lazily. **DuckDB** is an in-process SQL engine: you write SQL against CSV, Parquet or pandas DataFrames, without a server.

Not run here (libraries not installed); syntax is standard.

```
import polars as pl
lf = (pl.scan_csv("orders.csv")                    # lazy: nothing is read
yet
        .filter(pl.col("status") == "delivered")
        .group_by("city")
        .agg(pl.col("amount").sum().alias("revenue"))
        .sort("revenue", descending=True))
print(lf.collect())                                # runs the whole plan
now
```

```
import duckdb
duckdb.sql("""
    SELECT city, SUM(amount) AS revenue
    FROM 'orders.csv'
    WHERE status = 'delivered'
    GROUP BY city
    ORDER BY revenue DESC
""").df()                                          # returns a pandas
DataFrame
```

Why they are fast. Both read only the columns and rows they need (a "plan" is optimised before running), use all cores, and avoid copying. DuckDB can query a Parquet file larger than memory.

Comparison

Tool	Best for	Style	Typical limit
pandas	Everything up to a few GB, richest ecosystem	Python, eager	Single core, memory heavy
Polars	Big data on one machine, speed	Python expressions, lazy or eager	Smaller ecosystem than pandas
DuckDB	SQL users, files too large for memory	SQL	Single machine
Spark	Data bigger than one machine	Python/SQL on a cluster	Overhead and cost for small data

Worked example 6.5.1 Choose the tool.

- **Step 1** A 40 MB sales file for a weekly report: pandas is fine (or Excel Power Query).
- **Step 2** A 20 GB folder of Parquet logs on a laptop with 16 GB RAM: DuckDB or Polars lazy, because pandas would run out of memory.
- **Step 3** 5 TB in a data lake: Spark or a warehouse (Data Engineering document).
- **Answer** Pick the simplest tool that fits the data; move up only when you hit a limit.
- **Check** Ask "does it fit in memory comfortably (under about a third of RAM)?". If yes, pandas.

Connection to the real job. Job posts list pandas first, SQL always, and Polars/DuckDB as a bonus. Knowing that they exist, and when to reach for them, is enough for most analyst interviews.

Applications

- Analytics: DuckDB to query CSV exports with SQL directly.
- Data Science: Polars for fast feature engineering.
- Data Engineering: Parquet plus DuckDB for local pipelines and testing.

COMMON MISTAKE

Rewriting a working pandas job in a new library before measuring. First find the slow step (`%timeit`, `time.perf_counter`) and fix that one.

Check yourself

1. What does "lazy" mean in Polars? Answer: It builds a plan and only runs it when you call `collect()`.
2. Which tool lets you write SQL on a CSV file without a server? Answer: DuckDB.
3. When is pandas enough? Answer: When the data fits comfortably in memory.

Chapter recap

- pandas columns are arrays; learn NumPy shape, axis and dtype.
- `axis=0` collapses rows (down), `axis=1` collapses columns (across).
- Broadcasting stretches size-1 axes; it replaces many loops.
- Small int types overflow silently; floats are inexact, so use `np.isclose`.
- Seed random generators for repeatable results.

- Vectorised code was 120 times faster than a loop, and 2,000 times faster than `apply(axis=1)` in our runs.
- Use `category` for repeated text (54 times smaller in our test), smaller ints, `usecols`, `chunksize`, Parquet.
- Know Polars and DuckDB exist; pick the simplest tool that fits the data.

Task	Code
Column / row totals	`a.sum(axis=0)` / `a.sum(axis=1)`
Centre columns	`a - a.mean(axis=0)`
If / else	`np.where(c, x, y)`; many: `np.select([c1, c2], [v1, v2], default)`
Compare floats	`np.isclose(a, b)`
Seeded random	`rng = np.random.default_rng(42)`
Memory check	`df.memory_usage(deep=True).sum()`
Shrink	`df["c"].astype("category")`, `astype("int16")`
Time a step	`time.perf_counter()` before and after

Practice set

Problem 1 `b = np.array([[1, 2], [3, 4], [5, 6]])`. What are the shape and `b.sum(axis=0)`?

Answer: Shape `(3, 2)`. `sum(axis=0)` adds down the columns: 1 + 3 + 5 = 9 and 2 + 4 + 6 = 12, so `[9 12]`.

Problem 2 What is `b[b > 3]`?

Answer: `[4 5 6]`.

Problem 3 What is `b + np.array([10, 20])`?

Answer: `[[11 22] [13 24] [15 26]]`: the pair is added to every row.

Problem 4 Compute the mean of each row of `b`.

Answer: `b.mean(axis=1)` gives `[1.5 3.5 5.5]`.

Problem 5 What does `np.array([100], dtype=np.int8) * 2` return?

Answer: `[-56]`. 200 exceeds 127, so it wraps (200 − 256 = −56).

Problem 6 Find the bug: `if 0.1 * 3 == 0.3:` never runs. Why, and what is the fix?

Answer: `0.1 * 3` is 0.30000000000000004 in binary floating point. Use `np.isclose(0.1 * 3, 0.3)` (or `math.isclose`).

Problem 7 A text column has 2,000,000 rows and 5 distinct values. Estimate the saving from `category`, with codes of 1 byte.

Answer: The category stores about 2,000,000 bytes (1 per row), while text costs many times more (tens of MB, roughly 8 or more bytes per row for pointers alone, plus string content). Expect a saving of the order of 10 to 50 times. Measure with `memory_usage(deep=True)`.

Problem 8 Rewrite using vectorised code: `out = []; for i in range(len(df)): out.append("big" if df.amount[i] > 1000 else "small")`.

Answer: `df["size"] = np.where(df["amount"] > 1000, "big", "small")`.

Problem 9 What is the output shape of `np.ones((4, 1)) + np.ones((1, 3))` and why?

Answer: `(4, 3)`. Both size-1 axes are stretched: the column of 4 across 3 columns, the row of 3 down 4 rows. Every element is 2.

Problem 10 (interview) "Your pandas script takes 40 minutes on 30 million rows. How do you speed it up?"

Answer: First profile to find the slow step. Replace row-wise `apply` and Python loops with vectorised operations. Read only the needed columns with smaller dtypes. Convert repeated text to `category`. Use Parquet. Pre-aggregate in SQL if the data starts in a database. If still slow, try Polars or DuckDB. Measure each change.

Problem 11 (interview) "What is broadcasting and give a practical use."

Answer: NumPy's rule for combining arrays of different shapes by stretching size-1 axes. Example: `X - X.mean(axis=0)` subtracts each column's mean from every row without a loop.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in India moving into data roles, and English is my second language. Teach me Chapter 6 topic: NumPy arrays, shape, axis, indexing, boolean masks and broadcasting. Use tiny arrays and explain axis=0 versus axis=1 with a picture in text. Ask me to predict outputs, then verify them. Finish with a 5-question check.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving into data roles. Give me 8 graded exercises on vectorisation: replace Python loops and DataFrame.apply(axis=1) with NumPy/pandas operations (np.where, np.select, cumsum, shift, groupby transform). For each, include a small dataset and the expected output but hide the solution. After I attempt them I will paste my code for review and a speed comparison.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving into data roles. Explain how to reduce pandas memory: usecols, dtype downcasting, category, chunksize, Parquet, pyarrow strings. Include how to calculate the expected bytes by hand and when each trick backfires. Then quiz me one question at a time with small numbers.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving into data roles. Compare pandas, Polars, DuckDB and Spark for analysts: when to use each, how lazy evaluation works, and a tiny example of the same aggregation in each. Be honest about learning cost and ecosystem. Then give me a decision exercise with 5 scenarios and check my choices.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving into data roles. Run a mock interview on performance in data code:
```

slow loops, apply versus vectorised, float comparison, integer overflow, memory errors, and choosing tools. Ask 6 questions one at a time, grade my answers from 1 to 5, and give a model answer in simple English each time. My first answer will follow your first question.

CHAPTER 7

EXPLORATORY DATA ANALYSIS, STEP BY STEP

A complete worked analysis on a dataset you generate yourself, from first look to written insights.

Someone hands you a file and says, "Tell me what is in here." You do not yet know what the columns mean, which values are errors, or what questions are worth asking. Exploratory Data Analysis (EDA) is the habit of looking at the data carefully, in a fixed order, before you trust it or model it. In this chapter you will generate an online-shop dataset with known dirt in it (duplicates, a unit mistake, missing values, impossible quantities), then explore it step by step with real code and real output, exactly as you would on the job. By the end you will have a repeatable EDA routine and a list of insights written the way a senior analyst writes them.

We use the same shop as Chapter 1. Because we built the data ourselves, we know the truth behind it. That lets us catch the traps that real data hides: at the end you will see an "insight" that turns out to be a false alarm.

PROFESSOR'S NOTE

Everything here was run with pandas 3.0.6, NumPy 2.5.3 and SciPy 1.18. The generator uses a fixed seed, so you will get the same numbers. `ydata-profiling`, `sweetviz`, `missingno`, `pandera`, `great_expectations` and `statsmodels` were not installed here, so they are only described, not run.

7.1 Context, Grain and the First Look

INTUITION

Before you measure a room, ask what the room is for. Before you open a table, ask where it came from, what one row means, and what the columns mean. A table that was exported before payment confirmation is a different table from one exported after. Most wrong analyses start with a wrong guess about what a row is.

Questions to answer first (write them in your notebook)

Question	Our dataset
Where does the data come from?	A simulated online shop (we generate it)
What is one row? (the **grain**)	One order in `orders`; one customer in `customers`
What period does it cover?	Customers who signed up between 1 January and end of June 2025
What are the columns?	`order_id`, `customer_id`, `order_date`, `category`, `qty`, `amount`, `status`
Which column is the key?	`order_id` (should be unique)
What decision will it support?	Which categories and channels deserve attention, and is data quality good enough to report

The generator. Save this as `gen.py`. It builds two tables and then **injects dirt**: 60 missing amounts, 15 impossible quantities (−1), 8 amounts multiplied by 100 (as if entered in paise instead of rupees), 12 duplicate rows, and some upper-case category names.

```
import numpy as np, pandas as pd

def make_orders(seed=7, n_customers=1500):
    rng = np.random.default_rng(seed)
    cust = pd.DataFrame({
        "customer_id": [f"C{i:04d}" for i in range(n_customers)],
        "signup_date": pd.Timestamp("2025-01-01") +
pd.to_timedelta(rng.integers(0, 180, n_customers), unit="D"),
        "channel": rng.choice(["organic", "ads", "referral", "email"],
n_customers, p=[.4, .3, .2, .1]),
        "city": rng.choice(["Pune", "Mumbai", "Delhi", "Bengaluru",
"Chennai"], n_customers, p=[.15, .25, .25, .25, .10]),
    })
    rows = []
    oid = 1000
    for r in cust.itertuples():
        k = rng.poisson(1.6 if r.channel != "ads" else 1.1)
        for _ in range(k):
            od = r.signup_date +
pd.Timedelta(days=int(rng.exponential(30)))
            if od > pd.Timestamp("2025-12-31"): continue
            cat = rng.choice(["electronics", "fashion", "grocery",
"home"], p=[.2, .35, .3, .15])
            base = {"electronics": 6.8, "fashion": 6.0, "grocery": 5.0,
"home": 6.3}[cat]
            amount = round(float(rng.lognormal(base, 0.5)), 2)
            qty = int(rng.integers(1, 5))
            status = rng.choice(["delivered", "returned", "cancelled"], p=
[.86, .08, .06])
            rows.append((oid, r.customer_id, od, cat, qty, amount,
status)); oid += 1
    o = pd.DataFrame(rows, columns=["order_id", "customer_id",
"order_date", "category", "qty", "amount", "status"])
    idx = rng.choice(o.index, 60, replace=False); o.loc[idx, "amount"] =
np.nan
    idx = rng.choice(o.index, 15, replace=False); o.loc[idx, "qty"] = -1
    idx = rng.choice(o.index, 8, replace=False); o.loc[idx, "amount"] *=
100
    o = pd.concat([o, o.sample(12, random_state=1)], ignore_index=True)
    o["category"] = o["category"].where(rng.random(len(o)) > .05,
o["category"].str.upper())
    return cust, o.sample(frac=1, random_state=3).reset_index(drop=True)
```

(A few terms: a *seed* fixes the random numbers. `lognormal` makes right-skewed positive values, like order amounts. `poisson(k)` gives a random count with average k, here the

number of orders per customer.)

Step 1: shape and types

```
import pandas as pd, numpy as np
from gen import make_orders
cust, o = make_orders()
print(o.shape, cust.shape)
print(o.dtypes)
```

Predict: how many rows are in `orders`?

```
(2244, 7) (1500, 4)
order_id               int64
order_date     datetime64[us]
...
```

(`customer_id`, `category`, `status` are `str`; `qty` is `int64`; `amount` is `float64`.) There are 2,244 order rows and 1,500 customers. The types look right, so no conversion is needed. `qty` is an integer, which means it has no missing values yet. That will change after cleaning.

The picture

```
customers (1500 rows)  1 ──── * orders (2244 rows)
customer_id, signup_date,        order_id, customer_id, order_date,
channel, city                    category, qty, amount, status
```

Connection to the real job. In the first meeting about a dataset, ask for the grain and the date range. Writing these down in a data dictionary (Chapter 14) prevents a lot of rework.

Applications

- Analytics: understanding a new extract from a data warehouse.
- Data Science: deciding which rows are training examples.
- Data Engineering: checking that a pipeline output matches its documented grain.

COMMON MISTAKE

Skipping the grain question. If `orders` had one row per order item, then counting rows would count items, not orders, and "average order value" would be wrong.

Check yourself

1. What is the grain of `orders`? Answer: One order per row.
2. Why record the date range? Answer: Trends and totals depend on it; the last period may be incomplete.
3. What does a seed do? Answer: It makes random generation repeatable.

7.2 Data Quality Scan and Cleaning

INTUITION

A building inspector walks around with a checklist before painting. Our checklist for data: missing values, duplicates, impossible values, inconsistent categories, unit mistakes. Run it before any chart, because a chart of dirty data only looks professional.

The scan

```
print(o.isna().sum())
print("exact duplicate rows:", o.duplicated().sum())
print("negative or zero qty:", (o.qty <= 0).sum())
print(o.category.value_counts())
```

```
order_id        0
customer_id     0
order_date      0
category        0
qty             0
amount         61
status          0
dtype: int64
exact duplicate rows: 12
negative or zero qty: 15
category
fashion        735
grocery        661
electronics    437
home           310
GROCERY         31
FASHION         31
ELECTRONICS     21
HOME            18
Name: count, dtype: int64
```

- 61 amounts are missing (we injected 60; one more appeared because a duplicated row repeated a missing value).
- 12 duplicate rows, exactly the 12 we added.
- 15 quantities are zero or negative, impossible for an order.
- The category has eight spellings for four categories, because of upper-case copies. Without a fix, a `groupby` would report eight categories.

Step 2: clean, with a log of every decision

```
o = o.drop_duplicates()
o["category"] = o["category"].str.lower()
o.loc[o.qty <= 0, "qty"] = np.nan
cat_median = o.groupby("category")["amount"].transform("median")
o["ratio"] = o["amount"] / cat_median
print(o.loc[o.ratio >= 15, ["order_id", "category", "amount",
"ratio"]].sort_values("ratio").round(1))
o.loc[o.ratio >= 15, "amount"] /= 100
o = o.drop(columns="ratio")
print(len(o), o.amount.isna().sum(), o.qty.isna().sum())
```

```
      order_id     category    amount  ratio
2226      2633      grocery    3566.0   23.4
944       2072      fashion   21933.0   53.0
752       1675         home   47982.0   92.3
345       3043      grocery   15105.0   99.0
595       2282  electronics  111473.0  131.6
892       1796         home   89659.0  172.5
659       1870      grocery   30409.0  199.4
2232 60 15
```

Why this method? A simple fixed threshold ("amount above 20,000 is an error") would be wrong. A grocery order of 3,566 is also impossible (grocery's median is about 152), but a threshold of 20,000 would miss it. Instead we compare each amount with the **median of its own category**. Anything at least 15 times its category median is flagged. We found 7 rows (we injected 8; we checked that one of the 8 landed on a row whose amount was already missing, so there is nothing left to see there).

Worked example 7.2.1 Check one repair by hand.

- **Formula** repaired amount = recorded amount / 100 (a paise to rupees correction)
- **Plug in** Order 3043 (grocery) has amount 15,105.0 and ratio 99.0.
- **Compute** 15,105.0 / 100 = 151.05. The grocery median is 152.4, so 151.05 / 152.4 = 0.99, a typical grocery order. Also 15,105.0 / 152.4 = 99.1, close to the ratio 99.0 printed.
- **Answer** After the fix order 3043 has amount 151.05, an ordinary value.
- **Check** The ratio of 99 (almost exactly 100) is the signature of a paise error: a 100 times mistake gives ratios near 100 times the ordinary ratio of about 1.
- **Why it matters** The raw mean amount was 641.31 (with the errors). After repair it is 495.6. A 30% error in the average, from 7 rows out of 2,232.

Decisions to write in your log

Problem	Count	Decision
Duplicate rows	12	Dropped (identical in all columns)
Category case	8 spellings	Lower-cased
Quantity of −1	15	Set to missing, not guessed

Amount 23 to 199 times its category median	7	Divided by 100 (assumed paise), flagged for the data owner
Missing amount	60	Left missing for now (see 7.3)

The picture

```
amount / category median
1x ........ typical orders (cluster near 1)
4–7x ...... large but plausible (kept)
23x–200x .. data errors (fixed or flagged)
```

Connection to the real job. The log above is what you attach to your analysis ("I made these 5 cleaning decisions"). A reviewer can disagree with a decision only if you wrote it down. In SQL the same scan is a few `COUNT` queries (the SQL document covers data quality checks).

Applications

- Analytics: every report starts with this scan.
- Data Engineering: the same rules become automated tests in a pipeline (`pandera` , `great_expectations` , dbt tests).
- Finance: unit mistakes (thousands vs units, paise vs rupees) are a classic audit finding.

COMMON MISTAKE

Silently deleting "weird" rows. A big order might be real (a bulk purchase). Flag, ask, then decide. Never change data without a record.

Check yourself

1. Why use the category median instead of one global threshold? Answer: Categories have very different normal sizes.
2. What is the signature of a paise-for-rupee error? Answer: A value about 100 times too large.
3. Why write a cleaning log? Answer: So others can review and reproduce your decisions.

7.3 Missing Data: Pattern, Mechanism, Treatment

INTUITION

A missing value is a clue, not just a hole. A blank salary on a form may be a forgotten field, or it may be hiding because the salary is very high. The reason a value is missing decides how safely you can ignore it or fill it.

Three mechanisms (learn the names, interviewers ask)

Name	Meaning	Example	Safe to drop rows?
MCAR: Missing Completely At Random	Missingness has nothing to do with any data	A sensor drops 2% of readings at random	Yes, with a small loss of data
MAR: Missing At Random	Missingness depends on *other observed* columns	Younger customers skip the income field more often	Not blindly; use the other columns to fill or model
MNAR: Missing Not At Random	Missingness depends on the *missing value itself*	Rich customers hide their income	No; any fill or drop biases results

You cannot prove which one you have; you look for evidence.

Step 3: does missing amount depend on other columns?

```
print(o.assign(miss=o.amount.isna()).groupby("category")
["miss"].mean().round(3))
```

```
category
electronics    0.029
fashion        0.022
grocery        0.030
home           0.028
Name: miss, dtype: float64
```

Worked example 7.3.1 Interpret the table.

- **Formula** missing rate = number of missing amounts in the group / number of orders in the group

- **Plug in** For example, fashion has 743 non-missing amounts out of 760 orders, hence 17 missing. 17 / 760 = 0.022.
- **Compute** The rates are 2.9%, 2.2%, 3.0%, 2.8%, all close to the overall 60 / 2232 = 2.7%.
- **Answer** Missing amounts show no clear link with category, which is consistent with MCAR.
- **Check** A chi-square test of missingness versus category gives p = 0.81 (not significant), supporting the same conclusion.
- **Interpretation** Consistent with MCAR does not prove it. We also checked status (cancelled 1.5%, delivered 2.8%, returned 2.3%), which looks similar given the small counts. In real data you would also check missing versus date, channel and city, and ask the system owner why a value could be absent.

Treatment options

Option	When	Cost
Leave as missing and exclude from amount calculations	MCAR, small share	Slightly less data
Drop the rows	Row is useless without the value	Lose other columns' information
Fill with median of the group	Few missing, numeric, skewed	Shrinks variance
Predict (model-based imputation)	MAR, important column	Complex
Add a flag column `amount_missing`	MNAR suspected	None, and keeps the clue

Here we choose "leave missing and exclude": `mean` , `sum` and `describe` skip `NaN` by default, and 2.7% is small. Always report it ("amount is missing for 2.7% of orders").

The picture

```
 missing rate by group:   electronics ##  fashion ##  grocery ##  home ##
(flat -> MCAR-like)
                         young       ########   older ##
(varies -> MAR)
```

Connection to the real job. When survey, healthcare or banking data has missing values, the mechanism is the first question, because a wrong choice biases decisions. `missingno` (a plotting library, not run here) draws the missing pattern as a matrix, which helps you see this quickly.

Applications

- Analytics: deciding whether a KPI can ignore blank rows.
- Data Science: imputation strategy before modelling.
- Healthcare and banking: missing lab values or income often carry meaning (MNAR).

COMMON MISTAKE

Filling every missing value with the mean "to keep rows". It shrinks the spread, creates a fake spike at the mean and hides the reason for the gap.

Check yourself

1. What does MAR mean? Answer: Missingness depends on other observed columns.
2. Why add a flag column? Answer: To keep the information that the value was missing.
3. Does a flat missing rate across groups prove MCAR? Answer: No, it is only consistent with MCAR.

7.4 Univariate Analysis: One Column at a Time

INTUITION

Look at one column alone before relating it to others, like checking each ingredient before cooking. Ask: where is the middle, how spread out, is it symmetric, what are the extremes?

Words you need

Word	Meaning
Distribution	How often each value (or range) occurs
Skewness	Lopsidedness. Positive: long tail to the right (a few huge values). 0: symmetric.

Kurtosis (excess)	How heavy the tails are compared with a bell curve. 0 for a normal bell, positive means more extreme values.
Quantile / percentile	The value below which a share of data falls. The 0.99 quantile is the 99th percentile.

```
a = o["amount"].dropna()
print(a.describe().round(1))
print("skew", round(a.skew(), 2), "log-skew", round(np.log(a).skew(), 2),
"kurtosis", round(a.kurt(), 2))
print(a.quantile([.5, .9, .99]).round(1))
print(pd.cut(a, bins=[0, 100, 250, 500, 1000,
5000]).value_counts().sort_index())
```

```
count    2172.0
mean      495.6
std       410.3
min        33.3
25%       201.7
50%       383.8
75%       654.7
max      3866.0
Name: amount, dtype: float64
skew 2.09 log-skew -0.16 kurtosis 7.42
0.50     383.8
0.90    1013.7
0.99    1927.5
Name: amount, dtype: float64
amount
(0, 100]        130
(100, 250]      588
(250, 500]      631
(500, 1000]     598
(1000, 5000]    225
Name: count, dtype: int64
```

Worked example 7.4.1 Read the summary like an analyst.

- **Formula** Compare the mean with the median. If mean is clearly above median, the distribution has a right tail.
- **Plug in** Mean 495.6, median 383.8.
- **Compute** 495.6 − 383.8 = 111.8, so the mean is 29% above the median (111.8 / 383.8 =

0.291). Skewness is 2.09 (positive, strong). After `np.log` the skew is −0.16, almost symmetric.

- **Answer** Order amounts are right-skewed: most orders are modest and a few are large. The log of the amount is close to a bell shape, which is typical for money (the values multiply, they do not add).
- **Check** The bins add up: 130 + 588 + 631 + 598 + 225 = 2,172, equal to the count of non-missing amounts.
- **Interpretation** The typical order is about Rs 384. The mean (496) is pulled up by big orders, so for "typical" use the median. The 99th percentile is Rs 1,927: 1% of orders are above that. The kurtosis 7.42 says the tails are heavy compared with a normal curve.

Outlier check after the repair. The IQR fence is Q3 + 1.5 × IQR = 654.7 + 1.5 × (654.7 − 201.7) = 654.7 + 679.5 = 1,334.2, and 99 values (4.6%) lie above it. Are they errors? No. A right-skewed distribution naturally has many values above a bell-curve-style fence. These are the electronics orders, which we see next. The lesson is that an "outlier by the IQR rule" is only a candidate, not a mistake. Look at where they come from.

Left: histogram of order amount with a long right tail. Right: the same data on a log10 scale, close to symmetric

How to read it: each bar counts how many orders fall in an amount range (horizontal axis). The left chart has a long tail to the right. On the right, the horizontal axis is log10(amount), so each step is a multiplication by ten, and the shape becomes nearly symmetric.

Connection to the real job. Skew explains why dashboards show "average order value" that nobody can reproduce by eye. Hospitals report the 90th-percentile waiting time rather than the average, for the same reason.

Applications

- Product: session durations and order values are right-skewed; report median and percentiles.
- Finance: log returns instead of raw returns.
- Data Science: log-transform skewed features before linear models.
- Operations: delivery times and appointment waiting times.

COMMON MISTAKE

Using the mean for skewed money data and presenting it as "typical". Show the median and a percentile too.

Check yourself

1. What does positive skew mean? Answer: A long tail of large values to the right.
2. Why take the log of money values? Answer: It often makes the distribution close to symmetric.
3. Is every value above the IQR fence an error? Answer: No; it is only a candidate to check.

7.5 Bivariate and Multivariate Analysis

INTUITION

One column tells you what is typical. Two columns tell you whether they move together. Does the amount depend on the category? Does the channel change the return rate? This is where insights appear.

Tools

Question	Tool
Numeric by category	`groupby(...).agg` (count, median, mean)
Category by category	`pd.crosstab(a, b, normalize="index")`
Numeric by numeric	`df.corr()` (Pearson), `method="spearman"`
Attach another table	`merge` then group
Test a difference	`scipy.stats`: `ttest_ind`, `mannwhitneyu`, `f_oneway`, `chi2_contingency`

```
print(o.groupby("category")["amount"].agg(["count", "median",
"mean"]).round(1))
print(pd.crosstab(o.category, o.status, normalize="index").round(3))
print(o[["qty", "amount"]].corr().round(3))
j = o.merge(cust, on="customer_id", how="left", validate="many_to_one")
print(j.groupby("channel").agg(orders=("order_id", "count"), aov=
("amount", "mean")).round(1))
```

```
             count  median   mean
category
electronics    443   847.2  967.9
fashion        743   413.6  461.8
grocery        668   152.4  169.9
home           318   517.7  600.8
status       cancelled  delivered  returned
category
electronics      0.081      0.851     0.068
fashion          0.054      0.846     0.100
grocery          0.054      0.871     0.075
home             0.064      0.893     0.043
          qty  amount
qty     1.000   0.011
amount  0.011   1.000
          orders    aov
channel
ads          475  489.7
email        249  502.7
organic     1013  499.5
referral     495  490.0
```

Worked example 7.5.1 Insight 1: category drives amount.

- **Formula** ratio = median of one category / median of another
- **Plug in** Electronics median 847.2, grocery median 152.4.
- **Compute** 847.2 / 152.4 = 5.56.
- **Answer** A typical electronics order is about 5.6 times a typical grocery order. A Mann-Whitney test (a test that compares two groups without assuming a bell curve) on the 443 electronics and 668 grocery amounts gave a p-value near zero (8×10^{-171}), so this is not chance.
- **Check** The means (967.9 and 169.9) give 5.7, consistent.
- **Interpretation** Category is the main driver of order size. Any revenue forecast should be

built per category.

Worked example 7.5.2 Insight 2: channel does not change the order size.

- **Plug in** Average order value by channel: ads 489.7, email 502.7, organic 499.5, referral 490.0. The largest gap is 502.7 − 489.7 = 13.0 (2.6%).
- **Compute** One-way ANOVA across the four channels gave F = 0.116, p = 0.95 (a p of 0.95 means differences this small would appear by chance 95% of the time).
- **Answer** No evidence that the acquisition channel affects order size.
- **Check** If channel mattered, we would expect gaps of tens of percent, not 2.6%.
- **Interpretation** Do not spend effort optimising order value by channel. Differences in channel *volume* (organic brings 1,013 orders versus email 249) matter more. (Of course, volume partly reflects the shares we set in the generator: 40% organic, 10% email.)

Worked example 7.5.3 Insight 3: correlation between qty and amount is about zero.

- **Formula** Pearson correlation r measures straight-line association, from −1 to +1.
- **Compute** r = 0.011.
- **Answer** Quantity and amount are unrelated here. This looks odd (more units should cost more), and it is a **data quality clue**. In our generator the amount was drawn without using the quantity, so `amount` is the order total regardless of `qty`. In real data you would ask the owner: is `amount` per unit or per order? Never assume.
- **Check** `df[["qty", "amount"]].corr(method="spearman")` gave 0.02, the same conclusion.

Worked example 7.5.4 Insight 4 (a trap): returns by category.

```
from scipy import stats
ct = pd.crosstab(o.category, o.status == "returned")
chi2, p, dof, exp = stats.chi2_contingency(ct)
print(round(chi2, 2), round(p, 4), dof)
```

```
11.5 0.0093 3
```

- **Formula** chi-square = Σ (observed − expected)² / expected, where expected = group size × overall return rate.
- **Plug in** The overall return rate is 173 / 2,232 = 7.75%. For fashion, expected returns = 760 × 0.0775 = 58.9, but we observed 76. For home, expected = 327 × 0.0775 = 25.3, observed

14.

- **Compute** Contributions: fashion (76 − 58.9)² / 58.9 = 4.96; home (14 − 25.3)² / 25.3 = 5.08. With the other small terms the total is 11.5 with 3 degrees of freedom, p = 0.0093.
- **Answer** The test says return rates differ by category (fashion 10.0% versus home 4.3%) with p = 0.0093.
- **Check** We can check the truth: in our generator every order is returned with probability 8% whatever the category. So this "significant" finding is a **false alarm**. A p of 0.009 happens by chance about 1 time in 100 tests, and when you look at many cuts of the data (category, channel, city, month, status), some will look significant by luck.
- **Interpretation** Treat this as a hypothesis to confirm on new data (next quarter), not as a fact to act on. This is the "multiple comparisons" problem, which Chapter 13 shows again for A/B tests.

Time and customers

```
m = o.groupby(o.order_date.dt.to_period("M")).agg(orders=("order_id",
"count"), revenue=("amount", "sum")).round(0)
per = o.groupby("customer_id")["order_id"].count()
print(m)
print(per.value_counts().sort_index())
print("customers with >=1 order:", len(per), "of", len(cust))
print("repeat rate:", round((per >= 2).mean(), 3))
```

```
            orders  revenue
order_date
2025-01        144  76045.0
2025-02        233  115867.0
2025-03        348  162274.0
2025-04        364  164132.0
2025-05        390  194888.0
2025-06        393  186105.0
2025-07        222  114926.0
2025-08         95   39809.0
2025-09         31   16305.0
2025-10          9    3485.0
2025-11          2     648.0
2025-12          1    2022.0
order_id
1    521
2    358
3    178
4     77
5     19
6      5
7      4
Name: count, dtype: int64
customers with >=1 order: 1162 of 1500
repeat rate: 0.552
```

Worked example 7.5.5 Insights 5 and 6.

- **Compute (customers)** 1,162 of 1,500 customers ordered at least once: 1162 / 1500 = 77.5%. Among them, 358 + 178 + 77 + 19 + 5 + 4 = 641 placed 2 or more orders: 641 / 1162 = 55.2% (matches the printed 0.552).
- **Compute (months)** Orders rise from 144 in January to 393 in June, then fall to 1 in December. This is **not** a business collapse. Our customers all signed up between January and June (the latest signup date is 29 June 2025), and each customer's orders happen within weeks of signing up. There are no new customers after June, so the orders fade out.
- **Answer** 77.5% of customers ordered, and 55.2% of those ordered again. The monthly decline after June is an artefact of how the data was collected, not a trend.
- **Check** Sum of monthly orders = 144 + 233 + 348 + 364 + 390 + 393 + 222 + 95 + 31 + 9 + 2 + 1 = 2,232, equal to the number of unique orders.
- **Interpretation** Always ask "does the data cover everyone for the whole period?" Here it covers a *cohort* (a group that started together). A cohort view (Chapter 12) is the right way

to read it.

Joining tables. We joined `orders` to `customers` with `validate="many_to_one"` (many orders per customer, one customer row). The channel column had 0 missing after the join, so every order matched a customer. That integrity check should appear in every EDA.

The picture

```
median amount by category:
electronics ############################ 847
home        ################             518
fashion     #############                414
grocery     #####                        152
```

Connection to the real job. Steps 5 to 8 are what you present: "category drives order size; channel does not; return rate differences are probably noise; the monthly drop is a data coverage effect." Every one of those four statements saves the business from a wrong decision.

Applications

- Marketing: which channel produces valuable customers?
- Product: which segment retains better?
- Finance: which category contributes to revenue and returns?
- Operations: returns by supplier or warehouse.

COMMON MISTAKE

Testing many slices and reporting the one with p below 0.05. If you run 20 independent tests at 5%, the chance of at least one false alarm is $1 - 0.95^{20} = 64\%$ (we computed it: 0.6415). Decide your questions first, or correct for multiple testing (Chapter 13).

Check yourself

1. What does r = 0.011 say? Answer: Almost no straight-line relationship.
2. Why is the monthly decline after June not a business trend? Answer: No customers signed up after June; the data stops adding new customers.
3. What did the paise error do to the mean? Answer: It raised it from about 496 to 641.

7.6 From Hypotheses to Documented Insights (and

Profiling Tools)

INTUITION

A detective writes down each claim, the evidence for it, and how sure she is. Your EDA should end the same way, not with "I made 30 charts".

The routine (copy this into every analysis)

1. Context: source, grain, period, owner.
2. Shape, dtypes, memory.
3. Quality scan: missing, duplicates, impossible values, inconsistent categories, units.
4. Cleaning log.
5. Missing-data mechanism.
6. Univariate: distribution, skew, outliers.
7. Bivariate: groups, cross-tabs, correlations, tests.
8. Time and customer views.
9. Hypotheses, checks, insights with confidence.

Insight table (what you would send)

#	Insight	Evidence	Confidence	Action
1	Seven amounts were 23 to 199 times their category median (paise error). Mean falls from 641 to 496 after repair	Ratio table	High	Ask data owner to fix the source; add a validation rule
2	12 duplicate orders and 8 category spellings	Counts	High	Deduplicate on `order_id`; standardise at entry
3	Amount is missing for 2.7% of orders, no link to category or status	Missing rates, chi-	Medium	Exclude from totals;

		square p = 0.81		report the share
4	Order amount is right-skewed (skew 2.09); typical order Rs 384, mean Rs 496	describe, log skew	High	Report median and p90, not just mean
5	Category drives order size: electronics median 847 vs grocery 152	Medians, Mann-Whitney	High	Forecast per category
6	Channel does not change order size (ANOVA p = 0.95)	AOV by channel	High	Do not tune AOV by channel
7	`qty` and `amount` are uncorrelated (r = 0.011): the meaning of `amount` is unclear	Correlation	Medium	Confirm definition (per unit or per order)
8	Return rate varies by category (4.3% to 10.0%, p = 0.009)	chi-square	Low	Treat as hypothesis; re-test next quarter
9	77.5% of customers ordered; 55.2% of them reordered	Counts	High	Use as cohort baseline (Chapter 12)
10	Monthly orders fall after June because the dataset has no signups after June	Max signup date 29 June	High	Do not report as a trend; use cohort views

Notice how insight 8 is labelled **Low** confidence: honesty about uncertainty is part of the craft.

Automated profiling tools (awareness, not run here).

Tool	What it does
`ydata-profiling` (`ProfileReport(df)`)	One-click HTML report: types, missing, distributions, correlations, duplicates
`sweetviz`	Compare two datasets (train vs test)
`missingno`	Matrix and heatmap of missing values
`pandera` / `great_expectations`	Write data rules ("amount > 0", "order_id unique") and fail the pipeline if broken
`statsmodels`	OLS and logistic regression with full statistical output, ANOVA, time series, power analysis

Use profilers as a first pass to save time, never as a substitute for thinking. They cannot tell you the grain, or that 99,999 is a placeholder.

Worked example 7.6.1 Turn the routine into a reusable function.

```
def quick_scan(df, key):
    return {
        "rows": len(df),
        "dup_keys": int(df[key].duplicated().sum()),
        "missing_pct": (df.isna().mean() * 100).round(1).to_dict(),
    }
```

- **Compute** Run it on the raw `orders` table with `key="order_id"` . Real output: `{'rows': 2244, 'dup_keys': 12, 'missing_pct': {'order_id': 0.0, 'customer_id': 0.0, 'order_date': 0.0, 'category': 0.0, 'qty': 0.0, 'amount': 2.7, 'status': 0.0}}` (61 / 2244 = 0.0272).
- **Answer** A three-line health report you can paste at the start of any analysis.
- **Check** 61 / 2244 = 2.72%, which rounds to 2.7.
- **Why it matters** Automating the scan makes you faster and consistent. On a team, it becomes a shared standard.

The picture

```
 Question → Scan → Clean (log) → Explore → Hypothesis → Test → Insight
(with confidence) → Decision
              ↑                                                    │
              └──────────────── new question appears ──────────────┘
```

Connection to the real job. The mastery test for this skill: "given an unknown messy dataset, produce a clean notebook with profiling, cleaning, 10 insights, charts and a one-page summary in under four hours". The ten insights above are the shape of that deliverable.

Applications

- Analytics: the handover document for every new dataset.
- Data Science: EDA precedes every model; mistakes found here save weeks.
- Data Engineering: rules found in EDA become data quality tests.

COMMON MISTAKE

Reporting everything you found. Ten insights with evidence and an action are good; 40 charts without a message are noise. Rank by impact and confidence.

Check yourself

1. Why label confidence for each insight? Answer: Some findings are solid and some are only hypotheses; readers need to know the difference.
2. What can a profiler not tell you? Answer: The business meaning of columns, the grain, or whether a value is a placeholder.
3. What should follow a low-confidence insight? Answer: A planned re-test on new data.

Chapter recap

- Start with context: source, grain, period, key, decision.
- Quality scan before charts: missing, duplicates, impossible values, categories, units.
- Compare each value with its own group's median to catch unit mistakes.
- Missing data mechanisms: MCAR, MAR, MNAR; look at missing rates by other columns; keep a flag if unsure.
- Univariate: describe, skew, log, quantiles; mean greater than median means a right tail.
- Bivariate: grouped summaries, cross-tabs, correlation, tests; one significant test among

many can be a false alarm.

- Check the coverage of time: an incomplete period or a cohort-only dataset creates fake trends.
- Write insights with evidence, confidence and action; label uncertain ones honestly.

Step	Code
Shape, types	`df.shape` , `df.dtypes` , `df.info()`
Missing / duplicates	`df.isna().sum()` , `df.duplicated().sum()`
Group-relative outlier	`df.amount / df.groupby("g").amount.transform("median")`
Distribution	`df.x.describe()` , `df.x.skew()` , `df.x.quantile([.5,.9,.99])`
Group summary	`df.groupby("g").x.agg(["count","median","mean"])`
Cross-tab	`pd.crosstab(a, b, normalize="index")`
Test of independence	`scipy.stats.chi2_contingency(table)`
Join check	`merge(..., validate="many_to_one")`

Practice set

Problem 1 A column has values 10, 12, 11, 13, 200. Mean and median? Which is typical?

Answer: Mean 49.2, median 12. The median is typical; 200 pulls the mean up.

Problem 2 After cleaning, `orders` has 2,232 rows and 60 missing amounts. What share is missing?

Answer: 60 / 2232 = 2.69%, about 2.7%.

Problem 3 The grocery median is 152.4. An order has grocery amount 3,566. Ratio and verdict?

Answer: 3566 / 152.4 = 23.4. Far above a plausible range for grocery, so flag it (likely paise

error: 35.66, which is plausible).

Problem 4 Skewness is +2.1 and the mean is above the median. Describe the shape.

Answer: Right-skewed: a few large values stretch the right tail.

Problem 5 Fashion: 76 returns of 760 orders. Overall return rate 7.75%. What are the expected returns?

Answer: 760 × 0.0775 = 58.9. Observed 76 is 17.1 higher.

Problem 6 Find the bug: `df.groupby("category")["amount"].mean()` shows 8 categories for 4 real ones.

Answer: Inconsistent case. Normalise with `.str.lower()` (and `.str.strip()`) before grouping.

Problem 7 Find the bug: a report says "order volume collapsed 99% from June to December". The data holds only customers who signed up before July. What is wrong?

Answer: The dataset stops adding new customers; later months only contain orders from old signups. The "collapse" is a coverage artefact. Compare cohorts at the same age, not calendar months.

Problem 8 You run 10 independent tests at the 5% level when all nulls are true. Probability of at least one false alarm?

Answer: $1 - 0.95^{10} = 1 - 0.5987 = 0.401$, about 40%.

Problem 9 Missing income is more common for customers under 25 (observed age). Which mechanism?

Answer: MAR: the missingness depends on another observed column (age).

Problem 10 (interview) "You find a column where 5% of values are exactly 0 but the column is a price. What do you do?"

Answer: Check if 0 means "missing", "free" or a placeholder. Look at related columns (status, category, date) for patterns. Ask the data owner. Do not average the zeros as real prices. Report the count and treat them as missing if confirmed.

Problem 11 (interview) "Walk me through your EDA process on a new dataset in five minutes."

Answer: Ask the grain, key and period. Check shape, dtypes, missing, duplicates, impossible

values, category spellings and units. Write a cleaning log. Look at the distribution of key numbers (skew, percentiles). Compare groups and test differences. Check the time coverage. Summarise insights with evidence and confidence, and list open questions for the data owner.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in India moving into data roles, and English is my second language. Teach me Chapter 7 topic: Exploratory Data Analysis as a fixed routine (context and grain, quality scan, cleaning log, missing data, univariate, bivariate, time coverage, documented insights). Walk me through it on a small made-up orders dataset, step by step, and ask me to decide what to do at each step before you reveal the answer.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving into data roles. Explain missing data mechanisms MCAR, MAR and MNAR with three everyday examples each, how to look for evidence in a DataFrame with pandas, and what to do for each. Then give me 6 short scenarios and ask me to classify them one at a time, with feedback.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving into data roles. Give me a messy dataset described in words (columns, a few sample rows, known problems) and ask me to plan an EDA. Then give me 8 specific "what does this output mean" questions about skewness, medians versus means, outliers by IQR, correlations and chi-square p-values, graded easy to hard. No solutions until I answer.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving into data roles. Review my EDA notebook summary for mistakes: missing checks, hidden assumptions, false alarms from multiple testing, incomplete time coverage, and weak insight wording. Rewrite my top three insights as: claim, evidence, confidence, action. My summary: [PASTE HERE]
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. Run a mock interview on EDA and data quality:
grain, duplicates, outliers, unit mistakes, missing data, skewed
distributions and false discoveries. Ask 6 questions one at a time, grade
each answer from 1 to 5, and show a model answer in simple English with
one number-based example.
```

CHAPTER 8

DATA VISUALIZATION: PRINCIPLES, MATPLOTLIB, SEABORN AND PLOTLY

Choose the right chart, remove the noise, and make one picture carry one message.

Your manager will not read your pandas code. She will look at one chart for ten seconds and decide whether to trust you. A good chart makes the answer obvious in those ten seconds; a bad one hides it, or tells a false story. In this chapter you learn which chart fits which question, how to use colour, what to delete, and then the three libraries that draw charts: matplotlib (the base), seaborn (statistical charts in one line) and plotly (interactive). By the end you can turn the Chapter 1 question "which city sells the most?" into a chart that answers it at a glance.

PROFESSOR'S NOTE

Output and images were produced with pandas 3.0.6, matplotlib 3.11.2 and seaborn 0.13.2. plotly, Streamlit and the other libraries in 8.5 were **not installed here**, so that code is standard but marked "not run here".

We reuse the 8-row `Sales` table from Chapters 2 and 3 (total 9,340; Delhi 3,298, Mumbai 3,047, Pune 2,995), an 18-month series of monthly sales in Rs thousand (Jan 2024 to Jun 2025: `310, 295, 340, 360, 385, 400, 380, 410, 430, 470, 560, 620, 350, 330, 375, 405, 430, 455`), and in 8.4 a synthetic table of 300 orders with a fixed random seed.

8.1 Why Charts Work and Which Chart to Choose

INTUITION

Your brain reads some things instantly and other things slowly. Which of two lines is longer? Instant. Which of two slices of a pie is bigger? Slow, and you often get it wrong. A chart is a way of turning numbers into the things your eyes read fastest. Think of a hospital waiting room board: nobody reads 5,000 appointment records, everybody reads the one chart of waiting time through the day.

Words you need

Word	Plain meaning	Example
Encoding	How a number is drawn (length, position, colour, area)	Bar length = sales
Distribution	How values spread out	Most orders are small, a few are huge
Chart junk	Anything on the chart that carries no information	3D effects, heavy gridlines

How it works. People judge encodings with different accuracy. From most accurate to least:

Most to least accurate: position on a common scale, length, angle and slope, area, colour shade.

So put the important number on **position or length**. That is why bars beat pies and why we sort bars.

Choosing the chart. Start from the question, not from the chart menu.

Question you are answering	Best chart	Avoid
Compare a number across categories	Bar chart (horizontal if labels are long), sorted	Pie with more than 3 slices
How does a number change over time	Line chart	Bars for dense time series
How are values spread	Histogram, box plot, violin	Average alone
Relationship between two numbers	Scatter plot	Dual-axis lines
Part of a whole (few parts)	Stacked bar, or a pie with 2 to 3 slices	Donut with 8 slices
Two categories and a number	Heatmap or grouped bar	3D bars
Exact values matter	A table	Chart with 40 labels
Funnel steps	Funnel (bar) chart, see Chapter 12	Pie

The picture. The same data (sales by city), drawn two ways. First, the default from `by_city.plot.bar()` : correct, but you must hunt for the message.

Default pandas bar chart: vertical bars, generic title, no message

Second, the fixed chart: a **headline title** (the finding, not the topic), one highlighted bar, values on the bars instead of axis ticks. Each bar is a city, length is total sales in rupees, blue is the answer.

Fixed chart: sorted horizontal bars, one highlight colour, direct labels, headline title

Worked example 8.1.1 The truncated axis. A bar chart of the three city totals starts the y-axis at 2,900 instead of 0. How big do the bars look compared with how big the numbers really are?

- **Formula** visual ratio = (bar A − axis start) / (bar B − axis start); real ratio = A / B. Lie factor = visual ratio / real ratio.
- **Plug in** Delhi 3,298 and Pune 2,995, axis start 2,900.
- **Compute** Visible heights: Delhi 3,298 − 2,900 = 398, Pune 2,995 − 2,900 = 95. Visual ratio = 398 / 95 = 4.19. Real ratio = 3,298 / 2,995 = 1.10. Lie factor = 4.19 / 1.10 = 3.8.
- **Answer** Delhi's bar looks 4.2 times taller than Pune's, but Delhi really sold only 10% more. The chart exaggerates the gap by about 3.8 times.
- **Check** With the axis at 0 the heights are 3,298 and 2,995: ratio 1.10, the true value.
- **Why it matters** Bars encode value by **length from zero**; cutting the baseline breaks that. (Line charts may use a non-zero axis: they encode position, not length.)

Connection to the real job. Interviewers show a chart and ask "what is wrong?". Usual answers: wrong type, truncated axis, too many colours, no message in the title, unsorted bars.

Applications

- Data Analytics: line charts for trends, sorted bars for comparisons.
- Machine Learning: loss curves (line), confusion matrices (heatmap).
- Operations: waiting time over the day (line), delivery-time spread (histogram).

COMMON MISTAKE

Choosing the chart because it looks impressive (3D, donut, radar) instead of because it answers the question. If you cannot say the message in one sentence, you are not ready to draw the chart.

Check yourself

1. Which is read more accurately: bar length or pie angle? Answer: Bar length (position and length beat angle).
2. Why must bar charts start at zero? Answer: Bars encode value as length from the baseline, so cutting it exaggerates differences.
3. Which chart for "how does revenue change by month"? Answer: A line chart.

8.2 Clutter, Colour and Storytelling

INTUITION

A chart is like a shop window. Every item you add draws a little attention away from the main one. If you put 40 items in the window, nobody sees the product you wanted to sell. Good design is mostly **deleting**.

Formal idea. Edward Tufte's idea of the **data-ink ratio** says: most of the ink on a chart should show data, not decoration. Delete everything that does not help the reader, then check if the chart still works. Four steps, in order:

1. **Remove** borders, heavy gridlines, 3D, shadows, and the legend when direct labels do the job.
2. **Label directly** on or next to the marks.
3. **Colour with purpose**: grey for context, one strong colour for the message.
4. **Write a headline title**: the finding, in a sentence.

Colour rules.

Use	Palette type	Example
Categories with no order	Qualitative: at most 5 distinct colours	City A, B, C

A number from low to high	Sequential: one colour from light to dark	Median amount heatmap
A number with a meaningful middle	Diverging: two colours meeting at a neutral centre	Growth above and below 0%
Highlight one thing	Grey plus one accent	The best city

About 1 in 12 men has some colour blindness (mostly red-green). Never use red versus green alone to carry meaning: add labels or use a safe palette (`palette="colorblind"` in seaborn). Keep text contrast high and text size at least 10 to 12 points.

Storytelling structure. Business stories follow three beats: **headline** → **evidence** → **recommendation**, with one message per chart (Chapter 14 goes deeper).

Worked example 8.2.1 Write a headline from the data. Use the monthly series: Jan 2024 = 310, Jan 2025 = 350, Jun 2024 = 400, Jun 2025 = 455.

- **Formula** growth % = (new − old) / old × 100
- **Plug in** January: (350 − 310) / 310. June: (455 − 400) / 400.
- **Compute** January: 40 / 310 = 0.129, so 12.9%. June: 55 / 400 = 0.1375, so 13.75%.
- **Answer** Headline: "Sales are up about 13% on last year in January and June." (All six months of 2025 so far are between +10% and +14%, as 8.3 shows.) (A weak title would be "Monthly sales.")
- **Check** 310 × 1.129 = 350.0 and 400 × 1.1375 = 455. Correct.
- **Why it matters** The reader gets the conclusion from the title and uses the chart to verify it.

Connection to the real job. Stakeholders judge you by how fast they "get it"; the same applies to Power BI reports (Chapter 11).

Applications

- Analytics: weekly business review slides, one message each.

COMMON MISTAKE

Using the default colour cycle (blue, orange, green, red, purple...) for everything. Colour is a scarce signal. Use it only to say "look here".

Check yourself

1. What is a headline title? Answer: A title that states the finding, not just the topic.
2. When should you use a diverging palette? Answer: When the data has a meaningful centre, such as growth above and below 0%.
3. Why avoid red versus green only? Answer: Many people have red-green colour blindness.

8.3 matplotlib: The Figure and Axes Model

INTUITION

Think of a figure as a sheet of paper and the axes as a frame you glue onto it. The frame contains the actual drawing area with its x-axis, y-axis, title and data. One sheet can hold many frames (subplots). Everything you do is either "work on the sheet" or "work on a frame".

Formal definition. matplotlib has two objects you must know:

Object	Plain meaning	Example call
`Figure`	The whole canvas (size, resolution, saving)	`fig.savefig("a.png", dpi=150)`
`Axes`	One plot with its two axes	`ax.plot(x, y)`, `ax.set_title("...")`

The shortcut style `plt.plot(...)` works on a hidden "current axes" and breaks with several plots. **Always use** `fig, ax = plt.subplots()`. Because pandas (`df.plot`) and seaborn also draw on matplotlib axes, learning this once lets you adjust any chart.

Worked example 8.3.1 A figure with two subplots: the line for monthly sales on top, year-on-year growth bars below, an annotation on the peak.

```
import pandas as pd
import matplotlib
matplotlib.use("Agg")   # draw to files, no screen needed
import matplotlib.pyplot as plt

months = pd.period_range("2024-01", "2025-06", freq="M").to_timestamp()
sales = [310,295,340,360,385,400,380,410,430,470,560,620,
         350,330,375,405,430,455]
m = pd.Series(sales, index=months, name="sales_k")
print(m.tail(3))

fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(7, 5), sharex=True,
                               height_ratios=[2, 1], layout="constrained")
ax1.plot(m.index, m.values, color="#1f6feb", marker="o", linewidth=2)
peak = m.idxmax()
ax1.annotate(f"Festive peak: {m.max()}k", xy=(peak, m.max()),
             xytext=(peak - pd.Timedelta(days=150), m.max() - 5),
             arrowprops=dict(arrowstyle="->"))
ax1.set_title("Monthly sales (Rs thousand)", loc="left",
fontweight="bold")
ax1.set_ylabel("Rs thousand")
growth = m.pct_change(12) * 100
ax2.bar(growth.index, growth.values, width=20, color="#9aa5b1")
ax2.axhline(0, color="black", linewidth=0.8)
ax2.set_ylabel("YoY %")
fig.savefig("sales_two_panels.png", dpi=150)
print(growth.dropna().round(1).to_string())
```

Predict: `pct_change(12)` compares each month with the one 12 rows earlier. Which months get a value?

```
2025-04-01    405
2025-05-01    430
2025-06-01    455
Freq: MS, Name: sales_k, dtype: int64
2025-01-01    12.9
2025-02-01    11.9
2025-03-01    10.3
2025-04-01    12.5
2025-05-01    11.7
2025-06-01    13.7
Freq: MS
```

Two stacked panels: a line of monthly sales with an annotated festive peak of 620k, and bars of year-on-year growth below

- `plt.subplots(2, 1, ...)` returns the figure and two axes. `sharex=True` aligns the months, `height_ratios` makes the top panel twice as tall, `layout="constrained"` stops labels overlapping.
- `annotate` writes text and an arrow: `xy` is the point, `xytext` is where the text sits. Methods that set properties start with `set_` (`set_title`, `set_ylabel`).
- `pct_change(12)` is (this month − same month last year) / same month last year. Only Jan to Jun 2025 have a partner 12 rows back.
- `savefig(..., dpi=150)` saves a PNG; `dpi` (dots per inch) is resolution. Use 150 to 300 for reports.
- **Check** Jan 2025: 350 / 310 − 1 = 0.129, printed as 12.9. March: 375 / 340 − 1 = 0.103, printed as 10.3. Correct.
- **Interpretation** Sales are 10% to 14% higher than a year ago in every month where we can compare, and the November and December peak (560k and 620k) is the biggest event of the year.

A common error is `fig.set_title("x")`, which gives `AttributeError: 'Figure' object has no attribute 'set_title'`. Call it on the axes (`ax.set_title`) or use `fig.suptitle`.

Worked example 8.3.2 Build the fixed chart of 8.1 (`df` is the 8-row Sales table).

```
by_city = df.groupby("City")["Amount"].sum().sort_values(ascending=False)
fig, ax = plt.subplots(figsize=(6, 3.2))
ax.barh(by_city.index[::-1], by_city.values[::-1], color=["#9aa5b1",
"#9aa5b1", "#1f6feb"])
ax.set_title("Delhi sold the most: Rs 3,298 of Rs 9,340", loc="left",
fontweight="bold")
for y, v in enumerate(by_city.values[::-1]):
    ax.text(v + 30, y, f"{v:,}", va="center")
ax.set_xticks([]); ax.set_xlim(0, 3800)
```

- **Step 1** `barh` draws bottom up, so `[::-1]` puts Delhi on top (colours reversed too). `ax.text` writes each value; `set_xlim(0, ...)` keeps the zero baseline.
- **Check** 3,298 + 3,047 + 2,995 = 9,340, the number in the title.

Connection to the real job. matplotlib is for custom annotations, report images and fixes to seaborn charts. `fig.savefig` is how charts reach Word, PowerPoint and e-mail.

Applications

- Machine Learning: training and validation loss on the same axes.

COMMON MISTAKE

Mixing `plt.` and `ax.` calls and getting titles on the wrong plot. Use `fig, ax = plt.subplots()` and only `ax.` methods.

Check yourself

1. What is the difference between a Figure and an Axes? Answer: The figure is the whole canvas, an axes is one plot inside it.
2. What does `dpi` control? Answer: The resolution (dots per inch) of the saved image.
3. Why `sharex=True` in subplots? Answer: Both panels use the same x-axis so the months line up.

8.4 seaborn: Statistical Charts in One Line

INTUITION

matplotlib is a blank canvas and a box of pens. seaborn is a set of stamps for the things analysts draw all day: histograms, box plots, relationship plots. You give it a DataFrame and column names; it does the grouping, the maths and the colours.

Formal definition. seaborn is a library built on matplotlib. Most functions take `data=` (a DataFrame) and column names for `x`, `y`, `hue` (colour by a column). It returns an Axes (or a grid of them) so you can still fix things with matplotlib.

Our bigger dataset. 300 synthetic orders with a fixed seed. The code to create it:

```
import numpy as np, pandas as pd, seaborn as sns

rng = np.random.default_rng(7)
n = 300
orders = pd.DataFrame({
    "city": rng.choice(["Pune", "Mumbai", "Delhi"], n, p=[0.3, 0.4, 0.3]),
    "category": rng.choice(["Fashion", "Electronics", "Grocery"], n, p=
[0.4, 0.2, 0.4]),
})
base = {"Fashion": 900, "Electronics": 2400, "Grocery": 400}
orders["amount"] = [round(rng.lognormal(np.log(base[c]), 0.35)) for c in
orders.category]
orders["delivery_days"] = np.clip(rng.normal(3.5, 1.2, n)
                                  + (orders.city == "Delhi") * 0.8, 1,
None).round(1)
orders["rating"] = np.clip(np.round(5 - 0.35 * orders.delivery_days
                                    + rng.normal(0, 0.6, n)), 1,
5).astype(int)
print(orders.head(2))
print(orders.groupby("category")["amount"].median())
print(orders[["amount", "delivery_days", "rating"]].corr().round(2))
```

Real output:

```
     city category  amount  delivery_days  rating
0  Mumbai  Grocery     459            3.2       4
1   Delhi  Fashion     935            4.4       4
category
Electronics    2377.0
Fashion         870.0
Grocery         395.0
Name: amount, dtype: float64
              amount  delivery_days  rating
amount          1.00          -0.00   -0.08
delivery_days  -0.00           1.00   -0.54
rating         -0.08          -0.54    1.00
```

We built slower delivery to mean lower rating; the correlation table shows it (−0.54), and about 0 for amount against delivery days.

Worked example 8.4.1 Four charts in one figure.

```
import matplotlib.pyplot as plt
sns.set_theme(style="whitegrid")
fig, axes = plt.subplots(2, 2, figsize=(9, 6.5), layout="constrained")
sns.histplot(data=orders, x="amount", bins=30, kde=True, ax=axes[0, 0])
sns.boxplot(data=orders, x="category", y="amount", ax=axes[0, 1])
sns.violinplot(data=orders, x="category", y="delivery_days", ax=axes[1,
0])
sns.scatterplot(data=orders, x="delivery_days", y="rating", hue="city",
                alpha=0.6, ax=axes[1, 1])
fig.savefig("seaborn_four.png", dpi=130)
```


Four panels: a right-skewed histogram of amount with a smooth curve, box plots of amount by category, violin plots of delivery days, and a scatter of rating against delivery days

How to read each panel:

- **histplot**: bar height is the count of orders in each range; `kde=True` adds a smooth curve (**kernel density estimate**). Most orders are 300 to 1,000 rupees with a long tail to about 4,800: **right-skewed**, so the mean sits above the median and you report the median (Chapter 7).
- **boxplot**: the box spans the 25th to 75th percentile, the line is the median, dots are outliers. Medians 2,377 / 870 / 395 match the table.
- **violinplot**: a box plot with a mirrored smooth shape; wider means more orders there. Delivery time looks similar across categories, as we built it.
- **scatterplot**: rating is a whole number, so dots sit on lines and overlap (**overplotting**); `alpha=0.6` helps. The drift down to the right is the −0.54 correlation.
- **Check** The panels agree with the printed tables: category medians (2,377 / 870 / 395) and the correlation sign.

Other seaborn functions, one line each

- `sns.kdeplot(data=orders, x="amount", hue="category")` : smooth density curves per group.
- `sns.pairplot(df)` : scatter grid of all numeric pairs, histograms on the diagonal; the first look at new data.
- `sns.catplot(..., kind="box", col="city")` : one category chart per city.
- `sns.lmplot(x=..., y=..., data=...)` : scatter plus a fitted line.
- `sns.FacetGrid(df, col="city")` : one panel per group; map any plot into it

`( g.map_dataframe(sns.histplot, x="delivery_days") )`.

I ran all of these on the 300 orders in seaborn 0.13.2. `pairplot`, `catplot`, `lmplot` and `FacetGrid` return grid objects, not one Axes, so save with `grid.savefig(...)`.

Worked example 8.4.2 A heatmap of the median amount by city and category.

```
piv = orders.pivot_table(index="city", columns="category",
                         values="amount", aggfunc="median")
print(piv)
sns.heatmap(piv, annot=True, fmt=".0f", cmap="Blues")
```

```
category  Electronics  Fashion  Grocery
city
Delhi          2364.0    873.5    448.0
Mumbai         2454.0    878.0    374.0
Pune           2477.5    851.0    431.0
```


Heatmap of median order amount: rows are cities, columns are categories, darker blue means higher

- **Reading** `pivot_table` (Chapter 3's idea) puts cities in rows and categories in columns with the median in each cell; `heatmap` shades by value. Colour changes across columns, hardly across rows: category drives order size, not city.
- **Check** 873.5 is the average of two middle values, normal for a median of an even count.

Connection to the real job. In EDA (Chapter 7) you run `histplot`, `boxplot` and `pairplot` on every new dataset; `heatmap` suits correlations and cohort tables (Chapter 12).

Applications

- Data Science: pairplot and correlation heatmap before modelling.
- Healthcare: violin plots of appointment waiting times per clinic.

COMMON MISTAKE

Reading a bar of an average as the whole story. A bar chart of mean amount hides the skew and outliers that the box plot and histogram show. Check the distribution before you trust the mean.

Check yourself

1. What does `hue="city"` do? Answer: Colours the points or bars by the city column.
2. Which seaborn function gives a grid of scatter plots for all numeric pairs? Answer: `pairplot` .
3. Why use `alpha` in a scatter plot? Answer: To see overlapping points (overplotting).

8.5 plotly, Streamlit and Specialised Plots

INTUITION

A matplotlib chart is a photograph: you can look but not touch. A plotly chart is a small web page: hover to see exact values, zoom, click a legend item to hide a series. For charts people explore, that is worth a lot.

Formal definition. plotly express (`px`) is the short plotly interface, one function per chart type, with the same data-first style as seaborn. It returns a `Figure` that you can show, or save as a self-contained web page. **Streamlit** turns a Python script into a web app with widgets (dropdowns, sliders); **Dash** is plotly's more controlled app framework; **Altair** is a declarative chart library. Not run here (not installed), but standard:

```
import plotly.express as px
fig = px.bar(by_city.reset_index(), x="City", y="Amount",
             title="Delhi sold the most", text_auto=True)
fig.write_html("by_city.html", include_plotlyjs=True)   # open in any
browser
```

- `px.bar(df, x=..., y=...)` draws bars; `text_auto=True` writes the value on each bar.
- `write_html` makes a file you can e-mail. `include_plotlyjs=True` embeds the library so the file also opens offline (it is a few megabytes larger).

Specialised charts (awareness). `geopandas` and `folium` draw maps, `networkx` draws network graphs, `wordcloud` draws word clouds (frequency only, so prefer a bar of the top words). Normalise maps per 1,000 customers, or you mostly map population, and treat precise personal locations as personal data.

Worked example 8.5.1 Choose the tool. A sales head wants a page where she picks a region

and sees monthly sales, shared with 40 colleagues who log in with company accounts and each see only their own region.

- **Step 1** A region selector: plotly, Streamlit or Power BI can all do it.
- **Step 2** Each user seeing only their region is **row-level security** (Chapter 11), a Power BI Service feature. In Streamlit you would build it yourself.
- **Step 3** Automatic refresh from the company database is also built into Power BI (Chapter 11).
- **Answer** Power BI is the better fit. If the page also needed a Python forecast computed on demand, Streamlit would win.
- **Check** The decision follows requirements (security, refresh, custom logic), not taste.

Connection to the real job. Interactive charts help exploration; a published dashboard plus a short write-up is the stronger portfolio piece.

COMMON MISTAKE

Building an interactive chart where a still image would do. Interaction costs the reader effort. If the message is one sentence, a still chart wins.

Check yourself

1. What does `fig.write_html` produce? Answer: A web page file with the interactive chart.
2. Which tool for a fast internal Python app: Streamlit or Power BI? Answer: Streamlit (Power BI for managed, secured business dashboards).
3. Why not map raw totals? Answer: They mostly show population; normalise per customer.

Chapter recap

- Put the important number on position or length. Bars beat pies. Start bars at zero.
- Pick the chart from the question: compare, trend, distribution, relationship, part of whole.
- Delete clutter first. Then label directly, use grey plus one accent colour, and write a headline title.
- Never rely on red versus green alone, and use sequential or diverging palettes for the matching data.
- Tell stories as headline, evidence, recommendation, with one message per chart.

- matplotlib: always `fig, ax = plt.subplots()` ; the figure is the sheet, the axes is a plot; save with `dpi` .
- seaborn takes tidy DataFrames and draws distributions, categories, relationships and heatmaps in one line; grid functions return grid objects.
- plotly gives interactive charts, Streamlit and Dash make web apps, and Power BI is the managed option when you need security and refresh.

Item	Pattern
Figure and axes	`fig, ax = plt.subplots(figsize=(7, 4), layout="constrained")`
Save	`fig.savefig("a.png", dpi=150, bbox_inches="tight")`
seaborn	`sns.histplot / boxplot / violinplot / scatterplot / heatmap(data=..., x=..., y=...)`
Interactive	`px.bar(df, x=..., y=...)` , `fig.write_html("a.html")`
Lie factor	visual ratio / real ratio

Practice set

Problem 8.1 Which chart type fits: "how do daily active users change across 90 days"?

Answer: A line chart: time on the x-axis, users on the y-axis.

Problem 8.2 A pie has 8 slices of 11% to 14%. What is wrong and what replaces it?

Answer: Nearly equal angles cannot be compared by eye. Use a sorted horizontal bar with labels.

Problem 8.3 A bar chart shows two products, A = 52 and B = 48, with the y-axis starting at 46. What are the visible bar heights, the visual ratio and the lie factor?

Answer: Heights 52 − 46 = 6 and 48 − 46 = 2. Visual ratio 6 / 2 = 3. Real ratio 52 / 48 = 1.083. Lie factor 3 / 1.083 = 2.77. A gap of 8% looks like 200%.

Problem 8.4 What does `pd.Series([100, 110, 121]).pct_change().round(2).tolist()` print?

Answer: `[nan, 0.1, 0.1]`. The first value has no predecessor; 110 / 100 − 1 = 0.10 and 121 / 110 − 1 = 0.10.

Problem 8.5 Find the bug: `fig, ax = plt.subplots(); fig.set_title("Sales")`.

Answer: `AttributeError: 'Figure' object has no attribute 'set_title'`. Use `ax.set_title("Sales")` (for one plot) or `fig.suptitle("Sales")` (for the whole figure).

Problem 8.6 Write a headline for this data: Q1 revenue 4.0M, Q2 revenue 4.6M, Q2 returns rose from 5% to 9% of orders.

Answer: Revenue grew 15% (0.6 / 4.0 = 0.15) but returns nearly doubled from 5% to 9%, so check product quality before adding spend. The headline gives growth, risk and action.

Problem 8.7 Write a headline: Q1 revenue 4.0M, Q2 revenue 4.6M, Q2 returns rose from 5% to 9% of orders.

Answer: Revenue grew 15% (0.6 / 4.0) but returns nearly doubled from 5% to 9%: check product quality before adding spend.

Problem 8.8 (Interview) "Explain Figure versus Axes in matplotlib, and why `plt.subplots()` is preferred over `plt.plot()`."

Answer: The Figure is the canvas, an Axes is one plot with its axes. `plt.subplots()` returns both objects explicitly, so with several plots you decide exactly which one you modify. `plt.plot()` works on a hidden "current axes", which causes mistakes in scripts with multiple plots.

Problem 8.9 The monthly series gives Nov 2024 = 560 and Dec 2024 = 620. The average of the other ten months of 2024 is (310+295+340+360+385+400+380+410+430+470)/10. How many times bigger is the Dec peak than that average?

Answer: The sum is 310+295 = 605; +340 = 945; +360 = 1,305; +385 = 1,690; +400 = 2,090; +380 = 2,470; +410 = 2,880; +430 = 3,310; +470 = 3,780. Average 378. 620 / 378 = 1.64. December is about 1.6 times a normal month, so stock accordingly.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
```

moving into data roles, and English is my second language, so use simple words. Teach me "choosing the right chart" for analytics: bar, line, histogram, box plot, scatter, heatmap, pie. Start from the question being asked, not the chart. Use one running example of an online shop's orders and monthly sales. Explain step by step with small numbers, then check my understanding with 3 questions, one at a time, and wait for my answer before the next.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Give me 8 graded exercises (easy to hard) on chart design principles: truncated axes and the lie factor, clutter removal, colour choice for categorical, sequential and diverging data, headline titles, and colour-blind safety. Use real numbers I can compute by hand. Do not give solutions yet. After I answer, mark my work and explain every mistake in simple English.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Explain the matplotlib Figure and Axes model with a working example: two stacked subplots sharing an x-axis, an annotation on the peak, and savefig with dpi. Explain every line. Then show the three most common beginner errors with their real error messages and fixes. Finish with a 5-question quiz, one question at a time, waiting for my answers.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Review my chart code and the picture it produces for a data analyst job portfolio. Check: is the chart type right, does the title state a finding, is the axis honest, is colour used with purpose, is it accessible to colour-blind readers, and is the code clean (explicit fig and ax)? Give a numbered list of fixes in priority order. My code: [PASTE HERE]. What it shows and who the audience is: [PASTE HERE].

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Run a mock interview on data visualization for a junior data analyst role: ask me one question at a time, covering chart

```
choice, misleading charts, matplotlib versus seaborn versus plotly,
storytelling, and when to use Streamlit versus Power BI. After each
answer, score it out of 5, give a better model answer in simple English,
then ask the next question. Do 8 questions in total.
```

CHAPTER 9

POWER BI I: POWER QUERY (M) AND DATA MODELING

Prepare messy data once, then model it as a star so every report is fast and correct.

A Power BI report is only as good as what is under it. Two skills sit there. First, **Power Query** cleans and shapes raw files and databases into tidy tables, with a recorded recipe that re-runs on every refresh. Second, **data modeling** decides how those tables are split and connected, so that one click on a slicer filters the right rows. Most wrong numbers in dashboards come from a weak model, not from a weak chart. In this chapter you will clean a dirty orders file with Power Query and its language M, design a star schema for our shop, connect the tables with relationships, and build a date table. DAX (the formula language) comes in Chapter 10.

PROFESSOR'S NOTE

Power BI Desktop and Power Query cannot run in this environment. M and DAX code is standard and minimal but **not run here**; check function names in your editor, as menus and defaults change by release. To make numbers trustworthy I reproduced every transformation and relationship in pandas 3.0.6 and show its real output. Chapter 3 introduced Power Query and Power Pivot in Excel; Power BI uses the same engine.

9.1 Power Query and M: Cleaning with a Recipe

INTUITION

Imagine a cooking show. The chef does not hand you the cooked meal; she hands you the recipe, and the kitchen cooks it fresh each time. Power Query is that recipe. Each click (trim spaces, fix a type) becomes one numbered step. When next week's file arrives, Refresh re-cooks the whole recipe. The raw file is never touched. It is a saved washing routine for data, replayable on demand.

Words you need

Word	Plain meaning

Query	One recipe that produces one table
Applied step	One recorded transformation inside a query
M	The language behind the steps (a "formula language" for data shaping)
Connector	A plug for a source: Excel, CSV, SQL Server, SharePoint, Web, API

How it works. In Power BI Desktop, Home > Transform data opens the Power Query Editor. A query is a `let ... in` block: each line names a step, the next line uses that name, and the name after `in` is the result. M is **case sensitive**: `Table.AddColumn` works, `table.addcolumn` fails.

Connectors. Excel and CSV files, SQL Server and other databases, SharePoint folders, Web and APIs (`Web.Contents` fetches, `Json.Document` parses). The first connection asks for credentials and a **privacy level**; never paste real passwords into shared files.

Our dirty file. Four raw rows with typical problems: spaces and mixed case in City, dates written day-month-year, and a text `n/a` where a price should be.

```
OrderID,OrderDate,City,Category,Qty,Price,Status
1001,03-01-2025,  pune ,Fashion,2,499,Delivered
1002,03-01-2025,MUMBAI,Electronics,1,1299,Delivered
1003,04-01-2025,Pune,Grocery,3,199,Returned
1004,05-01-2025,Delhi,Fashion,1,n/a,Delivered
```

Worked example 9.1.1 The full cleaning query in M (not run here):

```
let
    Source   = Csv.Document(File.Contents("C:\data\orders_raw.csv"),
                  [Delimiter=",", Columns=7, Encoding=65001,
QuoteStyle=QuoteStyle.Csv]),
    Promoted = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
    Cleaned  = Table.TransformColumns(Promoted,
                  {{"City", each Text.Proper(Text.Trim(_)), type text}}),
    Typed    = Table.TransformColumnTypes(Cleaned,
                  {{"OrderID", Int64.Type}, {"OrderDate", type date},
                   {"Qty", Int64.Type}, {"Price", type number}}, "en-GB"),
    NoErrors = Table.ReplaceErrorValues(Typed, {{"Price", null}}),
    WithAmt  = Table.AddColumn(NoErrors, "Amount", each [Qty] * [Price],
type number)
in
    WithAmt
```

Predict: what happens to the `n/a` price, and to the date `03-01-2025`?

The same steps in pandas (`str.strip().str.title()`, `to_datetime(format="%d-%m-%Y")`, `to_numeric(errors="coerce")`) give the real output:

```
   OrderID  OrderDate    City     Category  Qty   Price     Status  Amount
0     1001 2025-01-03    Pune      Fashion    2   499.0  Delivered   998.0
1     1002 2025-01-03  Mumbai  Electronics    1  1299.0  Delivered  1299.0
2     1003 2025-01-04    Pune      Grocery    3   199.0   Returned   597.0
3     1004 2025-01-05   Delhi      Fashion    1     NaN  Delivered     NaN
```

- **Step Cleaned** `Text.Trim` removes spaces around `pune`; `Text.Proper` makes `MUMBAI` into `Mumbai`. Without this, `Pune` and `pune` are two cities in a slicer.
- **Step Typed** the extra argument `"en-GB"` is the **culture** (locale). It tells M the date is day-month-year, so `03-01-2025` is 3 January. With a US culture it would be read as 1 March, silently. pandas needs the same hint, `format="%d-%m-%Y"`.
- **Step Typed (price)** the text `n/a` cannot become a number, so that **cell** becomes an Error value; the rest survives. **NoErrors** (`Table.ReplaceErrorValues`) turns it into `null` (the `NaN` above). A row-level alternative: `each try Number.From([Price]) otherwise null`.
- **Step WithAmt** `[Qty] * [Price]` multiplies column values row by row. A `null` price gives a `null` amount (`NaN` in pandas): 2 × 499 = 998, 1 × 1299 = 1,299, 3 × 199 = 597.
- **Check** 998, 1,299 and 597 match the Chapter 2 Sales table; the fourth is blank on purpose. Missing is better than a made-up zero.

Worked example 9.1.2 The core transformations and their M functions.

Task	M function (not run here)
Change type	`Table.TransformColumnTypes(t, {{"Qty", Int64.Type}})`
Split / merge columns	`Table.SplitColumn(t, "Code", Splitter.SplitTextByDelimiter("-"), {"Dept", "No"})`, `[First] & " " & [Last]`
Conditional column	`Table.AddColumn(t, "Size", each if [Amount] = null then "Unknown" else if [Amount] >= 1000 then "Large" else "Small")`
Filter rows	`Table.SelectRows(t, each [Status] = "Delivered")`
Group by	`Table.Group(t, {"City"}, {{"Orders", each Table.RowCount(_), Int64.Type}, {"Revenue", each List.Sum([Amount]), type number}})`
Unpivot / pivot	`Table.UnpivotOtherColumns(...)` / `Table.Pivot(...)` (Chapter 3, 3.5)
Error handling	`try <expr> otherwise <value>`, `Table.ReplaceErrorValues`, `Table.RemoveRowsWithErrors`

- **Group by check** On the 8 orders, grouping by City gives (pandas) Delhi 2 orders, 3,298; Mumbai 3 orders, 3,047; Pune 3 orders, 2,995; total 9,340.
- **Why `= null` first** In M, `[Amount] >= 1000` on a `null` does not give true or false, and `if` then raises an error.

Connection to the real job. Every dashboard project starts here, and the recipe documents your cleaning for the next person.

Applications

- Finance: monthly bank files with inconsistent headers, appended every month.
- HR: unpivot a wide leave report into one row per person per month.

COMMON MISTAKE

Doing the cleaning with DAX calculated columns or by editing the source by hand. Clean upstream in Power Query, once.

Check yourself

1. What does the name after `in` mean in an M query? Answer: It is the result of the query (the final step).
2. Why is `Text.Trim` before `Text.Proper` useful? Answer: Spaces must go first so the text is identical and cased consistently.
3. What happens to a cell that cannot be converted to a number? Answer: It becomes an Error value (only that cell), which you replace or remove.

9.2 Merge, Append, Parameters, Functions and Query Folding

INTUITION

Appending is stacking sheets of paper on top of each other (more rows). Merging is stapling a column from another list next to yours by matching a key (more columns), like a lookup. Folding is a smart waiter: instead of fetching every dish to the table so you can pick, he tells the kitchen "only the vegetarian ones" and brings only those.

Append (`Table.Combine`) stacks tables with the same columns (for example 12 monthly files). Columns match by **name**; a column missing in one table becomes `null` there.

Merge (`Table.NestedJoin` , then `Table.ExpandTableColumn`) is a join. It needs a **join kind**:

Join kind (M)	Keeps
Left outer (`JoinKind.LeftOuter`)	All rows of the first table, matches from the second
Inner (`JoinKind.Inner`)	Only rows with a match in both
Right outer, Full outer	All of the second, or all of both

Left anti (`JoinKind.LeftAnti`)	First-table rows with **no** match (an audit tool)

Worked example 9.2.1 Join the 8 sales rows (total 9,340) to a Geography table `Pune: West, Mumbai: West, Chennai: South` . Delhi is missing from Geography; Chennai has no orders. Counts from pandas:

```
left 8 9340
inner 6 6042
right 7 6042.0
outer 9 9340.0
   OrderID   City
3     1004  Delhi
6     1007  Delhi
```

- **Left** keeps all 8 orders (Delhi rows get a null Region), total 9,340.
- **Inner** drops the 2 Delhi orders (799 + 2,499 = 3,298): 9,340 − 3,298 = 6,042 over 6 rows.
- **Right** gives 6 matched orders plus the Chennai row (7 rows); **full outer** gives 9.
- **Anti** shows the 2 unmatched rows, orders 1004 and 1007 (Delhi).
- **Check** 6,042 + 3,298 = 9,340.
- **Interpretation** An inner join would have quietly deleted 35% of revenue (3,298 / 9,340 = 0.353). After any merge, compare row counts and totals before and after.

Parameters and functions. A **parameter** is a named value you change without editing steps (Home > Manage Parameters), e.g. `File.Contents(FolderPath & "orders_raw.csv")` . A **custom function** wraps a recipe for reuse:

```
let
    CleanCity = (txt as text) as text => Text.Proper(Text.Trim(txt))
in
    CleanCity
```

Call it from another query: `Table.TransformColumns(Promoted, {{"City", CleanCity, type text}})` . The same pattern applied to every file in a folder (Get Data > Folder > Combine) merges 12 monthly files.

Query folding. With a database source, Power Query can translate your steps into one SQL query that runs **on the server**, so only the small answer travels. Filter, choose columns and group usually fold. Right-click a step: if "View Native Query" is enabled, it folds. Custom row functions, `Table.Buffer` and merging two different sources break folding, and later steps then

run on your machine. Rule: **filter and remove columns first**. CSV files never fold.

Incremental refresh and dataflows. Incremental refresh reloads only recent partitions (say the last 7 days) and keeps older ones. Create two date/time parameters named exactly `RangeStart` and `RangeEnd`, filter your date column with them (`[OrderDate] >= RangeStart and [OrderDate] < RangeEnd`), then set the policy in the model; it runs after publishing and works best when the filter folds. A **dataflow** is Power Query in the Service: it prepares a table once and many datasets reuse it.

Worked example 9.2.2 A SQL refresh takes 40 minutes: it loads all 20 million rows, then filters to 2 years in Power Query. What changes?

- **Step 1** Make the date filter the first step so it folds into `WHERE OrderDate >= ...`, and remove unused columns early. Check "View Native Query".
- **Answer** The database filters; only needed rows and columns arrive. The gain depends on your system (not run here).

Applications

- Analytics: append monthly extracts; merge a lookup of region or segment.
- Data Engineering: dataflows as a shared, governed preparation layer.

COMMON MISTAKE

Merging with the wrong join kind and not checking counts. A merge on `City` also fails to match `"Pune "` with `"Pune"`, so clean keys first (trim, same case).

Check yourself

1. Which join keeps only unmatched rows of the first table? Answer: Left anti.
2. What is query folding? Answer: Power Query translating steps into one source query (like SQL) that runs on the server.
3. What names must the incremental refresh parameters have? Answer: `RangeStart` and `RangeEnd`.

9.3 Star Schema: Facts and Dimensions

INTUITION

A receipt lists what was bought and how much it cost: those are **events with numbers**. The shop's catalogue, the city list and the calendar describe **things**. If you wrote the full city name, the region and the full product description on every receipt line, you would repeat yourself millions of times, and one spelling mistake would split a city in two. So keep receipts in one table and the descriptions in small tables, linked by short keys.

Formal definition.

Table type	What a row is	Typical columns	Size
Fact	One event at a fixed **grain** (one order, one click)	Keys to dimensions, numbers to add up (Qty, Amount)	Large: millions of rows
Dimension	One thing (a city, a product, a day)	A unique key, plus descriptive text	Small: tens to thousands

The **grain** is what one fact row means; decide it first. Ours: one row per order (the 8-row Sales table). Because the fact table is surrounded by dimensions, the diagram looks like a star.

*Star schema for the shop: FactSales in the middle linked to DimDate, DimCity and DimProduct; 1 marks the dimension side, * marks the fact side*

Read it: each arrow starts at a dimension (the **1** side: each key appears once) and ends at the fact table (the * side: a key can appear in many rows). A filter on a dimension flows down the arrow to the fact table.

Why not one flat table? It repeats text in every row (big, slow to refresh), one typo makes two categories, and DAX and relationships are fastest on the star shape. The engine compresses small integer keys far better than long repeated text.

Worked example 9.3.1 Split our flat 8-row Sales into a star. We build `DimCity` (3 rows: Delhi 1, Mumbai 2, Pune 3, with Region), `DimProduct` (Electronics 1, Fashion 2, Grocery 3) and `DimDate` (every day of January 2025, 31 rows). The fact table keeps keys and numbers. Real pandas output:

```
 OrderID  DateKey  CityKey  ProductKey    Status  Qty  Amount
    1001 20250103        3           2 Delivered    2     998
    1002 20250103        2           1 Delivered    1    1299
    1003 20250104        3           3  Returned    3     597
    1004 20250105        1           2 Delivered    1     799
    1005 20250106        2           3 Cancelled    5     450
    1006 20250107        2           2 Delivered    2    1298
    1007 20250109        1           1 Delivered    1    2499
    1008 20250110        3           2 Delivered    4    1400
```

- **Step 1** Replace each city name by its `CityKey` (Pune becomes 3); same for product; turn the date into an integer key `YYYYMMDD` (20250103).
- **Step 2** Keep `Qty`, `Amount` (numbers we sum) and the small `Status` column in the fact.
- **Check** 8 fact rows, and the `Amount` total is still 9,340, so nothing was lost or duplicated in the split.
- **Interpretation** The text "Pune" now lives in exactly one row of `DimCity`; to rename it, you change one cell.

Worked example 9.3.2 Classify `Qty`, `Category`, `Region`, `Amount`, `Weekday`, `OrderID`. Ask: do I sum it, or slice by it?

- **Answer** Fact: Qty, Amount, OrderID (an ID kept on the fact is a "degenerate dimension"). Dimensions: Category (DimProduct), Region (DimCity), Weekday (DimDate).

Connection to the real job. In an interview, "explain star schema" is a standard question. In Power BI work it is the first review a senior makes: if your model is a star, DAX stays simple and fast. A snowflake (dimensions split further, City to State) adds joins; flatten it when you can.

Applications

- Retail: Sales fact, Product, Store, Date, Customer dimensions.
- Finance: GL transactions fact, Account and Cost Centre dimensions.

COMMON MISTAKE

Building visuals on the flat export. It works for a demo and breaks at the first "total by region excluding cancelled". Model first.

Check yourself

1. What is the grain of a fact table? Answer: What one row represents, for example one order.

2. Which side of a relationship is the dimension? Answer: The 1 side (unique keys).
3. Is `City` a fact or a dimension column? Answer: Dimension: you slice by it.

9.4 Relationships: Cardinality, Direction, Active and Role-Playing

INTUITION

A relationship is a pipe between two tables. When you pick "West" in a slicer, the filter must flow through the pipe from `DimCity` to `FactSales` so that only West orders remain. Cardinality says how many rows match on each end. Direction says which way water flows.

Formal definition. A **relationship** links one column in each of two tables (the **key**). Its properties:

Property	Options	Default and advice
Cardinality	One-to-many (1:\), *many-to-one, one-to-one, many-to-many* (\:*)	Use **1:*** from dimension to fact almost always
Cross-filter direction	Single (dimension filters fact), Both	**Single**; Both only with a clear reason
Active	One active relationship per pair of tables; others are dashed and inactive	Role-playing dimensions use inactive ones

The key on the 1 side **must be unique**. Create relationships in Model view by dragging a key, or Home > Manage relationships.

Worked example 9.4.1 Filter flows from dimension to fact. Select Region = West. West contains Mumbai (key 2) and Pune (key 3). Which fact rows remain, and what is their total?

- **Step 1** West means `CityKey` 2 and 3, so the fact rows are orders 1001, 1002, 1003, 1005, 1006, 1008 (6 rows).
- **Step 2** Add Amount: 998 + 1,299 + 597 + 450 + 1,298 + 1,400 = 6,042.
- **Answer** pandas confirms 6 rows and 6,042. A sum measure under this slicer shows 6,042.
- **Check** West = Mumbai 3,047 + Pune 2,995 = 6,042. Delhi (North) 3,298 is excluded.
- **Interpretation** You wrote no formula for "West": the relationship did the filtering.

Worked example 9.4.2 What if the "1" side is not unique? Someone appends Pune twice to `DimCity` (11 rows after joining, instead of 8).

- **Step 1** Each Pune fact row (orders 1001, 1003, 1008) matches two dimension rows and appears twice.
- **Step 2** The Amount of Pune is counted twice: 9,340 + 2,995 = 12,335. pandas output: 11 rows, total 12,335.
- **Answer** Duplicates on the 1 side inflate totals by 32%: 2,995 / 9,340 = 0.32. Power BI would stop you from making a 1:* relationship here, which is exactly the protection you want; check for duplicate keys in Power Query with Remove Duplicates or a Group By count.
- **Check** 12,335 − 9,340 = 2,995, the Pune total, so the inflation is exactly the duplicated rows.

Cross-filter direction. With **single** direction, a filter on `DimCity` reaches `FactSales` but not the other way. **Both** lets filters travel upstream too, which can create ambiguous paths, slow DAX and surprising totals. Use `CROSSFILTER` inside one measure when you need it once (Chapter 10).

Many-to-many. If a key repeats on both sides (students and courses), use a **bridge table** with one row per valid pair, linked 1:* to both; native \:\ makes totals hard to reason about.

Role-playing dimensions. One dimension used for different purposes. An orders table has `OrderDate` and `ShipDate`. Both point to the **same** `DimDate`. Only one relationship can be active (usually order date); the ship date relationship stays inactive, and a measure activates it with `USERELATIONSHIP` (Chapter 10).

Connection to the real job. Most "my total is wrong" tickets are a duplicate key, an accidental many-to-many, a missing relationship (the slicer does nothing) or a Both filter. Learn to read the 1 and * symbols in Model view.

Applications

- Sales: Order Date versus Ship Date versus Delivery Date, one calendar.
- HR: Hire date versus Leave date.

COMMON MISTAKE

Setting every relationship to Both "to make the filter work". Fix the model instead.

Check yourself

1. Which table has the unique key in a 1:* relationship? Answer: The dimension (the 1 side).
2. How many relationships between two tables can be active at once? Answer: One.
3. What do you use when a key repeats on both sides? Answer: A bridge table (or a deliberate many-to-many with care).

9.5 Date Table, Hierarchies, Calculated Columns vs Measures and Aggregations

INTUITION

Time intelligence ("same period last year") needs a calendar that has **every day**, with no gaps, just as a ruler needs every centimetre marked. Your sales table has dates only on days with orders. A separate calendar fills the gaps and holds Month, Quarter, Weekday, and so on.

Date table. One row per day from the first to the last date you need (at least whole years), with a unique `Date` column of type date and no gaps. Build it in DAX: `DimDate = CALENDAR(DATE(2024,1,1), DATE(2025,12,31))` (or `CALENDARAUTO()`), then add columns:

```
DimDate =
ADDCOLUMNS(
    CALENDAR(DATE(2025,1,1), DATE(2025,1,31)),
    "Month", FORMAT([Date], "MMM"),
    "Weekday", FORMAT([Date], "dddd"),
    "IsWeekend", WEEKDAY([Date], 2) > 5)
```

Not run here. Then **Mark as date table** (Table tools) using the `Date` column, and relate `DimDate[Date]` to the fact date column (1:*). Sort `Month` by `MonthNo` (Column tools > Sort by column), or the months appear alphabetically. **Worked example 9.5.1** How many rows should a calendar for 2024 and 2025 have, and how many weekend days are in January 2025?

- **Step 1** 2024 is a leap year: 366 days. 2025: 365 days. Total 366 + 365 = 731 rows.
- **Step 2** January 2025 starts on Wednesday (pandas: `Wednesday`). Saturdays fall on 4, 11, 18, 25, Sundays on 5, 12, 19, 26.

- **Step 3** Count: 4 + 4 = 8 weekend days.
- **Answer** 731 rows; 8 weekend days in January 2025.
- **Check** The pandas calendar (`IsWeekend.sum()`) also gives 8.
- **Why it matters** A calendar with a missing day silently breaks time-intelligence functions in Chapter 10.

Hierarchies. An ordered drill path such as Year > Quarter > Month > Day: drag one field onto another in the Fields pane, then drill with the chart arrows.

Calculated column vs measure. This is the most important modeling distinction.

	Calculated column	Measure
When it computes	At refresh, for each row	At query time, for the current filter
Stored in the file	Yes (takes memory)	No (only the formula)
Used for	Slicers, row-level labels, grouping	Totals, ratios, KPIs
Reacts to slicers	No	Yes
Example	`Size = IF(FactSales[Amount] >= 1000, "Large", "Small")`	`Total Sales = SUM(FactSales[Amount])`

Worked example 9.5.2 Where do these belong? (a) "Revenue" shown in cards, (b) "Large order" label per order, (c) "Average order value".

- **Step 1** (a) Revenue must change when a slicer changes, so it is a measure: `Revenue = SUM(FactSales[Amount])` .
- **Step 2** (b) A label per row, usable in a slicer: calculated column (or better, make it in Power Query upstream).
- **Step 3** (c) A ratio: it must be a measure, because averages of row or group averages are not the overall average. `AOV = DIVIDE([Revenue], DISTINCTCOUNT(FactSales[OrderID]))` gives 9,340 / 8 = 1,167.5.
- **Answer** (a) measure, (b) column, (c) measure.

Performance impact. Calculated columns add memory and refresh time. Prefer Power Query

columns for row-level logic and measures for aggregates; remove unused columns, use integer keys, avoid high-cardinality columns like timestamps with seconds. **Performance Analyzer** (View > Performance analyzer) times each visual; Chapter 10 mentions DAX Studio.

Aggregations (awareness). For billion-row facts, a small **aggregation table** answers most visuals. You do not need this yet.

Connection to the real job. The date table, one-to-many keys, measures instead of columns: these three habits solve most modeling issues in the PL-300 exam and in real projects.

Applications

- Finance: fiscal calendars (year starting April, as in India) as extra columns in `DimDate`.
- Retail: hierarchy Category > Product for drill-down.

COMMON MISTAKE

Using the auto date/time feature on every date column. It creates hidden tables and bloats the model. Turn it off (Options > Current file > Data load) and use one shared `DimDate`.

Check yourself

1. Why must a date table have no gaps? Answer: Time-intelligence functions assume every day exists.
2. Which is stored in memory: a calculated column or a measure? Answer: The calculated column (a measure stores only the formula).
3. Is "average order value" a column or a measure? Answer: A measure.

Chapter recap

- Power Query records cleaning as steps in M (`let ... in`); refresh replays them, the source is untouched.
- Set types with the right culture, and replace errors or remove them deliberately.
- Merge joins add columns (choose the join kind, compare counts); append stacks rows.
- Parameters and functions make queries reusable; query folding pushes work to the database, so filter early.
- Incremental refresh needs `RangeStart` and `RangeEnd`; dataflows share cleaned tables across reports.

- A star schema has a few large fact tables (events, numbers) and small dimension tables (descriptions) linked by keys; decide the grain first.
- Use 1:* relationships, single direction, a unique key on the 1 side; use inactive relationships for role-playing dates.
- Build a gapless `DimDate`, mark it as a date table, and prefer measures over calculated columns for aggregates.

Item	Pattern
Clean text	`Table.TransformColumns(t, {{"City", each Text.Proper(Text.Trim(_)), type text}})`
Set types	`Table.TransformColumnTypes(t, {{"Qty", Int64.Type}}, "en-GB")`
Handle errors	`try ... otherwise null`, `Table.ReplaceErrorValues`
Join	`Table.NestedJoin(a, {"k"}, b, {"k"}, "B", JoinKind.LeftOuter)`
Stack	`Table.Combine({a, b})`
Calendar	`CALENDAR(start, end)`, then Mark as date table
Model rule	Dimension (1) filters Fact (*), single direction

Practice set

Problem 9.1 Is `table.addcolumn` valid M?

Answer: No. M is case sensitive: it must be `Table.AddColumn`.

Problem 9.2 Orders: Amount 100, 200, 300. After a left outer merge to a lookup, the total is 600. After an inner merge it is 300. How much revenue was unmatched, and what share?

Answer: 600 − 300 = 300 unmatched; share 300 / 600 = 50%.

Problem 9.3 What does `Table.Combine({a, b})` do with a column that exists only in `b`?

Answer: The result has that column, with `null` for the rows that came from `a`.

Problem 9.4 Find the bug: a date `13-02-2025` loads as an error or blank in your query, while `03-01-2025` loads fine.

Answer: The culture is US (month-day-year): month 13 does not exist. Set the type with culture `en-GB`, or split and rebuild the date. The `03-01-2025` row was silently wrong (March 1 instead of 3 January).

Problem 9.5 Product P1 appears twice in `DimProduct`; the fact has one P1 row with Amount 500. What does a join give, and what does Power BI do?

Answer: The join returns that row twice, 1,000 in total. Power BI refuses a 1:* relationship until the duplicate key is fixed.

Problem 9.6 A fact has 3 rows: DateKey 20250103 (Amount 998), 20250103 (1,299), 20250104 (597). A slicer selects 4 Jan 2025. What does the total show?

Answer: 597: only the rows whose key matches the selected date remain.

Problem 9.7 Scenario: a slicer on Region does nothing to a card showing Total Sales. List three things to check.

Answer: (1) Is there a relationship between `DimCity` and `FactSales`, and is it active? (2) Does the key have the same data type on both sides, with the dimension key unique? (3) Is the slicer built from the dimension column and the card a measure on the fact table (not a disconnected table)?

Problem 9.8 (Interview) Explain star schema versus one flat table and why Power BI prefers the star.

Answer: A star keeps events and numbers in a fact table and descriptions in small dimension tables joined by integer keys. It avoids repeated text (smaller, faster, compressed), prevents spelling splits, gives one place to maintain each attribute, and matches how DAX filter context flows (dimension to fact).

Problem 9.9 (Interview) What is query folding, and how do you check it?

Answer: Power Query turns your steps into one source query (like SQL) that runs on the server. Check with "View Native Query" on a step; keep filters and column removal first so folding is not broken.

Prompt pack

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles, and English is my second language. Teach me Power
Query and the M language in Power BI: queries as let/in steps, changing
types with culture, trimming text, handling errors, merge versus append
and the join kinds. Use a small dirty orders CSV as the running example.
Explain each step in simple words, show what each M line does to the
table, then ask me 4 questions one at a time and wait for my answers.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. Explain star schema for Power BI: fact versus
dimension tables, grain, keys, and why a flat table is worse. Take a flat
orders table with columns OrderID, Date, City, Category, Qty, Price and
design the star step by step, listing every table and its columns. Then
give me 6 exercises where I classify columns or design a model for a new
business (a hospital, a call centre), without solutions, and mark my
answers afterwards.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. Teach Power BI relationships: cardinality (1:*,
many-to-many), cross-filter direction, active versus inactive
relationships and role-playing date dimensions (OrderDate and ShipDate).
Use tiny tables with numbers I can check by hand, show how a filter on a
dimension travels to the fact table, and show what goes wrong with a
duplicate key. End with a debugging scenario: "my slicer does nothing".
Quiz me one question at a time.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. Review my Power BI data model design and my Power
Query steps for a junior analyst portfolio project. Check: grain of the
fact table, star shape, unique keys on the one side, relationship
directions, date table (no gaps, marked), calculated columns versus
measures, query folding and error handling. Give a numbered list of
problems and fixes by priority. My model (tables, columns, relationships):
[PASTE HERE]. My M code or steps: [PASTE HERE].
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. Run a mock interview for a Power BI analyst role
on Power Query and data modeling. Ask one question at a time: star schema,
query folding, incremental refresh, dataflows, many-to-many, bidirectional
filters, calculated column versus measure, date tables. After each answer,
score it out of 5, show a model answer in simple English, and then ask the
next question. Do 8 questions.
```

CHAPTER 10

POWER BI II: DAX, CONTEXT, CALCULATE AND TIME INTELLIGENCE

Write one formula that gives the right answer in every cell of every visual.

In Chapter 9 you built a clean star schema. Now you must ask it questions: "total sales", "delivered share", "same month last year", "running total". The language for this is **DAX** (Data Analysis Expressions). It looks like Excel, but it behaves differently: one formula is evaluated again for every cell of a table or chart, and each cell has its own filters. Understand that one idea, called **filter context**, and DAX stops being mysterious. By the end you can write measures for totals, shares, ranks, year-on-year growth, running totals and what-if scenarios, and explain why each gives its number.

PROFESSOR'S NOTE

DAX and Power BI cannot run in this environment. The DAX below is standard syntax, written carefully, **not run here**. To keep the numbers honest I computed every expected result with pandas (3.0.6) on the same small tables, and each example states a number you can verify by hand. Paste the code into Power BI Desktop and compare.

Our model. The star of Chapter 9: `FactSales` (8 orders with `OrderID`, `Date`, `ShipDate`, `CityKey`, `Status`, `Qty`, `Price`, `Amount = Qty × Price`), `DimCity` (`City`, `Region`: Delhi North, Mumbai West, Pune West) and `DimDate`. Totals: all 9,340; Delhi 3,298, Mumbai 3,047, Pune 2,995. For time intelligence (10.4) we use the 18-month series of Chapter 8 in `FactMonthly[SalesK]`, dated on the first of each month.

10.1 Measures, Row Context and Filter Context

INTUITION

Picture a school register. A **row context** is a teacher reading one student's line: "this student, marks 72". A **filter context** is the principal saying: "only Class 7B, only girls". A measure is a question like "average marks"; the answer depends on which students survive the filters. A PivotTable cell is exactly this: each cell carries its own filters (row = Pune, column = Delivered), and the measure runs once per cell.

Words you need. Calculated column: a formula stored per row, computed at refresh. **Measure**: a formula computed at query time for the current filters, stored as formula only. **Row context**: "the current row", available in calculated columns and iterators. **Filter context**: the filters from slicers, rows, columns and other visuals active for a cell. **Context transition**: turning the current row into filters (done by `CALCULATE`).

Rules.

1. A measure such as `Total Sales = SUM(FactSales[Amount])` has no row context. Aggregators (`SUM`, `AVERAGE`, `COUNT`) look at the rows the filter context leaves.
2. A calculated column has row context, but **no** filter context from a visual.
3. Row context does not filter other tables by itself; `CALCULATE` converts it into filter context.
4. Filters flow along relationships from the 1 side to the * side (Chapter 9).

Worked example 10.1.1 `Total Sales` in a matrix with `DimCity[City]` in rows.

- **Step 1** The Delhi row sets `DimCity[City] = "Delhi"`. It flows to `FactSales`, leaving orders 1004 (799) and 1007 (2,499).
- **Step 2** `SUM` adds them: 799 + 2,499 = 3,298. Mumbai: 1,299 + 450 + 1,298 = 3,047. Pune: 998 + 597 + 1,400 = 2,995.
- **Step 3** The Total row has no city filter, so all 8 rows: 9,340.
- **Answer** 3,298 / 3,047 / 2,995, total 9,340. One formula; the context did the splitting.
- **Check** 3,298 + 3,047 + 2,995 = 9,340, as in Chapter 3's PivotTable.

Worked example 10.1.2 Row context and context transition. In `DimCity` we add two calculated columns:

```
City Sales A = SUM(FactSales[Amount])
City Sales B = CALCULATE(SUM(FactSales[Amount]))
```

- **Step 1** In column A each `DimCity` row has a row context, but `SUM` ignores it: nothing filters `FactSales`. Every city shows **9,340**.
- **Step 2** In column B, `CALCULATE` turns the current row (Delhi) into the filter `City = "Delhi"`. It flows to `FactSales`: **3,298**, then 3,047 and 2,995.
- **Answer** A: 9,340, 9,340, 9,340. B: 3,298, 3,047, 2,995.
- **Check** B adds to 9,340. A adds to 28,020 (3 × 9,340), absurd for a total, so A is the bug.
- **Why it matters** This is the most tested DAX idea. A **measure** referenced inside a row context gets an implicit `CALCULATE`, so `[Total Sales]` inside an iterator behaves like B, not A.

The picture

```
Visual cell (City = Pune, Status = Delivered)
       |  filter context = { City = Pune, Status = Delivered }
       v
DimCity --1:*--> FactSales   rows left: 1001 (998), 1008 (1,400)
       v
SUM(Amount) = 998 + 1,400 = 2,398
```

Connection to the real job. When a total is "wrong", the cause is nearly always a filter context you did not expect: a slicer, a relationship direction, a removed filter. Debug by asking "what are the filters in this cell?".

Applications

- Analytics: one `Total Sales` measure serves every chart.
- Operations: one `Avg Delivery Days` measure sliced by city, category and month.

COMMON MISTAKE

Making a calculated column for something that must react to slicers (a ratio, a total). Columns freeze at refresh; use a measure.

Check yourself

1. Which has row context: a measure or a calculated column? Answer: A calculated column.
2. What does `CALCULATE` do to a row context? Answer: Converts it into an equivalent filter context (context transition).
3. What filter context does the Total row of a matrix have? Answer: None from rows or

columns, so all data (except slicers).

10.2 CALCULATE, Filters and the Core Functions

INTUITION

`CALCULATE` is the principal with a pen. You hand it a question ("total sales") and rule changes ("but only Delivered", "ignore the city filter"). It changes the filters first, then asks the question. Almost every serious DAX measure is a `CALCULATE`.

How it works. `CALCULATE(<expression>, <filter1>, ...)` takes the current filter context, applies each filter argument (a filter on a column **replaces** an existing filter on that same column; filters on different columns combine with AND), then evaluates the expression.

Function	What it does
`FILTER(table, condition)`	Rows of a table meeting a condition (an iterator; use when the condition needs a measure or several columns)
`ALL(table or column)`	Removes filters on that table or column
`ALLEXCEPT(table, col, ...)`	Removes all filters on the table except the listed columns
`ALLSELECTED(...)`	Removes the visual's own filters but keeps outer ones (slicers)
`VALUES(col)` / `DISTINCT(col)`	Distinct values visible now (`VALUES` may include a blank row)
`SELECTEDVALUE(col, alt)`	The one selected value, or `alt` if none or many
`RELATED(col)` / `RELATEDTABLE(table)`	In row context, fetch one value (many to 1) or the related rows (1 to many)
`USERELATIONSHIP(c1, c2)`	Switches on an inactive relationship for this calculation

`VAR ... RETURN`	Names an intermediate value, computed once

Worked example 10.2.1 Delivered sales and delivered rate, with variables.

```
Delivered Sales = CALCULATE([Total Sales], FactSales[Status] =
"Delivered")
Delivered Rate =
    VAR Del = [Delivered Sales]
    VAR Tot = [Total Sales]
    RETURN DIVIDE(Del, Tot)
```

- **Step 1** Pune row: filter City = Pune, plus Status = Delivered: orders 1001 (998) and 1008 (1,400) = 2,398.
- **Step 2** Total: delivered orders 1001, 1002, 1004, 1006, 1007, 1008: 998 + 1,299 + 799 + 1,298 + 2,499 + 1,400 = 8,293.
- **Step 3** `DIVIDE(8293, 9340) = 0.8879`. `DIVIDE` gives blank, not an error, when the denominator is 0.
- **Answer** Delhi 3,298, Mumbai 2,597, Pune 2,398, total 8,293; delivered rate 88.8% overall and 2,398 / 2,995 = 80.1% for Pune.
- **Check** 3,298 + 2,597 + 2,398 = 8,293, matching Chapter 3.
- **Interpretation** Nearly 9 of 10 rupees are delivered; Pune has the only returned order (597).

Worked example 10.2.2 Percent of total with `ALL`, and how `ALLSELECTED` differs.

```
% of All Cities     = DIVIDE([Total Sales], CALCULATE([Total Sales],
ALL(DimCity)))
% of Visible Cities = DIVIDE([Total Sales], CALCULATE([Total Sales],
ALLSELECTED(DimCity)))
```

- **Step 1** `ALL(DimCity)` removes every city filter inside `CALCULATE`, so the denominator is always 9,340: Delhi 3,298 / 9,340 = 35.3%, Mumbai 32.6%, Pune 32.1%.
- **Step 2** A slicer selects Region = West. `% of All Cities` still divides by 9,340: Mumbai 3,047 / 9,340 = 32.6%.
- **Step 3** `ALLSELECTED(DimCity)` removes the row filter but keeps the slicer, so the denominator is 3,047 + 2,995 = 6,042: Mumbai 50.4%, Pune 49.6%.
- **Answer** Under West, the first measure shows 32.6% and 32.1% (not adding to 100%), the second 50.4% and 49.6%.
- **Check** 50.4 + 49.6 = 100.0.

- **Why it matters** `ALL` is "share of the company", `ALLSELECTED` is "share of what I filtered". `ALLEXCEPT(DimCity, DimCity[Region])` would give "share within its region".

Worked example 10.2.3 `USERELATIONSHIP`. `FactSales[ShipDate]` has an inactive relationship to `DimDate[Date]` (role-playing, Chapter 9). Ship dates are order date plus 2, 1, 2, 3, 3, 2, 3, 3 days.

```
Sales by Ship Date =
    CALCULATE([Total Sales], USERELATIONSHIP(FactSales[ShipDate],
DimDate[Date]))
```

- **Step 1** On 9 Jan 2025, `Total Sales` uses the active order-date relationship: order 1007 = 2,499.
- **Step 2** Ship date 9 Jan belongs to orders 1005 (6 Jan + 3) and 1006 (7 Jan + 2): 450 + 1,298 = 1,748.
- **Answer** On 9 Jan: by order date 2,499, by ship date 1,748.
- **Check** pandas grouping by ship date gives 1,748 for 9 Jan, and by order date 2,499.

Other details. `CALCULATE(..., FactSales[Status] = "Delivered")` is shorthand for a `FILTER` over `ALL(FactSales[Status])`, which is why it **overrides** a Status slicer; wrap it in `KEEPFILTERS` to respect the slicer. `Region = RELATED(DimCity[Region])` in a calculated column on `FactSales` returns "West" for Pune orders. `Orders = COUNTROWS(RELATEDTABLE(FactSales))` in `DimCity` returns 2, 3, 3 for Delhi, Mumbai, Pune. `SELECTEDVALUE(DimCity[City], "All cities")` gives dynamic titles.

Connection to the real job. `CALCULATE` with `ALL` / `ALLSELECTED` sits behind percent-of-total, share and "vs all" in nearly every dashboard. Interviews ask you to explain `CALCULATE` and `ALL` versus `ALLEXCEPT` versus `ALLSELECTED`.

Applications

- Sales: share of category, delivered rate, returns rate.
- Finance: actual versus budget by switching a relationship.

COMMON MISTAKE

Expecting a `CALCULATE` filter to add to an existing filter on the same column. It **replaces** it. Also, `FILTER(FactSales, ...)` everywhere is slower than a simple column filter.

Check yourself

1. What does `ALL(DimCity)` do inside `CALCULATE`? Answer: Removes all filters on `DimCity` for that calculation.
2. Which function keeps a slicer but ignores row filters? Answer: `ALLSELECTED`.
3. What does `DIVIDE` do when the denominator is 0? Answer: Returns blank (or an alternate result you give), not an error.

10.3 Iterators: SUMX, AVERAGEX, COUNTX, RANKX, TOPN

INTUITION

`SUM` adds a finished column. An iterator is a clerk who walks through a table row by row, does a small calculation on each row (row context), then adds or averages the results. You need the clerk when the quantity you want is not a column: revenue is `Qty × Price`.

How it works. An iterator takes `(table, expression)`: for each row it evaluates the expression in row context, then aggregates (`SUMX`, `AVERAGEX`, `COUNTX`, `MINX`, `MAXX`). `RANKX` ranks; `TOPN` returns the top N rows of a table.

Worked example 10.3.1 Simple versus weighted average price. `Qty` totals 19.

```
Simple Avg Price   = AVERAGE(FactSales[Price])
Weighted Avg Price = DIVIDE(SUMX(FactSales, FactSales[Qty] *
FactSales[Price]),
                            SUM(FactSales[Qty]))
```

- **Step 1** Simple: (499 + 1,299 + 199 + 799 + 90 + 649 + 2,499 + 350) / 8 = 6,384 / 8 = 798.
- **Step 2** `SUMX` computes `Qty × Price` per row (998, 1,299, 597, 799, 450, 1,298, 2,499, 1,400) and adds: 9,340.
- **Step 3** Weighted: 9,340 / 19 = 491.58.
- **Answer** The simple average is 798, but the average price **per unit actually sold** is 491.58.
- **Check** 491.58 × 19 = 9,340. The simple average is higher because the expensive items (2,499; 1,299) sold 1 unit each while cheap ones sold 5 and 3.
- **Interpretation** Averaging a price column is a classic error: weight by quantity.

Worked example 10.3.2 Iterating over a list of values.

```
Avg Sales per City = AVERAGEX(VALUES(DimCity[City]), [Total Sales])
```

- **Step 1** `VALUES(DimCity[City])` lists the visible cities: Delhi, Mumbai, Pune.
- **Step 2** For each city, `[Total Sales]` runs with an implicit `CALCULATE` (context transition): 3,298, 3,047, 2,995.
- **Step 3** Average: 9,340 / 3 = 3,113.33.
- **Answer** 3,113.33 per city. Under the West slicer: (3,047 + 2,995) / 2 = 3,021.
- **Check** 3,113.33 × 3 = 9,340.

Ranking and top N.

```
City Rank          = RANKX(ALL(DimCity[City]), [Total Sales])
Top 2 Cities Sales = CALCULATE([Total Sales], TOPN(2, ALL(DimCity[City]),
[Total Sales]))
```

`RANKX` ranks the current value among all cities (`ALL` , so the list does not shrink to one row). Default is descending: Delhi 3,298 is rank 1, Mumbai 2, Pune 3. Optional arguments set the order and tie handling (`SKIP` or `DENSE`). `TOPN` returns the 2 largest cities, Delhi and Mumbai, so the measure is 3,298 + 3,047 = 6,345. Check: 9,340 − 2,995 = 6,345.

Connection to the real job. Iterators handle "row by row, then total": revenue from quantity and price, margin from costs, weighted averages. `RANKX` and `TOPN` power "Top 10 customers" visuals.

Applications

- Retail: weighted average price, basket revenue.
- Healthcare: average waiting time weighted by number of patients (never average the clinic averages).

COMMON MISTAKE

Using `SUMX` with a heavy expression over a huge table when a stored `Amount` column exists. If the column exists, plain `SUM` is faster.

Check yourself

1. When do you need `SUMX` instead of `SUM` ? Answer: When you must calculate per row first and no column exists.

2. Why `ALL(DimCity[City])` inside `RANKX` ? Answer: So the ranking list contains every city, not only the current one.
3. What does `TOPN(2, ...)` return? Answer: The 2 rows with the highest value of the expression.

10.4 Time Intelligence

INTUITION

Business questions are about time: "this month versus last month", "this year so far", "same month last year". Time intelligence functions shift the date filter. If the cell is June 2025, `SAMEPERIODLASTYEAR` slides the window back to June 2024 and asks the same question there.

Requirements. (1) A gapless date table marked as date table (Chapter 9). (2) A 1:* relationship from `DimDate[Date]` to the fact table's date. (3) Use the date table's column inside these functions, never the fact's date column.

Data: `FactMonthly[SalesK]` (Rs thousand, dated the 1st). 2024: 310, 295, 340, 360, 385, 400, 380, 410, 430, 470, 560, 620. 2025 Jan to Jun: 350, 330, 375, 405, 430, 455. `Sales K = SUM(FactMonthly[SalesK])` .

Function	What it gives
`SAMEPERIODLASTYEAR(dates)`	The same dates, one year earlier
`DATEADD(dates, n, MONTH)`	Dates shifted by n intervals (DAY, MONTH, QUARTER, YEAR)
`PARALLELPERIOD(dates, n, MONTH)`	The full period n steps away
`DATESYTD(dates)` , `TOTALYTD(expr, dates)`	Dates, or the total, from the start of the year to now
`DATESINPERIOD(dates, end, n, MONTH)`	A window of n intervals ending at `end` (rolling windows)

Worked example 10.4.1 Year on year.

```
Sales LY = CALCULATE([Sales K], SAMEPERIODLASTYEAR(DimDate[Date]))
YoY %    = DIVIDE([Sales K] − [Sales LY], [Sales LY])
```

- **Step 1** June 2025: `Sales K` = 455. **Step 2** the filter shifts to June 2024: `Sales LY` = 400.
- **Step 3** `YoY %` = (455 − 400) / 400 = 0.1375.
- **Answer** June 2025 is 13.75% above June 2024.
- **Check** 400 × 1.1375 = 455; Chapter 8's chart showed the same bar.

Worked example 10.4.2 Year to date, month on month, rolling average.

```
Sales YTD      = TOTALYTD([Sales K], DimDate[Date])
Sales FYTD     = TOTALYTD([Sales K], DimDate[Date], "3/31")
Sales PM       = CALCULATE([Sales K], DATEADD(DimDate[Date], −1, MONTH))
MoM %          = DIVIDE([Sales K] − [Sales PM], [Sales PM])
Rolling 3M     = CALCULATE([Sales K],
                    DATESINPERIOD(DimDate[Date], MAX(DimDate[Date]), −3,
MONTH))
Rolling 3M Avg = DIVIDE([Rolling 3M], 3)
```

- **Step 1** YTD at June 2025: 350 + 330 + 375 + 405 + 430 + 455 = 2,345. At June 2024: 310 + 295 + 340 + 360 + 385 + 400 = 2,090. Growth: 2,345 / 2,090 − 1 = 12.2%.
- **Step 2** `FYTD` with year end `"3/31"` (Indian fiscal year) starts 1 April 2025: 405 + 430 + 455 = 1,290.
- **Step 3** `Sales PM` at June = May = 430. MoM = (455 − 430) / 430 = 0.0581, so 5.8%.
- **Step 4** Rolling 3M at June covers April to June: 405 + 430 + 455 = 1,290; average 1,290 / 3 = 430.
- **Answer** YTD 2,345 (+12.2% vs 2,090), fiscal YTD 1,290, MoM +5.8%, rolling 3-month average 430.
- **Check** pandas gives 2,345, 2,090, 0.0581 and a rolling mean of 430.0.
- **Interpretation** Growth is steady at +10% to +14% year on year every month, not a one-month jump.

Notes. `PARALLELPERIOD(DimDate[Date], −3, MONTH)` from June 2025 returns the whole of March 2025 (375). The rolling average divides by 3, which is wrong for the first two months of data; there, divide by the number of months present. A date table with gaps gives silently wrong results. Year-end strings like `"3/31"` can depend on locale, so test them.

Connection to the real job. YoY, MTD, YTD and rolling averages are the most requested measures in finance, sales and operations. Built once on a good date table, they serve every report.

Applications

- Finance: fiscal YTD, quarter on quarter, budget variance.
- Operations: week-on-week change in average delivery days, rolling cancellation rate.

COMMON MISTAKE

Using the fact table's date column inside `SAMEPERIODLASTYEAR`, or a date table that contains only dates with sales. Results become blank or wrong with no error message.

Check yourself

1. Which date column goes inside time intelligence functions? Answer: The one from the marked date table.
2. What does `TOTALYTD(..., "3/31")` set? Answer: The fiscal year end (31 March).
3. What does `DATESINPERIOD(d, MAX(d), -3, MONTH)` return? Answer: The 3-month window ending on the last visible date.

10.5 Patterns and Tools

INTUITION

Once you know the building blocks, most requests are one of a few recipes. Learn the recipes and you answer them in minutes.

Worked example 10.5.1 Running total by order date.

```
Running Total =
    CALCULATE([Total Sales],
        FILTER(ALLSELECTED(DimDate[Date]), DimDate[Date] <=
MAX(DimDate[Date])))
```

- **Step 1** In the 5 Jan cell, `MAX(DimDate[Date])` is 5 Jan; `ALLSELECTED` lists the dates in view; `FILTER` keeps those up to 5 Jan.

- **Step 2** Daily totals: 3 Jan 2,297 (998 + 1,299), 4 Jan 597, 5 Jan 799. Running: 2,297, 2,894, 3,693.
- **Answer** The full column: 2,297, 2,894, 3,693, 4,143, 5,441, 7,940, 9,340 (pandas `cumsum`). The last value equals the total.
- **Check** 3,693 + 450 = 4,143; + 1,298 = 5,441; + 2,499 = 7,940; + 1,400 = 9,340.

Worked example 10.5.2 Dynamic segmentation with a disconnected table. Create `Bands`, related to nothing: Small (Min 0, Max 500), Medium (500, 1,000), Large (1,000, 1,000,000).

```
Orders in Band =
    VAR lo = SELECTEDVALUE(Bands[Min])
    VAR hi = SELECTEDVALUE(Bands[Max])
    RETURN COUNTROWS(FILTER(FactSales, FactSales[Amount] >= lo &&
FactSales[Amount] < hi))
```

- **Step 1** With `Bands[Band]` in rows, each row selects one band, so `lo` and `hi` are single numbers.
- **Step 2** Small (below 500): 450 gives 1 order. Medium (500 to below 1,000): 998, 597, 799 give 3. Large (1,000 and above): 1,299, 1,298, 2,499, 1,400 give 4.
- **Answer** Small 1, Medium 3, Large 4.
- **Check** 1 + 3 + 4 = 8 orders.
- **Why it matters** The user can edit the bands and the visual updates; the fact table never changes.

What-if parameter. Modeling > New parameter creates a table `GENERATESERIES(0, 0.3, 0.05)` and a `SELECTEDVALUE` measure. Then `Scenario Revenue = [Total Sales] * (1 + [Price Change Value])`. At +10%: 9,340 × 1.10 = 10,274, assuming the same volume (rarely true; say so).

Calculation groups (awareness). One group applies a transformation (Current, YTD, YoY) to **any** measure, so you do not write 3 versions of 20 measures. Built in Tabular Editor; recent Desktop versions also offer them in Model view (check yours).

Tool	Use
DAX Studio	Run DAX queries, see timings and model memory (VertiPaq Analyzer)
Tabular Editor	Edit measures and calculation groups in bulk, best-practice

	rules
Performance Analyzer (in Desktop)	Time each visual's query, copy it to DAX Studio

Debugging method. (1) Put the measure and the filter columns in a table visual. (2) Test one filter at a time. (3) Split the measure into `VAR`s and return each. (4) For slowness: Performance Analyzer, then DAX Studio; remove iterators over big tables.

Connection to the real job. These recipes are the PL-300 exam's DAX questions and a BI analyst's daily asks.

Applications

- Operations: running total of tickets closed, SLA band counts.
- Sales: what-if on discount or price.

COMMON MISTAKE

Relating the `Bands` table to the fact table. A disconnected table is related to nothing on purpose.

Check yourself

1. Why is `Bands` disconnected? Answer: So its selection does not filter the fact table; the measure reads the selection instead.
2. What does `ALLSELECTED(DimDate[Date])` give in a running total? Answer: The dates in view, so the window respects slicers.
3. Which tool times each visual's DAX query inside Desktop? Answer: Performance Analyzer.

Chapter recap

- A measure runs once per cell with that cell's filter context; a calculated column has row context only.
- `CALCULATE` changes the filters first, then evaluates; a column filter replaces an existing filter on that column.
- Context transition turns the current row into filters; measures inside iterators get it automatically.

- `ALL` is share of company, `ALLSELECTED` keeps slicers, `ALLEXCEPT` keeps chosen columns.
- Iterators calculate row by row first; use weights, never average averages.
- Time intelligence needs a gapless marked date table and its date column.
- Debug with VARs, one filter at a time, Performance Analyzer and DAX Studio.

Task	DAX pattern
Filtered total	`CALCULATE([Total Sales], FactSales[Status] = "Delivered")`
Percent of total	`DIVIDE([M], CALCULATE([M], ALL(DimCity)))`
Weighted value	`DIVIDE(SUMX(T, T[a] * T[b]), SUM(T[b]))`
Rank	`RANKX(ALL(DimCity[City]), [M])`
Last year	`CALCULATE([M], SAMEPERIODLASTYEAR(DimDate[Date]))`
YTD	`TOTALYTD([M], DimDate[Date])`
Rolling window	`CALCULATE([M], DATESINPERIOD(DimDate[Date], MAX(DimDate[Date]), -3, MONTH))`
Inactive relationship	`CALCULATE([M], USERELATIONSHIP(FactSales[ShipDate], DimDate[Date]))`

Practice set

Problem 10.1 `Total Sales = SUM(FactSales[Amount])`. What does the Delhi row show, given Delhi orders of 799 and 2,499?

Answer: 799 + 2,499 = 3,298.

Problem 10.2 What is `DIVIDE(8293, 9340)` as a percent, to one decimal?

Answer: 8,293 / 9,340 = 0.8879, so 88.8%.

Problem 10.3 Find the bug: in `DimCity`, the calculated column `=SUM(FactSales[Amount])` shows the same number on every row.

Answer: Row context does not filter `FactSales`, so every row sums the whole table (9,340). Use `=CALCULATE(SUM(FactSales[Amount]))` (context transition), or better, a measure.

Problem 10.4 Under a Region = West slicer, `% of All Cities` shows 32.6% for Mumbai. What does `% of Visible Cities` show?

Answer: 50.4%: 3,047 / (3,047 + 2,995) = 3,047 / 6,042.

Problem 10.5 Three orders: 1 unit at 100, 2 units at 400, 1 unit at 200. Simple and weighted average price?

Answer: Simple: (100 + 400 + 200) / 3 = 233.33. Weighted: (100 + 800 + 200) / 4 = 1,100 / 4 = 275. The weighted value is the real average per unit sold.

Problem 10.6 Sales (Rs thousand): Mar 2025 = 375, Apr = 405, May = 430. Rolling 3-month average at May, and MoM % for May?

Answer: Rolling: (375 + 405 + 430) / 3 = 1,210 / 3 = 403.33. MoM: (430 − 405) / 405 = 0.0617, so 6.2%.

Problem 10.7 YTD at March 2025 (350, 330, 375) versus YTD March 2024 (310, 295, 340): growth?

Answer: 1,055 versus 945; 1,055 / 945 − 1 = 110 / 945 = 0.1164, so 11.6%.

Problem 10.8 Scenario: your `YoY %` shows blank for every month of 2024. Two likely reasons?

Answer: There is no 2023 data (blank is then correct), or the date table starts in 2024, is not marked, or is not related, so the shifted dates do not exist. Check that the date table covers the earlier year.

Problem 10.9 (Interview) Explain `ALL`, `ALLEXCEPT` and `ALLSELECTED`.

Answer: `ALL` removes all filters on a table or column. `ALLEXCEPT(table, cols)` removes all filters on the table except the listed columns. `ALLSELECTED` removes the filters from the visual's own rows and columns but keeps slicer and outer filters, so shares add to 100% of what the user sees.

Problem 10.10 (Interview) What is context transition and where does it happen?

Answer: It converts the current row of a row context into an equivalent filter context. It happens when `CALCULATE` runs inside a row context, and implicitly when a measure is referenced in an iterator or calculated column. That is why `[Total Sales]` in

`AVERAGEX(VALUES(DimCity[City]), [Total Sales])` gives each city's own total.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles, and English is my second language. Teach me DAX
row context versus filter context and context transition using a tiny
orders table with 8 rows and a City dimension. Use an everyday analogy
first, then trace a measure through a matrix cell by cell with real
numbers. Show the classic calculated-column bug and its fix. Then ask me 4
questions one at a time and wait for each answer.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Teach me DAX CALCULATE in depth: how filter
arguments are applied, replacing versus combining filters, FILTER versus a
column filter, ALL, ALLEXCEPT, ALLSELECTED, KEEPFILTERS, VALUES and
SELECTEDVALUE. Use small tables with numbers I can calculate by hand and
state the expected result of each measure. Then give me 8 graded exercises
without solutions and mark my answers afterwards.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Explain DAX iterators (SUMX, AVERAGEX, COUNTX,
RANKX, TOPN) and why averaging an average is wrong. Use a weighted average
price example and a top-N example with real numbers. Show a wrong measure
and the right one, and explain row by row what the iterator does. End with
a debugging scenario where a ratio in a matrix total is wrong, and let me
find the cause before you reveal it.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Teach DAX time intelligence: the need for a
gapless marked date table, SAMEPERIODLASTYEAR, DATEADD, TOTALYTD with a
fiscal year end of 31 March, PARALLELPERIOD and DATESINPERIOD for rolling
averages, plus YoY and MoM. Use 12 months of small numbers and verify each
result by hand. Then review my measure and tell me what is wrong with it:
```

[PASTE HERE].

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Run a mock interview for a Power BI analyst role on DAX. Ask one question at a time: measure versus calculated column, row versus filter context, CALCULATE, ALL versus ALLSELECTED, iterators, time intelligence, running total, dynamic segmentation, performance tuning. After each answer, score it out of 5, give a model answer in simple English, then ask the next question. Do 8 questions.

CHAPTER 11

REPORT DESIGN, INTERACTIVITY, SECURITY, POWER BI SERVICE, GOVERNANCE AND OTHER BI TOOLS

Turn a correct model into a report people can read, explore, trust and safely share.

You can now clean data (Chapter 9) and write measures (Chapter 10). A correct number still fails if the report is cluttered, if the wrong person can see it, or if it refreshes at the wrong time. This chapter covers the last mile: designing the page, adding interactivity, restricting data with row-level security (RLS), publishing to the Power BI Service, and keeping the whole thing governed. We end by comparing Power BI with Tableau, Looker Studio and other tools, because job posts often name them. By the end you can design a three-page report, secure it by region, schedule its refresh, and explain which tool you would pick for a given team.

PROFESSOR'S NOTE

Power BI, Tableau and Looker Studio cannot run here, and cloud features (Service, Copilot, licences, limits) change often, so I describe stable ideas and say "check current documentation" for the rest. Formulas (DAX, Tableau) are standard but **not run here**. Numbers were computed in Python (pandas 3.0.6) on our 8-order shop table: total 9,340; Delhi (North) 3,298; Mumbai and Pune (West) 3,047 and 2,995.

11.1 Report Design: Layout, Visuals, Colour and Accessibility

INTUITION

A report page is a newspaper front page. The reader looks top-left first, wants the headline numbers at the top, details lower down, and no more than a handful of stories. Chapter 8 gave the chart rules; here we apply them to a whole page.

Words you need. Canvas: the page area (Power BI's default is 1280 x 720 pixels, a 16:9 shape). **Card**: one big number. **KPI visual**: a number with target and trend. **Theme**: a saved set of colours and fonts applied to all visuals. **Alt text**: a written description of a visual for screen-

reader users.

Layout rules.

1. Put the 3 to 5 most important numbers (cards) in a row at the top.
2. Left to right, top to bottom, from summary to detail.
3. Use a grid: equal margins and gaps, aligned edges. Misaligned visuals look untrustworthy.
4. At most 6 to 8 visuals per page. Use more pages (or drill-through) instead of crowding.
5. One page, one question. Title each page with its message.

Visual choice (the Chapter 8 table, in Power BI names).

Need	Visual
Compare categories / trend	Sorted bar or column / line (area for volume)
Two numbers / exact numbers	Scatter / table or matrix (matrix = pivot with drill)
One headline number	Card; KPI for target and trend; gauge sparingly
Geography	Map (normalise per customer)
Why did it change	Decomposition tree (break a measure down by chosen fields)
What drives an outcome	Key influencers (statistical; treat as leads, not proof)

Custom visuals come from AppSource inside Power BI. They add options but can add risk and slowness; prefer certified ones and check your company policy.

Worked example 11.1.1 Grid arithmetic. Three cards across a 1280-pixel-wide page, with a 20-pixel margin on the left and right and a 20-pixel gap between cards. How wide is each card?

- **Formula** card width = (page width − 2 × margin − number of gaps × gap) / cards
- **Plug in** (1280 − 2 × 20 − 2 × 20) / 3
- **Compute** 1280 − 40 − 40 = 1,200; 1,200 / 3 = 400
- **Answer** Each card is 400 pixels wide; they start at x = 20, 440 and 860.
- **Check** 20 + 400 + 20 + 400 + 20 + 400 + 20 = 1,280. The edges meet the margins exactly.
- **Why it matters** Exact numbers (Format > Position) give perfectly aligned pages without

eyeballing.

Worked example 11.1.2 Accessible colour. The WCAG guideline asks for a text contrast ratio of at least 4.5 to 1 (3 to 1 for large text). The ratio is `(L1 + 0.05) / (L2 + 0.05)`, where L is the relative luminance (a brightness number from 0 to 1) of the lighter and darker colour. I computed it in Python:

```
colour on white   ratio
#1f6feb (blue)    4.63
#9aa5b1 (grey)    2.5
#595959 (dark)    7.0
#767676 (mid)     4.54
```

- **Step 1** Blue (#1f6feb) scores 4.63: passes 4.5 for normal text, just barely.
- **Step 2** Light grey (#9aa5b1), the "context" colour of Chapter 8, scores 2.5: it fails for text and is weak even for graphics (3 to 1 is the usual target).
- **Answer** Use dark grey (#595959, 7.0) for text, keep light grey for background marks only, and add direct labels so colour is never the only signal.
- **Check** #767676 at 4.54 is the well-known lightest grey that still passes.

Accessibility checklist. Text contrast at least 4.5 to 1; alt text on key visuals; a sensible tab order; never colour alone (add labels or icons); fonts of 10 to 12 pt or more. Apply a **theme** (View > Themes, or a JSON file) so every visual shares fonts and a short palette.

Storytelling. Put the finding in the page title ("West region is 65% of sales": 6,042 / 9,340 = 0.647), the evidence in the visuals and the recommendation in a short text box (Chapter 14).

Connection to the real job. Stakeholders judge a dashboard in ten seconds; a theme, a grid and message titles make a portfolio look professional.

Applications

- Executive dashboards: 4 cards, 1 trend, 1 sorted bar.
- Hospital: one page for the daily decision (waiting time, no-show rate, bed use), another for diagnosis.

COMMON MISTAKE

Putting every available visual on one page "so nothing is missed". The reader then sees nothing. Split by question and link pages with drill-through.

Check yourself

1. Where do the headline numbers go on a page? Answer: In a row of cards at the top.
2. What contrast ratio does WCAG ask for normal text? Answer: At least 4.5 to 1.
3. What is a theme? Answer: Saved colours and fonts applied to all visuals.

11.2 Interactivity: Slicers, Drill, Bookmarks, Tooltips and Parameters

INTUITION

A good report answers the first question and invites the second. "Which region is down?" then "Which city?" then "Which orders?" Interactivity is that path: filter, drill, then see the details, without the reader asking you for a new report.

How it works.

Feature	What it does	Typical use
Slicer	A filter control on the page (list, dropdown, date range)	Region, date
Sync slicers	One slicer controls the same field on several pages	Region on all pages
Cross-filter / cross-highlight	Clicking a bar filters or highlights other visuals; control with Format > Edit interactions	Click Pune, see its trend
Drill down / up	Move down a hierarchy inside one visual (Year to Month to Day)	Category to Product
Drill through	Right-click a value to jump to a detail page filtered to it	City to order list
Bookmarks	Save a view (filters, visibility); buttons can switch bookmarks	Toggle chart or table, "reset filters"
Tooltips (report page)	A small page shown on hover	A mini trend on each bar

Buttons / navigation	Click to go to a page or run a bookmark	A menu bar
Field parameters	A slicer that swaps the measure or dimension shown	"Show: Sales / Orders / Qty"

Worked example 11.2.1 Drill-through. A summary page shows sales by city. You right-click Pune and choose Drill through to the "Orders" page, which holds a table of orders. Which rows appear, and what total?

- **Step 1** The drill-through passes the filter `City = Pune` to the detail page.
- **Step 2** The `FactSales` rows for Pune: 1001 (998), 1003 (597), 1008 (1,400).
- **Step 3** Total: 998 + 597 + 1,400 = 2,995.
- **Answer** Three rows totalling 2,995, equal to the Pune bar on the summary page.
- **Check** If the totals differed, the detail page would have a different filter or measure; always match them.
- **Interpretation** The reader goes from "Pune is lowest" to the three orders behind it in two clicks.

Field parameters. Modeling > New parameter > Fields; choose Total Sales, Orders, Qty. A slicer then picks which measure the chart shows, so one chart does the work of three (check your Desktop version).

Worked example 11.2.2 Edit interactions. Clicking the Pune bar should change the card (total) but not the daily line chart. Select the bar chart, Format > Edit interactions, set the card to Filter and the line to None. The card shows 2,995; the line stays at all-city daily totals (2,297, 597, 799, ...). Clicking empty space returns the card to 9,340.

Connection to the real job. Business users ask "can I see it by X?" constantly. Slicers, drill-through and field parameters answer that without extra pages. Too many slicers, however, confuse; choose 2 or 3 on the left or top.

Applications

- Sales: region slicer synced across pages; drill-through to customer detail.
- Operations: bookmark toggle between daily and weekly view.

COMMON MISTAKE

Hiding important filters (a slicer on a hidden page) so the same numbers look different on two pages. Show active filters on the page, or use a visible "filters applied" text.

Check yourself

1. What is the difference between drill down and drill through? Answer: Drill down moves within one visual's hierarchy; drill through jumps to another page filtered to the chosen value.
2. What do synced slicers do? Answer: Keep one slicer's selection in step across pages.
3. What do field parameters allow? Answer: The user switches which measure or field a visual shows.

11.3 Row-Level Security and the Power BI Service

INTUITION

Row-level security (RLS) is a lock on rows. One report, one dataset, but each viewer sees only their rows, like a bank statement that shows your account and nobody else's. Without it, you would build and maintain one copy of the report per region.

Words you need. Workspace: a shared folder in the Service for datasets and reports. **App**: a packaged read-only version for a wide audience. **Gateway**: software on a company server that lets the Service reach on-premises data. **Deployment pipeline**: Dev, Test, Prod stages. **Q&A**: ask in plain English, get a visual. **Copilot**: Microsoft's AI assistant in Power BI (availability and licensing change; check current documentation).

How RLS works. You define **roles**, each with DAX filters on tables. The filter on a dimension flows to the fact table through the relationship (Chapter 9), so one filter secures every related table.

Static RLS. One role per group.

```
Role "West":   DimCity[Region] = "West"
Role "North":  DimCity[Region] = "North"
```

Dynamic RLS. One role, driven by a security table `Security[Email], Security[Region]`,

not related to other tables:

```
Role "ByUser":
    DimCity[Region] IN CALCULATETABLE(VALUES(Security[Region]),
                                      Security[Email] =
USERPRINCIPALNAME())
```

`USERPRINCIPALNAME()` returns the signed-in user's login (an e-mail style name). Adding a user is then one new row in the table, not a new role. Test in Desktop with Modeling > View as, then assign people or groups to the role in the Service (dataset > Security).

Worked example 11.3.1 What does each user see? The security table has: asha West; ravi North; the CFO has two rows, North and West. Our 8 orders: West = Mumbai and Pune, North = Delhi. Real pandas output:

```
asha@corp.example 6 rows, total 6042
ravi@corp.example 2 rows, total 3298
cfo@corp.example 8 rows, total 9340
new.joiner@corp.example 0 rows, total 0
```

- **Step 1** Asha's rows: Region West means Mumbai (1002, 1005, 1006) and Pune (1001, 1003, 1008): 6 rows, 1,299 + 450 + 1,298 + 998 + 597 + 1,400 = 6,042.
- **Step 2** Ravi: North means Delhi (1004, 1007): 799 + 2,499 = 3,298.
- **Step 3** The CFO matches both regions: 6 + 2 = 8 rows, 6,042 + 3,298 = 9,340.
- **Step 4** A new joiner who is not in the table matches nothing: 0 rows.
- **Answer** 6,042; 3,298; 9,340; and an empty report for someone missing from the table.
- **Check** 6,042 + 3,298 = 9,340.
- **Why it matters** Empty reports for missing users are the most common RLS complaint; the fix is a row in the table, not a model change.

Important RLS facts.

- RLS applies to **viewers**. People with edit rights in the workspace (Admin, Member, Contributor) can see all data, so share reports to users as **Viewer** or through an **app**.
- Test with "View as"; a user in several roles gets the union (OR) of them.
- RLS hides rows, not columns (column-level needs object-level security) and does not replace least privilege at the source.

The Service workflow.

```
Desktop (.pbix) --Publish--> Workspace (dataset + report)
      |                          |-- Scheduled refresh (via gateway for
on-premises data)
      |                          |-- RLS roles assigned
      |                          |-- Share / Publish as App --> viewers
Deployment pipeline: Dev --> Test --> Prod
```

Refresh and gateways. Cloud sources refresh directly in the Service; on-premises sources need a **gateway** on a machine that stays on. Refresh limits depend on the licence (check current limits). **Import** mode copies data into the model (fast, needs refresh); **DirectQuery** queries the source live (current, slower, fewer DAX features).

Worked example 11.3.2 Plan a refresh. A file lands in the source system at 06:00 each day. The dataset refresh takes 25 minutes. Managers expect fresh numbers by 07:00. When should you schedule it, and how much slack remains?

- **Step 1** Start after the data lands, with a buffer: 06:15. Finish: 06:15 + 25 min = 06:40. Slack before 07:00: 20 minutes.
- **Answer** Schedule at 06:15; it finishes by 06:40 with 20 minutes for a retry.
- **Check** A 06:00 start would race the data and may load yesterday's file. At 45 minutes, 06:15 + 45 = 07:00 leaves no slack: add incremental refresh (Chapter 9).

Sharing and Copilot. Share a link (viewer needs a licence and dataset permission), publish an **app**, or embed in Teams. Q&A and Copilot work best on a clean model (friendly names, synonyms, hidden technical columns); always verify an AI-made visual against a number you trust, and check your data policy first.

Connection to the real job. In an interview, "how would you restrict data by region?" and "what happens on refresh?" are common. Explain dynamic RLS with a security table, and viewers versus members.

Applications

- Sales: regional managers see their own region.
- HR: managers see only their reports' leave data.

COMMON MISTAKE

Testing RLS as the report author (the author sees everything) and then finding that workspace members still bypass it. Test with "View as", and share to Viewers.

Check yourself

1. Who bypasses RLS in a workspace? Answer: Users with edit rights (Admin, Member, Contributor).
2. Why dynamic RLS? Answer: One role plus a security table scales to many users; adding a user is a data change.
3. What is a gateway for? Answer: To let the Service refresh on-premises data sources.

11.4 Governance, Documentation and Performance

INTUITION

In a company with 200 reports, "what is revenue?" soon has five answers. Governance is the set of habits that keep one answer: one trusted definition, one owner, clear names, controlled access.

Governance checklist.

- **Naming:** human names (`Total Sales` , `Delivered Rate`), `FactSales` / `DimCity` , hide keys and helper columns.
- **Documentation:** a data dictionary (name, definition, formula, owner, source, refresh) and a notes page in the report.
- **Certified datasets:** a reviewed shared dataset marked **Certified** or **Promoted** in the Service; reports reuse it.
- **Access:** RLS, least privilege, Viewer role, groups not individuals, periodic review.
- **Refresh SLA and lineage:** an agreed time, owner and failure alert; the lineage view shows what depends on what.
- **Versioning:** `.pbip` project files in Git where supported, deployment pipelines.

Semantic layer. A **semantic layer** is one shared place where business definitions (revenue, active customer) live as code or model, so every dashboard gets the same number. In Power BI it is the shared certified dataset. Elsewhere: LookML (Looker), dbt's semantic layer or metrics definitions, Cube. The idea: define once, reuse everywhere.

Performance tuning. Measure with Performance Analyzer (Chapter 10); cut visuals per page; keep a star schema with integer keys and no unused columns; replace slow DAX; aggregate at the source or use incremental refresh (Chapter 9); avoid slicers with huge lists.

Worked example 11.4.1 Write a data-dictionary entry and check the number.

Field	Entry
Name	Delivered Rate
Definition	Share of order value with status Delivered
Formula	`DIVIDE(CALCULATE([Total Sales], FactSales[Status] = "Delivered"), [Total Sales])`
Owner	Sales analytics
Source	`FactSales`, refreshed daily 06:15
Example	January 2025 sample: 8,293 / 9,340 = 88.8%

- **Step 1** Verify the example from the data: delivered orders sum to 8,293 (Chapter 10), total 9,340.
- **Step 2** 8,293 / 9,340 = 0.8879.
- **Answer** The dictionary entry gives anyone a definition, a formula and a number to test against.
- **Check** If a dashboard shows 90% for the same period, someone has a different definition (for example, excluding cancelled orders from the denominator: 8,293 / (9,340 − 450) = 93.3%). The dictionary settles the argument.

Worked example 11.4.2 Triage a slow page (12 s). Performance Analyzer shows a matrix at 9.5 s and five other visuals at 0.4 s each.

- **Step 1** Visual time: 9.5 + 5 × 0.4 = 11.5 s. The matrix is 9.5 / 11.5 = 83% of it.
- **Step 2** Copy its query into DAX Studio and look for an iterator over the fact table or a `FILTER` over the whole table.
- **Answer** Fix the matrix first; the five small visuals can save at most 2 s. If the matrix fell to 1.5 s the page would take about 1.5 + 2.0 = 3.5 s.

Connection to the real job. Governance is what distinguishes a senior analyst from someone who only builds charts. Roles such as "BI developer" and "analytics engineer" are largely about shared, certified models.

Applications

- Finance: certified "Revenue" dataset used by all finance reports.
- Data Engineering: the semantic layer as a contract between warehouse and BI.

COMMON MISTAKE

Every analyst making their own copy of the dataset "for convenience". Soon five copies disagree and nobody knows which is right. Build on one certified dataset.

Check yourself

1. What is a certified dataset? Answer: A reviewed shared dataset, marked in the Service, that reports should reuse.
2. What goes in a data dictionary? Answer: Name, definition, formula, owner, source and refresh time.
3. First step in tuning a slow report? Answer: Measure with Performance Analyzer to find the slowest visual.

11.5 Tableau, Looker Studio and Other BI Tools

INTUITION

Power BI, Tableau and Looker Studio are three kitchens making the same dishes. If you know one well, you can work in another after learning where the knives are. Employers name tools, but they hire for thinking: model, measure, chart, secure, explain.

Tableau. Strong, flexible visual exploration, popular in many large companies. Key ideas: **calculated fields** (formulas per row or aggregate), **parameters** (user inputs), **dashboard actions** (filter, highlight, URL), and **LOD (level of detail) expressions**: `{FIXED ...}`, `{INCLUDE ...}`, `{EXCLUDE ...}`. Live connections or extracts (Hyper files). Tableau Public is free (public data only); Tableau Desktop Specialist is an entry certificate.

Worked example 11.5.1 An LOD expression, checked with pandas. Each order's share of its city total in Tableau:

```
[Amount] / { FIXED [City] : SUM([Amount]) }
```

- **Step 1** `{ FIXED [City] : SUM([Amount]) }` computes the city total regardless of what

is in the view, and attaches it to each row: Pune 2,995, Mumbai 3,047, Delhi 3,298.

- **Step 2** Divide each order's amount by its city total. Order 1001 (Pune): 998 / 2,995 = 0.333. Order 1003: 597 / 2,995 = 0.199. Order 1008: 1,400 / 2,995 = 0.467.
- **Answer** pandas (`groupby("City").transform("sum")`) gives 0.333, 0.199, 0.467 for Pune; Pune's three shares add to 0.999 (rounding), that is 100%.
- **Check** 998 + 597 + 1,400 = 2,995.
- **Interpretation** `FIXED` does what `CALCULATE(..., ALLEXCEPT(...))` does in DAX (Chapter 10). `EXCLUDE [City]` drops City from the level (9,340 in a city view); `INCLUDE [OrderID]` adds a finer level.

Looker Studio (free, Google; not the same as Looker). Connects easily to Google Sheets, BigQuery, GA4 and CSV uploads. Calculated fields, **data blending** (joining sources inside the report), free sharing by link, scheduled e-mail delivery. Simple and quick; much weaker at modelling (no star schema with relationships like Power BI, no DAX-style measures, limited row-level security) and slower on large data.

Looker (LookML) is a paid platform whose **semantic layer** is code (dimensions, measures, joins) kept in Git. **Metabase** and **Apache Superset** are open source and SQL-friendly; **Qlik** uses an in-memory associative engine (awareness). Excel Power Pivot uses the same DAX as Power BI. In short: Power BI has the strongest modelling (DAX, RLS), Tableau the most flexible visuals, Looker Studio the lowest cost; pricing changes, so check current pages.

Worked example 11.5.2 Choose the tool. (a) A startup with data in Google Sheets and GA4, no budget. (b) A bank on Microsoft 365 needing RLS by branch and a governed model. (c) A team that wants deep visual exploration and has Tableau already.

- **Step 1** (a) Looker Studio: free, connects to Google sources in minutes.
- **Step 2** (b) Power BI: RLS, Microsoft 365 integration, certified datasets, DAX modelling.
- **Step 3** (c) Tableau: skills and licences exist; its visual flexibility fits the need.
- **Answer** (a) Looker Studio, (b) Power BI, (c) Tableau.
- **Check** Each answer follows the constraint named in the scenario (budget, security and governance, existing skills), not personal taste.

Not covered here. Product and web analytics tools (GA4, Mixpanel, tracking plans, UTM tags) and survey analysis; Chapter 12 builds funnels and cohorts. **Certification:** Microsoft **PL-300** (Power BI Data Analyst) tests Chapters 9 to 11; use Microsoft Learn's free practice material and build one full project.

Connection to the real job. Learn one BI tool deeply (Power BI), and know the vocabulary of the others. In interviews, show a project where you modelled, secured and explained a dashboard.

Applications

- Marketing: Looker Studio dashboards over GA4 and Sheets.
- Enterprise: Power BI or Tableau on a governed warehouse.

COMMON MISTAKE

Treating tool names as the skill. A resume line "Tableau, Power BI, Looker" is weaker than one project that shows a clean model, a correct measure and a clear message.

Check yourself

1. What does `{ FIXED [City] : SUM([Amount]) }` compute? Answer: The sum of Amount per city, regardless of the view's own level.
2. Which tool is free and Google-based? Answer: Looker Studio.
3. What is a semantic layer? Answer: One shared place where business definitions live, so every report uses the same numbers.

Chapter recap

- Design for ten seconds: headline cards on top, a grid, one question per page, message titles.
- Choose visuals by question; check contrast (4.5 to 1 for text), add alt text, use a theme.
- Slicers, drill-through, bookmarks, tooltips and field parameters let readers explore; set interactions on purpose.
- RLS locks rows by role; dynamic RLS uses a security table and `USERPRINCIPALNAME()`; viewers obey it, editors bypass it.
- The Service gives workspaces, apps, scheduled refresh (gateways for on-premises data), pipelines, Q&A and Copilot.
- Govern with names, a data dictionary, certified datasets, access reviews, refresh SLAs and a shared semantic layer.
- Tune slow reports by measuring first, then fixing the biggest visual.
- Tableau (LOD, parameters, actions) and Looker Studio (free, blends) do the same job with

different strengths; pick by constraints.

Item	Pattern
Card width	(page − 2 × margin − gaps × gap) / cards
Contrast	`(L1 + 0.05) / (L2 + 0.05)`, at least 4.5 for text
Static RLS	`DimCity[Region] = "West"`
Dynamic RLS	`DimCity[Region] IN CALCULATETABLE(VALUES(Security[Region]), Security[Email] = USERPRINCIPALNAME())`
Tableau LOD	`{ FIXED [City] : SUM([Amount]) }`
Refresh plan	start = data arrival + buffer; finish = start + duration; slack = deadline − finish

Practice set

Problem 11.1 Four cards on a 1280-pixel page with margins of 20 and gaps of 20. How wide is each card?

Answer: Gaps: 3 × 20 = 60; margins 40; (1280 − 40 − 60) / 4 = 1,180 / 4 = 295 pixels.

Problem 11.2 Under RLS, Ravi (North) sees how many orders and what total in our sample?

Answer: 2 orders (1004, 1007), 799 + 2,499 = 3,298.

Problem 11.3 A manager is a workspace Member and says RLS "does not work" for her. Why?

Answer: Members have edit rights and bypass RLS. Give her the Viewer role or share an app, then test with "View as".

Problem 11.4 A new user opens the report and sees nothing. The report is fine for others. What is the likely cause and fix?

Answer: With dynamic RLS, the user is not in the security table, so no rows match. Add a row

(e-mail, region) and refresh or reload as needed.

Problem 11.5 Contrast of text colour #9aa5b1 on white is 2.5. Does it pass for normal text? What would you do?

Answer: No (needs 4.5). Use a darker colour such as #595959 (7.0), keep light grey for non-text marks.

Problem 11.6 A page of 8 visuals takes 10 s. One visual takes 6.4 s, the other seven 0.5 s each. What share is the slow visual, and what is the best-case page time if you fix it to 1 s?

Answer: Total visual time 6.4 + 7 × 0.5 = 9.9 s. Share 6.4 / 9.9 = 65%. After the fix: 1 + 3.5 = 4.5 s.

Problem 11.7 (Interview) How would you implement row-level security by region for 200 users in Power BI?

Answer: Build a security table of e-mail and region (several rows per person if needed), create one role with the DAX filter `Region IN CALCULATETABLE(VALUES(Security[Region]), Security[Email] = USERPRINCIPALNAME())` on the region dimension, test with "View as", assign the group to the role in the Service, share to Viewers, and keep the security table updated from HR data.

Problem 11.8 (Interview) Power BI or Tableau for a new company? Give the questions you ask first.

Answer: What is the existing stack (Microsoft 365 or not), who builds and who consumes, how much governance and RLS is needed, the budget and licence model, and what skills exist. Power BI usually fits Microsoft-centric companies needing governed, shared models and low cost; Tableau fits teams that value visual flexibility and already have skills. A good analyst can work in both.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles, and English is my second language. Teach me how to
design a one-page Power BI report: layout grid, which visual for which
question, theme, colour contrast and accessibility, and message titles.
Use a small online-shop sales example with real numbers. Show a bad layout
```

and the fixed layout in words, then give me a checklist and quiz me with 4 questions, one at a time.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Teach me Power BI interactivity: slicers, synced slicers, cross-filter and edit interactions, drill down versus drill through, bookmarks, report-page tooltips, buttons and field parameters. For each, say when to use it and when it hurts the reader. Use a tiny dataset so I can verify what each click shows by hand. Then give me 6 design scenarios to solve, without answers first.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Explain row-level security in Power BI: static versus dynamic RLS, the security table with USERPRINCIPALNAME(), roles and Viewers versus workspace Members, View as, and common failures such as an empty report for a missing user. Use an 8-row sales table with regions and 4 users and compute what each user sees. Then ask me 5 debugging questions, one at a time.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Review my Power BI governance setup for a team: naming rules, data dictionary, certified datasets, workspace roles, deployment pipeline, refresh schedule and ownership, plus how I find and fix a slow report. Give a numbered list of risks and fixes in priority order, in simple English. My setup and report description: [PASTE HERE].

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Compare Power BI, Tableau, Looker Studio, Looker (LookML), Metabase and Superset for an analyst: data prep, modelling, measure language, security, sharing and cost model (no exact prices). Show how a Tableau FIXED LOD maps to a DAX pattern. Then run a mock interview of 6 tool-choice scenario questions, one at a time, scoring each answer out of 5 with a model answer.

CHAPTER 12

BUSINESS ANALYTICS: KPIS, FUNNELS, COHORTS, RETENTION, CHURN, RFM AND DOMAIN CASES

The handful of calculations that run almost every analytics team in the world.

Open any analytics job post and you will see the same words: funnel, cohort, retention, churn, segmentation, LTV, CAC. These are not advanced statistics. They are careful ways of counting people and money, and once you can do them, you can answer most of the questions a business asks. In this chapter you will design KPIs that cannot be gamed, build a funnel and a cohort retention table in pandas, calculate churn, segment customers with RFM, and walk through short cases from product, marketing, sales, finance and operations. By the end you will be able to take an events or orders table and produce the five tables that a weekly business review is built from. We reuse the Indian online shop from Chapter 1. All code below was run, and the outputs are real. Where the data is simulated, we say so.

12.1 KPI and Metric Design

INTUITION

A KPI is the dashboard of a car. Speed, fuel and engine temperature each tell you something different. If the driver is told only "get there fast", he may burn the engine. A good set of KPIs has one number that shows the goal, a few numbers that drive it, and a few warning lights that stop you from damaging something else.

Words you need first

Word	Plain meaning	Example
North-star metric	The one number that best shows the value customers get	Orders delivered per week
Output metric	A result you want but cannot directly push	Monthly revenue

Input metric	A thing you can push that moves the output	Number of product pages with reviews
Guardrail metric	A metric that must NOT get worse while you chase the goal	Return rate, complaint rate
SMART	Specific, Measurable, Achievable, Relevant, Time-bound	"Raise repeat purchase rate from 22% to 26% by 31 March"
ARPU	Average revenue per user = revenue / active users	Rs 150 per month
CAC	Customer acquisition cost = marketing spend / new customers	Rs 600
LTV	Lifetime value: total profit one customer brings before leaving	Rs 1,200
MRR / ARR	Monthly / annual recurring revenue (subscriptions)	ARR = MRR × 12
NPS	Net Promoter Score = % promoters − % detractors	+30

How it works. Output metrics (revenue) tell you if you won. Input metrics (traffic, conversion, delivery speed) tell you what to do on Monday. Guardrails protect you from "winning" the wrong way: if you raise sales by 20% with huge discounts, the margin guardrail turns red.

Why it is called "north-star". Sailors steer by the north star because it does not move. A north-star metric should stay meaningful even as the product changes. "Page views" is a poor north star because you can raise it by making navigation worse. "Orders delivered" is better because it needs a happy customer.

Real-world mapping

Item	Example value	Where it appears in the job
North star	Weekly delivered orders	Top of the company dashboard
Input metric	Search-to-cart rate	Product team's weekly goal

Guardrail	Return rate under 8%	Every A/B test (Chapter 13)
SMART goal	Cut delivery delay from 12% to 8% by March	Quarterly planning

Worked problem 12.1.1 (unit economics, small numbers) The shop earns Rs 6,00,000 per month from 4,000 active customers. Its gross margin (share of revenue left after the cost of goods) is 40%. Monthly churn is 5%. It spent Rs 3,00,000 on marketing and gained 500 new customers. Find ARPU, expected lifetime, LTV, CAC, LTV/CAC and payback time.

- **Formula** ARPU = revenue / customers. Lifetime (months) = 1 / churn. LTV = ARPU × margin × lifetime. CAC = spend / new customers. Payback = CAC / (ARPU × margin).
- **Plug in** ARPU = 600000 / 4000. Lifetime = 1 / 0.05. CAC = 300000 / 500.
- **Compute** ARPU = 150. Lifetime = 20 months. LTV = 150 × 0.40 × 20 = 1,200. CAC = 600. LTV/CAC = 1200 / 600 = 2.0. Monthly profit per customer = 150 × 0.40 = 60, so payback = 600 / 60 = 10 months.
- **Answer** Each customer is worth Rs 1,200 of profit and costs Rs 600 to win. The ratio is 2.0, and the money comes back in 10 months.
- **Check** 10 months × Rs 60 = Rs 600, which equals CAC. A common rule of thumb says LTV/CAC around 3 is healthy, so 2.0 is acceptable but thin. It is a rule of thumb, not a law.
- **Why it matters** If a marketing manager proposes a Rs 900 CAC channel, you can say immediately it loses money (1,200 / 900 = 1.3, payback 15 months).

Worked problem 12.1.2 (NPS) A survey of 100 customers asks "How likely are you to recommend us, 0 to 10?" Scores 9-10 are promoters, 7-8 passives, 0-6 detractors. Result: 50 promoters, 30 passives, 20 detractors. Find NPS.

- **Formula** NPS = %promoters − %detractors (passives are ignored)
- **Plug in** 50% − 20%
- **Compute** 30 points
- **Answer** NPS is +30. Range is −100 to +100.
- **Check** If all 100 were promoters the formula gives 100 − 0 = 100, the maximum. Good.
- **Why it matters** NPS is a *survey* metric. Always report the sample size and who answered. Unhappy customers often do not respond.

The picture: a KPI tree

```
North star: weekly delivered orders  (output)
  ├── New-customer orders  <- input: ad clicks x conversion
  ├── Repeat orders        <- input: repeat rate, email reach
  └── Guardrails: return rate < 8%, delivery delay < 10%, margin > 35%
```

Read it from the top. The north star is split like the metric tree of Section 1.4. Guardrails sit beside the tree and are checked every time.

Connection to the real job. When a stakeholder says "improve engagement", your first task is to ask "which metric, defined how, and which guardrail?" Chapter 14 shows how to write the definitions into the memo.

Applications

- Data Analytics: dashboard design, goal tracking, A/B test metrics.
- Product: north star plus input metrics per team.
- Finance: unit economics for each channel before scaling spend.
- Operations: SLA (service-level agreement) metrics with guardrails on quality.

COMMON MISTAKE

Choosing a metric that is easy to count rather than one that matters (page views, app downloads). Also forgetting to define it: "active user" must say active doing what, in which window. Write the definition once, in a data dictionary.

Check yourself

1. Is "return rate" an input, output or guardrail? Answer: Usually a guardrail.
2. If churn is 4% monthly, what is the expected lifetime? Answer: 1 / 0.04 = 25 months.
3. If CAC is Rs 600 and monthly profit per customer is Rs 50, what is the payback? Answer: 600 / 50 = 12 months.

12.2 Funnels

INTUITION

A funnel is the queue at a metro station. 100 people enter the gate, 80 reach the platform, 60 board the train. At each stage some people drop out. If you want more people to board, you do not stare at the total. You find the stage where the most people are lost and fix that.

Formal definition. A **funnel** is an ordered list of steps a user should take (visit, view product, add to cart, pay). For each step, count the *distinct users* who reached it. Two rates matter:

```
step conversion    = users at this step / users at the previous step
overall conversion = users at this step / users at the first step
```

Remember the performance-testing parallel: a funnel is like a transaction with several samplers. Where the failure rate jumps is where you look.

Real-world mapping

Item	Example value	Where it appears in the job
Step	Add to cart	Event name in a tracking plan
Step conversion	50% cart to paid	Product weekly review
Drop-off	50% lost between cart and payment	Checkout redesign project

Worked example 12.2.1 (predict, run, explain) An event table has one row per user per step reached. Six users, four steps. Before you run it: how many users reach "paid"? (Look at the data: only user 1 has a "paid" row.)

```
import pandas as pd
ev = pd.DataFrame({
 "user":[1,1,1,1,2,2,2,3,3,4,5,5,6],
 "step":
["visit","product","cart","paid","visit","product","cart","visit","product
","visit","visit","product","visit"]})
order=["visit","product","cart","paid"]
f = ev.groupby("step")["user"].nunique().reindex(order).to_frame("users")
f["step_conv_%"]=(f["users"]/f["users"].shift(1)*100).round(1)
f["overall_%"]=(f["users"]/f["users"].iloc[0]*100).round(1)
print(f)
```

Real output:

```
         users  step_conv_%  overall_%
step
visit        6          NaN      100.0
product      4         66.7       66.7
cart         2         50.0       33.3
paid         1         50.0       16.7
```

Line by line: `groupby("step")["user"].nunique()` counts distinct users per step (`nunique`, not `count`, so a user who views two products is counted once). `.reindex(order)` forces the funnel order, because groupby would sort the names alphabetically (cart, paid, product, visit), which is wrong. `.shift(1)` moves the column down one row so each step is divided by the previous step. The first row has no previous step, hence `NaN` (not a number).

Answer Overall conversion is 1 of 6, or 16.7%. The biggest relative drop is visit to product (33.3% lost), but the cart-to-paid step loses half of those who were ready to buy. That is usually the more valuable one to fix because these users already wanted the product.

The typical wrong version. Using `.count()` instead of `.nunique()` counts rows. If user 1 viewed the cart page 3 times, the cart step would be inflated and step conversion could exceed 100%. Another wrong version: skipping `.reindex(order)`, which gives rows in alphabetical order and a meaningless `shift`.

Worked problem 12.2.2 (hand calculation) Last week: 20,000 visits, 8,000 product views, 1,600 carts, 800 payments. Find each step conversion and overall conversion. Then find the effect of lifting cart-to-pay from 50% to 60%.

- **Formula** step = this / previous. Overall = last / first.
- **Plug in** 8000/20000, 1600/8000, 800/1600, 800/20000.
- **Compute** 0.40, 0.20, 0.50, and 0.04.
- **Answer** Step conversions are 40%, 20% and 50%. Overall is 4%. If cart-to-pay becomes 60%, payments = 1600 × 0.60 = 960, which is 960 / 20000 = 4.8% overall, 160 more orders a week (a 20% relative lift).
- **Check** 20000 × 0.40 × 0.20 × 0.50 = 800. The steps multiply to the total.
- **Why it matters** A 10-point gain at the last step gave a 20% gain in orders. Late-funnel fixes are often cheapest.

Funnel pitfalls. Define the *window* (must all steps happen in one session, or within 7 days?). Decide whether order matters (strict funnel: step 3 must come after step 2). Watch for users who skip steps (paid without a cart event usually means a tracking bug).

Applications

- Product: signup flow, onboarding completion.
- Marketing: ad impression, click, landing, lead, customer.
- Sales: lead, qualified, demo, proposal, closed.
- Banking: loan application steps.

COMMON MISTAKE

Counting events instead of distinct users, and forgetting to fix the step order. Always check that each step count is less than or equal to the step before it. If not, either the data or the funnel definition is broken.

Check yourself

1. Step counts 1000, 600, 300. What is overall conversion? Answer: 300 / 1000 = 30%.
2. Why `nunique`, not `count`? Answer: We count people, not clicks.
3. What if a later step is bigger than an earlier step? Answer: Tracking bug or users can enter the funnel midway; fix the definition.

12.3 Cohorts, Retention and Churn

INTUITION

Imagine a school. Each year's admission is a batch (cohort). To learn whether the school is improving, you do not look at the total number of students, because that mixes batches of different ages. You follow each batch: how many of the 2023 batch are still here after 1 year, after 2 years? Retention does exactly that for customers.

Words you need first

Word	Plain meaning
Cohort	A group of users who started in the same period (signup month)

Offset	Months since the cohort started (0 = the start month)
Retention at offset k	Share of the cohort still active k months later
Churn	Share of customers lost in a period. For a given period, churn = lost / customers at the start
Active	You must define it: "placed an order in the month" or "opened the app"

Formal definition.

```
retention(cohort, k) = active users of the cohort at offset k / cohort
size
monthly churn        = customers active last month and not active this
month / customers active last month
```

Retention and churn are two sides of one coin only if "active" means the same thing in both.

Worked example 12.3.1 (predict, run, explain) The data below is **simulated** (a seeded random generator, 1,000 users signing up January to March 2025, with true retention set to 100%, 45%, 32%, 25% at offsets 0 to 3). It stands in for real data. Before you run it: will each cohort's offset-1 retention be exactly 45%? (No, random noise.)

```
import pandas as pd, numpy as np
rng=np.random.default_rng(42)
n=1000
signup=pd.to_datetime("2025-01-
01")+pd.to_timedelta(rng.integers(0,90,n),unit="D")
users=pd.DataFrame({"user_id":range(n),"signup":signup})
users["cohort"]=users.signup.dt.to_period("M")
rows=[]
for u,s in zip(users.user_id,users.signup):
    m0=s.to_period("M")
    for k in range(0,4):
        p=[1.0,0.45,0.32,0.25][k]
        if rng.random()<p: rows.append((u,m0+k))
act=pd.DataFrame(rows,columns=["user_id","month"])
act=act.merge(users[["user_id","cohort"]],on="user_id")
act["offset"]=(act.month-act.cohort).apply(lambda x:x.n)
act=act[act.month<=pd.Period("2025-06")]
tab=act.pivot_table(index="cohort",columns="offset",values="user_id",aggfu
nc="nunique")
print(tab)
print((tab.div(tab[0],axis=0)*100).round(1))
```

Real output:

```
offset     0    1    2    3
cohort
2025-01  342  165   99  93
2025-02  308  137  104  82
2025-03  350  165  112  91
offset       0     1     2     3
cohort
2025-01  100.0  48.2  28.9  27.2
2025-02  100.0  44.5  33.8  26.6
2025-03  100.0  47.1  32.0  26.0
```

Line by line: a `Period` is pandas' "whole month" type, so `m0+k` means "k months after". `offset` is the month difference. `pivot_table(... aggfunc="nunique")` builds the cohort grid: rows are cohorts, columns are offsets, values are distinct active users. Dividing each row by its offset-0 value (`tab.div(tab[0], axis=0)`) turns counts into retention percentages.

The picture: how to read the triangle. Each row is one cohort. Each column is time since signup. Reading across a row shows one batch ageing. Reading down a column compares batches at the same age, which is the fair comparison. Here offset-1 retention is 48.2%, 44.5%,

47.1%: all near the true 45%, and the spread is noise from random sampling. In real data the bottom-right corner is empty, because later cohorts have not had time to reach later offsets.

Answer About 45 to 48% of users return the month after signup, and about 26 to 27% are still active at month 3. The drop is steepest in the first month (about 53% lost), then it flattens. This "steep then flat" shape is the typical retention curve. The flat part is your loyal core.

Worked problem 12.3.2 (churn by hand) In March two customers were active, E and G. In April only G was active. What is the April churn? And what if 400 customers were active in March and 340 of them returned in April?

- **Formula** churn = (active last month and not active this month) / active last month
- **Plug in** First case: lost = {E}, so 1 / 2. Second case: lost = 400 − 340 = 60, so 60 / 400.
- **Compute** 0.50 and 0.15
- **Answer** First case: 50% churn (tiny sample). Second case: 15% monthly churn, which means retention of 85%.
- **Check** Real code on the RFM data of the next section gave churned = ['E'], rate 0.5. A 15% monthly churn means expected lifetime 1 / 0.15 = 6.7 months, consistent with Section 12.1.
- **Why it matters** Monthly churn of 15% is brutal. 0.85^{12} is about 14%, so only about 1 in 7 customers survives a year. Compounding makes small monthly numbers large.

Two kinds of retention. *N-day or classic retention* asks "active exactly on day 30". *Rolling retention* asks "active on day 30 or later". Rolling is always higher. Always say which one you use. For a subscription business, churn is cancellations. For a shop with no subscription, "churn" is a judgement call: a customer who has not ordered for, say, 90 days is called churned, and the 90 comes from the typical gap between orders.

Connection to the real job. Retention is the first chart a product or growth team asks for. In SQL you build the same grid with a self-join or window functions (your SQL document covers cohort tables). Chapter 13 uses retention as a guardrail in experiments.

Applications

- Product: day-1, day-7, day-30 app retention, and feature adoption curves.
- Subscriptions: MRR churn and net revenue retention.
- Healthcare: patient return rates.
- HR: employee retention by hiring batch.

COMMON MISTAKE

Averaging retention over all users regardless of age, or comparing an old cohort's month 6 with a new cohort's month 1. Also treating an incomplete cohort (this month's signups) as if the month was finished, which makes recent retention look bad.

Check yourself

1. Cohort of 500, 150 active at offset 2. Retention? Answer: 150 / 500 = 30%.
2. Monthly churn 10%. Share left after 3 months? Answer: $0.9^3 = 0.729$, about 73%.
3. Which is higher, rolling or classic retention? Answer: Rolling.

12.4 Segmentation and RFM

INTUITION

A shopkeeper knows that the lady who comes every week and spends Rs 2,000 needs a different greeting from the man who bought once last year. Treating all customers the same wastes money. Segmentation puts customers in groups so each group gets the right action.

Formal definition. RFM scores each customer on three things: **Recency** (days since last order, smaller is better), **Frequency** (number of orders), and **Monetary** (total spend). Each gets a score (here 1 to 3, with 3 best). Then rules map score combinations to named segments with an action.

Worked example 12.4.1 Eight customers, 15 orders, "today" is 2025-07-01. Thresholds (chosen by us, they depend on the business): Recency score 3 if 30 days or fewer, 2 if 31-90, 1 if over 90. Frequency 1 if one order, 2 if 2-3, 3 if 4 or more. Monetary 1 if under Rs 500, 2 if 500-1,999, 3 if Rs 2,000 or more. Before you run it: customer B spent Rs 5,500 and last ordered 11 days ago. What segment?

```
import pandas as pd
o = pd.DataFrame({
 "cust":["A","A","A","B","B","C","D","D","D","D","E","F","G","G","H"],
 "date":pd.to_datetime(["2025-06-25","2025-06-10","2025-05-02","2025-06-
20","2025-01-10","2025-02-01",
   "2025-06-28","2025-06-15","2025-06-01","2025-05-20","2025-03-05","2025-
06-30","2025-04-01","2025-03-01","2025-01-05"]),
 "amount":
[1200,800,500,3000,2500,400,300,350,250,400,900,150,700,600,100]})
today=pd.Timestamp("2025-07-01")
rfm=o.groupby("cust").agg(last=("date","max"),F=("date","count"),M=
("amount","sum"))
rfm["R"]=(today-rfm["last"]).dt.days
rfm["R_s"]=pd.cut(rfm.R,[-1,30,90,10**6],labels=[3,2,1]).astype(int)
rfm["F_s"]=pd.cut(rfm.F,[0,1,3,10**6],labels=[1,2,3]).astype(int)
rfm["M_s"]=pd.cut(rfm.M,[0,499,1999,10**9],labels=[1,2,3]).astype(int)
def seg(r):
    if r.R_s==3 and r.F_s>=2: return "Champion"
    if r.R_s==3: return "New"
    if r.R_s==1 and r.M_s>=2: return "At risk"
    if r.R_s==1: return "Lost"
    return "Steady"
rfm["segment"]=rfm.apply(seg,axis=1)
print(rfm[["R","F","M","R_s","F_s","M_s","segment"]].to_string())
print(rfm.groupby("segment").agg(customers=("M","size"),revenue=
("M","sum")).assign(share=lambda d:
(d.revenue/d.revenue.sum()*100).round(1)))
```

Real output:

```
     R  F     M  R_s  F_s  M_s  segment
cust
A    6  3  2500    3    2    3  Champion
B   11  2  5500    3    2    3  Champion
C  150  1   400    1    1    1      Lost
D    3  4  1300    3    3    2  Champion
E  118  1   900    1    1    2  At risk
F    1  1   150    3    1    1       New
G   91  2  1300    1    2    2  At risk
H  177  1   100    1    1    1      Lost
          customers  revenue  share
segment
At risk           2     2200   18.1
Champion          3     9300   76.5
Lost              2      500    4.1
New               1      150    1.2
```

Line by line: `agg(last=("date","max"), ...)` is named aggregation: new column names on the left, (column, function) on the right. `(today - last).dt.days` is recency in days. `pd.cut(values, bins, labels)` puts numbers into ranges: bins `[-1,30,90,10**6]` mean (−1,30], (30,90], (90,1000000]; labels go from the smallest range up, so for recency the labels are `[3,2,1]` because a small number of days is good. `.astype(int)` turns the category labels into plain integers. Customer G has R = 91, just over the 90-day edge, so R_s = 1: edges matter, so choose thresholds from the business, not by taste.

Answer B is a Champion. Three Champions (A, B, D) are 3 of 8 customers but 76.5% of revenue. Two "At risk" customers (E, G) are worth Rs 2,200 and should get a win-back offer. Lost customers (C, H) hold only 4.1% of revenue, so do not spend much on them.

Real-world mapping

Segment	Rule	Action
Champion	recent and frequent	Early access, loyalty points
At risk	not recent, decent spend	Win-back coupon, reminder
New	very recent, one order	Welcome series, second-purchase offer
Lost	not recent, low spend	Cheap email only

Other segmentation. *Behavioural* (what they do: browse-heavy vs buy-fast), *demographic*

(age, city), and *clustering*, where an algorithm such as k-means finds groups from numbers (Chapter on machine learning in the data-science document). Clustering suits many features, but its groups need names and actions or nobody uses them. Quartile-based RFM (`pd.qcut`) is common, but it produces unstable edges on small data with many ties, so we used fixed thresholds here. Clustering code was not run here.

Customer lifetime value (CLV) in pandas. Past CLV is simply the total spend per customer (the M column). Predicted CLV needs the formula of Section 12.1 or a model.

Connection to the real job. CRM and marketing teams use RFM tables every week for campaign lists. In an interview you may be asked to define RFM and write the SQL (`NTILE(3)` over each measure).

Applications

- Marketing: campaign targeting, coupon budgets.
- E-commerce: loyalty tiers.
- Banking: customer value tiers.
- Telecom: churn-risk lists.

COMMON MISTAKE

Scoring recency the wrong way round (a high number of days given a high score). And making 125 segments (5 × 5 × 5) that nobody can act on. Keep four to six segments, each with one action.

Check yourself

1. Is high R (days since last order) good? Answer: No, low is good.
2. Customer: R = 20, F = 1, M = Rs 300. Segment by our rules? Answer: New (R_s = 3, F_s = 1).
3. Why name segments? Answer: A name carries an action.

12.5 Domain Cases: Product, Marketing, Sales, Finance, Operations

INTUITION

The tools are the same in every domain; only the nouns change. A funnel in sales is "lead to deal". A cohort in HR is "hiring batch". When you join a company, learn its five or six domain metrics in the first week and you are useful from week two.

Each case below uses small numbers so you can check them by hand.

Product: engagement, DAU and MAU

DAU is daily active users (distinct users active on a day), **MAU** is monthly active users. **Stickiness** is average DAU / MAU: roughly how often a typical monthly user comes back daily.

Worked example 12.5.1 Ten events over four days.

```
import pandas as pd
d=pd.DataFrame({"user":[1,2,3,1,2,1,4,1,2,3],"day":[1,1,1,2,2,3,3,4,4,4]})
dau=d.groupby("day").user.nunique(); print(dau.to_dict())
mau=d.user.nunique(); print("avg
DAU",dau.mean(),"MAU",mau,"stickiness",round(dau.mean()/mau,3))
```

```
{1: 3, 2: 2, 3: 2, 4: 3}
avg DAU 2.5 MAU 4 stickiness 0.625
```

- **Answer** Average DAU is 2.5 and MAU is 4, so stickiness is 2.5 / 4 = 0.625. (Real products use a full month; four days is only for checking.) Feature adoption is computed like a funnel: users who used the feature / active users.

Worked example 12.5.2 (segmenting a drop) Platform DAU before and after a release:

```
import pandas as pd
w=pd.DataFrame({"platform":
["android","android","ios","ios","web","web"],"period":
["before","after"]*3,"dau":[50000,45000,30000,29000,20000,10000]})
p=w.pivot(index="platform",columns="period",values="dau")
[["before","after"]]
p["chg"]=p.after-p.before; p["pct"]=(p.chg/p.before*100).round(1)
p["share_of_drop_%"]=(p.chg/p.chg.sum()*100).round(1)
print(p); print(p[["before","after"]].sum())
```

```
period    before  after    chg   pct  share_of_drop_%
platform
android    50000  45000  -5000 -10.0             31.2
ios        30000  29000  -1000  -3.3              6.2
web        20000  10000 -10000 -50.0             62.5
period
before    100000
after      84000
dtype: int64
```

- **Answer** Total DAU fell 16% (100,000 to 84,000), but web fell 50% and accounts for 62.5% of the drop. The problem is web, not the whole product. Chapter 15 turns this into a full interview method.
- **Check** −5000 − 1000 − 10000 = −16000, and 84000 − 100000 = −16000.

Marketing: ROAS, attribution and UTM

ROAS (return on ad spend) = revenue from ads / ad spend. A **UTM** is a tag added to a link (`utm_source` , `utm_medium` , `utm_campaign`) so the analytics tool knows where a visit came from. **Attribution** decides which touchpoint gets credit for a sale: last-click gives all credit to the final click, first-click to the first, and multi-touch splits it.

Worked problem 12.5.3 Campaign A: spend Rs 60,000, revenue Rs 1,80,000. Campaign B: spend Rs 40,000, revenue Rs 2,40,000. Which is better, and what is overall ROAS?

- **Formula** ROAS = revenue / spend
- **Compute** A: 180000 / 60000 = 3.0. B: 240000 / 40000 = 6.0. Overall: 420000 / 100000 = 4.2.
- **Answer** B returns Rs 6 for every Rs 1, A returns Rs 3. Overall ROAS is 4.2. Moving budget from A to B looks attractive.
- **Check** 3.0 × 60000 + 6.0 × 40000 = 180000 + 240000 = 420000.
- **Why it matters** ROAS uses revenue, not profit. With a 40% margin, A's profit per rupee is 3 × 0.40 − 1 = 0.2 (barely positive) and B's is 6 × 0.40 − 1 = 1.4. Also, B may saturate: doubling its spend will not double revenue. Say so.

Sessions vs users: a **session** is one visit; one **user** can have many sessions. A report that says "10,000 sessions" and one that says "10,000 users" are different, so always say which.

Sales: pipeline, win rate and forecasting

Win rate = won / (won + lost). Open deals are not counted. A **weighted pipeline** is the sum of deal value × probability for its stage.

Worked problem 12.5.4 Twenty deals: 6 won, 9 lost, 5 open. Three open deals of Rs 10 lakh are at the "demo" stage (20% chance), two of Rs 20 lakh are at "proposal" (60%). Find win rate and weighted pipeline.

- **Compute** Win rate = 6 / (6 + 9) = 0.40. Weighted = 3 × 10 × 0.20 + 2 × 20 × 0.60 = 6 + 24 = 30 lakh.
- **Answer** Win rate is 40%. The open pipeline is worth an expected Rs 30 lakh, even though the raw sum is 70 lakh.
- **Check** Raw: 3 × 10 + 2 × 20 = 70. The weighted figure must be lower, and it is.

Finance: P&L and variance

P&L (profit and loss statement) lists revenue minus costs equals profit. **Variance** is actual minus budget, split into favourable (helps profit) and unfavourable (hurts profit).

Worked problem 12.5.5 Budget: revenue Rs 500, cost Rs 300 (in lakh). Actual: revenue 460, cost 330. Find profit variance.

- **Compute** Profit budget = 500 − 300 = 200. Actual = 460 − 330 = 130. Variance = 130 − 200 = −70. Revenue variance = 460 − 500 = −40 (unfavourable). Cost variance = 330 − 300 = +30, which is unfavourable because costs are higher.
- **Answer** Profit is Rs 70 lakh below budget: 40 from lower revenue and 30 from higher cost.
- **Check** −40 − 30 = −70. Price-volume variance uses the same split as Section 1.4.

Operations: SLA and inventory

SLA compliance = tickets resolved within target time / total tickets. **Days of cover** = stock / average daily sales. **Reorder point** = daily demand × lead time + safety stock.

Worked problem 12.5.6 Support resolves 940 of 1,000 tickets within 24 hours; target is 95%. A product has 600 units, sells 40 per day, supplier lead time is 7 days, safety stock is 100.

- **Compute** SLA = 940 / 1000 = 94%, one point below target. Days of cover = 600 / 40 = 15. Reorder point = 40 × 7 + 100 = 380.
- **Answer** SLA is missed by 1 point (10 tickets). Stock lasts 15 days, and we must reorder when it falls to 380 units.
- **Check** At 40 per day, 380 units last 9.5 days, which covers the 7-day lead time with 2.5

days of buffer, matching the safety stock of 100 / 40.

Others in one table

Domain	Key metrics	Formula
HR	Attrition rate	leavers / average headcount: 12 / ((200 + 220)/2) = 5.7%
E-commerce	AOV, cart abandonment	1 − (orders / carts)
Healthcare	Readmission rate, length of stay	readmitted / discharged
Banking and risk	Default rate, NPA ratio	defaulted loans / total loans
Pricing	Revenue effect of a price change	(1 + price change) × (1 + volume change) − 1: +5% price and −3% volume gives 1.05 × 0.97 = 1.0185, so +1.85% revenue

For the pricing row, always check profit too, not only revenue. Forecasting basics and scenario analysis (best, base, worst) were practised in Chapter 3 with Goal Seek and Scenario Manager.

Real job tip, tooling. Product teams use event-tracking tools (GA4, Mixpanel, Amplitude, Segment) that need a **tracking plan**: a sheet listing each event name (use one style such as `order_completed`), when it fires and its properties. Messy event names break every funnel. A **semantic layer** (dbt metrics, LookML, Cube) stores each metric's definition once, so "revenue" means the same in every dashboard. Good dashboard governance means certified datasets, a data dictionary, a refresh schedule and controlled access. For surveys (NPS, Likert 1-5 scales), report the response rate and avoid averaging unless you state it is an approximation. Tableau, Looker and Metabase are BI tools in the same family as Power BI (the Power BI chapters cover the shared ideas). Not run here: any tool-specific steps.

Applications

- Product: DAU/MAU, feature adoption, onboarding funnels.
- Marketing: ROAS, CAC by channel, campaign lift (Chapter 13).
- Sales: forecast accuracy, rep performance.

- Finance: monthly variance reports, budget scenarios.
- Operations: SLA dashboards, stock-out alerts.

COMMON MISTAKE

Reading a metric without its denominator or time window. "Win rate 40%" with 5 deals is noise. Also comparing ROAS across channels without checking attribution rules, since last-click rewards the channel closest to payment.

Check yourself

1. Spend Rs 50,000, revenue Rs 2,00,000. ROAS? Answer: 4.0.
2. Budgeted cost 300, actual 270. Favourable or unfavourable? Answer: Favourable (lower cost raises profit), variance −30 on cost.
3. Why is last-click attribution biased? Answer: It gives all credit to the last touch and ignores earlier ones.

Chapter recap

- Design KPIs as a north star, input metrics you can push, and guardrails you must protect. Write each definition once.
- Unit economics: ARPU, CAC, LTV = ARPU × margin / churn, LTV/CAC, payback months.
- A funnel counts distinct users per ordered step. Step conversion and overall conversion multiply.
- A cohort table follows each signup batch across offsets. Compare cohorts at the same age.
- Churn = lost / active at start. Small monthly churn compounds into large yearly loss.
- RFM scores recency, frequency and monetary value, and each segment maps to an action. Fixed thresholds are stable and explainable.
- Segment any drop (platform, city, channel) before you theorise. The biggest share of the drop points to the cause.
- Domain metrics differ in names, not in thinking: always know the denominator and the window.

Pattern	Formula or code
Step	`users / users.shift(1)`

conversion	
Overall conversion	`users / users.iloc[0]`
Cohort grid	`pivot_table(index="cohort", columns="offset", values="user_id", aggfunc="nunique")`
Churn	lost / active last period
LTV	`ARPU × margin × (1 / churn)`
CAC, payback	`spend / new customers`, `CAC / (ARPU × margin)`
ROAS, win rate	`revenue / spend`, `won / (won + lost)`
Stickiness	`avg DAU / MAU`
NPS	`%promoters − %detractors`

Practice set

Problem 1 (easy). Funnel: 5,000 visits, 2,000 add to cart, 500 paid. Overall and step conversions? **Answer:** Visit to cart 40%, cart to paid 25%, overall 500 / 5000 = 10%.

Problem 2 (easy). NPS from 60 promoters, 25 passives, 15 detractors. **Answer:** 60% − 15% = +45.

Problem 3 (easy). Monthly churn is 8%. Expected lifetime and share left after 2 months? **Answer:** 1 / 0.08 = 12.5 months. $0.92^2 = 0.8464$, about 84.6%.

Problem 4 (medium). ARPU Rs 200, margin 50%, churn 4%, CAC Rs 1,500. LTV, LTV/CAC, payback? **Answer:** LTV = 200 × 0.5 × 25 = 2,500. Ratio = 2,500 / 1,500 = 1.67. Payback = 1500 / 100 = 15 months.

Problem 5 (medium). What is the output? `pd.Series([4,2]).shift(1)` **Answer:** `NaN, 4.0` . The first position has no previous value. This is why the first funnel step has NaN

conversion.

Problem 6 (find the bug). A funnel uses `ev.groupby("step")["user"].count()` and reports cart-to-pay of 120%. What is wrong? **Answer:** `count` counts events (rows), not people. A user who fires the "paid" event twice (a page refresh, a retry) is counted twice, so paid can exceed carts. Use `nunique` for distinct users, reindex to funnel order, and check each step is at most the previous step.

Problem 7 (medium). Customer: last order 45 days ago, 3 orders, total Rs 2,400. Scores with our thresholds and segment? **Answer:** R_s = 2, F_s = 2, M_s = 3. By our rules it is not Champion (R_s not 3), not New, and not At risk (R_s not 1), so "Steady". A good analyst would flag this edge case and add a rule.

Problem 8 (hard, scenario). Revenue is Rs 12 lakh, down from Rs 15 lakh. Orders fell from 3,000 to 2,500. How much of the fall is volume and how much is AOV? **Answer:** AOV_0 = 500, AOV_1 = 480. Volume effect = (2500 − 3000) × 500 = −2,50,000. AOV effect = 2500 × (480 − 500) = −50,000. Total −3,00,000. Mostly volume.

Problem 9 (interview). "Our new cohort has retention of 30% at month 1 but last year's had 40%. Is it worse?" **Answer:** Ask: same definition of active? Same acquisition channel mix (paid ads often bring weaker users)? Enough users for noise to be small? Compare at the same offset, split by channel, then decide. A mix shift can create a drop with no real decline in any channel (Simpson's paradox, Chapter 13).

Problem 10 (interview). "Which KPIs would you track for a food-delivery app, and what guardrails?" **Answer:** North star: completed orders per week. Inputs: restaurant availability, search-to-order conversion, delivery time. Output: revenue, repeat rate. Guardrails: cancellation rate, late-delivery rate, refund rate, rider earnings per hour.

Problem 11 (medium). Budget profit Rs 80, actual Rs 95. Revenue was +20 above budget. What happened to cost? **Answer:** Profit variance = +15, revenue variance = +20, so cost variance is +5 (costs were Rs 5 higher than budget, which is unfavourable), 20 − 5 = 15.

Prompt pack

✨ **AI prompt: copy and paste into any AI**

You are my patient, practical professor. I am a working professional in India moving into data roles, and English is my second language. Teach me Chapter 12 topic: KPI design, including north-star, input, output and guardrail metrics, plus LTV, CAC, ARPU and payback. Use an online shop with small numbers. Explain step by step, then ask me one calculation question at a time and wait for my answer. Check each of my numbers and explain mistakes in simple words.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional in India learning data analytics. Give me 8 graded exercises, easy to hard, on funnel analysis and cohort retention tables in pandas: groupby with nunique, shift, pivot_table, and reading a retention triangle. Include small data I can type in. Do not give solutions yet. When I paste my answers and code, review them line by line and correct my reasoning.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Explain RFM segmentation from scratch: recency, frequency, monetary, thresholds, segment names and the action for each. Then give me a table of 10 customers and ask me to score and segment them by hand. After I answer, compare with the correct result and explain any difference. Finally explain when k-means clustering is better than RFM rules.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Review my analysis of retention and churn for a business: [PASTE HERE]. Check that my definition of active is clear, cohorts are compared at the same age, incomplete cohorts are handled, and churn and retention numbers agree. List every problem, rank them by importance, and show how to fix the top two.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional preparing for data analyst interviews. Run a mock interview on business analytics: KPIs, funnels, cohorts, churn, RFM, ROAS, win rate and variance analysis. Ask six questions one at a time, with realistic numbers. After each answer, score me 1 to 5, give a model answer in simple English, and

```
name one concept I should revise.
```

CHAPTER 13

A/B TESTING FOR ANALYSTS: DESIGN, SAMPLE SIZE, ANALYSIS AND PITFALLS

How to find out whether a change really caused an improvement, and not just luck.

Your team changes the checkout button from grey to green. Conversion goes up the next week. Was it the button? Or was it a salary-week effect, a festival, or plain luck? You cannot tell by looking at before and after, because many things change at once. An A/B test solves this by running the old and the new version at the same time on randomly chosen users, so the only systematic difference between the two groups is the change. In this chapter you will design a test, calculate how many users you need, analyse the result with a z-test and a confidence interval, and spot the classic traps (peeking, sample ratio mismatch, Simpson's paradox). All numbers below were computed with Python (`numpy` , `scipy`). We reuse the online shop and the checkout conversion rate from Chapter 12. For the basics of p-values and distributions, see the learner's Math PDF and the statistics chapters of the data-science document.

13.1 Designing a Test

INTUITION

You want to know if a new fertiliser works. If you give it to your best field and old fertiliser to the worst field, you learn nothing. So you flip a coin for every plot of land. Randomness makes the two groups equal in everything else (soil, sun, luck), on average. An A/B test flips a coin for every user.

Words you need first

Word	Plain meaning	Example
Control (A)	The current version	Grey button
Treatment (B)	The new version	Green button
Hypothesis	A sentence you can prove wrong	"Green button raises checkout conversion"

Null hypothesis (H_0)	"There is no difference"	Conversion is the same in A and B
Primary metric	The one number that decides the test	Checkout conversion rate
Guardrail metric	Must not get worse	Return rate, page load time
Randomization unit	The thing you flip the coin for	User, not page view
MDE	Minimum detectable effect: the smallest change worth detecting	+1 percentage point

How it works. Before the test you write down four things: the hypothesis, the primary metric and guardrails, the randomization unit, and the MDE. Writing them first stops you from choosing the favourable metric after you see the results.

Randomization unit. Randomize by *user*, not by page view. If one user sees both versions, their behaviour is contaminated. Assignment is usually done with a hash: take the user ID plus the experiment name, compute a hash, and use `hash % 100 < 50` for control. The same user always gets the same arm, and no table of assignments is needed. The analysis unit should match the randomization unit; otherwise your standard errors are too small.

Duration. Run for full weeks (weekdays and weekends behave differently), and at least until you reach the sample size from Section 13.2. Do not stop because the result looks good (see peeking in 13.4).

Real-world mapping

Item	Example value	Where it appears in the job
Hypothesis	New checkout raises conversion by at least 1 point	Experiment brief
Primary metric	Orders per visitor	Experiment platform setup
Guardrail	Refund rate under 3%	Automatic stop rule

Unit	user_id	Assignment service

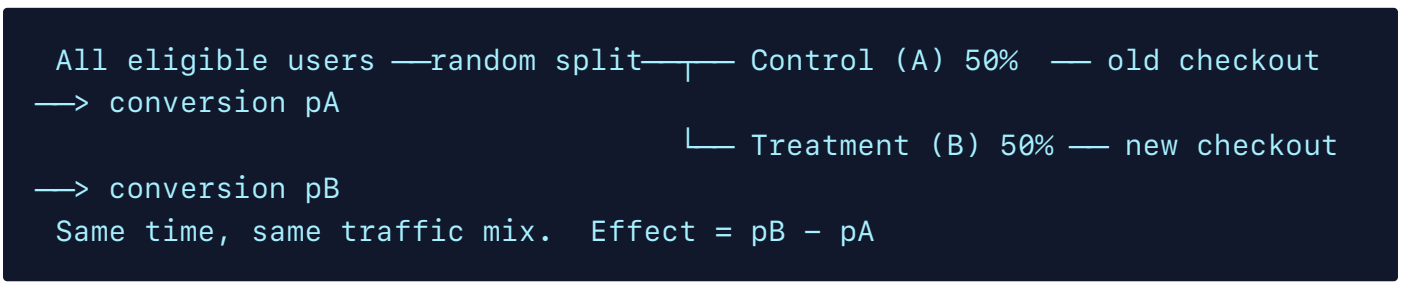
Worked problem 13.1.1 An experiment shows a new product page to users. Analysts propose "page views per session" as the primary metric. Is this a good choice?

- **Step 1** Ask what the business wants: more orders, not more page views.
- **Step 2** A confusing page could raise page views (people hunting for information) while lowering orders.
- **Step 3** Choose orders per user as primary. Keep page views as a diagnostic and add a guardrail such as bounce rate.
- **Answer** No. Page views can go up for bad reasons. The primary metric must be close to the real goal.
- **Check** Ask "can this metric go up while the business gets worse?" If yes, it cannot be the only decision metric.

Worked problem 13.1.2 Why must an analyst write the hypothesis and MDE before launching?

- **Step 1** If you check 20 metrics after the test, about one will look significant by luck (computed in 13.4).
- **Step 2** If you set the MDE after seeing the result, you can always say "this effect is what we hoped for".
- **Answer** Writing both first fixes the rules before data can tempt you. This is called pre-registration.
- **Check** A one-page experiment brief with date, metric definitions and the decision rule is the proof.

The picture



Connection to the real job. Product and marketing teams ask for tests constantly: subject lines, prices, layouts. The analyst usually writes the plan, checks the data, and writes the result memo (Chapter 14). A bakery does something similar: it sells two recipes of the same loaf on alternate days and compares the sales.

Applications

- Product: layouts, onboarding flows, recommendations.
- Marketing: email subject lines, ad creatives, landing pages.
- Pricing: price points or shipping thresholds.
- Operations: the effect of a new routing rule on delivery time.

COMMON MISTAKE

Randomizing sessions when you want to learn about users. A user with 5 sessions then lands in both arms. Another: changing the test (new traffic, new rules) in the middle and still reading it as one experiment.

Check yourself

1. What does randomization buy you? Answer: The two groups are the same on average in everything except the change.
2. Why hash the user ID? Answer: Each user gets a stable arm with no assignment table.
3. Why write the metric before launch? Answer: To avoid choosing the favourable result later.

13.2 Sample Size and Power

INTUITION

Suppose you flip a coin 10 times and get 6 heads. Is the coin unfair? Surely not: that is common luck. 600 heads in 1,000 flips would be surprising. A small difference needs many flips to be seen above the noise. Sample size is about "how many flips do I need to see a difference of this size?"

Words you need first

Word	Plain meaning	Usual value
α (alpha), significance level	Chance of a false alarm (we say "effect" when none exists)	0.05
β (beta)	Chance of a miss (an effect exists but we say none)	0.20

Power = 1 − β	Chance of catching a real effect of size MDE	0.80
$z_{0.975}$	Normal-curve cut-off for a two-sided 5% test	1.96
$z_{0.80}$	Cut-off for 80% power	0.8416

Formula (two proportions, equal groups). For baseline rate p_1 and target rate p_2, with $\bar{p} = (p_1 + p_2) / 2$ and $\delta = p_2 - p_1$:

```
n per group = ( z_α/2 × √(2 p̄ (1−p̄))  +  z_β × √(p₁(1−p₁) + p₂(1−p₂)) )²
/  δ²
```

Worked problem 13.2.1 The shop's checkout conversion is 5%. We want to detect a rise to 6% (MDE = 1 point) with α = 0.05 and power 80%. How many users per group?

- **Formula** the formula above
- **Plug in** $p_1 = 0.05$, $p_2 = 0.06$, $\bar{p} = 0.055$, $\delta = 0.01$, $z_{\alpha/2} = 1.96$, $z_\beta = 0.8416$
- **Compute** 2 × 0.055 × 0.945 = 0.10395, and √0.10395 = 0.32241, then × 1.96 = 0.63195. Next, 0.05 × 0.95 = 0.0475 and 0.06 × 0.94 = 0.0564, sum 0.1039, √0.1039 = 0.32234, then × 0.8416 = 0.27129. Add: 0.63195 + 0.27129 = 0.90324. Square: 0.81584. Divide by δ^2 = 0.0001: 8,158.
- **Answer** About 8,158 users in each group, 16,316 in total.
- **Check** Code with the exact normal quantiles gave 8157.7, rounded up to 8,158.

```
import numpy as np
from scipy.stats import norm
def n_per_group(p1,p2,alpha=0.05,power=0.8):
    za=norm.ppf(1-alpha/2); zb=norm.ppf(power)
    pbar=(p1+p2)/2
    num=(za*np.sqrt(2*pbar*(1-pbar))+zb*np.sqrt(p1*(1-p1)+p2*(1-p2)))**2
    return num/(p2-p1)**2
for p1,p2 in [(0.05,0.06),(0.05,0.055),(0.10,0.11),(0.05,0.0625)]:
    print(p1,p2,round(n_per_group(p1,p2),1))
```

```
0.05 0.06 8157.7
0.05 0.055 31233.4
0.1 0.11 14750.8
0.05 0.0625 5332.1
```

Line by line: `norm.ppf(q)` returns the normal-curve cut-off with probability q to its left. `za` is 1.96 and `zb` is 0.8416. The rest is the formula typed as written. Predict first: if you halve the MDE from 1 point to 0.5 point, will n double? It does not double. The second line shows 31,233, almost four times larger. The rule: n grows with $1/\delta^2$, so half the effect needs four times the users.

Worked problem 13.2.2 (duration) The shop has 2,000 eligible users per day. How long must the test run for the 5% to 6% test?

- **Formula** days = total users needed / users per day
- **Plug in** 16,316 / 2,000
- **Compute** 8.16 days
- **Answer** At least 9 days, but run 14 days (two full weeks) so every weekday appears twice.
- **Check** 14 days × 2,000 = 28,000 users, more than 16,316.
- **Why it matters** If the product manager wants an answer in 3 days, you can show that 6,000 users give only a power far below 80%, and a real effect would probably be missed.

Other metrics. For a mean metric (revenue per user), n per group = $2 \times (z_{\alpha/2} + z_{\beta})^2 \times \sigma^2 / \delta^2$, where σ is the standard deviation. Revenue is skewed with many zeros, so σ is large and the needed n is large, as we see in 13.3.

The picture (power). Draw two bell curves: one centred at "no effect", one at the true effect. The curves overlap. Power is the share of the second curve to the right of the cut-off. More users make both curves narrower, so the overlap shrinks and power rises.

Connection to the real job. In planning meetings, "how long will this take?" is your sample-size answer. Analysts often use a calculator, but you must know what goes in: baseline rate, MDE, α, power. In Python, `statsmodels.stats.power` has ready functions, which were not run here (the library was not installed).

Applications

- Marketing: how many emails per arm to see a 0.5-point click lift.
- Product: planning how many weeks a feature test needs.
- Healthcare: trial size for a new treatment.
- Pricing: sample needed for a revenue-per-user test.

COMMON MISTAKE

Choosing the MDE after the test so that "the result is enough". And running an under-powered test: a non-significant result from a tiny sample does not mean "no effect", only "not enough data to tell".

Check yourself

1. If you want to detect half the effect, n changes by what factor? Answer: About 4.
2. What does 80% power mean? Answer: If a true effect of the MDE size exists, we catch it 80% of the time.
3. What are the four inputs of the formula? Answer: Baseline rate, MDE, α, power.

13.3 Analysing the Result

INTUITION

Two groups never show exactly the same rate, even when nothing changed. The analysis asks: "If there were truly no difference, how surprising is the gap we saw?" If very surprising (a small p-value), we believe the change worked. The confidence interval then says how big the gain might be.

Words you need first

Word	Plain meaning
p-value	If H_0 (no effect) were true, the chance of seeing a gap at least this big. NOT the chance that H_0 is true
Standard error (SE)	Typical size of the random error in a number
Confidence interval (CI)	A range of effects consistent with the data. A 95% CI is built so that 95% of such intervals would contain the true value
Practical significance	Whether the effect is big enough to matter for business

Formulas (two proportions).

```
pooled p = (xC + xT) / (nC + nT)
SE (for the test) = √( pooled p × (1 − pooled p) × (1/nC + 1/nT) )
z = (pT − pC) / SE
two-sided p-value = 2 × (1 − Φ(|z|)),  Φ = normal cumulative probability
95% CI for the difference = (pT − pC) ± 1.96 × √( pC(1−pC)/nC +
pT(1−pT)/nT )
```

We "pool" because the test assumes H_0 is true, so both groups share one rate. The CI uses the unpooled SE because it describes the observed difference.

Worked problem 13.3.1 Control: 410 conversions out of 8,200 users. Treatment: 492 out of 8,200. Is the change real? Give the CI and a business reading.

- **Formula** the formulas above
- **Plug in** pC = 410 / 8200 = 0.05, pT = 492 / 8200 = 0.06, pooled p = 902 / 16400 = 0.055
- **Compute** SE = √(0.055 × 0.945 × (2 / 8200)) = 0.00356. z = 0.01 / 0.00356 = 2.809. p-value = 0.0050. SE for the CI is also 0.00356, so CI = 0.01 ± 1.96 × 0.00356 = 0.01 ± 0.00698, which is (0.003, 0.017).
- **Answer** The conversion rose by 1.0 point (a 20% relative lift), p = 0.005. The 95% CI is 0.3 to 1.7 points. We reject "no effect".
- **Check** Code below gives z = 2.809, p = 0.0050 and a chi-square test gives 7.888, equal to z^2 ($2.809^2 = 7.89$). Two ways, same answer.
- **Why it matters** The CI is the honest message: "somewhere between +0.3 and +1.7 points". If +0.3 points is not worth the engineering cost, say so.

```
import numpy as np
from scipy import stats
nC,xC=8200,410; nT,xT=8200,492
pC,pT=xC/nC,xT/nT
pp=(xC+xT)/(nC+nT)
se_pool=np.sqrt(pp*(1-pp)*(1/nC+1/nT))
z=(pT-pC)/se_pool
p=2*(1-stats.norm.cdf(abs(z)))
se=np.sqrt(pC*(1-pC)/nC+pT*(1-pT)/nT)
lo,hi=(pT-pC)-1.96*se,(pT-pC)+1.96*se
print(round(se_pool,5),round(z,3),round(p,4),round(lo,4),round(hi,4))
```

```
0.00356 2.809 0.005 0.003 0.017
```

p-value versus practical significance. With 10 million users, a 0.01-point gain can have p < 0.001 but be worth nothing. With 500 users, a big gain can have p = 0.2 and still be real. Always read the p-value together with the CI and the cost.

Worked problem 13.3.2 (a continuous metric that fails to reach significance) Revenue per user, simulated: 5,000 users per arm. Only 20% of users buy, order sizes follow a skewed exponential shape with mean Rs 300, and we apply a true 6% lift to the treatment. Is it detected?

```
rng=np.random.default_rng(7)
c=rng.exponential(300,5000)*(rng.random(5000)<0.2)
t=rng.exponential(300,5000)*(rng.random(5000)<0.2)*1.0
t=t*1.06
print(round(c.mean(),2),round(t.mean(),2),stats.ttest_ind(t,c,equal_var=Fa
lse))
print(stats.mannwhitneyu(t,c))
```

```
60.14 63.31 TtestResult(statistic=np.float64(0.8552077181849908),
pvalue=np.float64(0.39245672014246635), df=np.float64(9967.322625953564))
MannwhitneyuResult(statistic=np.float64(12522922.0),
pvalue=np.float64(0.8198981241835632))
```

- **Answer** Mean revenue per user is Rs 60.14 in control and Rs 63.31 in treatment (about +5%), but p = 0.39. We cannot tell the lift from noise. The standard deviation of this metric is large (most users spend 0), so 5,000 per arm is not enough. `equal_var=False` is Welch's t-test, the safe default when the arms may have different variance. Mann-Whitney compares ranks instead of means, and it answers a different question: whether one arm tends to have larger values. For a revenue decision, the mean is what the business cares about, so report the t-test and the CI.
- **Check** The mean difference 63.31 − 60.14 = 3.17 is only about 0.86 standard errors, below the 1.96 needed.
- **Why it matters** "Not significant" here is not "no effect". It is "this test was too small for this noisy metric". That is a sample-size problem (13.2).

The picture. A CI picture is a line with a dot: the dot is the observed lift, the line is the CI, and a vertical mark at zero is "no effect". If the line crosses zero, we cannot rule out no effect. In Problem 13.3.1 the line runs from 0.3 to 1.7 and stays right of zero.

Connection to the real job. The result memo (Chapter 14) should say: lift, CI, p-value, sample size, duration, guardrails, and the recommendation. Do not report only "p < 0.05".

Applications

- Marketing: subject-line tests on click-through rate.
- Product: checkout, signup and onboarding conversion.
- Operations: mean delivery time, using a t-test on the difference.
- Pricing: revenue per visitor (high variance, large samples).

COMMON MISTAKE

Saying "the p-value is the probability the new version is better" (wrong), or "p > 0.05 means no difference" (wrong). Also using a z-test for proportions when counts are tiny (a rule of thumb: need at least about 10 successes and 10 failures in each arm).

Check yourself

1. A 95% CI for the lift is (−0.2, +1.4) points. Significant at 5%? Answer: No, it contains zero.
2. Why use `equal_var=False` ? Answer: It does not assume equal variances in both arms.
3. z = 2.0. Roughly what is the two-sided p-value? Answer: About 0.046.

13.4 Pitfalls and Beyond the Basic Test

INTUITION

A test is like a medical scan. It is only useful if it is done carefully. A wrong procedure can show disease in a healthy person or hide it in a sick one. The pitfalls below are the common ways to run the scan wrongly.

Peeking (looking early and stopping when significant)

If you check the result every day and stop as soon as p < 0.05, your false-alarm rate is much higher than 5%. We tested this with a simulation: 2,000 A/A tests (both arms identical, 5% conversion, no real effect), 10,000 users per arm, either one look at the end or ten looks after every 1,000 users.

```
one look 0.053 peek 10x 0.2035
```

- **Answer** With one look, 5.3% of the tests said "significant" (close to the promised 5%). With ten looks, 20.4% did. Four times more false alarms. The fix: decide the sample size first and look once, or use a *sequential* testing method (designed for repeated looks), which the experiment platform may offer.
- **Check** Random numbers are seeded in the script, so the result can be reproduced, and 0.053 is close to 0.05 as theory says.

Multiple testing

If you test 20 metrics at α = 0.05 and none truly changes, the chance that at least one shows "significant" is $1 - 0.95^{20} = 1 - 0.358 = 0.642$, which is 64%. **Bonferroni correction**: divide α by the number of tests. For 20 metrics, use 0.05 / 20 = 0.0025. Simple and safe but conservative. Better: name one primary metric and treat the rest as exploratory.

Sample ratio mismatch (SRM)

If you planned a 50/50 split, you should get about 50/50. A mismatch signals a bug (a redirect that loses users, bots in one arm), and then the whole result is untrustworthy. Test it with a chi-square goodness-of-fit test:

```
print(stats.chisquare([50400,49600],[50000,50000]))
print(stats.chisquare([50800,49200],[50000,50000]))
```

```
Power_divergenceResult(statistic=np.float64(6.4),
pvalue=np.float64(0.011411203638600166))
Power_divergenceResult(statistic=np.float64(25.6),
pvalue=np.float64(4.200393976022009e-07))
```

- **Answer** 50,400 vs 49,600 gives p = 0.011: mildly suspicious. 50,800 vs 49,200 gives p = 4×10^{-7}, which is a clear SRM. Many teams alarm at p < 0.001. Fix the bug and restart. Never "adjust" for SRM and read the result.
- **Check** $(50800 - 50000)^2 / 50000 + (49200 - 50000)^2 / 50000 = 12.8 + 12.8 = 25.6$. Same as the statistic.

Simpson's paradox

The overall result can reverse the result inside every subgroup when the arms have different group mixes. Data (users and conversions by device):

```
import pandas as pd
d=pd.DataFrame({"device":["mobile","mobile","desktop","desktop"],"arm":
["A","B","A","B"],"users":[1000,4000,4000,1000],"conv":[50,220,400,110]})
d["rate_%"]=(d.conv/d.users*100).round(2)
print(d)
t=d.groupby("arm")[["users","conv"]].sum(); t["rate_%"]=
(t.conv/t.users*100).round(2); print(t)
```

```
    device arm  users  conv  rate_%
0   mobile   A  1000    50     5.0
1   mobile   B  4000   220     5.5
2  desktop   A  4000   400    10.0
3  desktop   B  1000   110    11.0
     users  conv  rate_%
arm
A     5000   450     9.0
B     5000   330     6.6
```

- **Answer** B wins on mobile (5.5% vs 5.0%) and on desktop (11.0% vs 10.0%), yet loses overall (6.6% vs 9.0%). Arm B had mostly low-converting mobile users, arm A mostly high-converting desktop users. In a proper test, randomization should give the same device mix in both arms. If it does not, randomization failed (a form of SRM by device). Also stratify and check balance.
- **Check** 330 / 5000 = 6.6% and 450 / 5000 = 9.0%. The subgroup weights differ (80% mobile in B, 20% in A).

Novelty and primacy effects

Users often click a new design simply because it is new (novelty effect), so the gain fades. Or they dislike change at first (primacy effect). Look at the effect by day. If it trends towards zero, wait longer. A fresh test with only new users avoids some of this.

Network effects and interference

On social networks, marketplaces and ride-sharing, one user's arm affects others (a discount to buyers in B changes prices for sellers in A). Then user-level randomization is biased. Options: randomize by cluster (city, school), or by time slot (switchback tests), and ask a specialist.

CUPED (variance reduction, awareness)

CUPED uses a pre-experiment measurement (such as last month's spend) to remove predictable noise. We tested it on simulated data with a true effect of +2: a pre-period value correlates with the outcome (correlation squared = 0.727).

```
raw 1.577 28.4 0.01303
cuped 2.028 14.85 0.0
theta 0.805 var reduction 0.727 corr^2 0.727
```

- **Answer** The raw difference is 1.577 with p = 0.013. After CUPED the estimate is 2.028 (closer to the true +2) and the spread of the metric falls from 28.4 to 14.85. Variance falls by about the squared correlation (72.7%). This is why platforms use it: the same users give a sharper answer, or fewer users are needed. The adjustment is `y − θ(pre − mean(pre))` with θ = cov(y, pre) / var(pre). This is awareness level: know what it is and why it helps.

Quasi-experiments: difference-in-differences

When you cannot randomize (a new rule launched in one city only), compare the *change* in the treated group with the *change* in a similar untreated group.

Worked problem 13.4.1 City X (treated) average delivery time went from 10 to 12 minutes after a launch in X. City Y (not treated) went from 8 to 9 minutes. What was the effect of the launch?

- **Formula** DiD = (treated after − treated before) − (control after − control before)
- **Plug in** (12 − 10) − (9 − 8)
- **Compute** 2 − 1 = 1
- **Answer** The launch added about 1 minute. The 1 extra minute in Y suggests a general trend (maybe weather) that would have hit X too.
- **Check** If there were no trend in Y, the formula would give the full 2 minutes. It relies on the "parallel trends" assumption: without the launch, X and Y would have moved the same way. Check it using several earlier weeks.

Connection to the real job. Peeking, SRM and multiple testing are the first things an experienced reviewer asks about. Add a checklist to every result: SRM check, one primary metric, planned duration reached, effect by day, guardrails.

Applications

- Product: experiment review before any launch.

- Marketing: geo experiments (a campaign in some cities only) analysed with DiD.
- Policy and finance: effect of a rule change in one branch or region.

COMMON MISTAKE

Reading the result before checking SRM. If the arms are unbalanced, nothing downstream is trustworthy. Also forgetting that "we saw B win in the dashboard on day 3" is not a result.

Check yourself

1. Why is stopping at the first $p < 0.05$ wrong? Answer: Repeated looks raise the false-alarm rate far above 5%.
2. What does SRM mean? Answer: The arm sizes differ from the planned split by more than chance allows.
3. What does DiD subtract? Answer: The control group's change, to remove the common trend.

Chapter recap

- An A/B test randomizes users to control and treatment at the same time, so the change is the only systematic difference.
- Write the hypothesis, primary metric, guardrails, unit and MDE before launch.
- Sample size depends on baseline, MDE, α and power. Halving the MDE needs about four times the users.
- Use the two-proportion z-test for rates and Welch's t-test for means. Report the lift, the CI and the p-value together.
- A p-value is not the chance that B is better. "Not significant" with a small sample is not "no effect".
- Peeking (about 20% false alarms in our simulation), multiple testing, SRM and Simpson's paradox are the classic failures.
- CUPED reduces noise using pre-experiment data. Difference-in-differences handles cases that cannot be randomized.
- Run for full weeks, look once, check SRM first.

Pattern	Formula or code

Sample size (rates)	$(z_{\alpha/2}\cdot\sqrt{(2\bar{p}(1-\bar{p}))} + z_{\beta}\cdot\sqrt{(p_1q_1+p_2q_2)})^2 / \delta^2$
Two-proportion z	$(pT-pC) / \sqrt{(p(1-p)(1/nC+1/nT))}$
95% CI of difference	$diff \pm 1.96\cdot\sqrt{(pC\ qC/nC + pT\ qT/nT)}$
Welch t-test	`stats.ttest_ind(t, c, equal_var=False)`
SRM check	`stats.chisquare(observed, expected)`
Bonferroni	α / number of tests
DiD	$(T_{after} - T_{before}) - (C_{after} - C_{before})$
CUPED	$y - \theta(pre - mean(pre)),\ \theta = cov(y,pre)/var(pre)$

Practice set

Problem 1 (easy). Control 200 of 4,000 convert, treatment 240 of 4,000. What are the two rates and the difference? **Answer:** 5.0% and 6.0%, a difference of 1.0 point (a 20% relative lift).

Problem 2 (easy). Which is the randomization unit when you want to compare two app onboarding flows, session or user? **Answer:** User, so that nobody sees both flows.

Problem 3 (medium). For Problem 1, find z and the p-value. **Answer:** Pooled p = 440 / 8000 = 0.055. SE = $\sqrt{(0.055 \times 0.945 \times 2 / 4000)} = 0.005098$. z = 0.01 / 0.005098 = 1.962. p = 0.0498 (checked by code). Just under 0.05.

Problem 4 (medium). Give the 95% CI for Problem 1 and interpret. **Answer:** SE = $\sqrt{(0.05 \times 0.95 / 4000 + 0.06 \times 0.94 / 4000)} = 0.005097$. CI = 0.01 ± 0.00999 = (0.00001, 0.0200), about 0.0 to 2.0 points. The lower end touches zero, so the evidence is borderline. A careful analyst says "likely positive, but could be near zero; consider a larger test".

Problem 5 (medium). What sample size per group for 10% to 11% at $\alpha = 0.05$, power 80%? How does it compare with 5% to 6%? **Answer:** 14,751 per group (code: 14,750.8, rounded up). Higher baselines have a bigger variance $p(1-p)$, so it needs more users than the 8,158 for 5% to 6%.

Problem 6 (medium). With 2,000 eligible users a day and 14,751 per group, how many days? **Answer:** 2 × 14,751 = 29,502 users, 29,502 / 2,000 = 14.75, so at least 15 days; round up to 3 full weeks (21 days).

Problem 7 (what is the output?). `stats.chisquare([5100,4900],[5000,5000])` gives which statistic and is it an SRM? **Answer:** Statistic = $100^2/5000 \times 2 = 4.0$ with p = 0.0455. Mildly suspicious but not below the usual 0.001 alarm line. Investigate, and check for a recurring pattern.

Problem 8 (find the bug). An analyst tests 10 metrics, finds one with p = 0.03, and announces a win. What is wrong, and what is the corrected threshold? **Answer:** Ten tests produce a $1 - 0.95^{10} = 40\%$ chance of at least one false positive. Bonferroni threshold is 0.05 / 10 = 0.005, and 0.03 does not pass. The metric should be labelled exploratory and re-tested.

Problem 9 (scenario). Treatment: 5,200 users, control: 4,800. Planned 50/50, total 10,000. Is this an SRM? **Answer:** Expected 5,000 each. Chi-square = $200^2/5000 + 200^2/5000 = 8 + 8 = 16$, p about 0.00006, well below 0.001. Yes, SRM. Stop and find the bug before reading any result.

Problem 10 (interview). "Your test shows +3% conversion with p = 0.04, but the guardrail refund rate rose from 2% to 2.6%. Do you ship?" **Answer:** Check whether the refund change is significant and what it costs. If refunds rose by 0.6 points and each refund costs more than the gain from the extra conversions, shipping loses money. Quantify both in rupees, show the CI of each, and recommend iterating or a partial rollout. Do not read the primary alone.

Problem 11 (interview). "Explain a p-value to a product manager in two sentences." **Answer:** "If the new design made no difference, we would see a gap this big only about 4 times in 100 by luck. That is why we believe the change helped, but the size of the gain is better read from the interval."

Problem 12 (hard). The first example of DiD: treated 10 to 12 minutes, control 8 to 9. If control had instead gone 8 to 7, what is the DiD effect? **Answer:** $(12 - 10) - (7 - 8) = 2 - (-1) = 3$ minutes. The launch effect is larger because the control improved while the treated city got worse.

Prompt pack

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional in India moving into data roles, and English is my second language. Teach me Chapter 13 topic: how to design an A/B test, including hypothesis, primary and guardrail metrics, randomization unit, MDE and duration. Use an online shop checkout example and compare it with a bakery selling two loaf recipes on alternate days. Explain step by step, and after each part ask me one question and wait for my answer.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data roles. Give me 8 graded exercises, easy to hard, on A/B test sample size and power for conversion rates: baseline, MDE, alpha, power, and converting the sample size to days. Use realistic Indian e-commerce numbers. Do not give solutions yet. After I answer, I will paste my work, and you will check every step and every number.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data roles. Teach me, step by step with small numbers, how to analyse an A/B test of two proportions: pooled z-test, p-value, 95% confidence interval, and practical significance. Then give me a new dataset and let me do it by hand before you reveal the result. Explain what a p-value does NOT mean, in simple English.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data roles. Review my A/B test report: [PASTE HERE]. Check for peeking, multiple testing, sample ratio mismatch, Simpson's paradox, novelty effect and missing guardrails. Tell me which checks the report passes or fails, rank the problems by importance, and rewrite the conclusion section in honest, clear language.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional preparing for data analyst interviews. Run a mock interview on A/B testing: choosing metrics, sample size, interpreting p-values and confidence intervals, peeking, SRM, Simpson's paradox and difference-in-
```

```
differences. Ask six questions one at a time with realistic numbers. After each answer, score me 1 to 5, give a model answer, and name one thing to revise.
```

CHAPTER 14

STORYTELLING, STAKEHOLDER COMMUNICATION AND WRITING AN ANALYSIS MEMO

Great analysis that nobody understands changes nothing. This chapter makes sure yours gets used.

Two analysts get the same question and find the same answer. One sends a 14-slide deck with 9 charts and the sentence "please find the analysis attached". The other sends six lines: what happened, why, and what to do on Monday. Only the second analyst gets a promotion. In this chapter you will learn how to clarify what a stakeholder really needs (before you open any data), how to tell the story in the order busy people think (headline, evidence, recommendation), how to write an analysis memo and an executive summary, how to present and answer hard questions honestly, and how to document your work so others can trust it. We use the Delhi revenue drop from the shop in Chapter 1. Everything here is practice you can start tomorrow, and most of it needs no tools at all. Data analysts are often strong at numbers and weak at writing, so this is where you can pull ahead.

14.1 Ask the Right Questions: Stakeholder Management

INTUITION

A doctor does not prescribe medicine the moment you say "I feel bad". She asks where it hurts, since when, and what you did yesterday. An analyst is a doctor for business questions. The first request is a symptom, not a diagnosis.

Words you need first

Word	Plain meaning	Example
Stakeholder	Anyone who needs, pays for or is affected by your analysis	Head of Marketing, CFO
Decision owner	The person who will act on your answer	The one who signs the budget

Scope	What is in and out of the work	Delhi only, January to February
Requirement	What the answer must contain to be useful	"A number by city, with a reason"
Scope creep	New requests slowly added until the work is endless	"While you are there, also check returns"

How it works: the clarifying conversation. Before starting, get answers to these (Chapter 1's SPARC list in conversation form):

1. What decision will this support, and who makes it?
2. What would you do if the answer were A? If B? (If both answers lead to the same action, the analysis is not needed.)
3. How do you define the metric (for example, "revenue": delivered orders only? before or after refunds?)
4. What time period and which segments?
5. What have you already tried or believe? (Their hypothesis saves you hours, and tells you what they will find convincing.)
6. By when, and how precise? (A rough number today may beat an exact number next week.)

Setting expectations. Say what you will deliver, by when, and what you will not cover. Send it back in writing in two lines. When asked "can you also...", reply: "Yes, that will add two days. Do you want it before or after the current question?" This turns scope creep into a visible trade-off, and the stakeholder chooses.

Saying "it depends, and here is why". A bare "it depends" sounds evasive. Instead: "It depends on whether you count cancelled orders. If you do, revenue is down 12%. If not, 10%. I recommend the second, because cancelled orders are not revenue." Name the condition, give both numbers, and recommend one.

Real-world mapping

Item	Example value	Where it appears in the job
Decision owner	Head of Marketing	First line of the brief and the memo
Definition	"Revenue = delivered orders, after	Data dictionary (14.5)

	refunds"	
Deadline plus precision	"Tuesday, rough within 5% is fine"	Request ticket
Scope note	"Delhi, Jan-Feb; returns excluded"	Top of the notebook and memo

Worked problem 14.1.1 Your manager says "Can you look at why sales are down?" Write the clarifying questions you would ask, and the one-sentence brief you send back.

- **Step 1** Decision: "Is this for the weekly review, or to decide on a budget change?"
- **Step 2** Definition: "Sales means delivered-order revenue after refunds, correct?"
- **Step 3** Scope and benchmark: "January versus February, all cities, and versus the same months last year?"
- **Step 4** Deadline and precision: "By Thursday; is a direct explanation within 10% enough?"
- **Answer** Brief: "I will explain the change in delivered-order revenue, February versus January, by city and category, and recommend one action, by Thursday. I will not cover returns or marketing spend unless you want them."
- **Check** Could someone else read the brief and know when the work is finished? Yes.
- **Why it matters** Two minutes of questions saves two days of rework.

Worked problem 14.1.2 A stakeholder asks to add three more breakdowns the day before the deadline. Write a reply that keeps trust.

- **Step 1** Acknowledge: the breakdowns are reasonable.
- **Step 2** State the cost: each needs about half a day.
- **Step 3** Offer a choice.
- **Answer** "Good ideas. Each adds about half a day. I can deliver the city analysis on Thursday as agreed and the extra three on Monday, or deliver everything Monday. Which do you prefer?"
- **Check** The reply contains a time, a cost and a choice, and no blame or refusal.

The picture: the loop

```
 Request ──> Clarify ──> Brief (written back) ──> Analyse ──> Draft ──>
Review with owner ──> Final
   ^
|
   └──────────────────── new questions go through the same clarify step
<──────┘
```

Read it as a loop, not a line: at the review step, new questions arrive. Treat them as a new request, with its own clarify and brief.

Connection to the real job. In interviews, the first thing a good candidate does with "DAU dropped 10%, why?" is ask clarifying questions (Chapter 15). On the job, this habit is the main difference between junior and senior analysts.

Applications

- Data Analytics: every ticket and ad-hoc request.
- Product: aligning on a metric definition before a launch.
- Finance: agreeing what is "in" the monthly close.
- Consulting-style work: scoping a project in a one-page note.

COMMON MISTAKE

Starting the analysis to look busy and asking questions later. Also taking requests literally. "Pull the list of customers" usually means "tell me which customers to call".

Check yourself

1. Why ask "what would you do if the answer were A or B"? Answer: If the action is the same, no analysis is needed.
2. How do you handle a new request near the deadline? Answer: State the time cost and offer a choice.
3. What is wrong with a bare "it depends"? Answer: It gives no condition and no recommendation.

14.2 Data Storytelling: Headline, Evidence, Recommendation

INTUITION

A news story starts with the headline ("Delhi orders collapse, rest of country steady"), then the key facts, then the details. Readers can stop at any point and still have the main message. Most analysts write like a detective novel: background, method, long journey, and the answer on page 12. Your readers will not reach page 12.

Formal structure (pyramid principle). Start with the answer, then support it.

```
 Level 1: Headline        One sentence with the finding AND the action
 Level 2: Evidence        Two to three facts that prove it, each with a
number
 Level 3: Detail          Method, caveats, appendix (only for those who
want it)
```

Rules for each chart.

- One message per chart, and the title *is* the message: "Delhi revenue fell 33% while other cities held steady" (an action title), not "Revenue by city".
- Choose the chart for the question: comparison (bars), change over time (line), share (stacked bar, or a table for few categories), relationship (scatter). Chapter 8 has the full chart guide.
- Remove clutter: no 3D, no heavy gridlines, no legend if you can label lines directly.
- Use colour with purpose: grey for context, one strong colour for the thing you want the reader to see.
- Show the comparison: a number without a benchmark is not a story (Chapter 1).

Worked example 14.2.1 (predict, run, explain) First, compute the facts you will tell. Before you run it: which city carries the whole drop?

```
import pandas as pd
r=pd.DataFrame({"city":["Mumbai","Delhi","Bengaluru","Pune"],"jan":
[40000,30000,20000,10000],"feb":[41000,20000,19000,10000]})
r["chg"]=r.feb-r.jan; r["pct"]=(r.chg/r.jan*100).round(1)
print(r.to_string(index=False));
print(r[["jan","feb","chg"]].sum().to_dict())
```

Real output:

```
      city   jan   feb    chg   pct
    Mumbai 40000 41000   1000   2.5
     Delhi 30000 20000 -10000 -33.3
 Bengaluru 20000 19000  -1000  -5.0
      Pune 10000 10000      0   0.0
{'jan': 100000, 'feb': 90000, 'chg': -10000}
```

Line by line: `chg` is February minus January in rupees. `pct` is the change relative to January, rounded to one decimal. The final line sums the three numeric columns: total revenue fell from Rs 1,00,000 to Rs 90,000, a fall of Rs 10,000 or 10%.

- **Answer** Delhi alone lost Rs 10,000, which equals the whole net fall. Mumbai (+1,000) and Bengaluru (−1,000) cancel each other out and Pune is flat.
- **Check** 1,000 − 10,000 − 1,000 + 0 = −10,000, which equals 90,000 − 100,000.

Worked problem 14.2.2 Turn the output into a headline, evidence and recommendation.

- **Headline** "Revenue fell 10% in February, and the entire fall is Delhi (−33%). Fixing Delhi is the priority."
- **Evidence 1** Delhi: Rs 30,000 to Rs 20,000 (−33.3%). Other three cities combined: Rs 70,000 to Rs 70,000 (0.0%).
- **Evidence 2** (illustrative, not computed here) Delhi late-delivery rate rose from 8% to 19%.
- **Recommendation** "Review Delhi courier capacity this week; re-check revenue by city on 14 March."
- **Answer** One paragraph a director can read in 20 seconds.
- **Check** Other cities: 41,000 + 19,000 + 10,000 = 70,000 in February and 40,000 + 20,000 + 10,000 = 70,000 in January, so "0.0%" is true.
- **Why it matters** The reader leaves knowing what happened, why we think so, and what to do.

Bad title versus good title

Weak title	Strong title	Why
Revenue by city	Delhi lost 33% of its revenue in February	States the finding
Orders over time	Orders dipped after 12 Feb, the day the courier changed	Points to the cause

Customer segments	3 customers produce 76% of revenue	Gives the so-what (our RFM example, Chapter 12)

Storytelling techniques that work. *Context, conflict, resolution:* "Revenue was steady (context). In February it dropped 10% (conflict). The cause is one city, and a fix is available (resolution)." *Show before and after, not just after. Use a single number big on the slide* when the message is one number. *Annotate the chart* with the event ("courier change") instead of leaving the reader to guess.

The picture. Imagine a line chart: x-axis is days of February, y-axis is Delhi orders per day. Grey line for the other cities, a thick coloured line for Delhi, and a vertical dashed line labelled "12 Feb: courier change". The viewer sees the problem in 3 seconds, without reading a legend.

Connection to the real job. Every dashboard, every deck, every email follows this. Power BI visuals (see the Power BI chapters) are the same: one message, a title that states it, and a purposeful colour.

Applications

- Executive reviews: one slide per decision.
- Product: launch retrospectives.
- Marketing: campaign post-mortems.
- Finance: monthly variance commentary.

COMMON MISTAKE

Showing everything you did. The reader does not want your journey, they want your destination. Put the journey in the appendix. Another mistake is a dual-axis chart that makes two unrelated lines look correlated.

Check yourself

1. What goes first in a story for executives? Answer: The headline with the finding and the action.
2. What is an action title? Answer: A chart title that states the message in a sentence.
3. Why use grey for most lines? Answer: So the one highlighted series draws the eye.

14.3 Writing: Executive Summaries, Emails and the

Analysis Memo

INTUITION

A memo is a flight briefing. The pilot hears the destination, the weather problem and the decision needed in 30 seconds. The detailed charts of every route sit in a folder if anyone wants them.

BLUF. Military writers call it *Bottom Line Up Front*: the first lines hold the conclusion and the ask. A busy reader may only read those lines.

The one-page analysis memo template

```
TITLE:       Delhi revenue drop, February 2025
TO / FROM / DATE
1. BOTTOM LINE (2–3 sentences)   What happened, why, what I recommend,
what I need from you.
2. KEY NUMBERS (3 bullets)       Each with a comparison.
3. WHY (short paragraph + 1 chart)  Evidence for the cause, with the
confidence level.
4. RECOMMENDATION / NEXT STEPS   Action, owner, date. If options: A, B
with trade-offs.
5. RISKS AND LIMITS              What could make this wrong. What I did
not check.
6. APPENDIX                      Definitions, data sources, method, links
to code.
```

Worked problem 14.3.1 Write the Bottom Line and Key Numbers of the memo for our Delhi case.

- **Bottom line** "February revenue fell 10% (Rs 1,00,000 to Rs 90,000). The whole fall is Delhi (−33%); the other cities together were flat. The likely cause is a rise in late deliveries after the courier change on 12 February. I recommend adding courier capacity in Delhi this week and re-checking on 14 March. I need approval for the courier budget by Friday."
- **Key numbers** (1) Total revenue: Rs 90,000 vs Rs 1,00,000 in January (−10%). (2) Delhi: Rs 20,000 vs Rs 30,000 (−33%). (3) Other cities: Rs 70,000 vs Rs 70,000 (flat).
- **Risks** "Only two months of data; the link to the courier is a correlation, not proven. Next step: compare orders for couriers before and after, by pincode."
- **Answer** The memo fits on one page and leads with the decision.
- **Check** Does the bottom line include all four: what, why, what to do, what I need? Yes.

- **Why it matters** A manager can forward the first paragraph to their boss unchanged. That is the highest compliment for a memo.

Calibrated language. Match words to the strength of evidence:

Evidence	Say
Tested with A/B test, tight CI	"The change increased conversion by 1.0 point (95% CI 0.3 to 1.7)"
Strong pattern, no experiment	"The data strongly suggests"
Weak or single source	"We saw ... but cannot yet say why"
Nothing	"I do not know yet. Next step: ..."

Avoid "proves" for observational data and avoid "significant" without saying statistical or practical.

Executive summary versus email. An *executive summary* (3 to 5 sentences) tops a long report and mirrors the memo's bottom line. An *email* should have a subject line with the point ("Delhi caused the Feb revenue drop: decision needed by Fri"), the bottom line in the first sentence, and attachments only as support.

Worked problem 14.3.2 Improve this email subject and opening: "Analysis attached. Subject: Report." Write a version that works.

- **Step 1** Put the finding in the subject.
- **Step 2** First sentence: the answer and the ask.
- **Step 3** Details below the line.
- **Answer** Subject: "Delhi caused February's 10% revenue drop. Courier budget decision needed by Fri". Opening: "Revenue fell 10% in February; all of it is Delhi, likely due to late deliveries after the courier change. Can you approve extra courier capacity by Friday? Details and chart below."
- **Check** A reader who reads only the subject gets the message.

Writing tips for non-native speakers. Short sentences. One idea per sentence. Active voice ("Delhi lost 33%", not "A loss of 33% was observed in Delhi"). Concrete numbers instead of "significant" or "large". Read it aloud. AI tools can polish grammar. You still own the facts, so

check every number it returns.

Connection to the real job. The memo is the permanent artifact. Slides disappear, but a clear memo in a shared folder gets searched for months later and builds your reputation.

Applications

- A/B test result memos (Chapter 13): lift, CI, guardrails, decision.
- Monthly business reviews.
- Project closing notes for the portfolio (Chapter 15).
- Incident or root-cause write-ups, such as why a hospital clinic ran late or why a delivery route failed.

COMMON MISTAKE

Burying the recommendation at the bottom, or giving no recommendation ("further analysis is needed" every time). Always recommend something, even if it is "do not ship yet, and here is what would change my mind".

Check yourself

1. What does BLUF stand for? Answer: Bottom Line Up Front.
2. Why include a "Risks and limits" section? Answer: It builds trust and shows what could make the conclusion wrong.
3. Replace "proves" for an observational finding. Answer: "strongly suggests".

14.4 Presenting and Handling Questions

INTUITION

In a live presentation you are the pilot announcing from the cockpit. Passengers do not need the engine details; they need to know the route, the problem and that you are in control. Questions are not attacks, they are the audience trying to land the plane with you.

Structure of a 10-minute presentation.

```
1 min   Context and the question (one sentence on why this matters)
1 min   The answer (headline + recommendation)
5 min   Two or three pieces of evidence (one chart each)
2 min   Risks, what we are unsure about
1 min   The ask: decision, owner, date
```

Tell them what you will say, say it, then say what you said. Keep one idea per slide. Rehearse out loud once.

Handling questions.

Situation	What to do	Example
You know the answer	Answer in one sentence, then stop	"Delhi was 33% down; the others were flat."
You do not know	Say so, and promise a time	"I have not split that by category. I will send it by 5 pm tomorrow."
The question is out of scope	Note it, park it	"Good question, outside today's scope; I will add it to the follow-ups."
Someone challenges a number	Define the number	"Revenue here means delivered orders after refunds. With cancelled orders the fall is 12%."
Hostile or confused	Restate the question, stay calm	"If I understand, you are asking whether the courier is the only cause. We have not proven that."

Admitting uncertainty honestly. Say how sure you are and why: "I am fairly confident about the Delhi finding because the numbers are large and the timing matches. I am less sure about the cause. Two things would raise my confidence: courier data by pincode, and checking the same effect in Pune, which uses a different courier." People trust analysts who show their limits more than those who bluff, and those who bluff are caught eventually.

Worked problem 14.4.1 In a review, the CFO asks: "Are you sure it is the courier?" You are not. Give an answer that is honest and useful.

- **Step 1** Say what you know: "Delhi is the only city down, and the drop started the week of

the courier change."

- **Step 2** Say what you do not know: "I have not yet proven the courier is the cause, only the timing."
- **Step 3** Say the test: "A check on late deliveries by courier and pincode would settle it, which takes one day."
- **Answer** "Not yet certain. The timing fits, other explanations are not excluded, and I will confirm by Wednesday with courier-level data."
- **Check** The answer has a fact, a limit and a next step. It would also hold up if the courier turned out not to be the cause.

Worked problem 14.4.2 Rewrite a slide title "Findings" and its four bullet points so the room can follow in 10 seconds.

- **Step 1** Pick the single most important finding.
- **Step 2** Put it as the title. Put the supporting number on a chart.
- **Step 3** Move other bullets to the appendix.
- **Answer** Title: "Delhi is the only city that fell: -33% versus flat elsewhere", one bar chart with Delhi coloured, one line of text about the likely cause.
- **Check** Cover the slide body: does the title alone give the message? Yes.

Stakeholder types. *Executives* want the answer and the decision. *Managers* want the answer plus the plan. *Engineers* want the method and the data. *Peers* want the code. Prepare one deck and tailor the first 60 seconds to the audience.

Connection to the real job. Weekly business reviews, quarterly planning, sprint demos, and interviews (you will present a portfolio project to a panel). The structure above works for all.

Applications

- Business reviews and steering meetings.
- Interview presentations of a case study.
- Customer-facing analytics (reports to clients).

COMMON MISTAKE

Reading the slide aloud, answering a question you did not understand, and guessing a number on the spot. A wrong number stated confidently damages trust much more than "I will get back to you".

Check yourself

1. What do you do when you do not know the answer? Answer: Say so, and promise a time to answer.
2. How long should the evidence part of a 10-minute talk be? Answer: Around five minutes, two to three charts.
3. Why define the number when challenged? Answer: Most disagreements over numbers are really disagreements over definitions.

14.5 Reproducibility and Documentation

INTUITION

A recipe that works only in your kitchen, with ingredients you did not write down, is not a recipe. If someone cannot repeat your analysis next month (or you cannot, after a holiday), the numbers cannot be trusted.

What to document.

Artifact	What it holds	Example
Data dictionary	Each column: meaning, type, units, allowed values, source	`order_value`: rupees, after discount, before shipping
Metric definitions	The formula and filters for each KPI	"Revenue = SUM(order_value) WHERE status = 'delivered'"
Assumptions log	Judgement calls you made	"Orders under Rs 10 treated as test orders and removed"
README	How to run the analysis from scratch	"Run `01_clean.py`, then `02_analysis.ipynb`"
Version control	History of changes to SQL and notebooks	Git commits with messages
Change log	What changed in a dashboard or number, and why	"14 Mar: switched to delivered-only revenue"

Worked example 14.5.1 Build the first draft of a data dictionary from a DataFrame, so you only need to type the meanings. Before you run it: what dtype will the date column show?

```
import pandas as pd
d=pd.DataFrame({"city":["Delhi"]*3+["Pune"],"delay_flag":
[1,0,0,1],"order_value":
[450.0,None,520.5,300.0],"order_date":pd.to_datetime(["2025-02-01"]*4)})
dd=pd.DataFrame({"column":d.columns,"dtype":d.dtypes.astype(str).values,"n
ulls":d.isna().sum().values,"example":[d[c].dropna().iloc[0] for c in
d.columns]})
print(dd.to_string(index=False))
```

Real output:

```
     column          dtype  nulls             example
       city            str      0               Delhi
 delay_flag          int64      0                   1
order_value        float64      1               450.0
 order_date datetime64[us]      0 2025-02-01 00:00:00
```

Line by line: `d.dtypes.astype(str)` turns types into text. `d.isna().sum()` counts missing values per column. The list comprehension picks the first non-missing value from each column as an example. The text type shows as `str` in recent pandas versions and as `object` in older ones, so do not be surprised if your screen differs. The date type shows `datetime64[us]` here and may show `[ns]` elsewhere, depending on the version.

- **Answer** You get a four-row skeleton with names, types, missing counts and examples. You then add two columns by hand: "meaning" and "units". The output flags that `order_value` has one missing value, which belongs in the assumptions log ("1 of 4 rows has no order value; excluded from revenue").
- **Check** The `nulls` column sums to 1, matching the single `None` in the input.

Reproducible notebook habits. Run "Restart and run all" before sharing; put paths and parameters at the top; set random seeds (as in Chapter 13's simulations); do not edit raw data by hand, always clean in code; keep SQL in files, not only in a BI tool; and save a copy of the exact query result for the memo. See Chapter 1 for the toolbox map.

Worked problem 14.5.2 A colleague asks, "Why is revenue Rs 4,50,000 in your report but Rs 4,72,000 in the dashboard?" How does documentation answer this in 5 minutes?

- **Step 1** Open the metric definition of both: the dashboard includes cancelled orders; the

report does not.

- **Step 2** Check: difference = 4,72,000 − 4,50,000 = Rs 22,000, and cancelled orders in the period sum to Rs 22,000.
- **Step 3** Decide: the official definition is delivered-only, so update the dashboard and add a change-log line.
- **Answer** Two definitions caused the gap. A written definition ended the debate in minutes instead of days.
- **Check** 450,000 + 22,000 = 472,000. The arithmetic ties out.

Connection to the real job. Governance of certified datasets, metric stores and code review all rely on these artifacts. In job interviews, mentioning "I keep a data dictionary and an assumptions log" signals seniority.

Applications

- Teams: onboarding a new analyst in a day.
- Audit and finance: showing where a number came from.
- Portfolio: a GitHub project with README and dictionary stands out (Chapter 15).

COMMON MISTAKE

Documenting after the project is "done", when the details are forgotten. Write the assumptions as you go, in one running file.

Check yourself

1. Why seed random numbers? Answer: So anyone can reproduce the same results.
2. Name two things a data dictionary holds. Answer: Meaning and units (also type, allowed values, source).
3. Where do judgement calls go? Answer: The assumptions log.

Chapter recap

- Before opening data, ask what decision the answer supports, how the metric is defined, the scope, and the deadline. Send the brief back in writing.
- Handle scope creep by stating the time cost and offering a choice.
- Tell the story from the top: headline, evidence, recommendation. Details go in the appendix.

- Use action titles, one message per chart, grey for context and one colour for the point.
- A memo has a bottom line, key numbers, why, recommendation, risks and an appendix. Always recommend something.
- Match your words to the strength of evidence: "increased", "suggests", "unclear".
- In presentations, answer briefly, say "I do not know, I will confirm by [time]", and define numbers when challenged.
- Reproducibility means a data dictionary, metric definitions, an assumptions log, a README and version control.

Pattern	Use
SPARC question	Scope, Purpose, Audience, Rules, Clock
Pyramid	Headline, evidence, detail
BLUF memo	Bottom line, numbers, why, recommendation, risks, appendix
Action title	Finding as a sentence, with the number
Honest uncertainty	What I know, what I do not, how I will find out
Metric tie-out	Explain a gap as definition A minus definition B

Practice set

Problem 1 (easy). Rewrite the chart title "Monthly orders" as an action title. Orders were 250 in January and 240 in February. **Answer:** "Orders fell 4% in February (250 to 240)". Check: (240 − 250) / 250 = −4%.

Problem 2 (easy). What are the three levels of the pyramid? **Answer:** Headline, evidence, detail.

Problem 3 (easy). A stakeholder asks for "a customer list". Name two clarifying questions. **Answer:** "What decision will the list support (for example, who to call)?" and "How should customers be selected (spend, recency, city)?"

Problem 4 (medium). Revenue in the report is Rs 8,20,000 and in the dashboard Rs 8,45,000.

Cancelled orders total Rs 25,000. Is the gap explained, and what do you do? **Answer:** 8,20,000 + 25,000 = 8,45,000, so the dashboard includes cancelled orders. Agree one official definition (usually delivered only), fix the dashboard, and log the change.

Problem 5 (medium). Find the weakness: "Our sophisticated analysis proves that the new campaign caused a significant increase." **Answer:** "Proves" is too strong for observational data. "Significant" is unspecified. It gives no number, comparison or interval. Better: "Orders were 12% higher during the campaign than in the prior four weeks; a holdout test would confirm that the campaign caused it."

Problem 6 (medium). Using the Delhi data, what share of the total fall is Delhi's, and what percent did the other cities combined change? **Answer:** Delhi fell 10,000 of the total 10,000 fall, which is 100%. Others: (70,000 − 70,000) / 70,000 = 0%.

Problem 7 (medium). Write a subject line for an email telling the head of marketing that campaign B has ROAS 6.0 versus A's 3.0 (Chapter 12). **Answer:** "Shift budget from Campaign A (ROAS 3.0) to B (6.0)? Decision needed by Friday." It has the finding and the ask.

Problem 8 (find the bug). A presenter reads a 7-bullet slide aloud, then answers a question with "I think it was around 15%". What are the two errors and the fixes? **Answer:** (1) Slide is a script, not a visual: one message, a chart, a title. (2) A guessed number: say "I will confirm the exact figure by tomorrow".

Problem 9 (scenario). Your manager adds three new analyses on Wednesday; the deadline is Thursday and each takes half a day. What do you say? **Answer:** "Each adds half a day, so 1.5 days in total. I can deliver the original on Thursday and the extras on Monday, or everything Monday. Which do you prefer?" Time cost plus a choice.

Problem 10 (interview). "Tell me about a time you had to explain a complex analysis to a non-technical stakeholder." **Answer:** Use STAR (Situation, Task, Action, Result). Example: the Delhi case: the situation was a 10% revenue fall; the task was to explain it to the director; the action was a one-page memo with the headline "Delhi alone", one chart, and a recommendation; the result was an approved courier budget and recovered revenue. Mention that you checked definitions first and stated what was uncertain. Use your own real example with real numbers in the interview.

Problem 11 (interview). "What would you do if you found an error in a report that was already

sent to the CEO?" **Answer:** Tell the owner immediately (do not hide it), quantify the impact (what changed, by how much), send a short correction with the right numbers and cause, then fix the process (a check, a test, a reviewer) so it cannot recur.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in India moving into data roles, and English is my second language. Teach me Chapter 14 topic: how to clarify a stakeholder request before analysing it, including decision owner, metric definition, scope, deadline and scope creep. Give me three realistic vague requests, one at a time. For each, ask me to write my clarifying questions, then give feedback and a model set of questions in simple English.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving into data roles. Teach me data storytelling with the pyramid principle: headline, evidence, recommendation, and action titles for charts. Give me five weak chart titles and a small data table for each, and ask me to rewrite them as strong titles. After each answer, show a better version and explain why in simple words.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving into data roles. Review my analysis memo using the BLUF structure: bottom line, key numbers, why, recommendation, risks, appendix. Check whether the first three lines alone give the finding, the action and the ask, whether numbers have comparisons, and whether the language matches the strength of evidence. Give a rewrite of the bottom line. My memo: [PASTE HERE]
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving into data roles, and English is my second language. Run a practice presentation Q and A with me. I will paste a 5-line summary of an analysis: [PASTE HERE]. Ask me one hard stakeholder question at a time (a
```

challenged number, a causation question, a scope request, a question I cannot answer). After each reply, rate it 1 to 5 and show a calm, honest model answer.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Help me build reproducible habits: a data dictionary, metric definitions, an assumptions log, a README and version control for an analysis project. Explain each artifact with a small example, then ask me to draft a data dictionary and assumptions log for a dataset I describe. Review my draft and tell me what a senior analyst would add.

CHAPTER 15

CAPSTONE PROJECTS AND THE ANALYST INTERVIEW BANK

Build proof that you can do the job, then practise saying it out loud.

Hiring managers for analyst roles rarely hire on certificates. They hire on evidence: a project you can walk them through, and an interview where you think clearly under pressure. This chapter gives you both. First, eight portfolio projects with a clear scope, a checklist and a way to present each one, including a project on order-processing times, a classic operations dataset. Then the interview side: a step-by-step method for the famous case question "DAU dropped 10%, why?" with a worked example, followed by an interview bank covering metrics cases, SQL, Excel and Power BI, statistics, A/B testing, data cleaning and behavioural questions. Everything numeric below was run or checked by hand. By the end you should have a plan for your next eight weeks and a method you can use in any case interview. This is the last teaching chapter: it pulls together Chapters 1 to 14.

15.1 Capstone Projects: Your Proof of Skill

INTUITION

A cook is hired after you taste the food, not after you read the recipes. A portfolio project is a dish. It shows you can pick ingredients (data), cook (analysis), and serve (communication). Three excellent dishes beat ten half-cooked ones.

The eight projects.

#	Project	Skills shown	Chapter to reuse	Deliverable
1	Order-processing time analysis	SQL, pandas, charts, operations insight	4, 5, 8	Notebook and 1-page memo
2	Excel dashboard (Power Query, PivotTables)	Excel, refresh without copy-paste	2, 3	Workbook and 2-minute screen

				recording
3	Power BI sales dashboard	Star schema, DAX time intelligence, RLS	Power BI chapters	.pbix and screenshots
4	SQL case study: funnel and cohort retention	SQL, product thinking	12	SQL file and write-up
5	A/B test analysis	Statistics, honesty	13	Report with CI, SRM check
6	RFM customer segmentation	pandas, segmentation, actions	12	Notebook and segment playbook
7	Marketing or finance case with recommendations	Domain, storytelling	12, 14	Memo
8	A written case study for each, on GitHub or LinkedIn	Communication	14	Published posts

How to scope a project (so you finish it). Use this recipe: (1) one business question in one sentence; (2) a dataset you can legally use (public datasets, simulated data you generate, or your own anonymised work records); (3) a time box (one to two weeks at one hour a day); (4) a definition of "done": a memo and a repository others can run.

Project 1 in detail: the order-processing time analysis (a signature piece). Operations questions such as "which kind of order is slow, and how often is it late?" show the whole analyst loop with numbers a manager cares about: percentiles, orders per minute and where the delay sits. The code below runs on **simulated** order data (log-normal processing times, which is how real waiting and processing times look: a body of quick orders and a long tail). In real life you would load your own CSV with `pd.read_csv`. Before you run it: which order type do you expect to have the worst 95th percentile?

```
import pandas as pd, numpy as np
rng=np.random.default_rng(11)
rows=[]
for kind,med,sig,n,err in [("standard",40,0.5,600,0.01),
("express",30,0.6,1000,0.02),("international",90,0.7,400,0.06)]:
    t=np.exp(rng.normal(np.log(med),sig,n)).round(0)
    ok=rng.random(n)>err

rows.append(pd.DataFrame({"order_type":kind,"minutes":t,"on_time":ok}))
o=pd.concat(rows,ignore_index=True)
o["ts"]=pd.Timestamp("2025-03-01
10:00:00")+pd.to_timedelta(np.sort(rng.integers(0,600,len(o))),unit="m")
g=o.groupby("order_type").agg(orders=("minutes","size"),avg=
("minutes","mean"),median=("minutes","median"),
   p95=("minutes",lambda s:s.quantile(0.95)),late_pct=("on_time",lambda s:
(1-s.mean())*100)).round(1)
print(g)
dur=(o.ts.max()-o.ts.min()).total_seconds()/60;
print("orders/min",round(len(o)/dur,2),"duration",dur)
```

Real output:

```
               orders    avg  median    p95  late_pct
order_type
express          1000   35.9    30.0   77.0       1.7
international     400  106.4    86.0  246.2       5.0
standard          600   45.3    41.5   87.0       1.0
orders/min 3.34 duration 599.0
```

Line by line: `np.exp(rng.normal(np.log(med), sig, n))` draws log-normal values whose typical (median) size is `med` minutes. `on_time` is True with probability `1 - err`. The named aggregation builds one row per order type: order count, mean, median, a 95th percentile through `quantile(0.95)` inside a `lambda` (a small unnamed function), and late percent. Throughput is total orders divided by the duration in minutes: 2,000 orders / 599 min = 3.34 per minute.

Answer International orders are the bottleneck: the 95th percentile is 246 minutes against 87 (standard) and 77 (express), and the late rate (5.0%) is the highest. Their mean (106.4 minutes) is well above their median (86.0 minutes), the sign of a long tail. The memo: "International orders take 4 hours at the 95th percentile, with 5% late, at 3.3 orders per minute overall. Target: 95th percentile under 2.5 hours and late rate under 1%. Recommendation: review the customs and packing steps first." **Check** 400 + 600 + 1000 = 2,000 orders, and 2,000 / 599 = 3.34.

This project shows SQL (load the same data to a table), pandas, a chart (95th-percentile bar with a target line), and a recommendation. Do not publish confidential company data: use public or fully anonymised or simulated data, and say so in the README.

Checklist for every capstone (the "done" test).

- A one-sentence business question and a named decision owner at the top.
- Data source, licence or "simulated" label, and a data dictionary.
- Cleaning steps logged in an assumptions file.
- A notebook that runs top to bottom ("Restart and run all").
- One memo (Chapter 14 template) and 3 to 5 charts with action titles.
- A README: what, how to run, key findings, limits.
- A 150-word LinkedIn or blog post with one chart.

Real-world mapping

Item	Example value	Where it appears in the job
Public dataset	Online retail transactions, NYC taxi trips	Case study fuel
Time box	10 days at 1 hour a day	Sprint planning
README	"Run `make all`"	Code review
Memo	One-page recommendation	Stakeholder review

Worked problem 15.1.1 You have 8 weeks and about 1 hour a day (56 hours). Plan the portfolio.

- **Step 1** Choose three flagship projects that show different skills: #1 order-processing analysis (operations insight), #4 or #6 (SQL and product thinking) and #3 Power BI dashboard (BI tool).
- **Step 2** Budget hours: 15 hours each for three flagship projects = 45 hours.
- **Step 3** Use the remaining 11 hours for write-ups (3 hours each = 9) and a 2-hour buffer.
- **Answer** 45 + 9 + 2 = 56 hours: three finished projects and three published case studies in 8 weeks.
- **Check** 15 × 3 = 45, 3 × 3 = 9, 45 + 9 + 2 = 56.
- **Why it matters** Finished and published beats started. Put the best project first on your GitHub profile.

Worked problem 15.1.2 Write the "story" for your order-processing project in 4 lines for the resume.

- **Step 1** Action verb, what, method, number.
- **Answer** "Analysed 2,000 order-processing records with pandas and SQL; found international orders at a 95th percentile of 4.1 hours and 5% late; recommended a review of customs and packing targeting under 2.5 hours. Published notebook and memo on GitHub."
- **Check** It has a method, numbers and an outcome. Replace the numbers with those of your real (anonymised) project.

Connection to the real job. Interviewers open your repository and ask "walk me through your project". They test whether you did it yourself (they ask why you chose that metric) and whether it was useful.

Applications

- Career switch: evidence replaces missing job titles.
- Internal move: a data project at your own company (anonymised) is the strongest proof.
- Freelance and contests: the same artifacts.

COMMON MISTAKE

Copying a tutorial project (Titanic, Iris) with no business question. Also a repository with no README, no explanation of choices, and charts with default titles. Hiring managers have seen 500 Titanic notebooks.

Check yourself

1. What is the "done" test for a capstone? Answer: A memo and a repository that someone else can run.
2. Why use simulated data openly? Answer: It avoids confidentiality problems if you label it clearly.
3. Which project shows operations insight with percentiles and a late rate? Answer: The order-processing time analysis.

15.2 The Metrics Case: "DAU Dropped 10%, Why?"

INTUITION

A doctor with a patient who says "I have a fever" does not guess the disease. She follows a routine: check the thermometer is right, check the history, run tests that split the possible causes into groups, then narrow. A metrics case is the same routine for a number that moved.

The method (eight steps). Remember it as **C-V-S-D-H-P-R**: Clarify, Verify, Segment, Decompose, Hypothesise, Prioritise, Recommend.

1. **Clarify** the metric and the time: What is DAU here (distinct users who did any action? a logged-in action)? Drop compared to when: yesterday, last week, same day last year? Sudden or gradual? Is it a complete drop or recoverable?
2. **Verify** the data before the business: a tracking bug, a late-loading pipeline, a changed definition, a timezone cut-off. A bank manager does the same: before blaming the cashiers for a long queue, check that the ticket counter was not reset.
3. **Segment** the drop to localise it: by time (what day did it start), platform (Android, iOS, web), geography, user type (new vs returning), acquisition channel, app version, feature.
4. **Decompose** the metric (Section 1.4 of Chapter 1): DAU = new-user DAU + returning-user DAU, or DAU = MAU × stickiness.
5. **Hypothesise** causes in two groups: *internal* (release, bug, pricing, outage, marketing change) and *external* (seasonality, festival, competitor, holiday, a news event).
6. **Prioritise** by checking the hypotheses that the segments point to, cheapest first.
7. **Recommend**: one action and one follow-up metric.
8. State uncertainty honestly.

Worked example 15.2.1 (predict, run, explain) DAU fell from 100,000 to 90,000 (−10%). You split by new versus returning users. Before you run it: could the drop be spread evenly across both groups?

```
import pandas as pd
df=pd.DataFrame({"segment":["new users","returning users"],"before":
[30000,70000],"after":[21000,69000]})
df["chg"]=df.after-df.before; df["pct"]=(df.chg/df.before*100).round(1);
print(df); print(df[["before","after","chg"]].sum().to_dict())
ch=pd.DataFrame({"channel":["paid ads","organic","referral"],"new_before":
[15000,10000,5000],"new_after":[6000,10000,5000]})
ch["chg"]=ch.new_after-ch.new_before; print(ch)
```

Real output:

```
           segment  before  after   chg   pct
0        new users   30000  21000 -9000 -30.0
1  returning users   70000  69000 -1000  -1.4
{'before': 100000, 'after': 90000, 'chg': -10000}
      channel  new_before  new_after   chg
0     paid ads       15000       6000 -9000
1      organic       10000      10000     0
2     referral        5000       5000     0
```

Line by line: the first table splits the total change into the two user types. `pct` is change relative to the before value. The second table splits the new users by acquisition channel.

- **Answer** New users fell 30% (−9,000), and returning users fell only 1.4% (−1,000). So 90% of the fall (9,000 of 10,000) is new users. Within them, paid ads account for all of the loss (15,000 to 6,000, −9,000), while organic and referral are flat. The product itself (returning users) is healthy. The problem is paid acquisition.
- **Check** −9,000 − 1,000 = −10,000 equals 90,000 − 100,000, and 15,000 + 10,000 + 5,000 = 30,000 new users before, 6,000 + 10,000 + 5,000 = 21,000 after.
- **Hypotheses to test next.** Internal: an ad campaign was paused or the budget was cut (check spend), tracking broke for ad-click landing pages (check the attribution tags and install events), the ad creative or targeting changed. External: festival season raised competitors' bids, an ad platform policy changed.
- **Recommendation.** Check ad spend by day against the drop date. If spend fell, the cause is budget; if spend is flat while installs fall, check tracking and cost per install. Re-check DAU after the fix, and monitor returning-user retention as a guardrail.

The picture: the issue tree

```
 DAU −10%
  ├── Data problem? (tracking, pipeline, definition, timezone)       <-
check first
  ├── Internal change? (release, outage, pricing, marketing spend)
  │     ├── by platform / app version
  │     └── by feature / funnel step
  └── External? (season, festival, competitor, news)
        └── compare with last year and with industry
 Segment: new vs returning, channel, geography, platform  --> find where
the drop lives
```

Read from the top down. At each level, do a quick segment cut to decide which branch deserves your next hour.

Worked problem 15.2.2 In a different case, DAU fell 10% (100,000 to 90,000) and the segmentation by platform shows: Android 50,000 to 45,000, iOS 30,000 to 29,000, web 20,000 to 16,000. Where is the problem?

- **Step 1** Compute changes: Android −5,000 (−10%), iOS −1,000 (−3.3%), web −4,000 (−20%).
- **Step 2** Sum: −5,000 − 1,000 − 4,000 = −10,000. Total matches.
- **Step 3** Shares of the drop: Android 50%, iOS 10%, web 40%.
- **Answer** No single platform explains it. Android is 50% of the drop, but it fell 10%, the same as the total, which means Android is just proportional; web fell 20%, twice the rate, and holds 40% of the drop although it has only 20% of users. Web is the unusual one. Check web releases, SEO traffic and tracking first, then look at Android more broadly.
- **Check** 5000 / 10000 = 50%, 1000 / 10000 = 10%, 4000 / 10000 = 40%.
- **Why it matters** Judge a segment by its rate of change compared with the total, and not only by the size of its change.

Words to use in the room. "Before I theorise, let me check the data is right." "I would like to split this by..." "The most likely explanation is X, because Y. I would confirm it by Z." The interviewer wants a structured thinker, not the right answer.

Connection to the real job. This is the real weekly routine of a product or growth analyst when a number falls. The same issue tree works for revenue, conversion, delivery time (!), errors, or churn.

Applications

- Product: DAU, retention, feature usage drops.
- E-commerce: conversion or revenue drops.
- Operations: SLA breaches.
- Operations: a rise in 95th-percentile delivery time, which follows the identical routine (data check, segment by city, courier, week).

COMMON MISTAKE

Jumping to a favourite hypothesis ("it must be the new release") before verifying the data and segmenting. Also forgetting to ask whether the drop is within normal variation: check the metric's usual daily range first. A 10% drop on a metric that normally moves 8% is a different story from 10% on a metric that moves 1%.

Check yourself

1. What do you check first? Answer: Whether the data and the definition are right.
2. Name three ways to segment a drop. Answer: Platform, new vs returning, channel (also geography, version, time).
3. Platform A is 50% of users and 50% of the drop. Is it the problem? Answer: No, it is proportional, so look for a segment whose share of the drop is bigger than its share of users.

15.3 More Case Questions, With Model Answers

INTUITION

Case questions come in a few families. Learn the shape of each family and every new question looks familiar.

Words you need first. *Fermi estimate*: estimating a number from simple known facts, with stated assumptions. *Guardrail*: a metric that must not worsen. *Trade-off*: gaining one thing at the cost of another.

Family	Typical question	Skeleton of the answer
Metric dropped or rose	"Orders are up but revenue is flat"	Decompose (orders × AOV), segment, check mix shift
Define success	"How would you measure success of a new Wishlist feature?"	Goal, north star, inputs, guardrails, how to test
Choose a metric	"What KPIs for a food-delivery app?"	North star, inputs, outputs, guardrails

Estimate	"How many chai cups are sold in Bengaluru per day?"	Fermi: population, share, frequency
Experiment	"A/B test shows +3% but one guardrail worsened"	CI of each, cost-benefit, rollout plan
Prioritise	"Three analysis requests, one analyst"	Impact, effort, deadline, owner

Worked problem 15.3.1 "Orders rose 20% but revenue was flat. Why?" January: 1,000 orders, AOV Rs 500. February: 1,200 orders, revenue Rs 5,00,000.

- **Formula** revenue = orders × AOV, so AOV = revenue / orders
- **Plug in** February AOV = 500,000 / 1,200
- **Compute** 416.67. January revenue = 1,000 × 500 = 5,00,000. The change in AOV = (416.67 − 500) / 500 = −16.7%.
- **Answer** Orders +20% but AOV −16.7%, so revenue is flat: 1.20 × 0.8333 = 1.00. Likely causes: discounting, a flood of low-value orders (a new cheap category, free-delivery threshold changes, or a new city with smaller baskets). Check the AOV by category and by acquisition channel, the discount rate and the order mix. Is margin still healthy? Flat revenue with a rise in order cost (delivery, support) is worse than it looks.
- **Check** 1,200 × 416.67 = 500,000 and 1.2 × 0.8333 = 1.0.
- **Why it matters** Volume alone is not success. Link to the metric split of Chapter 1.

Worked problem 15.3.2 (Fermi) "How many cups of chai are sold in a city of 12 million in a day?" State assumptions clearly.

- **Step 1** Adults and teens who might drink: 70% of 12 million = 8.4 million.
- **Step 2** Share who drink chai: 60%, so 5.04 million.
- **Step 3** Cups per drinker per day: 2, of which 40% are bought outside the home (the rest are made at home): 2 × 0.4 = 0.8 bought cups.
- **Compute** 5.04 million × 0.8 = 4.03 million.
- **Answer** About 4 million cups per day, as a rough estimate. The point is the structure and the stated assumptions, and each can be challenged and replaced with real data.
- **Check** Sanity: 4 million for 12 million people is 1 cup for every 3 people per day. Plausible, so the order of magnitude is reasonable.

Worked problem 15.3.3 "How would you measure success of a new Wishlist feature?"

- **Step 1** Goal: help users come back and buy things they saved.
- **Step 2** North star: orders per user within 30 days of first use. Inputs: adoption (share of active users who add an item), items per wishlist, wishlist-to-purchase rate. Guardrails: page speed, regular cart conversion, unsubscribe rate of wishlist emails.
- **Step 3** Test: an A/B test (Chapter 13), 50/50 by user, planned sample size, 2 full weeks; one primary metric.
- **Answer** Success is a significant, practically meaningful rise in orders per user, without worse guardrails, and adoption above a pre-set bar (for example 15% of weekly active users within 4 weeks).
- **Check** Does the answer name a metric, a guardrail, an experiment and a threshold? Yes.

Worked problem 15.3.4 (prioritising) You have three requests: (A) a weekly dashboard fix, 1 hour, used by 20 people; (B) a churn analysis, 3 days, the CEO wants it in two weeks; (C) an ad-hoc number for a VP in a meeting tomorrow, 30 minutes.

- **Step 1** C has a hard deadline and is cheap: do it first.
- **Step 2** A is cheap and has many users: do it second (1 hour).
- **Step 3** B is the largest, deadline in two weeks: schedule it starting tomorrow.
- **Answer** C, then A, then B. Tell all three owners the plan and the dates.
- **Check** Total today is 1.5 hours, leaving the day for starting B.

Connection to the real job. Product managers ask these questions daily. A structured, calm answer is the skill being hired.

Applications

- Product analytics interviews.
- Consulting-style case rounds.
- Stakeholder meetings.

COMMON MISTAKE

Answering with a list of every metric you know. Pick one north star, defend it, and name the guardrails. Another mistake in Fermi questions is silence: always speak your assumptions aloud.

Check yourself

1. Orders +10%, AOV −10%. Revenue change? Answer: 1.10 × 0.90 = 0.99, −1%.

2. What must a "define success" answer include? Answer: Goal, north star, input metrics, guardrails, and how to test.
3. Why state Fermi assumptions? Answer: So they can be challenged and replaced with data.

15.4 The Technical Interview Bank: SQL, Excel, Power BI, Statistics

INTUITION

Technical rounds test whether the basics are automatic. Think of it as a driving test: nobody expects racing skill, but you must handle the clutch without thinking.

SQL (the top skill). Four patterns come up in nearly every interview. All three queries below were run in SQLite on this small orders table (8 rows: `order_id, cust, city, amount, odate`).

Worked example 15.4.1 (top order per city, second-highest value, month-1 retention). Before you run it: how many Delhi orders will the first query return? (One per city.)

```
-- Top order per city
select city, order_id, amount from (
  select *, row_number() over (partition by city order by amount desc) rn
from orders
) where rn=1 order by city;

-- Second highest amount
select max(amount) as second_highest from orders where amount < (select
max(amount) from orders);

-- Month-1 retention
with f as (select cust, min(substr(odate,1,7)) m0 from orders group by
cust),
a as (select distinct cust, substr(odate,1,7) m from orders)
select f.m0, count(distinct f.cust) cohort,
 count(distinct case when a.m = strftime('%Y-%m', date(f.m0||'-01','+1
month')) then a.cust end) retained
from f join a on a.cust=f.cust group by f.m0 order by f.m0;
```

Real output:

```
city    order_id  amount
------  --------  ------
Delhi   4         900
Mumbai  6         800
Pune    8         400

second_highest
--------------
800

m0       cohort  retained
-------  ------  --------
2025-01  3       1
2025-02  2       0
```

Line by line. Query 1: `row_number() over (partition by city order by amount desc)` numbers the orders inside each city from the biggest, and `rn = 1` keeps the top one. Query 2: take the max of everything smaller than the overall max (900), which gives 800. Query 3: `f` finds each customer's first order month (the cohort), `a` lists the distinct customer-months, and the join counts cohort members who also ordered in the month after `m0`. January cohort: A, B, C joined; only A ordered in February (B and C returned in March, not in month 1), so retained = 1. The February cohort (D, E) had no March orders, so 0.

- **Answer** Delhi's top order is #4 for Rs 900; the second-highest amount overall is Rs 800; month-1 retention for January is 1 of 3 (33%).
- **Check** Hand check of A: orders on 5 Jan and 10 Feb, so retained in month 1.

Quick SQL Q&A.

Question	Short answer
`WHERE` vs `HAVING`?	WHERE filters rows before grouping; HAVING filters groups after
`RANK` vs `DENSE_RANK` vs `ROW_NUMBER`?	With ties 1,1,2: RANK gives 1,1,3; DENSE_RANK 1,1,2; ROW_NUMBER 1,2,3
How to find duplicates?	`GROUP BY key HAVING COUNT(*) > 1`
`INNER` vs `LEFT JOIN`?	LEFT keeps all left rows, with NULLs where there is no match

Why does `COUNT(col)` differ from `COUNT(*)`?	COUNT(col) skips NULLs
Running total?	`SUM(x) OVER (ORDER BY d ROWS UNBOUNDED PRECEDING)`

Excel and Power BI Q&A.

Question	Short answer
`XLOOKUP` vs `VLOOKUP`?	XLOOKUP looks left or right, defaults to exact match, and has an if-not-found argument
Fix `#N/A` from a lookup?	Check for trailing spaces and type mismatch (text vs number); wrap in `IFERROR` only after fixing the cause
PivotTable vs formulas?	Pivot for quick slicing; formulas for fixed reports; Power Query for repeatable cleaning
Measure vs calculated column (DAX)?	A measure is computed at query time in the current filter context; a column is stored per row at refresh
What is a star schema?	Facts (events, amounts) in the centre, dimensions (date, product, customer) around it, joined 1:many
Row-level security?	Rules that limit which rows each user sees, defined per role and tested with "View as"
`CALCULATE` does what?	Changes the filter context for an expression

Statistics and A/B Q&A.

Question	Short answer
Mean vs median?	Median is robust to outliers; waiting times and incomes are skewed, so use the median or percentiles
What is a p-value?	If no effect existed, the chance of seeing a result at least this extreme. It is not the chance the hypothesis is true

Correlation vs causation?	Two things moving together may have a third cause; only randomised experiments (or careful quasi-experiments) support causal claims
Type I vs Type II error?	Type I: false alarm (α). Type II: missed effect (β). Power = 1 − β
How to compute sample size?	Baseline, MDE, α, power: the 8,158 per group example of Chapter 13
What is SRM?	Arm sizes differ from the planned split more than chance allows; the test is invalid until fixed
Confidence interval in plain words?	The range of effects the data supports; if it excludes 0, the result is significant at that level

Data cleaning case. "You get a CSV of orders. What do you do?" Answer in order: (1) Understand the source and grain (one row per what?). (2) Shape, dtypes, memory (`df.info()`). (3) Missing values: how many, which columns, the pattern (Chapter 7). (4) Duplicates (`duplicated`). (5) Impossible values (negative prices, dates in 1970, future dates). (6) Inconsistent categories (`Delhi` , `delhi` , `New Delhi`). (7) Outliers: check, do not delete automatically. (8) Log each decision. Each step has a reason; say it.

Worked problem 15.4.2 A column `amount` has values [100, 120, 110, 95, 5000]. Is 5,000 an outlier by the IQR rule, and would you delete it?

- **Formula** IQR rule: outlier if value < Q1 − 1.5·IQR or > Q3 + 1.5·IQR
- **Plug in** Sorted: 95, 100, 110, 120, 5000. With the default `numpy.percentile` (which pandas `quantile` also uses), Q1 = 100 and Q3 = 120. In code: `np.percentile([100,120,110,95,5000],[25,75])` gave `100.0, 120.0` .
- **Compute** IQR = 120 − 100 = 20. Upper fence = 120 + 1.5 × 20 = 150.
- **Answer** 5,000 is far above the fence of 150, so it is flagged. Do not delete it automatically. Check the source: a bulk order (keep), a typo of 500.0 (fix after confirming), or a test order (remove). Record the decision in the assumptions log.
- **Check** The median is 110 and 5,000 is about 45 times that. With five rows the quartiles are unstable (a hand method that takes the median of each half gives Q3 = 2,560 and does NOT flag it), so the IQR rule needs enough data and a domain check.

Connection to the real job. Technical rounds are 40-60% SQL. Practise on a free site daily for

20 minutes (any SQL practice platform), and explain your answers aloud.

Applications

- SQL screens (live coding or take-home).
- Excel tests (a timed spreadsheet task).
- Power BI take-home dashboards.
- Statistics fundamentals in analyst and data-science screens.

COMMON MISTAKE

Writing the query without asking about ties, NULLs and duplicates. Say "assuming one row per order and no NULL amounts" and the interviewer sees your care.

Check yourself

1. Which function gives 1, 1, 3 for ties 1, 1, 2? Answer: RANK.
2. Why does COUNT(col) differ from COUNT(*)? Answer: COUNT(col) ignores NULLs.
3. A measure or a column for a ratio that depends on filters? Answer: A measure.

15.5 Behavioural Interviews, Your Story and Your Next 8 Weeks

INTUITION

Behavioural questions test whether you will be pleasant, honest and reliable to work with. A short, true story with a number in it beats a long general answer.

STAR. Situation (context in one sentence), Task (your responsibility), Action (what you did, step by step, "I" not "we"), Result (a number and a lesson).

Your bridge story. You are moving to analytics from another career. Do not hide it, use it:

```
 "For N years I worked in [your field]. I read reports with percentiles
and
  rates, found where problems sat, and explained them to colleagues and
managers.
  I now apply the same habit to business data: segment, find where it
hurts, recommend a fix.
  Proof: my project X, where I [one number and one result]."
```

Common behavioural questions.

Question	What they test	Skeleton
Tell me about yourself	Focus	Bridge story, 60 seconds
A time you disagreed with a stakeholder	Collaboration	Listen, show data, define metric, agree next step
A time your analysis was wrong	Honesty	Found it, told the owner, fixed it, added a check
A tight deadline with unclear requirements	Judgement	Clarify, scope, deliver the 80% version first
Why analytics?	Motivation	Decisions from data, plus the experience from your earlier career
A time you influenced a decision	Impact	Headline, evidence, recommendation (Chapter 14)

Worked problem 15.5.1 Convert this weak answer to STAR: "I helped the team with delivery delays and it went well."

- **Situation** "International orders were taking nearly 6 hours at the busiest time of the week."
- **Task** "I had to find the cause before the festival sale in 3 days."
- **Action** "I segmented the order log by order type, found international orders at a 95th percentile of 5.8 hours, correlated it with customs-hold time, and showed the warehouse team a chart."
- **Result** "They added a pre-clearance step. The 95th percentile fell to 1.9 hours, and the sale went on time. I learned to show the evidence, not just the number."

- **Answer** The rewritten answer has a number in every part and uses "I".
- **Check** Replace the numbers with your own real story; never invent achievements in an interview.

Your eight-week plan (1 hour a day).

Weeks	Focus	Output
1-2	SQL practice (20 min a day) and Project 1 (order-processing analysis)	Repository 1, case study 1
3-4	Project 4 or 6 (funnel, cohort or RFM)	Repository 2
5-6	Power BI dashboard (Project 3) or A/B analysis (Project 5)	Repository 3
7	Interview bank, mock interviews with the Prompt pack	Notes of weak areas
8	Resume, LinkedIn, 3 posts, apply to 10 roles	Applications sent

Honest note on the market. Hiring changes with time and place; I will not quote salaries or demand figures here because they go stale. Check current job posts for your target city and role to see which tools they ask for. Apply early, even before you feel ready, because interviews are also your best teacher.

Connection to the real job. The story you tell in interviews is the same you tell your manager every week: what I found, why it matters, what I suggest.

Applications

- Interviews at product companies, banks, e-commerce and IT services.
- Internal transfer pitches.
- LinkedIn profile summary.

COMMON MISTAKE

Answering behavioural questions with "we" and no number, or inventing a story. Interviewers probe details. A true, modest story withstands follow-up questions.

Check yourself

1. What does STAR stand for? Answer: Situation, Task, Action, Result.

2. Why use "I" and not "we"? Answer: They are hiring you, so they need to know what you did.
3. Your first fix when you do not know an answer in an interview? Answer: Think aloud with your method, say what you would check, and be honest about the gap.

Chapter recap

- Three finished, published projects beat many half-done ones. Make one of them the order-processing analysis, which shows operations insight.
- Every project needs a business question, a data dictionary, an assumptions log, a runnable notebook, a memo and a README.
- For "metric dropped" questions: Clarify, Verify data, Segment, Decompose, Hypothesise, Prioritise, Recommend.
- Judge a segment by its rate of change versus the total, not just by size.
- Case families (success metric, estimate, experiment, prioritisation) each have a standard skeleton.
- SQL basics (window functions, top-N, retention, duplicates, NULLs) are in nearly every technical round.
- Clean data in a fixed order and log each decision. Rules like IQR assist, but do not decide.
- Use STAR with real numbers and your bridge story from your earlier career.

Pattern	Use
C-V-S-D-H-P-R	Metric-drop case method
Issue tree	Data, internal, external; then segment
Share of drop	`segment change / total change` and compare with the segment's user share
`row_number() over (partition by g order by x desc)`	Top-N per group
`GROUP BY k HAVING COUNT(*) > 1`	Find duplicates
STAR	Situation, Task, Action, Result
Fermi	Population × share × frequency, with stated assumptions

Practice set

Problem 1 (easy). In Example 15.2.1, what share of the 10,000 fall is new users? **Answer:** 9,000 / 10,000 = 90%.

Problem 2 (easy). First thing to do when told "revenue dropped 15%"? **Answer:** Check the data and definitions (tracking, pipeline, timezone, metric definition), and clarify the period and the comparison.

Problem 3 (easy). From `p95 = 246 minutes` and `median = 86 minutes` for international orders, what does the gap say? **Answer:** A long tail: a few very slow orders, so the mean (106.4 minutes) and p95 are far above the median. Use percentiles, not averages, in the report.

Problem 4 (medium). DAU fell from 80,000 to 72,000. Android 40,000 to 38,000, iOS 25,000 to 24,000, web 15,000 to 10,000. Where is the problem? **Answer:** Changes: −2,000, −1,000, −5,000, total −8,000 (−10%). Web fell 33.3% and holds 62.5% of the drop with 18.75% of users. The problem is web.

Problem 5 (medium). What is the output? `select max(amount) from orders where amount < (select max(amount) from orders)` on amounts 900, 800, 800, 700. **Answer:** 800. It takes the maximum of values below 900. If you want the second distinct highest, this is correct; with ties it returns the tied value once.

Problem 6 (find the bug). A retention query counts a customer retained if they ordered "in a later month", but the report is titled "month-1 retention". What is wrong? **Answer:** It is rolling retention (any later month), which is higher than month-1 retention (exactly the next month). Match the join condition to the title, or retitle it.

Problem 7 (medium). Orders +25%, AOV −20%. Revenue change? **Answer:** 1.25 × 0.80 = 1.00, no change.

Problem 8 (medium). Fermi: how many cars are on the road in a city of 6 million with 1 car per 6 people, and 30% on the road at any moment? **Answer:** 6,000,000 / 6 = 1,000,000 cars. 30% on the road = 300,000. State that these are assumptions and show the structure.

Problem 9 (hard). After a pricing change, conversion fell from 5.0% to 4.5% but revenue per visitor rose from Rs 25 to Rs 27. Was the change good? **Answer:** AOV before = 25 / 0.05 = Rs 500; after = 27 / 0.045 = Rs 600. Revenue per visitor rose 8% (27 / 25 = 1.08) even though conversion fell 10% (0.045 / 0.05 = 0.9). Good for revenue if the effect is statistically solid, but check margin, repeat rate (customers may be lost long term) and returns. Ideally confirm with an A/B test.

Problem 10 (interview). "DAU on Android fell 10% yesterday only. What do you do?" **Answer:** Verify data (did logging or the SDK change?), check release history (a new app version yesterday), segment by app version and OS version, look at crash rates and Play Store ratings, compare with other days and weekdays. If only one version is down, it is probably a bug; recommend a hotfix and monitor.

Problem 11 (interview). "Which metric would you use to judge a recommendation feature, and what guardrails?" **Answer:** North star: orders per user, or revenue per user, from users who saw recommendations in an A/B test. Inputs: click-through rate on recommendations, add-to-cart rate. Guardrails: page load time, return rate, diversity of items bought (avoid over-focusing on bestsellers), and unsubscribe/complaint rates.

Problem 12 (interview, behavioural). "Tell me about a time you made a mistake in an analysis." **Answer:** A STAR answer with a real example: describe the error (for example a join that duplicated rows and inflated revenue 12%), how you found it (a reconciliation against the finance total), what you did (told the owner the same day, sent corrected numbers), and what you changed (a row-count check after every join, as in `validate=` in `merge`). End with the lesson.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in India moving into data roles, and English is my second language. Teach me Chapter 15 topic: the metrics case question "DAU dropped 10%, why?" using the steps Clarify, Verify data, Segment, Decompose, Hypothesise, Prioritise, Recommend. Walk me through one example with numbers. Then give me a new case with a small table, and let me solve it before you show your answer. Correct my structure gently.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data roles. Help me plan three portfolio projects for a data analyst job: one on order-processing or delivery times (percentiles, late rate), one on SQL funnels and cohorts, and one Power BI dashboard. For each project give a business question, a free dataset idea, a checklist, a README outline and a 150-word LinkedIn post outline. Ask me three questions first about my time and tools.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional preparing for data analyst interviews. Quiz me one question at a time on business case questions: metric dropped, define success for a feature, choose KPIs, Fermi estimation, and prioritising requests. Give a new question each time, wait for my answer, then score me 1 to 5, show a structured model answer, and tell me one habit to improve.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional preparing for data analyst interviews. Give me 10 SQL interview questions with a small table I can type into SQLite: top N per group, second highest, duplicates, running total, month-1 retention, and a window-function question with ties. Do not give solutions yet. When I paste my queries, run them mentally, find bugs, and explain the fix. My answers: [PASTE HERE]
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data roles, and English is my second language. Run a full mock analyst interview: two behavioural questions, one metrics case, one SQL question, one A/B testing question and one data cleaning case. Ask them one at a time. After each answer, score me 1 to 5 on structure, correctness and clarity, show a model answer in simple English, and at the end give me a ranked list of what to practise next.
```

CHAPTER 16

PROJECT LAB I: OPERATIONS ANALYZER, SALES DASHBOARD, EXCEL DASHBOARD

Three portfolio projects, built end to end, with the planning method that makes them finish.

Chapter 15 told you what a good portfolio looks like. This chapter shows you how to build it. A project is not "open a dataset and make charts". It is a question, a dataset you can defend, a repeatable method, a result you can measure, and a short story that a stranger can read in two minutes. In the first half of this chapter you learn the workflow that every project in this lab follows. Then you build three projects: an operations results analyzer (SQL and pandas), a sales dashboard design (Power BI), and an Excel dashboard (Power Query and PivotTables). Every Python and SQL block was run, and the output shown is the real output. Power BI and Excel steps are marked "not run here" because those tools are not available in this environment; for them I give you data that I generated and totals that I checked in pandas, so that you can verify your own dashboard against known numbers. Chapter 17 continues with three more projects.

16.1 How to Run a Project in This Area

INTUITION

Building a house without a plan gives you rooms nobody can enter. A project without a plan gives you charts nobody can use. Ten minutes of planning at the start (what is the question, what does "finished" look like) saves ten hours of rework at the end.

Plan the question and the success criteria

Write three lines before you open any tool.

1. **The question** in one sentence, with a decision owner. Weak: "Analyse orders." Strong: "Which order types are slow or failing, and is it getting worse, so the operations manager can decide where to add staff?"
2. **The output**: what the reader receives. A notebook, a dashboard, a one-page memo.
3. **Success criteria that can be measured.** Examples: "report matches an independent calculation to the last digit", "refreshes with one click", "a new reader can run it in 10 minutes".

PROFESSOR'S NOTE

If you cannot write the success criteria, you do not yet understand the question. Go back and talk to a (real or imagined) stakeholder. Chapter 1 (the analytics mindset) calls this "framing".

Repo layout

Use the same layout for every project. A reviewer who has seen one of your repos can then navigate all of them.

```
project-name/
├── README.md              what, why, how to run, key results, limits
├── requirements.txt       pinned libraries (pandas==..., numpy==...)
├── data/
│   ├── raw/               untouched inputs or generated files (never edit
by hand)
│   └── processed/         cleaned outputs you can recreate
├── src/
│   ├── make_data.py       data generator (if the data is simulated)
│   ├── load.py            raw -> database or clean table
│   └── analyze.py         the numbers and charts
├── sql/                   .sql files, one query per file, named by
purpose
├── notebooks/             exploration (optional, restart-and-run-all must
work)
├── tests/                 small checks, e.g. totals match, no duplicate
IDs
├── reports/               memo.md, charts (png), dashboard screenshots
└── .gitignore             large files, .env, *.db
```

Environment and dependencies

Create a virtual environment (an isolated folder of libraries for this project only), install a short list of libraries, and write them to `requirements.txt`.

```
python3 -m venv .venv
source .venv/bin/activate            # Windows: .venv\Scripts\activate
pip install pandas numpy matplotlib
pip freeze > requirements.txt        # not run here
```

`sqlite3` is built into Python, so you need nothing extra for a local SQL database. Library versions change over time, so recording them in `requirements.txt` is what lets someone else repeat your results.

Data sourcing or generation

You have three honest options: (1) a public dataset with a licence you can read (state the link and licence in the README); (2) data you generate with code and a fixed random seed (a seed is a starting number that makes "random" numbers repeat exactly, so everyone gets the same file); (3) your own records, fully anonymised. Never publish confidential data. Every project below uses option 2, so you can run it today, and says so in the README.

KEY INSIGHT

Simulated data has one big advantage for learning: you know the truth. If you build in a slow upward trend and your analysis finds it, you have proof that your method works. With real data you never get that proof.

Commit rhythm and good commit messages

Commit when one small thing works, not at the end of the week. A good rhythm is 6 to 12 commits per project. A commit message says what changed and why, in the present tense, in under 70 characters for the first line.

Weak message	Strong message
`update`	`Add nearest-rank percentile query in SQL`
`fix bug`	`Fix YoY: compare same period, not full year`
`final version`	`Add data checks: unique IDs, no negative times`
`changes`	`Document how to regenerate data with seed 42`

Definition of done

A project is done when all of these are true:

- The question and decision owner are stated at the top of the README.
- A stranger can follow the README and reproduce your numbers in under 10 minutes.

- At least one independent check passes (a second method gives the same total).
- Every chart has an action title ("International orders are slowest"), not a label ("Chart 1").
- Limits are stated: what the data cannot tell you.
- One screenshot or chart is in the README, and one post is drafted.

README template

```
# <Project title>

**Question:** <one sentence, with the decision it supports>
**Answer in one line:** <the main finding, with a number>

## Data
Source or "simulated with seed N". Columns and meaning (data dictionary).

## How to run
1. python3 -m venv .venv && source .venv/bin/activate
2. pip install -r requirements.txt
3. python3 src/make_data.py
4. python3 src/analyze.py

## Method
Short bullets: cleaning rules, definitions (for example how percentile is
computed).

## Results
Table or chart with an action title. Three bullets of insight.

## Checks
What I compared against and the result.

## Limits and next steps
What this cannot show. Extensions.
```

Demo and deploy options

- **GitHub repo** with the README and a screenshot: the minimum.
- **GitHub Pages or a static HTML export**: save a chart or table as HTML and publish it for free.
- **Short screen recording (1 to 2 minutes)** of a dashboard refreshing: ideal for Excel and Power BI, where a repository alone does not show the product.

- **Power BI publishing** needs an account type that your organisation or licence allows; if you cannot publish, use screenshots and a PDF export (not run here).

How to write the LinkedIn or blog post

Use five short parts: (1) the question in plain words; (2) the surprising number ("5% of orders fail, but on day 14 it was 4.7% against 1.0% on day 1"); (3) one chart; (4) what you did, in two lines; (5) the link, plus one limit you were honest about. 150 to 250 words. No jargon without a definition. Avoid the phrase "I am excited to share".

Testing and debugging checklist for analytics projects

Run this list before you call anything finished.

1. **Row counts**: rows in = rows out at every step (or you can explain every difference).
2. **Unique keys**: `df.order_id.is_unique` is True.
3. **Ranges**: no negative times, no dates in the future, no percentages above 100.
4. **Totals tie out**: a second method (SQL vs pandas, DAX vs pandas, PivotTable vs pandas) gives the same total.
5. **Join check**: after a merge, row count did not grow (a many-to-many join duplicates rows silently).
6. **Definitions written down**: "error rate = errors / all orders", "percentile = nearest rank".
7. **Time zones and date formats**: `01/02/2025` is 1 February in some countries and 2 January in others; set the format explicitly.
8. **Restart and run all** in a notebook, with a fresh kernel.
9. **Sample by eye**: print 5 random rows and check them against the source by hand.

COMMON MISTAKE

Trusting a number because the code ran without errors. Code that runs can still be wrong. Only a second method that agrees tells you the number is probably right.

Check yourself

1. Write the three planning lines for a project on hospital appointment waiting times. Answer: a sample is "Which clinics have the longest waits and are waits growing?"; output: notebook and memo; success: totals match an independent calculation and a stranger can run it in 10 minutes.

2. Why fix a random seed? Answer: so the same data is generated each time and results can be reproduced.
3. After a merge, rows went from 6,000 to 7,500. What happened? Answer: duplicate keys on the right table created extra rows (a many-to-many join).

16.2 Project: Operations Results Analyzer

Goal and why employers care

Goal: load a CSV of timed events (here: how many minutes each shop order took from payment to dispatch) into a database, then report the average, median, 90th, 95th and 99th percentile, the error rate and the trend over time, using SQL and pandas, and check that both give the same answer.

Employers care because almost every business has a table of "things that took some time and sometimes failed": orders, support tickets, loan approvals, deliveries. A candidate who reports only the average misses the slow tail that customers actually feel. A candidate who can compute percentiles in SQL, check them in pandas, and say "the error rate is rising 0.3 points a day" is doing real analyst work.

INTUITION

Think of a bank queue. The average wait is 10 minutes, but one customer in twenty waits 40. The average hides those people. The 95th percentile (the wait that 95 out of 100 customers do not exceed) shows them.

Skills used

Chapter 4 (pandas basics: `read_csv`, `groupby`, `agg`), Chapter 5 (cleaning and transforming: types, filtering), Chapter 6 (NumPy: `polyfit`), Chapter 7 (EDA step by step), Chapter 8 (data visualization), and the SQL window functions from Chapter 12 (business analytics).

Data

Generate it. This file has 3,000 orders over 14 days, three order types, a slow upward drift in processing time, and an error rate that rises by about 0.2 points a day, so we know the truth we

want to find.

```
# src/make_data.py
import numpy as np, pandas as pd
rng = np.random.default_rng(42)
n = 3000
start = pd.Timestamp("2025-03-01")
ts = start + pd.to_timedelta(np.sort(rng.integers(0, 14*24*60, n)),
unit="m")
kind = rng.choice(["standard", "express", "international"], n, p=[0.5,
0.35, 0.15])
med = pd.Series(kind).map({"standard": 40, "express": 30, "international":
90}).to_numpy()
day = (ts - start).days.to_numpy()
mins = np.exp(rng.normal(np.log(med), 0.5, n)) * (1 + 0.01 * day)   # slow
upward drift
status = np.where(rng.random(n) < (0.02 + 0.002*day), "error", "ok")
df = pd.DataFrame({"order_id": np.arange(1, n+1), "ts": ts, "order_type":
kind,
                   "minutes": mins.round(1), "status": status})
df.to_csv("orders.csv", index=False)
print(df.head(3)); print(df.shape)
```

Before you run it: how many columns will `df.shape` show? Real output:

```
   order_id                  ts     order_type  minutes status
0         1 2025-03-01 00:10:00        express     31.0     ok
1         2 2025-03-01 00:21:00  international     71.4     ok
2         3 2025-03-01 00:24:00       standard     24.1     ok
(3000, 5)
```

Why log-normal? Times are never negative and have a long right tail: many quick orders and a few very slow ones. A normal (bell) curve would give negative times.
`np.exp(rng.normal(np.log(med), 0.5, n))` produces values whose median is `med`.

Architecture

```
orders.csv --load.py--> ops.db (table: orders) --sql/*.sql--> numbers
     |                                                          |
     +---------------- pandas (analyze.py) ---------------------+-->
compare --> memo + charts
```

Step-by-step build

1. **Load into SQLite and get headline numbers.** Convert the timestamp to text for SQLite (it has no true datetime type), then run a summary query.

```
import sqlite3, pandas as pd
df = pd.read_csv("orders.csv", parse_dates=["ts"])
con = sqlite3.connect("ops.db")
df.assign(ts=df.ts.astype(str)).to_sql("orders", con, if_exists="replace",
index=False)
print(pd.read_sql("""
SELECT COUNT(*) AS n, ROUND(AVG(minutes),1) AS avg_min,
       ROUND(100.0*SUM(status='error')/COUNT(*),2) AS error_pct
FROM orders""", con))
```

```
      n  avg_min  error_pct
0  3000     53.3       3.7
```

In SQLite `status='error'` is 1 when true and 0 when false, so `SUM(...)` counts the errors. 111 errors in 3,000 orders is 3.7%.

1. **Percentiles in SQL.** SQLite has no percentile function, so use the **nearest-rank** method: sort the values, number them 1..n, and the p-th percentile is the first value whose rank is at least p × n. We use only successful orders, because failed orders have no meaningful processing time (write this rule in the README).

```
q_pct = """
WITH ranked AS (
  SELECT minutes, ROW_NUMBER() OVER (ORDER BY minutes) AS rn, COUNT(*)
OVER () AS n
  FROM orders WHERE status='ok')
SELECT
 MIN(CASE WHEN rn >= 0.50*n THEN minutes END) AS p50,
 MIN(CASE WHEN rn >= 0.90*n THEN minutes END) AS p90,
 MIN(CASE WHEN rn >= 0.95*n THEN minutes END) AS p95,
 MIN(CASE WHEN rn >= 0.99*n THEN minutes END) AS p99
FROM ranked"""
print(pd.read_sql(q_pct, con))
```

```
    p50   p90    p95    p99
0  42.3  99.0  125.1  211.0
```

Line by line: `ROW_NUMBER() OVER (ORDER BY minutes)` numbers the rows from the fastest; `COUNT(*) OVER ()` puts the total count on every row; `CASE WHEN rn >= 0.95*n THEN minutes END` keeps only values at or above the 95% position and gives NULL otherwise; `MIN` of those is the first such value.

1. **Check against pandas.** Same data, two methods. Predict first: will they match exactly?

```
ok = df[df.status=="ok"].minutes
print(ok.quantile([.5,.9,.95,.99]).round(1).to_dict())
print(ok.quantile([.5,.9,.95,.99],
interpolation="higher").round(1).to_dict())
```

```
{0.5: 42.3, 0.9: 98.9, 0.95: 124.9, 0.99: 210.4}
{0.5: 42.3, 0.9: 99.0, 0.95: 125.1, 0.99: 211.0}
```

The default pandas `quantile` **interpolates** (draws a line between the two neighbouring values), so it gives 124.9 where nearest rank gives 125.1. With `interpolation="higher"` pandas matches SQL exactly. Neither is wrong; they are two definitions. The lesson: **write the definition in your README**, and compare like with like.

COMMON MISTAKE

Reporting that "SQL and pandas disagree" and hunting for a bug when the real difference is the percentile definition. Always check the method before the data.

1. **Breakdown by order type.** The averages are meaningless unless you split by type.

```
by = df[df.status == "ok"].groupby("order_type")["minutes"].agg(
    n="size", avg="mean", median="median",
    p90=lambda s: s.quantile(.90), p95=lambda s: s.quantile(.95)).round(1)
by["error_pct"] = df.groupby("order_type")["status"].apply(lambda s:
(s=="error").mean()*100).round(2)
print(by)
```

```
                n    avg  median    p90    p95  error_pct
order_type
express      1003   36.9    31.9   63.7   76.5       3.84
international 428  108.8    95.5  183.6  218.1       3.82
standard     1458   48.0    42.4   80.3   98.7       3.57
```

International orders have a 95th percentile of 218 minutes against 77 for express. Error rates are close across types (about 3.6% to 3.8%), so slowness and failure are separate problems.

1. **Trend by day.** Group by calendar day and fit a straight line. `np.polyfit(x, y, 1)[0]` is the slope: how much the value changes per day.

```
daily = pd.read_sql("""SELECT DATE(ts) AS day, COUNT(*) AS n,
ROUND(AVG(minutes),1) AS avg_min,
 ROUND(100.0*SUM(status='error')/COUNT(*),1) AS err_pct FROM orders GROUP
BY DATE(ts)""", con)
import numpy as np
x = np.arange(len(daily))
for c in ["avg_min","err_pct"]:
    print(c, "slope per day", round(np.polyfit(x, daily[c], 1)[0], 3))
```

```
avg_min slope per day 0.45
err_pct slope per day 0.34
```

The daily table runs from 50.4 minutes on 1 March to 51.7 on 14 March, with noise in between (for example 59.3 on 10 March). The fitted slope says the average rises by about 0.45 minutes a day and the error rate by about 0.34 percentage points a day. Over 13 days that is about 6 minutes and 4.4 points. The error trend is the real alarm: 1.0% on day 1 and 4.7% on day 14.

1. **Add automated checks** (a tiny `tests` step).

```
assert df.order_id.is_unique and df.minutes.gt(0).all()
assert len(df[df.status=="ok"]) + (df.status=="error").sum() == len(df)
print("checks passed")
```

```
checks passed
```

Evaluation and acceptance criteria

- SQL nearest-rank percentiles equal pandas `interpolation="higher"` percentiles for p50, p90, p95, p99 (exact match).
- Error rate equals errors / total orders = 111 / 3,000 = 3.70%.
- The trend slope on the generated file is positive for both metrics (the truth we built in).
- The README states the percentile definition and that failed orders are excluded from time statistics.

- A bar chart with an action title: "International orders: 95th percentile 218 minutes, 3 times express".

Common pitfalls and how to detect them

1. **Reporting only the average.** Detect: mean far above median (here 53.3 vs 42.3 overall).
2. **Mixing failed and successful orders in time statistics.** Detect: unusually small times with status "error".
3. **Percentile definition mismatch** between tools. Detect: values differ in the first decimal while ranks agree.
4. **Trend from too few points or one noisy day.** Detect: plot daily values; look at whether one day drives the slope.
5. **Time stored as text and sorted wrongly.** Detect: `ORDER BY ts` puts "2025-03-10" before "2025-03-2" if the format is not zero-padded; always use ISO format `YYYY-MM-DD`.

Three extensions

1. Add a target ("95th percentile under 120 minutes") and compute the daily share of orders that meet it.
2. Rolling 7-day 95th percentile with window functions or `rolling(7)`.
3. Compare the error rate before and after day 7 with a two-proportion test (Chapter 13).

Interview talking points

Why do you report the 95th percentile and not just the average? Because the average hides the slow minority. If 95 orders take 30 minutes and 5 take 3 hours, the average looks fine but 5 customers are very unhappy. The 95th percentile shows what the slowest 5% experience.

SQLite has no percentile function. How did you do it? I numbered the sorted rows with `ROW_NUMBER`, counted rows with `COUNT(*) OVER ()`, and took the first value whose rank is at least p times n. This is the nearest-rank method. I checked it against pandas with `interpolation="higher"` and got the same numbers.

How do you know the trend is real and not noise? I look at the daily chart, fit a slope, and check that it is not driven by one day. With real data I would also test the difference between early and late periods. In this simulated data I built the trend in, so I could confirm the method finds it.

README outline

Question and one-line answer; data (simulated, seed 42, data dictionary); how to run (four commands); method (nearest rank, failed orders excluded); results table by order type; trend chart; checks; limits (simulated data, 14 days); extensions.

16.3 Project: Sales Dashboard Design in Power BI

> **Not run here.** Power BI Desktop is not available in this environment. The data below was generated and the expected totals were calculated in pandas, so you can check your DAX against them. The DAX is standard syntax; test it in your own model.

Goal and why employers care

Goal: design a one-page sales dashboard for a retail manager: a clean star schema, six DAX measures, and a layout that answers "how are we doing, against last year, and where?" in under ten seconds.

Employers care because the same job appears everywhere: turn tables into a page that a manager reads without help. A hiring manager looks for three things: a proper data model (not one giant flat table), correct measures (year-over-year on the right period), and restraint (few visuals, clear titles).

INTUITION

A dashboard is the dashboard of a car. Speed, fuel and warning lights are always visible. The engine diagram is in the manual. Put only the speedometer-level facts on page one.

Skills used

Chapter 9 (Power BI query and modeling: Power Query, star schema, relationships), Chapter 10 (DAX: measures, CALCULATE, time intelligence), Chapter 11 (reports, governance and tools: layout, row-level security), Chapter 8 (data visualization: choosing charts), Chapter 14 (storytelling: action titles).

Data

This generator makes a fact table (6,000 order lines over 2024 and 2025) and three dimension

tables (product, customer, date).

```
import numpy as np, pandas as pd
rng = np.random.default_rng(7)
products = pd.DataFrame({
  "ProductKey": range(1, 9),
  "Product": ["Notebook","Pen Pack","Backpack","Desk
Lamp","Headphones","Water Bottle","Calculator","Planner"],
  "Category":
["Paper","Paper","Bags","Home","Tech","Home","Tech","Paper"],
  "UnitPrice": [120, 60, 900, 650, 1500, 300, 450, 250],
  "UnitCost":  [70, 30, 520, 380, 1000, 150, 300, 120]})
customers = pd.DataFrame({
  "CustomerKey": range(1, 201),
  "Region": rng.choice(["North","South","East","West"], 200),
  "Segment": rng.choice(["Consumer","Business"], 200, p=[0.7, 0.3])})
dates = pd.DataFrame({"Date": pd.date_range("2024-01-01", "2025-12-31")})
n = 6000
fact = pd.DataFrame({
  "OrderID": np.arange(1, n+1),
  "Date": pd.to_datetime(rng.choice(dates.Date, n)),
  "CustomerKey": rng.integers(1, 201, n),
  "ProductKey": rng.integers(1, 9, n),
  "Qty": rng.integers(1, 6, n)})
fact = fact.sort_values("Date").reset_index(drop=True)
fact["OrderID"] = range(1, n+1)
fact.to_csv("fact_sales.csv", index=False);
products.to_csv("dim_product.csv", index=False)
customers.to_csv("dim_customer.csv", index=False);
dates.to_csv("dim_date.csv", index=False)
m = fact.merge(products, on="ProductKey")
m["Sales"] = m.Qty*m.UnitPrice; m["Cost"] = m.Qty*m.UnitCost
m["Year"] = m.Date.dt.year
print(len(fact), "rows;", fact.Date.min().date(), "to",
fact.Date.max().date())
print(m.groupby("Year")[["Sales","Cost"]].sum())
t = m.groupby("Year")["Sales"].sum()
print("YoY %", round((t[2025]/t[2024]-1)*100, 2))
print("Margin %", round((1-m.Cost.sum()/m.Sales.sum())*100, 2))
print(m.groupby("Category").Sales.sum().sort_values(ascending=False))
```

Real output (these are your expected values):

```
6000 rows; 2024-01-01 to 2025-12-31
        Sales     Cost
Year
2024  4882910  2971040
2025  4660090  2831930
YoY % -4.56
Margin % 39.19
Category
Tech     4499850
Home     2027150
Bags     2013300
Paper    1002700
Name: Sales, dtype: int64
```

Architecture

The data model is a **star schema**: one central fact table (events, one row per order line) surrounded by dimension tables (descriptions) joined on keys.

```
        dim_date (Date)
             |  1
             |  *
dim_customer ---- fact_sales ---- dim_product
 (CustomerKey) 1 * (OrderID,Date,  * 1 (ProductKey)
                   CustomerKey,
                   ProductKey, Qty)
```

Relationships are one-to-many from each dimension to the fact table, with single-direction filtering from the dimension to the fact. Mark `dim_date` as a date table (Table tools, Mark as date table, choose `Date`). Hide the key columns from the report view so report authors use dimension columns.

Folder layout:

```
sales-dashboard/
├── data/ fact_sales.csv dim_product.csv dim_customer.csv dim_date.csv
├── src/make_data.py
├── powerbi/ sales.pbix (or screenshots/ and a PDF export)
├── docs/ measures.md   layout.png   data_dictionary.md
└── README.md
```

Step-by-step build

1. **Load the four CSVs** with Power Query (Get Data, Text/CSV). Set types: `Date` as Date, keys as Whole number, `UnitPrice` and `UnitCost` as Fixed decimal. Check row counts: 6,000, 8, 200, 731 (2024 is a leap year: 366 + 365 days).
2. **Create relationships** as in the diagram. Confirm there are no many-to-many relationships.
3. **Write the measures** in a dedicated table called `_Measures` (an empty table that holds only measures, to keep the field list tidy).

```
Total Sales =
SUMX ( fact_sales, fact_sales[Qty] * RELATED ( dim_product[UnitPrice] ) )

Total Cost =
SUMX ( fact_sales, fact_sales[Qty] * RELATED ( dim_product[UnitCost] ) )

Margin % =
DIVIDE ( [Total Sales] - [Total Cost], [Total Sales] )

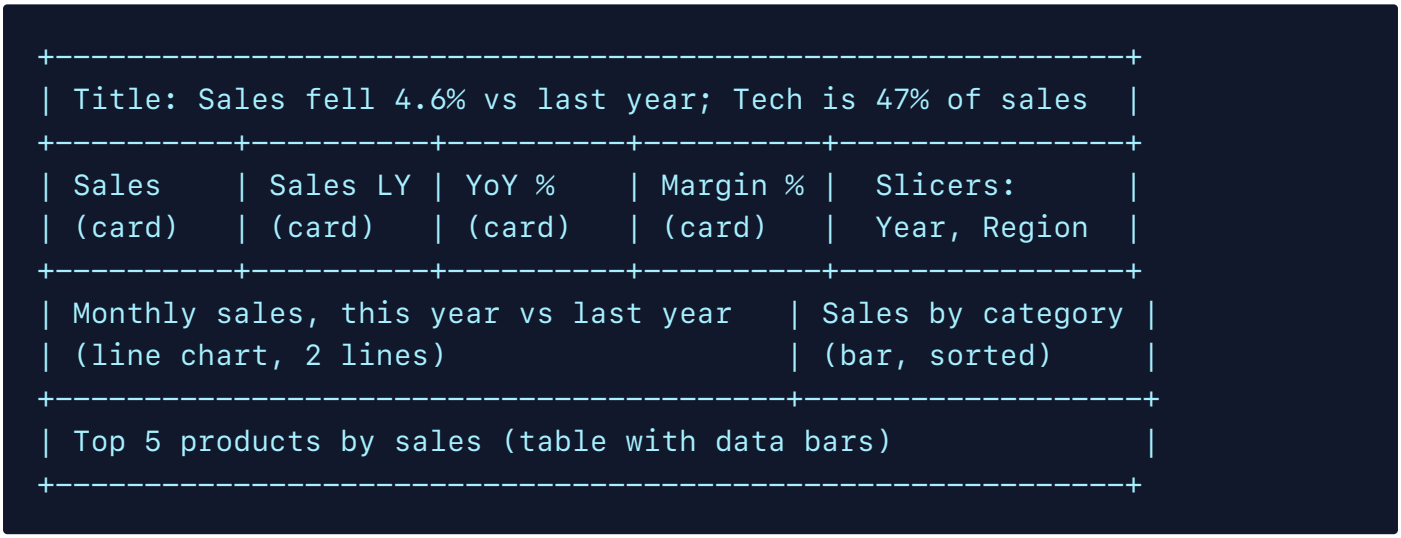
Sales LY =
CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( dim_date[Date] ) )

Sales YoY % =
DIVIDE ( [Total Sales] - [Sales LY], [Sales LY] )

Sales YTD =
TOTALYTD ( [Total Sales], dim_date[Date] )
```

`SUMX` goes row by row through `fact_sales` (an iterator, see Chapter 10) and `RELATED` fetches the price from the product dimension on the "one" side. `DIVIDE` returns blank instead of an error when the denominator is zero. `SAMEPERIODLASTYEAR` shifts the current date filter back one year, so it needs a proper date table.

1. **Test measures against the pandas numbers.** Put a card with `Total Sales` and a slicer on year. With the year slicer set to 2025 you should see 4,660,090; `Sales LY` should show 4,882,910; `Sales YoY %` should show -4.56% (with the format set to percentage, 2 decimals). `Margin %` for all years should show 39.19%.
2. **Design the layout** (one page, 16:9).



The title is an action title: a sentence with the finding. Across both years Tech is 4,499,850 of 9,543,000 total sales, which is 47.2%. If your page is filtered to one year, recalculate the share for that year before you write the title.

1. **Add row-level security (RLS)** if asked: a role "North manager" with the DAX filter `dim_customer[Region] = "North"` (Modeling, Manage roles). Test it with "View as". (Not run here.)
2. **Document the measures** in `docs/measures.md`: name, formula, meaning in one line, expected test value.

Evaluation and acceptance criteria

- All four expected values match: 2025 sales 4,660,090; LY 4,882,910; YoY -4.56%; margin 39.19%.
- No many-to-many relationships; date table is marked; keys are hidden.
- Page has at most 7 visuals; every visual has a title that states its point.
- A new user can answer "which category is largest?" in 5 seconds (test with a friend).
- Measures are in a measure table and documented.

Common pitfalls and how to detect them

1. **Flat file instead of star schema.** Detect: one table with 20 columns and repeated product names.
2. **YoY compared with the wrong period** (full last year against part of this year). Detect: sales LY larger than expected when the current year is incomplete; use `SAMEPERIODLASTYEAR` or `DATESYTD`.
3. **Calculated columns where measures are needed.** Detect: totals change wrongly when

you add slicers, or the file is large and slow.

4. **Using a table column for the date axis instead of the date table.** Detect: time intelligence returns blank or wrong values.
5. **Too many visuals and colours.** Detect: the 5-second test fails.

Three extensions

1. Add a drill-through page for one product with its monthly trend and top customers.
2. Add a "what if" parameter for price change and show the effect on margin.
3. Publish a PDF export and write a one-page "how to read this dashboard" note for managers.

Interview talking points

Why a star schema and not one big table? A star schema stores each fact once and each description once. It is smaller, faster, and filters flow cleanly from dimension to fact. One big table repeats product names thousands of times and makes errors easy.

What is the difference between a measure and a calculated column? A calculated column is computed row by row when the data loads and stored. A measure is computed when the report asks for it, in the context of the current filters. For totals and ratios you want measures, because a ratio of sums is not the sum of ratios.

How did you know your DAX was right? I computed the same totals independently in pandas on the same files and compared them: 2025 sales, last year, YoY percent and margin. They matched, and I wrote the expected values in the measure documentation.

README outline

Business question and screenshot; data (simulated, seed 7, dictionary); model diagram; measure table (name, DAX, expected test value); layout wireframe; how to open and refresh; RLS note; limits (simulated data, one currency).

16.4 Project: Excel Dashboard with Power Query and PivotTables

> **Not run here.** Excel is not available in this environment. I generated three monthly CSV files and calculated the totals in pandas, so you can check your PivotTable against them.

> The M code below is standard Power Query syntax.

Goal and why employers care

Goal: build a workbook that combines monthly sales files from a folder, cleans them with Power Query, summarises them with PivotTables, shows a small dashboard with slicers and charts, and refreshes with one click when next month's file is added.

Employers care because Excel is still the most common analytics tool in business, and the sign of a good analyst is a workbook that does not need copy-paste. "Refresh and done" saves hours every month and removes typing mistakes.

INTUITION

Copy-pasting each month is like washing dishes by hand every day. Power Query is a dishwasher: you load it the same way each time and press one button.

Skills used

Chapter 2 (Excel formulas: lookups, text, dates), Chapter 3 (Excel analysis: Power Query, PivotTables, slicers, charts), Chapter 8 (data visualization), Chapter 14 (storytelling).

Data

```
import numpy as np, pandas as pd
rng = np.random.default_rng(13)
tot = []
for m in ["2025-01","2025-02","2025-03"]:
    n = 120
    df = pd.DataFrame({
      "Date": rng.choice(pd.date_range(m+"-01", periods=28), n),
      "Store": rng.choice(["Pune","Chennai","Jaipur"], n),
      "Item": rng.choice(["Rice","Oil","Tea","Soap"], n),
      "Units": rng.integers(1, 20, n),
      "Price": rng.choice([60, 140, 250, 35], n)})
    df["Amount"] = df.Units*df.Price
    df.to_csv(f"raw/sales_{m}.csv", index=False)
    tot.append(df)
a = pd.concat(tot)
print(len(a), a.Amount.sum())
a["Month"] = a.Date.dt.month
print(a.pivot_table(index="Store", columns="Month", values="Amount",
aggfunc="sum", margins=True))
```

```
360 427270
Month         1       2       3     All
Store
Chennai   44875   43870   57525  146270
Jaipur    44870   38270   46670  129810
Pune      47015   50305   53870  151190
All      136760  132445  158065  427270
```

These are the numbers your PivotTable must reproduce: 360 rows and a grand total of 427,270.

COMMON MISTAKE

My first attempt at this pivot used `columns=a.Date.dt.month` directly together with `margins=True`, and pandas raised `TypeError: unhashable type: 'Series'`. Pass a column name (`"Month"`), not a computed Series, when you ask for margins. Create the helper column first.

Architecture

```
data/raw/sales_2025-01.csv  \
data/raw/sales_2025-02.csv   >-- Power Query (From Folder) --> table
"Sales" --> PivotTables --> Dashboard sheet
data/raw/sales_2025-03.csv  /                                  (typed,
clean)    + slicers + charts
```

Workbook sheets: `Sales` (query output), `Pivots` (3 PivotTables), `Dashboard` (charts and slicers), `Notes` (definitions, refresh steps).

Step-by-step build

1. **Put the files in one folder** and note the path. Use a relative or shared path if others will open the workbook; a path like `C:\Users\you\...` will break on another machine.
2. **Get Data from Folder** (Data, Get Data, From File, From Folder). Choose "Transform Data" (not "Combine" yet) so you control the steps. The Advanced Editor shows M code like this:

```
let
    Source   = Folder.Files("C:\data\raw"),
    OnlyCsv  = Table.SelectRows(Source, each Text.EndsWith([Name],
".csv")),
    Parsed   = Table.AddColumn(OnlyCsv, "Data", each
                 Table.PromoteHeaders(
                   Csv.Document([Content], [Delimiter = ",", Encoding =
65001]),
                   [PromoteAllScalars = true])),
    Combined = Table.Combine(Parsed[Data]),
    Typed    = Table.TransformColumnTypes(Combined,
                {{"Date", type date}, {"Units", Int64.Type},
                 {"Price", type number}, {"Amount", type number}})
in
    Typed
```

Read it as a recipe: list the files; keep only CSVs; read each file and promote its first row to headers; stack all tables; set the column types. `Table.Combine` is what makes a new month appear automatically.

1. **Clean**: remove blank rows, trim text, check for errors (Home, Keep Rows, Keep errors). Add a `Month` column (Add Column, Date, Month, Name of Month). Add a `Source` column from the file name so you can trace any row back to its file.
2. **Load to a Table** (Close and Load To, Table). Name it `Sales` in Table Design.
3. **Verify**: the table must have 360 rows. Put `=SUM(Sales[Amount])` in a cell; expect

427,270. This is your tie-out against the pandas total.

4. **Insert PivotTables** from the `Sales` table: (a) Amount by Store (rows) and Month (columns), which must equal the table above; (b) Amount by Item; (c) Units by Store. Show values as "% of grand total" in one of them.
5. **Add slicers** for Store and Month, and connect them to all PivotTables (Slicer, Report Connections). **Insert charts** from the PivotTables (PivotCharts): a clustered column by store and month, and a bar of items sorted by amount.
6. **Dashboard sheet**: put two KPI cells at the top (`=SUM(Sales[Amount])` for total sales and `=ROWS(Sales[Amount])` for the number of rows), the two charts, and the slicers. Add a title that states the finding, for example "Chennai grew most in March: 57,525 against 43,870 in February".
7. **Refresh test**: copy a fourth file `sales_2025-04.csv` into the folder, press Data, Refresh All, and check that the row count grows and the dashboard updates with no manual step. Then delete the extra file to restore the checked state.
8. **Notes sheet**: where the data comes from, how to refresh, what each pivot means, and the tie-out values.

Evaluation and acceptance criteria

- `Sales` table has 360 rows; `SUM(Sales[Amount])` is 427,270; the store by month pivot equals the pandas table cell for cell.
- Adding a new CSV and pressing Refresh All updates everything with no editing.
- Slicers control every pivot and chart.
- No hard-coded numbers on the dashboard; every number comes from a table or pivot.
- The workbook opens on a clean Dashboard sheet with a stated finding.

Common pitfalls and how to detect them

1. **Numbers stored as text** (left-aligned, pivot shows count instead of sum). Detect: the PivotTable field defaults to "Count of Amount".
2. **Date parsed in the wrong locale** (`03/04/2025`). Detect: months come out wrong or dates fail; set the column type with an explicit locale ("Using Locale").
3. **Pivot not refreshed** after the source changes. Detect: grand total differs from `SUM(Sales[Amount])` .
4. **Absolute file path** that fails on another computer. Detect: refresh error on a colleague's machine; use a shared or relative location.

5. **Pasting values over formulas** or editing the query output by hand. Detect: refresh overwrites your edits; make changes in Power Query steps, never in the output sheet.

Three extensions

1. Add a lookup table of store regions and merge it in Power Query (Merge Queries), then pivot by region.
2. Add month-over-month growth with a PivotTable "Show Values As, % Difference From".
3. Build a parameter (a cell holding the folder path) so the same workbook works on any computer.

Interview talking points

What is Power Query and why use it? It is the tool inside Excel that records the steps for loading and cleaning data. Because the steps are saved, I can press Refresh and run them again on new data. That removes manual copy-paste and the mistakes that come with it.

Your PivotTable total looks wrong. What do you check? First I refresh it. Then I compare its grand total with `SUM` of the source table. Then I check that the Amount column is a number, not text, and that no filters or slicers are hiding rows. Last, I look for duplicate rows in the source.

How do you prove your dashboard is correct? I calculated the same totals independently in Python: 360 rows, a grand total of 427,270, and the store by month table. The pivot matched cell for cell. I wrote these values on the notes sheet.

README outline

Question and screenshot; data (simulated, seed 13, three CSV files); folder and refresh instructions; Power Query steps list; pivot descriptions; tie-out values; limits; 90-second screen recording link.

Chapter recap

- A project starts with one question, one output, and measurable success criteria, written before any tool is opened.
- Use one repo layout and one README template every time: reviewers learn to trust your structure.
- Simulated data with a fixed seed lets you check that your method finds a truth you built in.

- Report percentiles, not just averages; the tail is where customers feel the problem.
- Percentile definitions differ between tools (nearest rank versus interpolation); write yours down and compare like with like.
- A number is believable only when a second method agrees: SQL against pandas, DAX against pandas, PivotTable against pandas.
- A Power BI model should be a star schema with measures in a measure table and a marked date table.
- Excel dashboards should refresh from Power Query with no copy-paste; test refresh by adding a new file.
- A finished project has a README, a tie-out check, an action-title chart and a short post.

Pattern	Code or rule
Nearest-rank percentile in SQL	`ROW_NUMBER() OVER (ORDER BY x)`, `COUNT(*) OVER ()`, `MIN(CASE WHEN rn >= p*n THEN x END)`
Same in pandas	`s.quantile(p, interpolation="higher")`
Error rate	`SUM(status='error') / COUNT(*)`
Trend slope	`np.polyfit(x, y, 1)[0]`
YoY in DAX	`DIVIDE([Total Sales]-[Sales LY],[Sales LY])` with `SAMEPERIODLASTYEAR`
Combine files in Power Query	`Folder.Files`, `Table.PromoteHeaders`, `Table.Combine`

Practice set

1. (Easy) Twenty order times in minutes, sorted: 5, 7, 8, 9, 10, 11, 12, 12, 13, 14, 15, 16, 18, 20, 22, 25, 30, 41, 60, 95. Find the mean, the median, and the nearest-rank 90th percentile.

Answer: The sum is 443, so the mean is 443 / 20 = 22.15. The median is the average of the 10th and 11th values: (14 + 15) / 2 = 14.5. For the 90th percentile, rank = ceil(0.90 × 20) = 18, and the 18th value is 41. The mean (22.15) is far above the median (14.5): a long right tail.

2. (Easy) 37 of 1,000 orders failed. What is the error rate, and how many would you expect in

8,000 orders at the same rate?

Answer: 37 / 1,000 = 3.7%. In 8,000 orders: 0.037 × 8,000 = 296.

3. (Easy) In the 20-value list above, what is the nearest-rank 95th percentile?

Answer: rank = ceil(0.95 × 20) = 19, and the 19th value is 60.

4. (Medium, what is the output?) `ok.quantile(0.95)` returns 124.9 but your SQL returns 125.1 on the same data. Is one of them wrong? How do you make pandas match SQL?

Answer: Neither is wrong. pandas interpolates between neighbours by default, while the SQL uses nearest rank. Use `ok.quantile(0.95, interpolation="higher")` to match.

5. (Medium, find the bug) `a.pivot_table(index="Store", columns=a.Date.dt.month, values="Amount", aggfunc="sum", margins=True)` fails with `TypeError: unhashable type: 'Series'` . Fix it.

Answer: Create `a["Month"] = a.Date.dt.month` first and use `columns="Month"` . With `margins=True` , pandas needs a column name.

6. (Medium) Your DAX `Sales YoY %` shows -4.56% for 2025. Total 2025 sales are 4,660,090. What must 2024 sales be, and verify the percentage.

Answer: 2024 = 4,882,910. Check: 4,660,090 / 4,882,910 = 0.95437, minus 1 is -0.04563, which is -4.56%.

7. (Interview style) A manager says "average processing time is fine, 53 minutes, so why are customers complaining?" Give a three-step reply.

Answer: (1) Show the median (42) and the 95th percentile (125): one order in twenty takes over two hours. (2) Split by type: international orders have a 95th percentile of 218 minutes. (3) Show the trend: errors rose from 1.0% to 4.7% in two weeks. The average hides all three.

8. (Interview style) Your Excel dashboard grand total does not match the source table total. List four causes in the order you would check them.

Answer: (1) The PivotTable was not refreshed. (2) A slicer or filter hides some rows. (3) Amount is stored as text in some rows, so they are counted, not summed. (4) Duplicate or blank rows in the source, or a new file in the folder that the query did not combine because of a name filter.

Prompt pack

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles, and English is my second language. Help me plan a
portfolio project in this chapter's area: an operations results analyzer
that loads a CSV of timed shop orders into SQLite and reports average,
median, 90th, 95th and 99th percentile, error rate and trend. Ask me three
questions about my goal, then help me write the one-sentence question, the
success criteria, the repo layout and a day-by-day plan for one week at
one hour a day. Check my answers before moving on.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. Review my project README for a data analysis
portfolio project: [PASTE HERE]. Check it against this template: question,
one-line answer, data source and licence, how to run, method and
definitions, results with action-title chart, checks, limits. List what is
missing, rewrite the weakest section, and tell me whether a stranger could
reproduce my numbers in 10 minutes.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. My SQL percentile and my pandas percentile
disagree on the same data: [PASTE BOTH OUTPUTS AND CODE HERE]. Teach me
the nearest-rank and interpolated definitions with a small 10-number
example, then help me find out which definition each of my tools used and
how to make them match. Do not guess: ask me for the exact code if you
need it.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. I am designing a Power BI sales dashboard with a
star schema and measures for Total Sales, Margin percent, Sales last year,
YoY percent and YTD. Review my model and DAX: [PASTE HERE]. Check
relationships, the date table, measure definitions and the layout. Give me
a test plan that compares each measure with an independent pandas
calculation, and explain any error in simple English.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. Run a mock interview about my Excel dashboard
project with Power Query and PivotTables. Ask me one question at a time:
how I combined files, how I cleaned data, how I checked totals, what
happens when a new file arrives, and what could go wrong. After each
answer, give feedback on clarity and correctness, then ask the next
question. End with a score out of 10 and two improvements.
```

CHAPTER 17

PROJECT LAB II: MARKETING ATTRIBUTION, P&L VARIANCE, PRODUCT FUNNEL

Three business-analytics projects where the maths is simple and the judgement is everything.

Chapter 16 gave you the project workflow and three tool-focused builds. This chapter uses the same workflow (question, success criteria, repo layout, tie-out check, README, post) for three projects that employers in marketing, finance and product teams ask about all the time: which marketing channel deserves credit for a sale, why profit missed budget even though sales beat it, and where users drop out of a purchase journey. The code is small. The hard part is definitions and honesty: what does "credit" mean, which sign means "bad", what counts as a "user in the funnel". Every Python and SQL block was run and the printed output is the real output. All data is simulated with a fixed seed and labelled that way, so you know the truth and can confirm that your method finds it.

17.1 Project: Marketing Campaign and Attribution Analysis

Goal and why employers care

Goal: from a table of customer journeys (the sequence of marketing channels each customer touched) and a spend table, compute conversion rate, cost per acquisition (CAC) and return on ad spend (ROAS) for each channel under three attribution rules, and explain why the answers differ.

Employers care because marketing budgets are large and every channel owner says their channel works. An analyst who can show that Search's CAC is 1,348 under last-click but 1,846 under first-click, and who explains that attribution is a rule for sharing credit and not proof of cause, is trusted by both marketing and finance.

INTUITION

A football goal is scored by the striker, but the defender started the move and the midfielder passed. "Last-click" gives all credit to the striker. "First-click" gives it to the defender. "Linear" splits it equally. None of these is the truth; they are three ways to share the credit, and each one makes different players look good.

Words you need

Word	Meaning	Example
Touchpoint	One contact between a customer and a marketing channel	An email opened
Journey	All touchpoints of one customer, in order	Social, then Search, then Email
Conversion	The goal event	A purchase
CAC	Customer acquisition cost = spend / conversions	120,000 / 76.3 = 1,573
ROAS	Return on ad spend = revenue / spend	190,750 / 120,000 = 1.59
Attribution	Rule for splitting credit for a conversion across touchpoints	Last-click

Skills used

Chapter 12 (business analytics: segmentation, funnels, marketing metrics), Chapter 4 and 5 (pandas groupby, transform, joins), Chapter 7 (EDA), Chapter 8 (charts), Chapter 13 (A/B testing, for the incrementality extension), Chapter 14 (storytelling).

Data

Generated: 2,000 journeys of 1 to 4 touchpoints across four channels. Email and Search are made to convert better when they are the last touch, so we know what a good method should notice. Spend per channel is in rupees.

Architecture

```
touchpoints.csv (journey, step, channel, converted)   spend (dict or
spend.csv)
        |                                                    |
        +--> conversions per journey --> credit rules --> credit per
channel
                                                             |
                                            join spend --> CAC, ROAS -->
table + chart + memo
```

```
marketing-attribution/
├── README.md
├── data/raw/touchpoints.csv   data/raw/spend.csv
├── src/make_data.py   src/attribution.py
├── tests/test_credit_sums.py       # credit totals must equal conversions
└── reports/memo.md   reports/channel_cac.png
```

Step-by-step build

1. **Generate data and compute the three rules.** Before you run it: the three credit columns each add up to what number? (Hint: each conversion hands out exactly one unit of credit.)

```
import numpy as np, pandas as pd
rng = np.random.default_rng(16)
channels = ["Search","Social","Email","Display"]
spend = pd.Series({"Search": 120000, "Social": 90000, "Email": 15000,
"Display": 60000}, name="spend")
rows = []
for j in range(1, 2001):
    k = rng.integers(1, 5)
    path = rng.choice(channels, k, p=[.35,.3,.15,.2])
    p = 0.05 + 0.08*(path[-1] in ("Email","Search")) + 0.02*(k>=3)
    conv = rng.random() < p
    for i, c in enumerate(path):
        rows.append((j, i, c, int(conv and i == len(path)-1)))
tp = pd.DataFrame(rows, columns=["journey","step","channel","converted"])
tp.to_csv("touchpoints.csv", index=False)
J = tp.groupby("journey").agg(n=("step","size"), conv=("converted","max"))
print("journeys", len(J), "conversions", J.conv.sum(), "rate",
round(J.conv.mean()*100,2))
won = tp[tp.journey.isin(J.index[J.conv==1])].copy()
last =
won.sort_values("step").groupby("journey").tail(1).channel.value_counts()
first =
won.sort_values("step").groupby("journey").head(1).channel.value_counts()
won["w"] = 1/won.groupby("journey").step.transform("size")
linear = won.groupby("channel").w.sum()
out =
pd.DataFrame({"first":first,"last":last,"linear":linear}).fillna(0).round(
1).loc[channels]
out["spend"] = spend
for m in ["first","last","linear"]:
    out["CAC_"+m] = (out.spend/out[m]).round(0)
print(out)
print(out[["first","last","linear"]].sum().round(1).to_dict())
```

```
journeys 2000 conversions 181 rate 9.05
         first  last  linear   spend  CAC_first  CAC_last  CAC_linear
channel
Search      65    89    76.3  120000     1846.0    1348.0      1573.0
Social      46    35    45.6   90000     1957.0    2571.0      1974.0
Email       39    33    35.7   15000      385.0     455.0       420.0
Display     31    24    23.4   60000     1935.0    2500.0      2564.0
{'first': 181.0, 'last': 181.0, 'linear': 181.0}
```

Line by line: `groupby("journey").agg(...)` gives one row per journey with its length and

whether it converted. `won` keeps only the touchpoints of converting journeys. After sorting by `step`, `.tail(1)` is the last touch and `.head(1)` is the first touch of each journey. For linear credit, `transform("size")` puts the journey length on every row, so `1/size` is each touchpoint's share (a 3-touch journey gives 0.333 to each). The last line is the **check**: each rule distributes exactly 181 credits, equal to the number of conversions.

1. **Read the table.** Search gets 89 credits under last-click but 65 under first-click, so its CAC is 1,348 or 1,846 depending on the rule: first-click is 37% higher for the same spend. Social and Display get more credit when we look at first touch than last touch: they *introduce* customers and others close the sale. Email is cheapest under every rule (385 to 455), but it is a small channel that mostly reaches people already in the funnel.

1. **ROAS with an assumed order value.** Assume each conversion is worth 2,500 (state the assumption in the README).

```
out["ROAS_linear"] = (out.linear*2500/out.spend).round(2)
print(out[["spend","linear","ROAS_linear"]])
```

```
          spend  linear  ROAS_linear
channel
Search   120000    76.3         1.59
Social    90000    45.6         1.27
Email     15000    35.7         5.95
Display   60000    23.4         0.98
```

For Search: 76.3 × 2,500 = 190,750 revenue, and 190,750 / 120,000 = 1.59. Display's 0.98 means it returns less than a rupee per rupee spent. Even under first-click, its friendliest rule (31 credits), ROAS is 31 × 2,500 / 60,000 = 1.29.

1. **Write the interpretation, not just the table.** "Under every rule Email has the lowest CAC, and Display and Social sit near or below break-even. Search's credit depends on the rule: 65 to 89 conversions. We recommend moving a small test budget out of Display (ROAS 0.98) and treating Email's result with care, because Email reaches people who already know us."

1. **Add the test** `tests/test_credit_sums.py`: `assert abs(out[["first","last","linear"]].sum() - conversions).max() < 1e-6`.

KEY INSIGHT

Attribution answers "who gets credit?" It does not answer "what would have happened without this channel?" That second question is **incrementality**, and it needs an experiment (for example switching a channel off for a random group). Say this in every memo.

Evaluation and acceptance criteria

- Credit under each rule sums to the number of conversions (181).
- CAC and ROAS tables reproduce exactly: Search linear CAC 1,573; Email linear ROAS 5.95.
- The memo shows at least two rules side by side and states which channel changes rank.
- Assumptions (order value 2,500, simulated data, one conversion type) are in the README.
- One chart with an action title, for example "Search CAC is 1,348 under last-click but 1,846 under first-click".

Common pitfalls and how to detect them

1. **Credit does not add up to conversions.** Detect: the sum check above; a typical cause is giving full credit to every touchpoint.
2. **Counting touchpoints as customers.** Detect: conversion rate over 100% or different denominators; always count journeys.
3. **Treating attribution as cause.** Detect: the memo says "Email generated 33 sales"; reword to "was credited with".
4. **Dividing spend by credit in a different period.** Detect: spend from a month, conversions from a quarter.
5. **Ignoring journeys that did not convert** when you want conversion rate by channel. Detect: only `won` journeys used in a rate; use all journeys.

Three extensions

1. Add a position-based rule (40% first, 40% last, 20% shared among the middle).
2. Design a holdout test for Display: sample size and duration with the method from Chapter 13.
3. Cohort view: conversion rate by journey length (1, 2, 3, 4 touches).

Interview talking points

What is the difference between first-click, last-click and linear attribution? They are three ways to share credit for a sale. First-click gives all credit to the first channel, last-click to the last, and linear splits equally. In my data Search got 89, 65 and 76.3 credits under the three rules, so the choice changes decisions.

Does attribution prove a channel works? No. It shares credit among touches that already happened. A customer who was going to buy anyway still gives credit to the last email. To know the real effect I need an experiment with a holdout group.

How would you check your numbers? The credit under each rule must add up to the number of conversions, 181 here. I also recompute one channel by hand: Search linear credit is the sum of 1/length over its touches in converted journeys.

README outline

Question; data (simulated, seed 16); definitions (touchpoint, journey, CAC, ROAS); results table with three rules; assumptions (order value); interpretation; incrementality warning; checks; extensions.

17.2 Project: Finance P&L Variance Dashboard

Goal and why employers care

Goal: compare budget with actual for a profit and loss statement (P&L: revenue minus costs equals profit), compute the variance for each line with a correct favourable/adverse sign, split a revenue variance into volume and price effects, and design a dashboard a finance manager can use in a monthly review.

Employers care because every finance team runs a monthly "budget versus actual" review. Sign errors are the classic mistake: a cost that is higher than budget is *bad*, even though the number is positive. An analyst who builds the sign rule into the model, and can explain why profit is flat although revenue beat budget, is useful on day one.

INTUITION

You plan to spend 5,000 on groceries this month and spend 5,600. The difference is +600, but it is bad news. If you planned to earn 5,000 from a side job and earned 5,600, the same +600 is good news. Same number, opposite meaning. The "type" of the line (income or expense) decides which.

Words you need

Word	Meaning	Example
Budget	The plan for the period	Revenue 1,000,000
Actual	What really happened	Revenue 1,060,000
Variance	Actual minus budget	+60,000
Favourable (FAV)	The variance helps profit	Revenue up, or cost down
Adverse (ADV)	The variance hurts profit	Revenue down, or cost up

Skills used

Chapter 2 and 3 (Excel formulas and analysis), Chapter 4 and 5 (pandas), Chapter 9 and 10 (Power BI model and DAX), Chapter 8 (visualization: waterfall idea), Chapter 14 (storytelling).

Data

A small P&L for one month, five lines (rupees).

Architecture

```
fact_pl (Month, Line, Actual, Budget)   dim_line (Line, Type:
Income/Expense, Order)
           \                                  /
            +--- model: variance, fav sign ---+
                              |
     dashboard: KPI cards, variance table (colour), waterfall, drivers
```

```
pl-variance/
├── README.md
├── data/pl_month.csv   data/dim_line.csv
├── src/variance.py     tests/test_signs.py
└── reports/dashboard.png   reports/memo.md
```

Step-by-step build

1. **Compute variance and the favourable sign.** Before you run it: which lines will be adverse, and will profit be above or below budget?

```
import pandas as pd
rows = [("Revenue","Income",1000000,1060000),
        ("Cost of goods","Expense",600000,650000),
        ("Salaries","Expense",150000,148000),
        ("Marketing","Expense",60000,72000),
        ("Rent","Expense",40000,40000)]
d = pd.DataFrame(rows, columns=["line","type","budget","actual"])
d["variance"] = d.actual - d.budget
d["fav"] = d.apply(lambda r: r.variance if r.type=="Income" else -
r.variance, axis=1)
d["var_pct"] = (d.fav / d.budget * 100).round(1)
d["flag"] = d.fav.apply(lambda v: "FAV" if v>0 else ("ADV" if v<0 else "-
"))
print(d.to_string(index=False))
inc = d[d.type=="Income"][["budget","actual"]].sum(); exp =
d[d.type=="Expense"][["budget","actual"]].sum()
np_b, np_a = inc.budget-exp.budget, inc.actual-exp.actual
print("Profit budget", np_b, "actual", np_a, "fav", np_a-np_b,
round((np_a-np_b)/np_b*100,1), "%")
print("sum of fav =", d.fav.sum())
```

```
         line    type  budget  actual  variance    fav  var_pct flag
      Revenue  Income 1000000 1060000     60000  60000      6.0  FAV
Cost of goods Expense  600000  650000     50000 -50000     -8.3  ADV
     Salaries Expense  150000  148000     -2000   2000      1.3  FAV
    Marketing Expense   60000   72000     12000 -12000    -20.0  ADV
         Rent Expense   40000   40000         0      0      0.0    -
Profit budget 150000 actual 150000 fav 0 0.0 %
sum of fav = 0
```

Line by line: `variance` is always actual minus budget, so it is simply a difference. `fav` flips

the sign for expenses: for an expense a positive variance (spent more) becomes a negative favourable number. `var_pct` divides the favourable variance by budget, so -20.0% for Marketing means "20% worse than plan". Profit is budget 150,000 and actual 150,000: revenue beat plan by 60,000, but costs ate all of it (+50,000 cost of goods, +12,000 marketing, -2,000 salaries: net 60,000).

Check: the favourable variances add to 0, which equals the profit variance (60,000 - 50,000 + 2,000 - 12,000 + 0 = 0).

1. **Explain the headline.** "Revenue beat budget by 6% but profit is flat. Cost of goods is 50,000 over (8.3% adverse) and marketing 12,000 over (20% adverse). Salaries saved 2,000."
2. **Split the revenue variance into volume and price.** Say budget was 10,000 units at 100 each (1,000,000). Actual was 9,600 units for 1,060,000, so the actual price is 1,060,000 / 9,600 = 110.4167.

```
bu, bp, au = 10000, 100, 9600
ap = 1060000/au
vol = (au-bu)*bp; price = (ap-bp)*au
print("volume effect", vol, "price effect", round(price), "total",
round(vol+price))
```

```
volume effect -40000 price effect 100000 total 60000
```

Volume effect = (actual units - budget units) × budget price = (9,600 - 10,000) × 100 = -40,000. Price effect = (actual price - budget price) × actual units = 10.4167 × 9,600 = 100,000. They add to the 60,000 revenue variance. Meaning: we sold 400 fewer units, but at a higher price. The story for the manager is not "revenue is up" but "we are selling fewer units; price rises are hiding a volume problem".

1. **Build the model in Power BI or Excel** (not run here). Keep a `dim_line` table with a `Type` column and a `Sign` column (+1 for income, -1 for expense). DAX (standard syntax):

```
Actual = SUM ( fact_pl[Actual] )
Budget = SUM ( fact_pl[Budget] )
Variance = [Actual] - [Budget]
Fav Variance =
SUMX ( fact_pl, ( fact_pl[Actual] - fact_pl[Budget] ) * RELATED (
dim_line[Sign] ) )
Fav Variance % = DIVIDE ( [Fav Variance], [Budget] )
```

Sums over a mix of lines are only meaningful for `Fav Variance`, because each row has its sign applied before summing. In Excel, use `=IF(type="Income", actual-budget, budget-actual)`.

1. **Dashboard layout.** Top row: KPI cards for Revenue, Total costs, Profit (actual, budget, fav variance). Middle: a variance table with red for adverse and green for favourable (do not use colour alone; also show the FAV/ADV text for colour-blind readers). A waterfall chart from budget profit through each favourable or adverse step to actual profit. Bottom: the volume/price bridge for revenue. Title: "Revenue beat plan by 6%, but profit is flat: costs absorbed every rupee".
2. **Write the tie-out test**: `assert d.fav.sum() == np_a - np_b`.

Evaluation and acceptance criteria

- Sum of favourable variances equals the profit variance (0 in this data).
- Marketing is flagged ADV at -20.0%; Salaries FAV at +1.3%.
- Volume plus price effects equal the revenue variance (60,000).
- Every colour has a text label; every variance column states its sign rule in a note.
- A reviewer can find which line caused the profit miss within 10 seconds.

Common pitfalls and how to detect them

1. **Sign errors on expenses.** Detect: the sum-of-favourable test fails against profit variance.
2. **Percent variance against the wrong base** (actual instead of budget). Detect: -8.3% becomes -7.7% (50,000 / 650,000).
3. **Mixing periods** (month actual, year budget). Detect: variance of 90% or more that nobody believes.
4. **Showing only percentages.** Detect: a 20% variance on a tiny line is less important than 8% on a big line; show both rupees and percent.
5. **Hard-coded flags** (typing "ADV" by hand). Detect: change a number and see if the flag

changes.

Three extensions

1. Add a 12-month trend with year-to-date and a rolling forecast.
2. Add cost-of-goods split into price and quantity variance.
3. Add a materiality rule (flag only variances over 5% and over 10,000) and explain it in the notes.

Interview talking points

What is a favourable variance? One that improves profit. Higher revenue is favourable, and lower cost is favourable. So for an expense I flip the sign: spending more than budget is adverse even though the raw difference is positive.

Revenue is 6% over budget but profit is flat. What happened? Costs rose by the same amount. Cost of goods was 50,000 over, marketing 12,000 over, salaries 2,000 under. 60,000 - 50,000 - 12,000 + 2,000 = 0. I would also check whether the revenue gain is price or volume: here it is price (+100,000) offset by lower volume (-40,000).

How do you know the dashboard is right? The favourable variances add up to the profit variance, which I check in code. I also recompute the volume plus price effect, which must equal the revenue variance.

README outline

Question; data (simulated); sign convention table; variance formulas; results table; volume/price bridge; dashboard screenshot; checks; limits; extensions.

17.3 Project: Product Analytics: Event Tracking Plan and Funnel Analysis

Goal and why employers care

Goal: design an event tracking plan for a shopping app (which actions we record and with what properties), then run a funnel analysis from the event log: how many users reach each step, where the biggest drop is, and whether it differs by platform.

Employers care because product teams cannot improve what they cannot measure, and most

funnel mistakes come from bad tracking, not bad maths. An analyst who writes the tracking plan, validates the data against it, and then reports "35% of cart users never start checkout" is a partner to the product manager.

INTUITION

A shop owner stands at the door with a clicker. Customers enter (app open), look at shelves (product view), put items in a basket (add to cart), go to the counter (checkout), and pay (purchase). The funnel is how many of the first group reach each next stage. You can only fix the stage where most people leave, and only if you counted every stage the same way.

Skills used

Chapter 12 (business analytics: funnels, retention), Chapter 4 and 5 (pandas, pivot, groupby), SQL conditional counting (`COUNT(DISTINCT CASE WHEN ...)`), Chapter 7 (EDA and data checks), Chapter 13 (testing whether a funnel change is real), Chapter 14 (storytelling).

The tracking plan

A tracking plan is a table that says what to record. Write it before you look at any data.

Event name	Fires when	Required properties	Owner
`app_open`	App is opened	user_id, platform, event_time	Mobile team
`product_view`	A product page is shown	user_id, product_id, platform, event_time	Mobile team
`add_to_cart`	An item is added	user_id, product_id, platform, event_time	Mobile team
`checkout_start`	Checkout page is shown	user_id, cart_value, platform, event_time	Payments team
`purchase`	Payment succeeds	user_id, order_id, amount, platform, event_time	Payments team

Rules: names are `lowercase_snake_case` , past or present action, one name for one

meaning; every event has the same `user_id` field; times in UTC; no personal data (name, phone, email) in event properties.

Data

Generated: 5,000 users over four weeks. Each user passes a step with a fixed probability (product view 60%, add to cart 35%, checkout 60%, purchase 75%), so we know what a correct analysis should find.

Architecture

```
events.csv --> validate vs tracking plan --> SQLite / pandas --> funnel
table --> by platform --> memo
 (user_id, event_name, event_time, platform)   (unknown names, nulls,
(distinct users per step)
                                               duplicates, order)
```

```
product-funnel/
├── README.md   docs/tracking_plan.md
├── src/make_events.py   src/validate.py   src/funnel.py
├── sql/funnel.sql
└── reports/funnel.png   reports/memo.md
```

Step-by-step build

1. **Generate events and count the funnel in SQL.** Before you run it: the step-to-step conversions should be close to 60%, 35%, 60%, 75%. Which will differ most from their planned value, and why?

```
import numpy as np, pandas as pd, sqlite3
rng = np.random.default_rng(18)
steps =
["app_open","product_view","add_to_cart","checkout_start","purchase"]
rates = [1, .6, .35, .6, .75]
rows = []
for u in range(1, 5001):
    plat = rng.choice(["android","ios","web"], p=[.5,.3,.2])
    t = pd.Timestamp("2025-04-01") +
pd.Timedelta(minutes=int(rng.integers(0, 28*1440)))
    for s, r in zip(steps, rates):
        if rng.random() > r: break
        rows.append((u, s, t.isoformat(sep=" "), plat))
        t += pd.Timedelta(minutes=int(rng.integers(1, 30)))
ev = pd.DataFrame(rows, columns=
["user_id","event_name","event_time","platform"])
ev.to_csv("events.csv", index=False)
print(len(ev), "events")
con = sqlite3.connect(":memory:"); ev.to_sql("events", con, index=False)
q = """
SELECT
 COUNT(DISTINCT CASE WHEN event_name='app_open' THEN user_id END) AS
s1_open,
 COUNT(DISTINCT CASE WHEN event_name='product_view' THEN user_id END) AS
s2_view,
 COUNT(DISTINCT CASE WHEN event_name='add_to_cart' THEN user_id END) AS
s3_cart,
 COUNT(DISTINCT CASE WHEN event_name='checkout_start' THEN user_id END) AS
s4_checkout,
 COUNT(DISTINCT CASE WHEN event_name='purchase' THEN user_id END) AS
s5_purchase
FROM events"""
r = pd.read_sql(q, con).iloc[0]; print(r.to_dict())
f = pd.DataFrame({"users": r.values}, index=steps)
f["step_conv_%"] = (f.users / f.users.shift(1) * 100).round(1)
f["overall_%"] = (f.users / f.users.iloc[0] * 100).round(1)
print(f)
```

```
10164 events
{'s1_open': 5000, 's2_view': 3019, 's3_cart': 1037, 's4_checkout': 628,
's5_purchase': 480}
                users  step_conv_%  overall_%
app_open         5000          NaN      100.0
product_view     3019         60.4       60.4
add_to_cart      1037         34.3       20.7
checkout_start    628         60.6       12.6
purchase          480         76.4        9.6
```

Line by line: `COUNT(DISTINCT CASE WHEN event_name='x' THEN user_id END)` counts each user once for step x (the `CASE` gives NULL for other events and `COUNT` ignores NULL). `shift(1)` moves the column down one row so each step is divided by the step before it. Step conversions land within a point of their planned values (60.4, 34.3, 60.6, 76.4 against 60, 35, 60, 75); the small gaps are random sampling noise.

1. **Find the biggest drop.** The largest *absolute* loss is product view to add to cart: 3,019 to 1,037, a loss of 1,982 users (65.7% leave). The weakest *rate* is also that step (34.3%). That is where a product team should look first: product pages, prices, photos.

1. **Compare platforms.**

```
g = ev.groupby(["platform","event_name"]).user_id.nunique().unstack()
[steps]
g["open_to_purchase_%"] = (g.purchase/g.app_open*100).round(1)
print(g[["app_open","purchase","open_to_purchase_%"]])
```

```
event_name  app_open  purchase  open_to_purchase_%
platform
android         2541       245                 9.6
ios             1467       139                 9.5
web              992        96                9.7
```

All three platforms convert near 9.6%. That is expected here, because the data generator did not give platforms different rates. In real data, a gap would need a significance test before you act (Chapter 13); with 992 web users the margin of noise is several percentage points on smaller steps.

1. **Validate the data against the tracking plan.** A funnel is only as good as the logging.

```
ev = pd.read_csv("events.csv", parse_dates=["event_time"])
plan =
{"app_open","product_view","add_to_cart","checkout_start","purchase"}
print("unknown events:", (~ev.event_name.isin(plan)).sum())
print("null user_id:", ev.user_id.isna().sum(), "null platform:",
ev.platform.isna().sum())
print("duplicates:", ev.duplicated(["user_id","event_name"]).sum())
p = ev.pivot_table(index="user_id", columns="event_name",
values="event_time", aggfunc="min")
print("cart before view:", (p.add_to_cart < p.product_view).sum())
d = (p.purchase - p.app_open).dropna().dt.total_seconds()/60
print("median minutes open to purchase:", round(d.median(),1), "n",
len(d))
```

```
unknown events: 0
null user_id: 0 null platform: 0
duplicates: 0
cart before view: 0
median minutes open to purchase: 61.0
```

(The last line also printed `n 480` .) The checks pass because this data is clean by construction. Real logs rarely are: expect unknown event names after an app release, duplicated events from retries, and missing user IDs for logged-out visitors. A failing check is a finding, not an embarrassment: report it before you report the funnel.

COMMON MISTAKE

Counting events instead of distinct users. If one user views five products, `COUNT(*)` counts five. A funnel that reports 120% of users reaching a step has this bug. Always count `DISTINCT user_id` per step, and decide in writing whether "step order" is enforced.

1. **Write the memo.** "9.6% of app openers purchase. The largest loss is product view to cart (34.3% continue). Raising that step from 34.3% to 40% would lift purchases by about 17%." Check: 40 / 34.3 = 1.166, so about 17% more purchases if later steps stay the same.

Evaluation and acceptance criteria

- Funnel counts reproduce: 5,000, 3,019, 1,037, 628, 480; overall conversion 9.6%.
- SQL and pandas counts match for every step.
- All four validation checks are shown, and each has a rule for what to do when it fails.

- The tracking plan is in the repo and every event in the data is in the plan.
- The memo names one step to fix and gives a numeric "if we improve it" estimate with its assumption.

Common pitfalls and how to detect them

1. **Counting events, not users.** Detect: a step with more users than the step before it.
2. **Ignoring event order.** Detect: users with a purchase but no cart event; check timestamps with the pivot above.
3. **Different time windows per step.** Detect: late steps look low because users had less time; fix by using a conversion window (for example, purchase within 7 days of first open).
4. **Bots or test accounts inflating the top of the funnel.** Detect: a few user IDs with hundreds of `app_open` events.
5. **Reading a platform gap that is noise.** Detect: compute a confidence interval before you recommend a change.

Three extensions

1. Add a time-to-convert distribution (median 61 minutes from open to purchase here) and a conversion window.
2. Add weekly retention (users who return in week 2) using the same event table.
3. Run a significance test for a step difference between two platforms (two-proportion test from Chapter 13).

Interview talking points

How do you build a funnel in SQL? For each step I count distinct users who did that event, using `COUNT(DISTINCT CASE WHEN event_name = 'x' THEN user_id END)`. Then I divide each step by the previous one and by the first step. In my project that gave 5,000, 3,019, 1,037, 628, 480.

Where would you look first, and why? At the step with the biggest absolute loss and the lowest rate. Here that is product view to cart: 34.3% continue. A gain there has the most room, and it affects the most users.

What is a tracking plan and why write it first? It lists each event, when it fires, and its required properties. It prevents two teams logging the same action under different names, and it lets me write checks: unknown event names, missing user IDs, duplicates. Most "analysis

problems" are logging problems.

README outline

Question; tracking plan table; data (simulated, seed 18); funnel table and chart; platform comparison with the noise warning; validation checks; memo; limits; extensions.

Chapter recap

- Attribution rules share credit for a conversion. First, last and linear give different answers for the same data, and each total equals the number of conversions.
- Attribution is not cause. Only an experiment with a holdout group measures real incremental effect.
- CAC = spend / conversions and ROAS = revenue / spend; both depend on the credit rule and on stated assumptions.
- In a P&L variance, favourable means "helps profit". Flip the sign for expense lines, then the favourable variances add up to the profit variance.
- Split a revenue variance into volume and price effects to see what is really happening.
- A tracking plan comes before analysis. Validate the event log against it before you trust a funnel.
- Funnels count distinct users per step, not events, and need a clear rule on step order and time window.
- The biggest absolute loss and the lowest step rate point to where to improve first.
- Every project here has a tie-out check: credits sum to conversions, favourable variances sum to profit variance, SQL equals pandas.

Pattern	Formula or code
CAC	`spend / conversions`
ROAS	`revenue / spend`
Linear credit	`1 / journey_length` per touch
Favourable variance	Income: actual - budget; Expense: budget - actual

Variance %	`fav variance / budget`
Volume effect	`(actual units − budget units) × budget price`
Price effect	`(actual price − budget price) × actual units`
Funnel step in SQL	`COUNT(DISTINCT CASE WHEN event_name='x' THEN user_id END)`
Step conversion	`users / users.shift(1)`

Practice set

1. (Easy) Email has linear credit 35.7 and spend 15,000. What is its CAC?

Answer: 15,000 / 35.7 = 420.2, so about 420. Each credited sale cost about 420 rupees.

2. (Easy) Display has linear credit 23.4, spend 60,000, and each conversion is worth 2,500. Find ROAS and say what it means.

Answer: Revenue = 23.4 × 2,500 = 58,500. ROAS = 58,500 / 60,000 = 0.975, about 0.98. Each rupee spent returned about 98 paise, so on this rule Display loses money.

3. (Easy) Marketing budget 60,000, actual 72,000. Give the variance, the flag and the percent.

Answer: Variance = 72,000 - 60,000 = +12,000. It is an expense, so it is adverse (ADV). Percent = -12,000 / 60,000 = -20.0%.

4. (Medium) Revenue budget 1,000,000, cost of goods budget 600,000, actual revenue 1,060,000, actual cost of goods 650,000. What is the gross profit variance?

Answer: Budget gross profit 400,000. Actual 410,000. Variance +10,000 favourable (revenue +60,000, cost -50,000 adverse).

5. (Medium, what is the output?) Budget 10,000 units at 100. Actual 9,600 units at revenue 1,060,000. What are the volume and price effects?

Answer: Volume effect = (9,600 - 10,000) × 100 = -40,000. Actual price = 1,060,000 / 9,600 = 110.4167, price effect = 10.4167 × 9,600 = 100,000. Total 60,000, equal to the revenue variance.

6. (Medium, scenario) Funnel: 5,000 open, 3,019 view, 1,037 cart, 628 checkout, 480

purchase. If cart to checkout improves from 60.6% to 70% and everything else stays the same, about how many purchases?

Answer: Users at checkout = 1,037 × 0.70 = 725.9. Checkout to purchase is 480 / 628 = 0.7643. Purchases = 725.9 × 0.7643 = about 555, up from 480 (about 15.6% more).

7. (Find the bug) A funnel query uses `COUNT(*)` for each step and shows 5,800 `product_view` against 5,000 `app_open`. What is wrong and how do you fix it?

Answer: It counts events, not users: users who view several products are counted several times. Use `COUNT(DISTINCT user_id)` per step (with the `CASE` filter). A later step can never have more users than an earlier step in a correct funnel.

8. (Interview style) A marketing head says "Email brought us 33 sales, so double the email budget." How do you respond?

Answer: Under last-click Email was credited with 33 of 181 sales, but credit is not cause. Email mostly reaches people who already know the brand, so many would have bought anyway. I would run a holdout test: send emails to a random 90% and withhold from 10%, and compare conversion. If the lift is small, doubling the budget will not double sales. Meanwhile Email's low CAC (about 420) is a good sign and worth a careful test.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles, and English is my second language. Teach me
marketing attribution for a portfolio project: first-click, last-click and
linear credit, CAC and ROAS. Use a small example of 5 customer journeys
with numbers and show every step. Then explain why attribution is not the
same as cause and what a holdout test is. After each part, ask me one
question and wait for my answer before continuing.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Review my attribution analysis: [PASTE CODE AND
OUTPUT HERE]. Check that credit under each rule adds up to the number of
conversions, that CAC and ROAS use the same period for spend and
conversions, and that my memo does not claim cause. List each error with
```

the line number, the effect on the numbers, and the fix, in simple English.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Teach me budget versus actual variance analysis for a finance dashboard: sign conventions for income and expense lines, favourable and adverse, percent variance, and splitting a revenue variance into volume and price. Use a monthly P&L with five lines and small numbers. Then give me 8 graded exercises without solutions, and wait for my answers before checking them step by step.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Help me plan a product analytics project: an event tracking plan for a shopping app and a funnel analysis from an event log. Ask me about my goal, then help me write the tracking plan table, the validation checks, the SQL for distinct users per step, and the success criteria. Warn me about counting events instead of users, event order, and conversion windows. Check each answer before moving on.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles. Run a mock interview for a data analyst role focused on my three projects: marketing attribution, P&L variance dashboard and funnel analysis. Ask one question at a time, such as how I checked my numbers, what the biggest limitation is, and what I would recommend. After each answer, give feedback on clarity and correctness and suggest a stronger version. End with a score out of 10 and two things to practise.

APPENDIX

CHEAT SHEET

One page per chapter: the formulas, commands and patterns you will use every week.

Use this appendix after you have worked through the chapters. If a line here does not feel obvious, go back to the chapter named in the heading and redo its worked problems. Excel, Power Query, DAX and Power BI lines were not run here (they need Microsoft tools); the pandas, NumPy, SQL and statistics lines were run in the chapters.

Chapter 1: The Analytics Mindset and Process

Pattern	Formula or rule
Percent change	`(new − old) / old × 100`
Metric split	`ΔM = ΔA × B₀ + A₁ × ΔB` (exact, no leftover)
Revenue tree	revenue = visitors × conversion × AOV
Insight	number + comparison + implication
Four analytics	descriptive (what), diagnostic (why), predictive (what next), prescriptive (what to do)
Five steps	ask, get data, prepare, analyse, communicate
Question checklist (SPARC)	Scope, Purpose, Audience, Rules, Clock

Chapter 2: Excel Formulas

Need	Formula pattern
Conditional total	`=SUMIFS(sum_range, crit_range, crit, ...)`

Lookup	`=XLOOKUP(key, keys, values, "Not found")`
Band lookup	`=XLOOKUP(v, starts, labels, , -1)`
Weighted average	`=SUMPRODUCT(w, x) / SUM(w)`
Safe error handling	`=IFERROR(expr, "")` (only after you understand the error)
Text clean	`=TRIM(PROPER(A2))`, `=SUBSTITUTE(A2, "-", "")`
Dates	`=EDATE(d, 1)`, `=EOMONTH(d, 0)`, `=DATEDIF(a, b, "m")`, `=TEXT(d, "yyyy-mm")`
Dynamic arrays	`=UNIQUE(r)`, `=FILTER(t, (a)*(b))`, `=SORTBY(t, col, -1)`, `=LET(x, expr, x*2)`
Absolute reference	`$A$1`, press F4

Chapter 3: Excel Analysis

Task	Tool or syntax
Summarise	PivotTable (rows, columns, values, slicers); refresh after data change
Share of total	Show Values As, % of Grand Total
Reverse solve	Goal Seek: set cell, to value, by changing cell
Monthly loan payment	`=PMT(rate/12, n_months, -loan)`
Unpivot (M)	`Table.UnpivotOtherColumns(src, {"key"}, "attr", "val")`
Join	Merge Queries, Left Outer
DAX measure in Power Pivot	`Delivered := CALCULATE(SUM(T[Amount]), T[Status] = "Delivered")`

Statistics	Analysis ToolPak: t-test, ANOVA, regression

Chapter 4: pandas I

Task	Code
First look	`df.head()`, `df.info()`, `df.describe()`, `df.isna().sum()`
Read file	`pd.read_csv(p, dtype=..., parse_dates=[...], usecols=[...], chunksize=n)`
Select columns	`df[["a", "b"]]`
Filter rows	`df[(df.a > 1) & (df.b == "x")]` or `df.query("a > 1 and b == 'x'")`
By label / by position	`df.loc[rows, cols]` (end included) / `df.iloc[rows, cols]` (end excluded)
Fix numbers	`pd.to_numeric(s, errors="coerce")`
Fix dates	`pd.to_datetime(s, dayfirst=True)`
Set values safely	`df.loc[mask, "col"] = v`

Chapter 5: pandas II (cleaning and transforming)

Task	Code
Duplicates	`df.drop_duplicates(subset="id")`
Text clean	`s.str.strip().str.title()`
Outlier fences	`q1 − 1.5 × iqr`, `q3 + 1.5 × iqr`
If / else	`np.where(cond, a, b)`

Bins	`pd.cut(x, bins, labels)`, `pd.qcut(x, q)`
Group summary	`df.groupby("k").agg(n=("id","count"), s=("x","sum"))`
Share in group	`x / df.groupby("k")["x"].transform("sum")`
Safe join	`a.merge(b, on="k", how="left", validate="many_to_one", indicator=True)`
Wide to long	`df.melt(id_vars="k", var_name="m", value_name="v")`
Time buckets	`df.set_index("ts").resample("D").sum()`
Rolling mean	`s.rolling(7).mean()`

Chapter 6: NumPy and Fast pandas

Task	Code
Column / row totals	`a.sum(axis=0)` / `a.sum(axis=1)`
Centre columns	`a - a.mean(axis=0)` (broadcasting)
Many conditions	`np.select([c1, c2], [v1, v2], default)`
Compare floats	`np.isclose(a, b)`
Seeded random	`rng = np.random.default_rng(42)`
Memory	`df.memory_usage(deep=True).sum()`, `astype("category")`, `astype("int16")`
Speed rule	vectorise first; avoid `apply(axis=1)`

Chapter 7: EDA Step by Step

Step	Code or rule
Context	source, grain, period, key, decision
Shape and types	`df.shape`, `df.dtypes`
Missing / duplicates	`df.isna().sum()`, `df.duplicated().sum()`
Group-relative outlier	`df.amount / df.groupby("g").amount.transform("median")`
Distribution	`df.x.describe()`, `df.x.skew()`, `df.x.quantile([.5, .9, .99])`
Cross-tab	`pd.crosstab(a, b, normalize="index")`
Independence test	`scipy.stats.chi2_contingency(table)`
Missing mechanisms	MCAR (random), MAR (depends on other columns), MNAR (depends on the value itself)

Chapter 8: Data Visualization

Item	Pattern
Chart choice	compare: bars; trend: line; distribution: histogram or box; relationship: scatter; part of whole: stacked bar or table
Figure and axes	`fig, ax = plt.subplots(figsize=(7, 4), layout="constrained")`
Save	`fig.savefig("a.png", dpi=150, bbox_inches="tight")`
seaborn	`sns.histplot / boxplot / violinplot / scatterplot / heatmap(data=..., x=..., y=...)`

Interactive	`px.bar(df, x=..., y=...)`, `fig.write_html("a.html")`
Design rule	bars start at zero; grey plus one accent colour; headline title

Chapters 9 and 10: Power BI (modelling and DAX)

These lines were not run here. Check them in Power BI Desktop.

Topic	Pattern
Star schema	one fact table in the centre, dimension tables (Date, Product, Customer) around it, 1-to-many relationships, filters flow from dimension to fact
Measure vs column	measure: computed at query time in the current filter context; column: stored per row at refresh
Base measure	`Total Sales = SUM(Sales[Amount])`
Change filter context	`CALCULATE([Total Sales], Product[Category] = "Fashion")`
Safe division	`DIVIDE([A], [B])`
Percent of total	`DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Product)))`
Year to date	`TOTALYTD([Total Sales], 'Date'[Date])`
Same period last year	`CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))`
Variables	`VAR x = ... RETURN x * 2`
Iterator	`SUMX(Sales, Sales[Qty] * Sales[Price])`

Power Query	transformations are recorded as M steps and re-run on refresh; prefer steps that fold back to the source

Chapter 11: Report Design, Governance and BI Tools

Topic	Rule
Layout	most important number top left; one message per page; consistent theme
Interactivity	slicers, drill-through, bookmarks, tooltips
Row-level security	roles with a filter rule; test with "View as"
Governance	certified datasets, data dictionary, refresh schedule, access control, naming standards
Tool map	Power BI (Microsoft stack), Tableau (calculated fields, LOD expressions), Looker Studio (free, Google), Metabase and Superset (open source)

Chapter 12: Business Analytics

Pattern	Formula or code
ARPU	revenue / active customers
CAC	marketing spend / new customers
LTV	`ARPU × margin × (1 / churn)`
Payback (months)	`CAC / (ARPU × margin)`
NPS	`%promoters − %detractors`
Step conversion	`users / users.shift(1)`
Overall	`users / users.iloc[0]`

conversion	
Cohort grid	`pivot_table(index="cohort", columns="offset", values="user_id", aggfunc="nunique")`
Retention	active at offset k / cohort size
Churn	lost / active at start of the period
RFM	R: days since last order (small is good); F: orders; M: total spend; score with `pd.cut` and fixed thresholds
Stickiness	average DAU / MAU
ROAS	revenue / ad spend
Win rate	`won / (won + lost)`
Weighted pipeline	`Σ value × stage probability`
Variance	actual − budget; mark favourable or unfavourable by effect on profit
SLA	within target / total
Reorder point	`daily demand × lead time + safety stock`

Chapter 13: A/B Testing

Pattern	Formula or code
Sample size per group (rates)	$(z\alpha/2\cdot\sqrt{(2\bar{p}(1-\bar{p}))} + z\beta\cdot\sqrt{(p_1q_1 + p_2q_2)})^2 / \delta^2$; 5% to 6%, α 0.05, power 80%: 8,158
Effect-size rule	halving the MDE needs about 4 times the users
Two-proportion z	$(pT - pC) / \sqrt{p(1-p)(1/nC + 1/nT)}$
95% CI of difference	$diff \pm 1.96\cdot\sqrt{pC\cdot qC/nC + pT\cdot qT/nT}$

Welch t-test	`stats.ttest_ind(t, c, equal_var=False)`
SRM check	`stats.chisquare(observed, expected)` ; alarm often at p < 0.001
Multiple tests	Bonferroni: α / number of tests
DiD	`(T_after − T_before) − (C_after − C_before)`
CUPED	`y − θ(pre − mean(pre))` , θ = cov(y, pre) / var(pre)
Checklist	one primary metric, SRM check, planned sample reached, look once, effect by day, guardrails

Chapter 14: Storytelling and Communication

Pattern	Use
Clarify	decision owner, definition, scope, deadline, what you would do under answer A or B
Pyramid	headline, evidence, detail
Action title	the finding as a sentence with a number
Memo (BLUF)	bottom line, key numbers, why, recommendation, risks, appendix
Calibrated words	"increased" (tested), "suggests" (pattern), "unclear" (weak), "I do not know yet"
Honest uncertainty	what I know, what I do not, how I will find out and by when
Reproducibility	data dictionary, metric definitions, assumptions log, README, version control
Metric tie-out	explain a gap as definition A minus definition B

Chapter 15: Capstones and Interviews

Pattern	Use
Metric-drop case	Clarify, Verify data, Segment, Decompose, Hypothesise, Prioritise, Recommend
Share of drop	segment change / total change; compare with the segment's share of users
Top-N per group	`row_number() over (partition by g order by x desc)`
Duplicates	`GROUP BY k HAVING COUNT(*) > 1`
Ties in ranking	values 1,1,2: RANK 1,1,3; DENSE_RANK 1,1,2; ROW_NUMBER 1,2,3
Fermi	population × share × frequency, with stated assumptions
Behavioural	STAR: Situation, Task, Action, Result, using "I" and a number
Capstone done-test	business question, data dictionary, assumptions log, runnable notebook, memo, README

An Honest Closing Note

If you have worked through this document, you can now do real analyst work. You can turn a vague request into a sharp question, and you can build Excel models and PivotTables. You can clean and combine messy data in pandas, explore it, and chart it with a clear headline. You can design a star schema and a DAX measure in Power BI. You can calculate funnels, retention, churn and RFM segments, and size and read an A/B test with a confidence interval instead of a bare p-value. You can write the one-page memo that makes a manager act, and you can walk through "DAU dropped 10%, why?" in a calm, structured way. That is a real, hireable skill set.

Be honest about what this document does not cover. It does not make you a statistician: regression, time-series forecasting and causal inference need the statistics and data-science documents. It does not teach SQL from scratch (the SQL document does), and the SQL here is interview-level, not warehouse-level. It does not cover big data tools, machine learning, or a deep treatment of Tableau, Looker or the Power BI Service. Power BI, Excel and DAX examples were written to be standard but were not run here, so test them in your own copy. Salaries, tool versions and the job market change, so check current job posts for your city and role.

What to practise next, in this order. First, SQL for 20 minutes a day: window functions, cohorts, retention. It is the most tested skill. Second, finish three portfolio projects (Chapter 15), starting with the order-processing time analysis, which shows percentiles and operations insight. Third, learn one business domain well (e-commerce, banking or healthcare), because domain knowledge is what separates a good analyst from a good technician. Fourth, practise writing: one memo a week, with a bottom line first. Fifth, do mock interviews with the Prompt packs and apply to roles before you feel ready.

Never move on until the current topic feels obvious, and never trust a number you have not checked. A good analyst habit is to measure, find where the problem sits, and prove the fix. Keep that habit, and use it.